

DATA ENGINEERING CASEBOOK

85 Workplace Dilemmas in Data Engineering

Scenario-based case studies across retail, finance, healthcare, energy, transport, the public sector and beyond. Each case pairs a realistic professional dilemma with real, publicly documented data sources and a full set of project deliverables.

Structure of every case

- The Scenario — a long-form workplace dilemma, told from inside the organisation
- The Decision in Front of You — the question the case forces you to answer
- Data Sources — real, hyperlinked datasets, APIs and standards to build the project with
- Deliverables — at least seventeen concrete, reviewable outputs that define “done”

How to Use This Casebook

Data engineering is rarely hard because of the technology alone. The difficult part is almost always the decision that sits around the technology: whether to fix the legacy pipeline or replace it, whether to ship the dashboard the executive asked for or the one the data can honestly support, whether to trade freshness for cost, or speed for auditability. The cases in this book are built around those decisions.

Each case opens with a long scenario written from inside a fictional organisation. The organisations, people and internal figures are invented, but the problems are drawn from patterns that recur across real data teams: schema drift, late-arriving data, regulatory deadlines, vendor lock-in, conflicting definitions of the same metric, under-funded platforms and over-promised timelines. The scenario ends with an explicit decision point. There is rarely a single correct answer; the aim is to reason about trade-offs as a senior engineer would, and to be able to defend that reasoning to a sceptical stakeholder.

After the scenario, each case lists real data sources that can be used to build a working version of the project. These are genuine public datasets, open APIs, reference standards and documentation pages, with hyperlinks. Some require free registration, some carry licence conditions (for example non-commercial use, attribution, or research-only access), and some large public datasets are best accessed through cloud-hosted copies. Always read the terms on the provider's page before using a source, and treat the internal company data described in each scenario as something you would simulate or substitute with the public sources listed.

Finally, every case defines at least seventeen deliverables. Together they describe what a complete, professional outcome looks like: not just pipelines and tables, but decision records, data contracts, quality checks, runbooks, cost models, security reviews and stakeholder communication. They can be used as a project brief, a portfolio checklist, an interview discussion guide, or a marking scheme for a course.

Suggested ways to work through a case: read the scenario and write a one-page recommendation before looking at the deliverables; then pick three to five deliverables and build them for real using the listed sources; finally, revisit your recommendation and note what changed once you touched the actual data. Doing that across a handful of cases from different industries is one of the fastest ways to develop the judgement that separates a pipeline builder from a data engineer.

Index of Case Studies

No.	Project title	Industry
01	Near Real-Time Store Inventory Synchronisation for Click-and-Collect	Grocery Retail
02	Modernising the Anti-Money Laundering Transaction Monitoring Pipeline	Retail Banking
03	Unified Emergency Department Patient Flow Data Platform	Healthcare (NHS Acute Trust)
04	Half-Hourly Smart Meter Data Lakehouse for Market-Wide Settlement	Energy Supply
05	Trip Data Platform and Dynamic Pricing Features for a Ride-Hailing Operator	Urban Transport and Mobility
06	Clickstream Sessionisation and Marketing Attribution After Cookie Loss	E-commerce and Digital Marketing
07	Motor Claims Fraud Detection Feature Store	General Insurance
08	Predictive Maintenance Data Pipeline for a Turbine Component Plant	Advanced Manufacturing
09	Call Detail Record Processing and Customer Churn Data Mart	Telecommunications
10	Operational Flight Disruption Data Hub	Aviation (Airline Operations)
11	Tick Data Lake and MiFID II Best Execution Reporting	Capital Markets (Brokerage)
12	Pharmacovigilance Adverse Event Signal Detection Pipeline	Pharmaceuticals
13	Real-Time Viewing Event Pipeline for Personalised Recommendations	Media and Video Streaming
14	Police Force Crime Data Quality and Open Data Publication Platform	Public Sector (Policing)
15	Last-Mile Delivery Telemetry and Route Performance Platform	Logistics and Parcel Delivery
16	Satellite and Weather Data Platform for Crop Yield Forecasting	Agriculture and Agri-Food
17	Residential Property Valuation Data Warehouse	Real Estate and Mortgage Lending
18	Water Network Leakage Detection from Sensor and Smart Meter Data	Water Utilities
19	Learning Analytics Early Warning System for Student Retention	Higher Education
20	Security Log Lake Cost Crisis and SIEM Re-Architecture	Cybersecurity (Financial Services Group)
21	Multi-Cloud Cost Data Normalisation and FinOps Platform	Technology (Software as a Service)
22	Hotel Revenue Management Data Platform and Demand Forecasting	Hospitality

No.	Project title	Industry
23	Match Event Data Platform for a Professional Football Club	Professional Sport
24	Mobile Game Telemetry Pipeline and Experimentation Platform	Video Games
25	Vessel Tracking and Port Congestion Intelligence Platform	Maritime and Port Operations
26	Humanitarian Needs Data Integration for a Refugee Response	Humanitarian and International Development
27	Real-Time Card Payment Fraud Scoring Stream	Financial Technology (Payments)
28	On-Chain Analytics and Crypto Compliance Data Platform	Digital Assets and Cryptocurrency
29	Store-Level Demand Forecasting Data Platform for Fresh Food	Food Retail and Supply Chain
30	People Analytics Platform and Statutory Pay Gap Reporting	Human Resources (Professional Services Firm)
31	Legal Document Ingestion and Case Law Intelligence Pipeline	Legal Services
32	City-Wide Air Quality Sensor Network Data Platform	Environmental Monitoring (Local Government)
33	Connected Vehicle Telemetry and Early Warranty Defect Detection	Automotive Manufacturing
34	Semiconductor Wafer Yield Analytics and Process Data Integration	Semiconductor Manufacturing
35	Product Traceability and Rapid Recall Data Platform for a Food Manufacturer	Food and Beverage Manufacturing
36	Integrated Infectious Disease Surveillance Data Pipeline	Public Health
37	Bike Share Demand Forecasting and Fleet Rebalancing Data Platform	Urban Mobility (Micromobility)
38	Programmatic Advertising Bid Log Pipeline and Supply Path Transparency	Advertising Technology
39	Public Procurement Spend Transparency and Supplier Risk Data Platform	Central Government
40	Donor Data Platform and Supporter Journey Analytics for a National Charity	Charity and Non-Profit
41	Clinical Trial Data Standardisation and Submission Pipeline	Biotechnology (Clinical Research)
42	Genomic Variant Processing Pipeline for a Clinical Diagnostics Laboratory	Genomics and Diagnostics
43	Physical Climate Risk Data Platform for a Commercial Lending Portfolio	Banking (Climate Risk)

No.	Project title	Industry
44	Scope 3 Supply Chain Emissions Data Platform for a Consumer Goods Company	Consumer Goods (Sustainability Reporting)
45	Supplier Risk and Shipping Disruption Early Warning Platform	Manufacturing Supply Chain (Electronics)
46	News Archive Digitisation and Content Enrichment Pipeline	News Media and Publishing
47	Trust and Safety Content Moderation Data Platform	Social Media and Online Communities
48	Streaming Royalty Calculation and Rights Data Pipeline	Music Publishing and Rights Management
49	Hospital Price Transparency Data Ingestion and Analytics	US Healthcare (Health Insurance)
50	Converting Health Insurance Claims to the OMOP Common Data Model for Outcomes Research	Health Insurance and Real-World Evidence
51	IFRS 9 Expected Credit Loss Data Lineage and Model Input Remediation	Banking (Credit Risk and Finance)
52	Open Banking Transaction Categorisation and Affordability Data Service	Consumer Lending (Fintech)
53	Renewable Generation Forecasting Data Feeds for a Grid-Scale Energy Trader	Energy Trading and Renewables
54	Offshore Well Sensor Historian Migration and Subsurface Data Platform	Oil and Gas (Upstream Production)
55	Haul Truck Telematics and Mine Safety Analytics Platform	Mining
56	Mobile Network Performance Monitoring and Customer Experience Data Platform	Telecommunications
57	Usage-Based Billing Data Pipeline for a B2B SaaS Platform	Software as a Service
58	Contact Centre Interaction Analytics and Complaint Root Cause Platform	Utilities (Customer Service)
59	Smart Building Energy and Occupancy Data Platform for a Commercial Property Portfolio	Commercial Real Estate
60	Aircraft Maintenance Records Digitisation and Reliability Analytics	Aviation Maintenance, Repair and Overhaul
61	Railway Punctuality and Delay Attribution Data Platform	Rail Transport
62	Corporate Customer KYC and Beneficial Ownership Data Platform	Banking (Corporate and Commercial)
63	Wealth Management Portfolio Consolidation and Instrument Reference Data	Wealth Management
64	Retail Loyalty Customer 360 and Identity Resolution Platform	Retail (Department Stores and Online)

No.	Project title	Industry
65	Competitive Price Intelligence Data Pipeline for a Grocery and General Merchandise Retailer	Retail (Grocery and General Merchandise)
66	Fashion E-Commerce Returns and Reverse Logistics Data Platform	Fashion E-Commerce
67	Used Vehicle Listings Aggregation and Valuation Data Platform	Automotive Retail (Online Marketplace)
68	Regional Visitor Economy Data Observatory for a Tourism Board	Tourism and Destination Management
69	Social Housing Repairs, Damp and Mould Risk Data Platform	Social Housing
70	Emergency Call and Ambulance Dispatch Performance Data Platform	Emergency Medical Services
71	Regional Traffic Monitoring and Congestion Data Platform for Transport Planning	Transport Planning (Local Government)
72	Smart Waste Collection and Recycling Performance Data Platform	Waste Management (Local Authority Services)
73	Laboratory Information and Instrument Data Integration for Early Drug Discovery	Biotechnology (Drug Discovery)
74	Livestock Disease Surveillance and Animal Movement Data Platform	Animal Health (Government Agency)
75	Deforestation-Free Supply Chain Monitoring with Earth Observation Data	Commodity Trading and Food Manufacturing
76	Catastrophe Exposure Data Pipeline for a Specialist Reinsurer	Reinsurance
77	Labour Market Intelligence from Job Postings and Skills Taxonomies	Workforce Development and Recruitment
78	University Admissions Funnel and Widening Participation Analytics Platform	Higher Education
79	Large Language Model Training Data Governance and Deduplication Pipeline	Artificial Intelligence (Model Development)
80	Enterprise Knowledge Ingestion for Retrieval-Augmented Generation	Engineering Consultancy
81	Data Mesh Migration for a Diversified Conglomerate	Diversified Industrial and Consumer Conglomerate
82	Regulatory Reporting Data Quality Incident and Risk Data Aggregation Remediation	Banking (Regulatory Reporting)
83	Right to Erasure and Data Subject Rights Across a Data Lake	Online Marketplace and Consumer Services
84	Election Night Live Results Data Service	Broadcasting and News Media

No.	Project title	Industry
85	Astronomical Survey Data Pipeline and Archive Modernisation	Scientific Research (Astronomy)

CASE STUDY 01 · GROCERY RETAIL

Near Real-Time Store Inventory Synchronisation for Click-and-Collect

Industry	Grocery Retail
Organisation	Harrowgate Foods, a UK grocery chain with 412 stores and a fast-growing online click-and-collect service
Your role	Lead Data Engineer, Commercial Data Platform
Core themes	Change data capture, legacy POS integration, data freshness versus data correctness, peak-trading risk

The Scenario

It is the first week of October and the trading calendar on the wall of the Leeds head office already has a red band running through the last six weeks of the year. Harrowgate Foods does almost a fifth of its annual revenue between mid-November and New Year, and this year, for the first time, online click-and-collect is forecast to account for eleven per cent of that. You joined eight months ago as Lead Data Engineer for the commercial data platform, and the problem that has landed on your desk this morning is one that everybody in the business has an opinion about: the stock figures that the website shows customers are wrong often enough that the online team has started to lose faith in them.

The mechanics are simple to describe and painful to live with. Every one of the 412 stores runs a point-of-sale and back-office system supplied by a vendor that Harrowgate has used for fourteen years. Each store holds its own Oracle database on a small server in the back office. Sales, deliveries, waste and manual stock adjustments are written locally throughout the day, and at around two in the morning a batch job extracts the day's movements, compresses them and sends them to the central merchandising system. From there, another overnight process calculates the expected stock position for each product in each store and publishes it to the e-commerce platform at roughly six in the morning. By the time a customer in Harrogate places an order at four in the afternoon for collection the next day, the stock figure they are looking at is at least ten hours old and ignores everything that happened on the shop floor since the store opened.

The consequence shows up in one number that the Online Director, Priya Venkataraman, now puts at the top of every weekly review: the

substitution and cancellation rate. Last quarter, fourteen per cent of click-and-collect order lines were either substituted with a different product or dropped altogether because the item was not on the shelf when the picker went to find it. Customer complaints mention it constantly, and the customer insight team has shown that shoppers who experience two or more substitutions in their first three orders are far less likely to place a fourth. Priya's position is clear and she has already taken it to the executive committee: Harrowgate needs real-time stock, and she wants it live before the Christmas peak. She has secured a budget, and she has told the committee that the data team can deliver it.

You have spent the last fortnight understanding what "real-time stock" would actually mean, and you have arrived at three uncomfortable observations. The first is technical. The most obvious architecture is change data capture: read the transaction logs of each store's Oracle database, stream every insert and update into a central Kafka cluster, and maintain a continuously updated stock position that the website can query. Your platform team has built CDC pipelines with Debezium before, against PostgreSQL. Oracle is a different animal. Reading redo logs requires supplemental logging to be switched on in every store database, and the infrastructure DBA group, led by a veteran called Graham Lister, has flatly refused. Graham points out that the store servers are nine years old, that several already run at high disk utilisation during trading hours, and that the last time anyone changed a database setting across the estate, a botched rollout took forty tills offline on a Saturday morning. He also reminds you that Oracle's own streaming product carries a licence cost that finance will not approve without a full business case, and that LogMiner-based approaches have known limits under heavy write load.

The second observation is commercial. The POS vendor has, somewhat conveniently, just released a REST API that exposes store-level stock movements. It is not a streaming interface; it supports polling every fifteen minutes per store, and it is rate-limited in a way that means polling all 412 stores at that cadence would need an enterprise tier that roughly doubles the annual support fee. The vendor's account manager has been in touch with Priya directly, and the API option is now being presented upstairs as the low-risk route. You are not convinced. The API is three months old, its documentation is thin, and the one pilot store you tested it against returned duplicate movements twice in a single afternoon with no idempotency key to deduplicate them.

The third observation is the one that worries you most, and it is not about latency at all. When you joined the store-level movement data to a sample of manual shelf counts carried out by the loss prevention team, you found that for roughly a quarter of product lines the system stock figure was wrong even immediately after the overnight recalculation. Some of this is well understood in retail: theft, damaged goods that are binned without being recorded as waste, delivery discrepancies that are never reconciled, products scanned under the wrong barcode at the till. Some of it is specific to Harrowgate, including a known defect in how multi-buy promotions decrement stock for bundled items. The implication is awkward. Making the stock figure fresher does not make it correct. A figure that is ten hours old and wrong by six units becomes a figure that is two minutes old and still wrong by six units. You suspect that a significant share of the fourteen per cent substitution rate is driven by this phantom inventory rather than by staleness, but you cannot prove the split without more analysis, and there is no time to do that analysis thoroughly before the peak.

Your head of data, Tom Okafor, is supportive but politically cautious. He agrees with your diagnosis in private but has told you that walking into the executive committee and saying "real-time will not fix your problem" is not an option when Priya has already committed the team. He wants you to find a way to deliver something visible for Christmas that also moves the business towards addressing accuracy. Meanwhile, the data science team has pitched an alternative: rather than chasing true real-time, build a probabilistic "available to promise" model that estimates the chance an item is actually on the shelf, using historical sales velocity, time since last delivery, known shrinkage rates by category and the gap between system stock and shelf counts. They argue that a model like this could run on the existing overnight data plus a lightweight intraday sales feed, and would directly target the substitution rate. Priya dislikes this idea. She thinks customers and store colleagues will not trust a number that is a probability, and she has been burned before by models that looked good in a slide deck and failed on the shop floor.

There are also constraints that nobody is arguing about but everybody must respect. There is a code freeze on all store systems from the fourteenth of November. The central Kafka cluster is shared with the logistics team, which runs its warehouse slotting and transport planning on it, and they have made it clear that any new high-volume producer needs load testing before peak. The e-commerce platform can only accept

stock updates through a batch API that allows a maximum of one full refresh per store every five minutes. And the information security team has asked that any new connection from the central cloud estate into store networks be reviewed, which usually takes three to four weeks.

So you have eleven weeks until the peak, six until the code freeze, a sponsor who wants one thing, a DBA team that will block another, a vendor offering a third, a data science team proposing a fourth, and evidence suggesting that the real problem is something none of them are directly addressing. You have been asked to present a recommendation and delivery plan to Tom and Priya on Friday.

THE DECISION IN FRONT OF YOU

Do you pursue CDC against the store Oracle databases, adopt the vendor's polling API, champion the probabilistic availability model, or design a phased hybrid, and how do you present the uncomfortable truth about stock accuracy without losing the sponsor you need to fix it?

Data Sources

#	Source and link	How it is used in this project
1	UCI Online Retail II Dataset https://archive.ics.uci.edu/dataset/502/online+retail+ii	Two years of real transactional line-level retail data used to simulate intraday sales movements, returns and cancellations for the prototype pipeline.
2	M5 Forecasting (Walmart) Dataset https://www.kaggle.com/c/m5-forecasting-accuracy	Store-item daily unit sales with calendar and price data across multiple stores, used to model sales velocity and availability probabilities by store.
3	Instacart Market Basket Analysis https://www.kaggle.com/c/instacart-market-basket-analysis	Three million grocery orders used to model online basket composition and estimate substitution exposure per product.
4	Debezium Oracle Connector Documentation https://debezium.io/documentation/reference/stable/connectors/oracle.html	Reference for LogMiner-based CDC, supplemental logging requirements and operational limits that inform the CDC option appraisal.
5	Apache Kafka Documentation https://kafka.apache.org/documentation/	Topic design, partitioning, idempotent producers and quota management for the shared streaming cluster.
6	GS1 UK Barcode and Product Data Standards https://www.gs1uk.org	GTIN standards for product identification and resolving barcode mismatches that drive phantom inventory.
7	ONS Retail Sales Index https://www.ons.gov.uk/businessindustryandtrade/retailindustry	Official UK retail seasonality benchmarks used to size peak-period volumes for load testing.

Deliverables (17)

- D1 Options appraisal paper.** A written comparison of CDC, vendor API polling, probabilistic availability and hybrid designs covering latency, accuracy impact, cost, risk and delivery time, ending in a clear recommendation.
- D2 Substitution root-cause analysis.** A quantified breakdown of the fourteen per cent substitution rate into staleness-driven versus accuracy-driven causes, using shelf-count samples and order-line outcomes.
- D3 Target-state architecture diagram.** A logical and physical diagram showing store sources, ingestion layer, Kafka topics, stock state store, availability service and e-commerce integration.
- D4 Store movement event schema.** A versioned Avro or JSON schema for stock movement events with event time, store, GTIN, movement type, quantity and an idempotency key.
- D5 Data contract with the POS vendor.** A written contract defining API fields, delivery semantics, deduplication rules, SLAs and change notification obligations.
- D6 Ingestion service prototype.** A working polling or CDC ingester for a pilot of ten stores with retries, back-off, rate-limit handling and exactly-once deduplication.
- D7 Streaming stock state computation.** A stateful stream job that maintains current stock per store and product, handling late and out-of-order events with watermarks.
- D8 Available-to-promise model feed.** A daily and intraday feature pipeline supplying sales velocity, delivery recency and shrinkage rates to the availability model.
- D9 Kafka capacity and load-test report.** Evidence that the new producers and consumers fit within the shared cluster's peak capacity, agreed with the logistics team.
- D10 E-commerce publishing adapter.** A component that batches stock state into five-minute refreshes per store within the platform's API limits.
- D11 Data quality rule set.** Automated checks for negative stock, duplicate movements, missing stores, stale feeds and implausible adjustments with alert thresholds.
- D12 Freshness and accuracy dashboard.** A monitoring view showing feed latency per store alongside shelf-count accuracy trends and substitution rates.
- D13 Security and network review pack.** Documentation submitted to information security covering connectivity, credentials, encryption and data classification.
- D14 Peak trading runbook.** Step-by-step operational procedures for feed failure, fallback to overnight stock, store outages and escalation during the freeze.
- D15 Rollout and rollback plan.** A phased store rollout schedule with go/no-go criteria and a tested rollback path before the code freeze.
- D16 Stakeholder briefing deck.** A concise executive presentation that explains the accuracy problem, the recommended plan and what success will look like after peak.
- D17 Post-peak benefits measurement plan.** A defined method, baseline and control stores for measuring the impact on substitutions and cancellations after Christmas.

CASE STUDY 02 • RETAIL BANKING

Modernising the Anti-Money Laundering Transaction Monitoring Pipeline

Industry	Retail Banking
Organisation	Northbridge Bank, a UK challenger bank with 2.3 million personal and small business customers
Your role	Senior Data Engineer, Financial Crime Technology
Core themes	Regulatory data lineage, batch-to-streaming migration, alert quality, auditability versus agility

The Scenario

Northbridge Bank grew quickly. Seven years after receiving its banking licence it has 2.3 million customers, a well-reviewed mobile app and a reputation for opening accounts in minutes rather than days. That speed of growth has a shadow, and the shadow now sits in a letter from the Financial Conduct Authority that has been circulated, in redacted form, to the senior leadership of the financial crime function. Following a thematic review of challenger banks, the regulator has concluded that Northbridge's transaction monitoring arrangements are "not commensurate with the scale and risk profile of the business" and has asked for a remediation plan within ninety days. You are the senior data engineer in the financial crime technology team, and the transaction monitoring data pipeline is, in large part, yours.

The current system was built when the bank had two hundred thousand customers. Every night, a set of stored procedures in the core banking data warehouse extracts the previous day's transactions, joins them to customer and account reference data, and writes a flat file that is picked up by a commercial rules engine. The rules engine applies around ninety scenarios: large cash deposits, rapid movement of funds in and out, transfers to high-risk jurisdictions, structuring just below reporting thresholds, and so on. Alerts are generated and loaded into a case management tool, where a team of forty analysts reviews them. The system works, in the sense that it produces alerts. The problem is what kind of alerts, and how many.

Last month the pipeline generated just over sixty-one thousand alerts. The analyst team closed ninety-seven per cent of them as false positives. The backlog of unreviewed alerts stands at nineteen thousand, some of which

are more than sixty days old, which is precisely the kind of detail the regulator noticed. When you dug into the causes, you found that a large share of the noise was not the rules themselves but the data feeding them. Customer occupation and expected activity fields, captured at onboarding, are blank or defaulted for about a third of customers because the onboarding journey was simplified two years ago to improve conversion. The counterparty country field on international payments is populated from a lookup on the bank identifier code, but the lookup table has not been refreshed in eighteen months. Transactions from the bank's new card-issuing partner arrive a day late and are sometimes processed twice when the partner resends a file. And the customer risk rating that scenarios use to set thresholds is recalculated only quarterly, so a customer who becomes high-risk in January is monitored as low-risk until April.

The Money Laundering Reporting Officer, Helen Carr, is a former police financial investigator who has little patience for technology explanations. Her priority is defensibility. Anything that changes must be documented, tested, explainable to the regulator and approved through the bank's model risk governance, which treats monitoring scenarios as models. She would, in her words, rather have a clunky system she can defend than a clever one she cannot. The Chief Technology Officer, Marcus Reid, sees the regulatory letter as the opportunity he has been waiting for. He wants to retire the nightly batch process entirely and move transaction monitoring onto the bank's event streaming platform, scoring every payment as it happens. He argues that real-time monitoring would also enable the bank to stop suspicious payments before they leave, which the fraud team has been requesting for years, and that the rules engine vendor's licence renewal in fourteen months is the natural moment to replace it with an in-house solution combining rules and machine learning.

You can see merit in both positions, and serious risk in both. Marcus is right that the architecture is outdated and that the nightly batch creates structural latency. But a wholesale rebuild during a regulatory remediation is exactly the kind of programme that goes wrong in public. If the new system misses a typology that the old one caught, or if the bank cannot show the regulator a complete, reproducible record of which transactions were monitored under which rules on which day, the consequences would be far worse than the current letter. Helen is right that defensibility matters above all, but her instinct to minimise change

risks leaving the underlying data problems untouched, and the ninety-day plan needs to show real improvement, not just more analysts clearing the backlog faster.

There is a further complication. The data science team has built a promising alert prioritisation model that scores each alert by its likelihood of leading to a suspicious activity report. In back-testing it would have allowed analysts to clear the lowest-scoring sixty per cent of alerts with light-touch review while catching almost all the historically escalated cases. Helen is wary. Suppressing alerts, even with a model, is something regulators have criticised elsewhere, and the model was trained on historical analyst decisions that may themselves have been inconsistent. The data science lead, Arjun Mehta, counters that the model does not suppress anything; it orders the queue. But his training data depends on precisely the reference fields you know to be unreliable, and the model's feature pipeline is a set of notebooks that nobody outside his team has reviewed.

The budget holders have made it clear that there is funding for one major initiative this year, not three. Your manager, Sophie Adeyemi, has asked you to produce a data-led remediation proposal that the MLRO, the CTO and the data science lead can all put their names to. It needs to address the data quality defects, the backlog, the regulatory expectation of timely and complete monitoring, and the longer-term architecture, while being realistic about what forty analysts and a data engineering team of six can deliver in ninety days and beyond. You also know that whatever you propose will be scrutinised by the bank's internal audit function, which has previously criticised the financial crime team for being unable to demonstrate end-to-end lineage from source transaction to alert.

There is one more dimension that nobody has raised in the steering meetings but that you think matters a great deal: the people who use the output. You spent a day sitting with the analyst team and watched how they work. Each alert arrives as a single screen showing the triggering transactions and a handful of customer fields. To understand context, analysts open four other systems, copy account numbers between them, and build a picture of the customer's behaviour manually. The average alert takes eleven minutes to close, and the analysts told you that most of that time is spent assembling information that the monitoring pipeline already joined together overnight and then discarded. Some of the most

experienced analysts keep personal spreadsheets of customers they have reviewed before, because the case management tool does not show prior alert history across scenarios. A better data product for investigators, with a consolidated customer view, prior alerts and network links between counterparties, might do as much to reduce the backlog as any change to rules or models, and it would be far less contentious with the regulator. But it is not what anybody asked you for, and it would compete for the same scarce engineering time.

As you sit down to plan, you keep returning to a single question. Is the right move to fix the data feeding the existing batch system first and earn the credibility to modernise later, or to use the remediation as the forcing function to build the streaming platform the bank will eventually need, accepting the risk that comes with change in a regulatory spotlight?

THE DECISION IN FRONT OF YOU

How do you sequence data remediation, alert prioritisation and a move to streaming so that the bank satisfies the regulator in ninety days, preserves full auditability, and still lays foundations for real-time monitoring, and which of the CTO's ambitions do you defer?

Data Sources

#	Source and link	How it is used in this project
1	PaySim Synthetic Mobile Money Transactions https://www.kaggle.com/datasets/ealaxi/paysim1	Large synthetic transaction dataset with labelled fraudulent behaviour, used to prototype monitoring scenarios and streaming throughput tests without real customer data.
2	IBM AML Transactions Dataset (Kaggle) https://www.kaggle.com/datasets/ealtman2019/ibm-transactions-for-anti-money-laundering-aml	Synthetic multi-bank transfer data with laundering typologies labelled, used to back-test rules and the prioritisation model.
3	UK Sanctions List (FCDO) https://www.gov.uk/government/publications/the-uk-sanctions-list	Authoritative UK sanctions designations ingested as reference data for counterparty screening enrichment.
4	OpenSanctions https://www.opensanctions.org	Consolidated sanctions and politically exposed persons data used to enrich counterparty risk and test entity matching.
5	FATF High-Risk and Monitored Jurisdictions https://www.fatf-gafi.org/en/countries/black-and-grey-lists.html	Reference list driving the high-risk jurisdiction lookup that is currently eighteen months stale.
6	SWIFT BIC and ISO 9362 Information https://www.swift.com/standards/data-standards/bic-business-identifier-code	Standard for bank identifier codes used to rebuild the counterparty country derivation.

#	Source and link	How it is used in this project
7	FCA Financial Crime Guide https://www.handbook.fca.org.uk/handbook/FCG/	Regulatory expectations for transaction monitoring that shape the remediation acceptance criteria.
8	OpenLineage Specification https://openlineage.io	Open standard for capturing lineage events from pipelines to evidence source-to-alert traceability for audit.

Deliverables (17)

- D1 Regulatory remediation data plan.** A ninety-day plan mapping each regulator concern to specific data fixes, owners, milestones and evidence of completion.
- D2 Data defect register.** A catalogue of every known data defect affecting monitoring, with impact on alert volumes, root cause, fix and verification status.
- D3 Reference data refresh pipelines.** Automated, scheduled ingestion of sanctions lists, FATF jurisdictions and BIC-to-country mappings with change logs.
- D4 Card partner file deduplication.** An idempotent ingestion process with file-level and record-level duplicate detection and reconciliation reports.
- D5 Event-driven customer risk rating feed.** A pipeline that recalculates risk ratings when triggering events occur rather than quarterly, with full history retained.
- D6 Monitoring coverage reconciliation.** A daily control proving every eligible transaction was presented to monitoring, with exceptions flagged and investigated.
- D7 End-to-end lineage map.** Documented and machine-captured lineage from source transaction through enrichment and scenarios to alert and case outcome.
- D8 Point-in-time reproducibility store.** Versioned snapshots of rules, thresholds and reference data allowing any historical alert decision to be recreated exactly.
- D9 Alert prioritisation feature pipeline.** A productionised, tested, version-controlled replacement for the notebook features used by the prioritisation model.
- D10 Model governance evidence pack.** Documentation, validation results and bias checks required for the model risk committee to approve prioritisation.
- D11 Streaming target architecture.** A future-state design for real-time monitoring and payment interdiction, with a phased migration path from batch.
- D12 Parallel-run framework.** Tooling to run new and legacy monitoring side by side and compare alerts transaction by transaction before any cutover.
- D13 Backlog reduction analytics.** Dashboards tracking backlog age, throughput, false-positive rate and suspicious activity report conversion by scenario.
- D14 Data quality monitoring suite.** Automated tests on completeness, validity and timeliness of monitoring inputs with alerting to owners.
- D15 Onboarding data capture proposal.** Recommendations for reinstating critical customer fields at onboarding and back-filling missing data for existing customers.
- D16 Audit and regulator briefing pack.** A clear narrative with evidence for internal audit and the FCA demonstrating control improvements.

- D17 Vendor exit and build appraisal.** A costed analysis of renewing, replacing or augmenting the rules engine ahead of the licence renewal.

CASE STUDY 03 · HEALTHCARE (NHS ACUTE TRUST)

Unified Emergency Department Patient Flow Data Platform

Industry	Healthcare (NHS Acute Trust)
Organisation	Pennine Valley NHS Foundation Trust, three hospital sites serving 780,000 people
Your role	Principal Data Engineer, Digital and Data Services
Core themes	Interoperability and FHIR, operational real-time reporting, information governance, clinical safety

The Scenario

At half past eight on a Monday morning in January, the site operations meeting at Pennine Valley NHS Foundation Trust begins, as it always does, with a room full of people holding different numbers. The emergency department matron says there are fifty-two patients in the department, eighteen of whom have been waiting more than twelve hours from arrival. The bed manager's spreadsheet shows forty-seven. The operational dashboard on the wall screen, which you and your team maintain, shows fifty-five, but its timestamp says it was last refreshed at seven forty. The chief operating officer, Dr Rachel Osei, has asked the same question at this meeting for three weeks running: why can a trust that spends millions on digital systems not tell her how many patients are in its own emergency department right now?

You are the principal data engineer in the trust's digital and data services directorate, and you know the answer, even if it is not one anybody wants to hear. The trust runs three hospital sites that were merged eight years ago, and each still uses a different emergency department system. The largest site uses a module of the main electronic patient record. The second uses a standalone ED product that the clinicians refused to give up during the merger. The third, a smaller site with a minor injuries unit, records attendances in a system that exports to CSV every hour. Bed management uses a fourth system. The data warehouse ingests all of these on different schedules, some every fifteen minutes, some hourly, some nightly, and the dashboard stitches them together with a set of views written by a contractor who left two years ago. Each source defines key events differently. "Arrival" might mean the time the patient walked in, the time they were booked in at reception, or the time they were triaged.

"Decision to admit" is recorded consistently at one site, inconsistently at another and not at all at the third, where it is inferred from a bed request.

The pressure to fix this has become acute. NHS England publishes monthly emergency care performance data for every trust, and Pennine Valley's four-hour performance has been in the bottom quartile nationally for most of the last year. The integrated care board has placed the trust in an enhanced support programme and expects weekly reporting on twelve-hour waits and ambulance handover delays. The national emergency care data set requires accurate, coded submissions, and the trust's data quality scores for that submission have been flagged as poor. Meanwhile, clinicians are frustrated by duplicate data entry, and the board has approved a business case for a new trust-wide electronic patient record, but its deployment is at least two and a half years away.

Two options have emerged, each with a champion. The Chief Clinical Information Officer, Dr Imran Siddiqui, wants to build a real-time patient flow platform using FHIR, the interoperability standard now expected across the NHS. His vision is an integration layer that subscribes to HL7 messages and FHIR resources from every clinical system, maintains a live, standardised picture of every patient's journey through the department, and exposes it to dashboards, the bed management tool and eventually the national submissions. He argues that this would also prepare the trust for the new electronic patient record, because the integration layer would outlive the current systems. The problem is that only the main EPR module can produce FHIR resources today. The standalone ED system produces HL7 version 2 messages but only for admissions, discharges and transfers, not for the clinical milestones that matter most in the department. The minor injuries system produces nothing but its hourly CSV.

The Director of Performance, Anne Whitfield, wants something far more modest. She would like the existing warehouse feeds speeded up to fifteen minutes across all sources, the definitions harmonised and documented, and a single, trusted dashboard delivered within three months so that the site meeting can stop arguing about numbers. She points out, reasonably, that the integration engine team is already overstretched supporting the pathology and radiology interfaces, that the new EPR procurement will change everything anyway, and that the trust does not need to build a national-grade interoperability platform to count patients in a corridor.

There are other voices. The trust's information governance lead, Lorraine Bell, has reminded everyone that patient-level operational data displayed on wall screens in shared spaces raises confidentiality issues, and that any new flow of identifiable data needs a data protection impact assessment. The clinical safety officer has noted that a live dashboard used to make operational decisions about moving patients is arguably a clinical risk management concern under the NHS clinical safety standards, which means hazard logs and a clinical safety case. The integration engine manager, Steve Harper, has warned that his team has capacity for one new interface this quarter, not five. And the finance director has asked why the trust would invest in integration now when the new EPR supplier will provide its own.

You have also spoken to the people who would actually live with the result. The nurse in charge at the largest site told you that she does not need a beautiful dashboard; she needs to know, at a glance, which patients are breaching, where they physically are, and whether a bed has been requested for them. The bed managers told you they already keep a whiteboard because the systems disagree, and that they will only abandon it when the screen is right every time they check it for a month. The performance analysts told you that they spend the first two days of every month manually correcting the national submission, and that they would welcome anything that reduces that burden, but not if it means a new system they have to learn during winter pressures. Each group defines success differently, and any proposal that serves only one of them will be quietly ignored by the others.

You also have your own technical worries. Harmonising definitions is not just a data engineering task; it requires clinical agreement, and the ED consultants at the three sites have historically disagreed about when a patient should be considered to have been seen. Any single number you publish will be contested unless the definitions are agreed and signed off. Late-arriving data is endemic: triage times are sometimes entered retrospectively when the department is overwhelmed, which means a patient can appear to have been waiting untriaged for three hours when they were in fact triaged at the door. And the national submission requires coded diagnoses and treatments using SNOMED CT, which the minor injuries system captures only as free text.

The chief operating officer has asked for a proposal that answers her question within weeks, not years, but the CCIO has asked you to make sure the proposal does not paint the trust into a corner. You have two weeks to prepare it.

THE DECISION IN FRONT OF YOU

Do you build the FHIR-based real-time integration layer, accelerate and harmonise the existing warehouse feeds, or design a staged approach that delivers a trusted operational view quickly while laying interoperability foundations, and how do you secure clinical agreement on definitions and the necessary governance approvals?

Data Sources

#	Source and link	How it is used in this project
1	NHS England A&E Attendances and Emergency Admissions https://www.england.nhs.uk/statistics/statistical-work-areas/ae-waiting-times-and-activity/	Monthly national performance data used to benchmark four-hour and twelve-hour wait metrics against peer trusts.
2	Emergency Care Data Set (ECDS) Technical Output Specification https://digital.nhs.uk/data-and-information/data-collections-and-data-sets/data-sets/emergency-care-data-set-ecds	National data definitions and coding requirements that drive harmonised event definitions and submission quality.
3	HL7 FHIR R4 Specification https://hl7.org/fhir/R4/	Standard resources such as Encounter, Patient and Location used to model the patient journey in the integration layer.
4	FHIR UK Core Implementation Guide https://simplifier.net/hl7fhirukcorer4	UK-specific FHIR profiles ensuring the integration layer aligns with national interoperability expectations.
5	Synthea Synthetic Patient Generator https://synthea.mitre.org	Generates realistic synthetic patient records in FHIR format for building and testing pipelines without identifiable data.
6	MIMIC-IV-ED (PhysioNet) https://physionet.org/content/mimic-iv-ed/	De-identified emergency department visit data with triage and disposition timestamps, used to model patient flow and late-arriving events (credentialed access).
7	NHS Digital Terminology Server and SNOMED CT UK https://digital.nhs.uk/services/terminology-server	Terminology services for coding diagnoses and treatments to national standards.
8	DCB0129 and DCB0160 Clinical Risk Management Standards https://digital.nhs.uk/data-and-information/information-standards/information-standards-and-data-collections-including-extractions/publications-and-notifications/standards-and-collections/dcb0129-clinical-risk-management-its-application-in-the-manufacture-of-health-it-systems	Clinical safety standards that define the hazard log and safety case for the operational dashboard.

Deliverables (17)

- D1** **Current-state source inventory.** A documented map of every ED, bed and minor injuries system, its data feeds, latency, formats and known quality issues.
- D2** **Harmonised event definitions.** Clinically signed-off definitions for arrival, triage, seen by clinician, decision to admit, departure and handover, with mapping from each source.
- D3** **Clinical definitions workshop report.** A record of the consultant workshop outcomes, disagreements resolved and decisions escalated for executive sign-off.
- D4** **Staged architecture proposal.** A design showing the rapid warehouse improvement phase and the later FHIR integration layer, with clear interfaces between them.
- D5** **Fifteen-minute ingestion pipelines.** Accelerated, monitored feeds from all three ED systems and bed management into the warehouse with latency alerting.
- D6** **Patient journey data model.** A canonical event-based model of each attendance that supports both real-time status and national submission derivation.
- D7** **Late-arriving event handling logic.** Rules and code for reconciling retrospectively entered timestamps so live counts and historical metrics remain consistent.
- D8** **Operational patient flow dashboard.** A trusted, refreshed view of department occupancy, waits, handovers and bed requests by site for the site meeting.
- D9** **Metric reconciliation report.** A daily check comparing dashboard counts with each source system and bed management, with discrepancies explained.
- D10** **ECDS submission quality improvement.** Improved coding completeness through SNOMED CT mapping of minor injuries free text and validation before submission.
- D11** **Data protection impact assessment.** A completed DPIA covering identifiable operational data flows, display in shared areas and role-based access.
- D12** **Clinical safety case and hazard log.** DCB0129 and DCB0160-aligned documentation identifying hazards from incorrect or delayed dashboard data and their mitigations.
- D13** **Role-based access design.** A design that restricts patient-identifiable detail to appropriate staff while showing aggregate views on public screens.
- D14** **FHIR integration pilot.** A proof of concept consuming FHIR Encounter resources from the main EPR and HL7 v2 ADT from the standalone ED system.
- D15** **EPR transition alignment note.** An analysis showing how the integration layer and data model will carry over to the new electronic patient record.
- D16** **Operational support runbook.** Procedures for feed failures, data corrections and out-of-hours escalation for the flow platform.
- D17** **Benefits and adoption review.** A measurement plan tracking whether the site meeting uses one agreed number and whether ECDS quality scores improve.

CASE STUDY 04 · ENERGY SUPPLY

Half-Hourly Smart Meter Data Lakehouse for Market-Wide Settlement

Industry	Energy Supply
Organisation	Brightwell Energy, a UK electricity and gas supplier with 1.1 million customer accounts
Your role	Data Engineering Manager, Energy Data Platforms
Core themes	Very large time-series ingestion, regulatory deadlines, cost of storage and compute, data quality at scale

The Scenario

For most of its eleven-year history, Brightwell Energy has settled the electricity its customers use in the same way as almost every other domestic supplier in Great Britain: using profiles. A customer's meter might be read once a quarter, or once a month, or not at all for a year, and the industry settlement process allocates their consumption across half-hour periods using standard load shapes. That era is ending. The industry programme known as market-wide half-hourly settlement is moving every meter in the country to settlement based on actual half-hourly consumption. For Brightwell, it means that roughly 700,000 smart electricity meters will each generate forty-eight readings a day that must be collected, validated, estimated where missing, and submitted into the new settlement arrangements on tight industry timetables. Suppliers that fail to meet those timetables face charges and, more importantly, financial exposure: under half-hourly settlement, a supplier's wholesale energy costs are reconciled against actual customer demand far more precisely than before.

You manage the energy data platforms team, and the half-hourly data programme sits squarely with you. The current architecture was never designed for this. Meter readings arrive through the national smart metering data communications infrastructure into a meter data management system Brightwell licenses from a specialist vendor. The vendor product holds readings in a relational database, performs validation and estimation using industry rules, and exposes an export for billing. It was sized for daily and monthly reads. When the vendor ran a load test with half-hourly volumes for the full customer base, the nightly processing window extended from three hours to more than fourteen, and

the vendor's proposed solution is a hardware and licence upgrade priced at a level that made the finance director, Oliver Grant, visibly wince.

Your own team has a different idea. Over the past year you have built a cloud lakehouse for trading and forecasting analytics, using open table formats on object storage and a distributed processing engine. You believe that half-hourly consumption data, which is essentially a vast, append-mostly time series, fits the lakehouse far better than a relational database. The volumes are significant but not frightening: 700,000 meters, forty-eight readings a day, around thirty-four million readings daily, twelve billion a year, plus reactive power and export readings for customers with solar panels. Stored in a columnar format with sensible partitioning, the storage cost is modest. The harder part is not storage; it is the industry rules for validation, estimation and submission, which the vendor product implements and your team would have to replicate and prove.

The industry timetable is fixed. Migration of meters into the new settlement arrangements is phased, and Brightwell's first cohort is due to transition in seven months. The regulator has made clear that delays will not be tolerated. The head of settlement operations, Karen Doyle, has been through three industry change programmes and is deeply sceptical of anything new. Her team knows the vendor system intimately, and every industry audit Brightwell has passed has relied on it. She would prefer to pay for the upgrade and avoid the risk. The chief data officer, Neil Fairbairn, strongly supports the lakehouse approach because the half-hourly data has enormous value beyond settlement: better demand forecasting, time-of-use tariffs, flexibility services where customers are paid to shift demand, and much more accurate hedging. He points out that the vendor's architecture keeps the data locked in a system that analysts cannot easily query, and that the upgrade simply buys more of the same constraint.

The trading desk has joined the argument from a third direction. The head of trading, Leila Hussain, cares less about where the data lives than about when she gets it. Under half-hourly settlement, knowing actual customer demand the day after delivery, rather than weeks later, would let her team correct hedging positions and reduce imbalance costs that currently run into millions of pounds a year. She wants validated consumption available to her models by nine each morning. The vendor's upgraded system would deliver it by midday at best.

Then there is the question of data quality, which is the part keeping you awake. In a pilot of fifty thousand meters, eleven per cent of meter-days had at least one missing half-hour reading, usually because of communication failures. Around two per cent of meters showed clock drift, with readings shifted by one or more periods. A small number of meters reported impossible values, such as consumption spikes of several hundred kilowatt-hours in a half hour, which turned out to be firmware defects. Meters that had been exchanged mid-month produced overlapping readings from old and new devices. Industry rules specify how to estimate missing data, but those rules assume the data has been cleaned in consistent ways first, and they have to be applied identically every time, because settlement runs are reconciled several times over the following months and each run must be reproducible.

Your team is small: seven engineers, two of whom are contractors whose contracts end in four months. Nobody on the team has implemented industry settlement validation rules before, although you have hired an analyst from another supplier who knows them well. The cloud provider has offered engineering support credits if Brightwell commits to a larger annual spend. Karen's team has formally asked that, whatever the architecture, they be able to run their existing operational reports without retraining. And internal audit has asked how Brightwell will demonstrate, for any historical settlement submission, exactly which readings, estimates and rule versions produced it.

Security and privacy have entered the conversation too. Half-hourly consumption data is personal data, and at that granularity it can reveal when a household wakes up, when it is empty, and sometimes which appliances it uses. Brightwell's data protection officer, Fiona Marsh, has reminded the programme that the data access and privacy framework governing smart meter data restricts how granular consumption can be used for purposes beyond settlement without customer consent. Settlement use is permitted, but Neil's plans for tariff design, flexibility products and analytics would require either aggregation, pseudonymisation or explicit consent, and the consent flags held in the customer relationship system are known to be incomplete. Any lakehouse design that makes granular data freely queryable by analysts would therefore need fine-grained access controls, consent-aware views and auditing from the first day, which adds scope your team has not yet estimated.

The steering committee meets in three weeks. Neil wants you to present a recommendation that commits to the lakehouse. Karen wants you to present the risks honestly. Oliver wants a cost comparison that holds up over five years, not just the first. And you know that the first cohort deadline does not move regardless of which option wins.

THE DECISION IN FRONT OF YOU

Do you recommend the vendor upgrade, a full lakehouse replacement for settlement data processing, or a hybrid where the lakehouse runs alongside the vendor system until it is proven, and how do you de-risk the first cohort deadline while unlocking the analytical value the business wants?

Data Sources

#	Source and link	How it is used in this project
1	Elexon Market-wide Half Hourly Settlement Programme https://www.elexon.co.uk/mhhs/	Industry programme documentation, timetables and design artefacts that set the regulatory requirements for the platform.
2	Elexon Insights Solution API (BMRS) https://developer.data.elexon.co.uk	Wholesale market and system data such as system prices and imbalance volumes used to value demand forecasting improvements.
3	Smart Meter Energy Use in London Households (London Datastore) https://data.london.gov.uk/dataset/smartmeter-energy-use-data-in-london-households	Real half-hourly consumption data from 5,567 households used to prototype ingestion, validation and estimation logic.
4	Low Carbon London Project Data https://innovation.ukpowernetworks.co.uk/projects/low-carbon-london/	Distribution network trial data providing context on half-hourly demand, heat pumps and solar export behaviour.
5	Apache Iceberg Documentation https://iceberg.apache.org/docs/latest/	Open table format features such as snapshots, time travel and partition evolution that underpin reproducible settlement runs.
6	Met Office Weather DataHub https://www.metoffice.gov.uk/services/data/met-office-weather-datahub	Weather observations and forecasts used as features for estimation plausibility checks and demand forecasting.
7	Carbon Intensity API (NESO) https://carbonintensity.org.uk	Half-hourly regional grid carbon intensity used to enrich consumption data for time-of-use and carbon reporting products.
8	Ofgem Market-wide Half-Hourly Settlement Decision https://www.ofgem.gov.uk/energy-policy-and-regulation/policy-and-regulatory-programmes/electricity-settlement-reform	Regulatory decisions and obligations for suppliers that frame the compliance case.

Deliverables (17)

- D1 Five-year total cost of ownership model.** A comparison of vendor upgrade, lakehouse and hybrid options including licences, compute, storage, staff and risk contingency.
- D2 Options and risk assessment.** A structured appraisal of delivery risk, regulatory risk, operational change and analytical value for each option.
- D3 Lakehouse reference architecture.** A design covering ingestion, raw and validated zones, settlement outputs, analytics access and security boundaries.
- D4 Half-hourly reading data model.** A partitioned table design for readings, estimates, meter configuration and exchanges supporting fast point lookups and bulk scans.
- D5 Ingestion pipeline.** A resilient pipeline landing daily meter data files and messages with schema validation, duplicate detection and completeness reporting.
- D6 Validation rule engine.** A test-driven implementation of industry validation checks, including clock drift, spikes, negative values and meter exchange overlaps.
- D7 Estimation rule implementation.** Code implementing industry estimation methods for missing periods with every estimate flagged, versioned and explainable.
- D8 Reproducibility framework.** Use of table snapshots and rule versioning so that any past settlement submission can be regenerated exactly.
- D9 Parallel-run comparison suite.** Automated reconciliation of lakehouse outputs against the vendor system meter by meter and period by period.
- D10 Settlement submission interface.** A controlled export meeting industry file or message formats and timetables, with submission receipts tracked.
- D11 Trading consumption feed.** A validated next-day demand dataset delivered to trading models by nine each morning with freshness monitoring.
- D12 Data quality observatory.** Dashboards tracking missing reads, drift, spikes, estimation rates and communication failures by meter type and region.
- D13 Operational reporting compatibility layer.** Views that reproduce settlement operations' existing reports on the new platform to minimise retraining.
- D14 Capacity and performance test report.** Evidence that processing completes within the required window at full customer base and projected growth.
- D15 Access control and data protection design.** Controls governing access to granular consumption data, which can reveal household behaviour, aligned with data protection law.
- D16 Skills and staffing plan.** A plan covering contractor knowledge transfer, hiring and training in settlement rules and lakehouse operations.
- D17 First cohort cutover runbook.** A detailed plan with go/no-go criteria, fallback to the vendor system and hypercare arrangements for the first transition.

CASE STUDY 05 · URBAN TRANSPORT AND MOBILITY

Trip Data Platform and Dynamic Pricing Features for a Ride-Hailing Operator

Industry	Urban Transport and Mobility
Organisation	Metroride, a ride-hailing and taxi dispatch operator active in eight cities in the UK and the United States
Your role	Staff Data Engineer, Marketplace Data
Core themes	Geospatial streaming, feature freshness, regulatory reporting, fairness and pricing transparency

The Scenario

Metroride started as a taxi dispatch app for licensed black cab and private hire drivers in Manchester and has since expanded into seven more cities, including New York, where it operates under the rules of the city's Taxi and Limousine Commission. The company's growth has been funded on the promise that its marketplace, the matching of riders with drivers at the right price, is smarter than its rivals'. At the centre of that promise is dynamic pricing: when demand in an area outstrips the available drivers, prices rise to attract drivers and ration demand. You are the staff data engineer for marketplace data, and the quality of the data feeding that pricing engine is now the subject of an uncomfortable internal investigation.

Three weeks ago, a local news outlet in New York ran a story about riders being charged surge prices in a neighbourhood where, according to drivers interviewed, there were plenty of empty cars. The story went viral. Metroride's pricing team investigated and found that the surge had been real in the sense that the pricing engine computed it correctly from its inputs, but the inputs were wrong. The demand and supply features that the pricing model uses are computed every minute by aggregating ride requests and driver locations into hexagonal geographic cells. During the incident, a mobile app release had introduced a bug that caused some drivers' location updates to be sent only every few minutes instead of every few seconds. Those drivers disappeared from the supply count in their cells, and the pricing engine saw a shortage that did not exist. The feature pipeline had no check for this. It computed exactly what it was told.

The chief executive, Daniel Brooks, has promised publicly that it will not happen again. The Head of Marketplace, Sarah Lindqvist, wants the

feature pipeline rebuilt with robust data quality controls and a kill switch that caps surge automatically when input data looks abnormal. The regulatory affairs lead, Marcus Webb, has a different concern: New York regulators require trip-level data submissions, and several UK licensing authorities are consulting on requirements for private hire operators to share trip and pricing data. Marcus wants a single trusted trip data platform from which both regulatory submissions and internal analytics are derived, because he is worried that the pricing pipeline and the regulatory submissions currently use different logic and could tell different stories about the same trip.

You have been asked to lead a rebuild, and as you map the existing estate, the scale of the tangle becomes clear. Driver location pings arrive at around sixty thousand events per second at peak across all cities and are processed by a streaming job that has accreted features over four years. Ride requests, acceptances, cancellations and completions are written to an operational database and replicated to the data warehouse hourly. The pricing features are computed in the stream; the regulatory submissions are computed in the warehouse from the replicated data; and the analytics team computes its own version of supply and demand in notebooks for reporting. Nobody has ever reconciled the three. When you compared them for a single day in Manchester, the warehouse showed four per cent more completed trips than the stream, mostly because of trips that were completed after a network reconnection and never re-emitted as events.

The obvious engineering approach is to build a single event-sourced trip data platform in which every trip lifecycle event is captured once, immutably, and every downstream product, whether pricing features, regulatory reports or analytics, derives from the same log. You are confident this is the right destination. The question is how to get there without disrupting the pricing engine, which runs the business and cannot be switched off, and within the timeframe the chief executive has implied publicly.

The tension is sharpened by a debate about freshness. Sarah's team insists that pricing features must be computed with under ten seconds of latency, because demand in a busy area can change in less than a minute. The data science team, led by Dr Hannah Cole, has shared analysis suggesting that the pricing model's performance barely changes whether features are ten seconds or sixty seconds old, but that it changes significantly when

features are computed from incomplete data. Hannah argues that the pipeline should wait for a short window to allow late location pings to arrive, trading a little freshness for much greater completeness. Sarah is unconvinced and worries that competitors will react faster.

Fairness adds another layer. After the news story, several city councillors have asked whether dynamic pricing disproportionately affects particular neighbourhoods. Metroride's legal team has asked whether the company can demonstrate, with data, that surge pricing is driven only by supply and demand and not by proxies for the demographics of an area. Answering that question requires joining trip and pricing data with public demographic and geographic datasets, retaining pricing decision inputs for long enough to audit them, and being able to reconstruct why any particular price was shown to any particular rider. At present, pricing inputs are retained for seven days.

Your engineering director, Tom Farrow, has given you a team of six engineers for two quarters. He wants a plan that addresses the incident quickly, satisfies regulators, and moves toward the unified platform, but he has told you that you cannot have all three in the first quarter. He has also warned you that the infrastructure budget is under scrutiny, because the streaming platform already accounts for a large share of cloud spend and the finance team has asked for it to be reduced, not increased.

There is also a people dimension that you cannot ignore. The streaming pipeline was written largely by two engineers who are still on your team and who feel, with some justification, that they are being blamed for an incident caused by a mobile release they had no visibility of. They are defensive about proposals to replace their work, and they know its quirks better than anyone. The analytics team, meanwhile, is excited by the idea of a unified platform but has made clear that it expects to keep its notebooks until the new datasets are proven. And the mobile engineering team, whose release caused the problem, has pushed back on the idea of a formal data contract for location events, arguing that it would slow down their release cadence at a time when the product roadmap is already behind schedule.

As you prepare your plan, you realise that the most important decision is not about technology but about sequence and about which definition of truth the company will adopt. If you harden the existing streaming pipeline first, you address the incident quickly but entrench the inconsistency with

regulatory data. If you build the unified event log first, you fix the root cause but leave the pricing engine exposed for months.

THE DECISION IN FRONT OF YOU

How do you sequence incident-driven data quality controls, a unified trip event platform and regulatory reporting so that pricing is safe quickly, regulators and analysts see consistent numbers, and pricing decisions become auditable for fairness, all within a reduced infrastructure budget?

Data Sources

#	Source and link	How it is used in this project
1	NYC TLC Trip Record Data https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page	Billions of real yellow, green and for-hire vehicle trip records used to build and test the trip data model and regulatory-style aggregations.
2	Chicago Transportation Network Providers Trips https://data.cityofchicago.org/Transportation/Transportation-Network-Providers-Trips/m6dm-c72p	Ride-hailing trip records including fares and surge indicators for studying pricing patterns by area.
3	H3 Hexagonal Geospatial Indexing https://h3geo.org	Indexing system for aggregating supply and demand into consistent hexagonal cells at multiple resolutions.
4	US Census Bureau American Community Survey API https://www.census.gov/data/developers/data-sets/acs-5year.html	Neighbourhood demographic data used in the fairness audit of surge pricing distribution.
5	ONS Census 2021 Data https://www.ons.gov.uk/census	UK small-area demographic data for equivalent fairness analysis in UK cities.
6	OpenStreetMap Extracts (Geofabrik) https://download.geofabrik.de	Road network and point-of-interest data for map matching and context features.
7	Apache Flink Documentation https://nightlies.apache.org/flink/flink-docs-stable/	Event-time processing, watermarks and state management reference for the feature pipeline.
8	Transport for London Open Data https://tfl.gov.uk/info-for/open-data-users/	Public transport disruption and status feeds used as demand context features in London operations.

Deliverables (17)

- D1 Incident post-mortem and data failure analysis.** A blameless analysis of the surge incident explaining the data defect, detection gaps and contributing causes.
- D2 Input anomaly detection controls.** Real-time checks on ping rates, driver counts and request volumes per cell with thresholds based on historical baselines.
- D3 Surge safety cap and kill switch.** A mechanism that caps or freezes surge when input quality checks fail, with logging and alerting.

- D4 Freshness versus completeness study.** An experiment quantifying model performance and business outcomes across different feature latency and lateness windows.
- D5 Unified trip event schema.** A canonical, versioned schema for every trip lifecycle event with unique identifiers and event-time semantics.
- D6 Event-sourced trip log.** An immutable, replayable log of trip events from which all downstream products are derived.
- D7 Late and reconnected event handling.** Logic ensuring trips completed after reconnection are captured once and reconciled.
- D8 Streaming feature pipeline rebuild.** A redesigned supply and demand feature job using event-time windows and watermarks on the unified log.
- D9 Cross-product reconciliation report.** Daily reconciliation of trip counts and fares across pricing, regulatory and analytics outputs.
- D10 Regulatory submission pipelines.** Automated, validated trip data submissions for New York and UK licensing authorities derived from the unified log.
- D11 Pricing decision audit store.** Long-term retention of pricing inputs, model version and output per quote to reconstruct any price shown.
- D12 Fairness analysis framework.** A repeatable analysis joining pricing outcomes with demographic data to test for disparate impact by neighbourhood.
- D13 Geospatial reference data service.** Managed H3 cell, zone and boundary reference data shared across pricing, regulation and analytics.
- D14 Streaming cost optimisation plan.** Measures such as state tuning, compaction and tiered storage that reduce streaming spend while adding new controls.
- D15 Mobile telemetry data contract.** An agreement with mobile engineering on ping frequency, schema and release testing for location events.
- D16 Operational runbook and on-call guide.** Procedures for data quality alerts, surge cap activation and regulatory feed failures.
- D17 Phased delivery roadmap.** A two-quarter plan sequencing safety controls, unified log, regulatory migration and fairness auditing with milestones.

CASE STUDY 06 · E-COMMERCE AND DIGITAL MARKETING

Clickstream Sessionisation and Marketing Attribution After Cookie Loss

Industry	E-commerce and Digital Marketing
Organisation	Lumen & Loom, an online homewares retailer with £180 million annual revenue and customers in the UK and EU
Your role	Analytics Engineering Lead
Core themes	Event data modelling, consent and privacy, attribution methodology, metric trust

The Scenario

Every Monday at ten, the marketing leadership team at Lumen & Loom meets to decide how to spend roughly three million pounds a month across paid search, social advertising, affiliates, email and a growing influencer programme. For five years, those decisions were made from a single report: last-click revenue by channel, pulled from the web analytics tool. It was never perfect, and everyone knew it flattered paid search and brand terms, but it was consistent and people trusted its direction of travel. Over the past eighteen months that consistency has quietly fallen apart, and the Chief Marketing Officer, Charlotte Byrne, has asked you, as analytics engineering lead, to tell her whether any of the numbers she is looking at can still be believed.

The reasons are an accumulation of changes rather than a single event. The company migrated to the current generation of its web analytics platform, which uses an event-based data model rather than sessions and pageviews, and exports raw events to a cloud data warehouse. The consent banner was redesigned after a regulatory complaint, and consent rates for analytics and marketing cookies fell from around eighty per cent to fifty-eight per cent in the UK and even lower in some EU markets. Browsers now restrict third-party cookies and shorten the life of some first-party cookies set by scripts, which means a customer who visits on Monday and returns on Friday is increasingly counted as two different people. The paid social platforms report conversions using their own modelling, and their combined claimed revenue last quarter exceeded the company's actual online revenue by forty per cent. The affiliate network claims credit for sales that the analytics tool attributes to email. And the finance team, which reconciles everything against the order management system, has

stopped attending the Monday meeting because, in the words of the finance director, the marketing numbers "are a work of fiction".

You have been looking at the raw event export in the warehouse for several weeks, and you have concluded that the data is more useful than the reports built on top of it suggest, but also more treacherous. Each event carries a pseudonymous client identifier, a session identifier, a timestamp, page and item details, and traffic source parameters, but the session identifier only resets under specific conditions, the traffic source is attached only to certain events, and events from users who declined consent arrive stripped of identifiers under the platform's consent mode, with modelled conversions added later in the interface but not in the export. When logged-in customers are identified, a user identifier is available, and for about thirty per cent of orders you can link the web journey to the customer record in the order system. For everyone else, the journey fragments.

Two camps have formed around what to do. The performance marketing manager, Jake Morrison, wants a multi-touch attribution model built in the warehouse: stitch sessions into user journeys, apply a data-driven model that distributes credit across touchpoints, and use the results to reallocate budget weekly. He has seen a vendor demonstration of exactly this and believes the company is leaving money on the table by over-crediting brand search. The head of data science, Dr Nadia Rahman, thinks multi-touch attribution built on fragmentary, consent-biased clickstream data is fundamentally unreliable. She argues that the customers who decline consent are systematically different from those who accept, that cross-device journeys are invisible, and that attribution models measure correlation, not the causal effect of spending. Her proposal is to invest instead in marketing mix modelling using aggregated weekly spend and revenue, combined with a programme of geographic holdout experiments to measure the true incremental effect of each channel.

Charlotte finds both arguments persuasive and neither satisfying. She points out that mix modelling produces answers quarterly at best, whereas her team makes decisions weekly, and that holdout experiments mean deliberately switching off spending in some regions, which the trading director will resist during peak season. She also notes that whichever approach is chosen, it will depend on data foundations that do not currently exist: a clean, consistent definition of a session and a conversion;

reliable mapping of campaign parameters to channels, which is currently a spreadsheet maintained by an agency; spend data from each advertising platform landed in the warehouse daily rather than downloaded manually; and a reconciliation between web conversions and actual orders, including returns, which run at twenty-two per cent for some categories.

Legal and privacy add further constraints. The data protection officer, Martin Kerr, has reviewed a proposal from Jake to use server-side tagging and hashed email matching to recover attribution for non-consenting users, and has said firmly that circumventing consent choices, even technically, is not acceptable. He is, however, open to modelling approaches that use aggregated or anonymised data, and he has asked that any customer-level journey data retained in the warehouse be subject to a defined retention period and purpose limitation. The warehouse currently retains raw events indefinitely.

There is also a cost question. The raw event export has grown to over four billion rows, and the warehouse bill for marketing queries has tripled because analysts repeatedly scan the full event tables to answer simple questions. The platform team has asked you to bring costs under control. Well-designed incremental models, partitioned by date and clustered by client identifier, would help significantly, but only if analysts are persuaded to use them instead of their own queries.

Your team consists of three analytics engineers and a part-time data engineer shared with the logistics function. You can build good models, but you cannot do everything at once. Charlotte has asked you to present an approach at the next quarterly planning session that restores the finance team's trust, gives marketing something useful to act on weekly, respects privacy, and controls cost. She has been explicit that she does not want a technology proposal; she wants to know what numbers the business will look at in six months and why it should believe them.

You have also discovered an organisational problem underneath the technical one. Three different teams currently produce revenue figures for the Monday meeting. Marketing uses the analytics interface, which reports revenue including tax and before returns. Ecommerce trading uses the order management system, which reports revenue excluding tax and including cancelled orders until they are processed. Finance uses the ledger, which recognises revenue on dispatch rather than on order. Each figure is correct by its own definition, and none of the teams has ever

written its definition down. When a channel manager claims that paid social drove two hundred thousand pounds of revenue last week, nobody in the room can say which of these three revenues is meant. You suspect that a surprising amount of the distrust between marketing and finance would disappear if the company simply agreed, documented and consistently applied one definition of net revenue per order, adjusted for returns once the return window has closed, and made every report state explicitly which version it uses. That work is unglamorous and would not appear in any vendor demonstration, but it may matter more than the choice of attribution model.

As you plan, you are aware that the most seductive option, a sophisticated journey-level attribution model, may also be the one most likely to produce confident-looking numbers that are wrong. But you are also aware that telling a marketing team that it cannot have weekly channel measurement is a quick way to be ignored, and that someone else will simply buy the vendor product Jake saw.

THE DECISION IN FRONT OF YOU

Do you build journey-level multi-touch attribution, back marketing mix modelling and incrementality testing, or design a layered measurement framework, and what data foundations must exist first for any of these answers to be trusted by marketing and finance alike?

Data Sources

#	Source and link	How it is used in this project
1	GA4 BigQuery Sample E-commerce Dataset https://developers.google.com/analytics/bigquery/web-e-commerce-demo-dataset	Real obfuscated GA4 event export from the Google Merchandise Store used to build sessionisation, journey and conversion models.
2	GA4 BigQuery Export Schema https://support.google.com/analytics/answer/7029846	Reference for event, user, traffic source and item fields including consent-related behaviour.
3	REES46 E-commerce Behaviour Data https://www.kaggle.com/datasets/mkechinov/e-commerce-behavior-data-from-multi-category-store	Hundreds of millions of view, cart and purchase events used to stress-test incremental models at scale.
4	Google Meridian (Marketing Mix Modelling) https://developers.google.com/meridian	Open-source Bayesian MMM framework used to prototype aggregated channel effect estimation.
5	Meta Robyn MMM https://facebookexperimental.github.io/Robyn/	Alternative open-source MMM package for comparing model outputs and assumptions.

#	Source and link	How it is used in this project
6	ICO Guidance on Cookies and Similar Technologies https://ico.org.uk/for-organisations/direct-marketing-and-privacy-and-electronic-communications/guide-to-pecr/cookies-and-similar-technologies/	UK regulatory guidance defining the consent boundaries for tracking and measurement.
7	dbt Documentation https://docs.getdbt.com	Framework for building tested, documented, incremental warehouse models for sessions, journeys and conversions.
8	ONS Internet Retail Sales https://www.ons.gov.uk/businessindustryandtrade/retailindustry	Market-level online retail trends used as an external control variable in mix modelling.

Deliverables (17)

- D1 Measurement strategy paper.** A clear explanation of which questions attribution, mix modelling and experiments can answer, and a recommended layered framework.
- D2 Canonical event and session model.** Tested, incremental warehouse models that define sessions, users and conversions consistently from raw events.
- D3 Conversion-to-order reconciliation.** A daily reconciliation between web conversions and order system sales including cancellations and returns.
- D4 Campaign taxonomy and channel mapping.** A governed, version-controlled mapping from campaign parameters to channels replacing the agency spreadsheet.
- D5 Advertising spend ingestion.** Automated daily ingestion of spend, impressions and clicks from each advertising platform into the warehouse.
- D6 Consent-aware data design.** Models that segregate consented and non-consented data correctly and never attempt to re-identify declining users.
- D7 Consent bias analysis.** A study comparing behaviour and basket characteristics of consenting versus modelled non-consenting traffic.
- D8 Journey stitching logic.** Rules linking sessions to logged-in customers where permitted, with coverage metrics reported transparently.
- D9 Rule-based attribution views.** First-click, last-click and position-based attribution views presented side by side with explicit caveats.
- D10 Marketing mix model prototype.** A weekly aggregated model with channel response curves, validated against historical budget shifts.
- D11 Incrementality test design.** A geographic holdout design with power calculations, region selection and timing agreed with trading.
- D12 Weekly marketing scorecard.** A single, reconciled report combining spend, orders, net revenue after returns and model-based channel effects.
- D13 Retention and purpose policy implementation.** Automated deletion or aggregation of event-level data after the agreed retention period.
- D14 Warehouse cost optimisation.** Partitioning, clustering and curated datasets that reduce marketing query costs, with before-and-after measurement.

- D15 Data quality test suite.** Tests on event volume anomalies, missing parameters, duplicate transactions and taxonomy gaps.
- D16 Metric definitions glossary.** Business definitions of sessions, users, conversions, revenue and return-adjusted revenue signed off by marketing and finance.
- D17 Stakeholder adoption plan.** Training, documentation and a transition timeline for retiring legacy last-click reporting.

CASE STUDY 07 • GENERAL INSURANCE

Motor Claims Fraud Detection Feature Store

Industry	General Insurance
Organisation	Castlegate Insurance, a UK motor and home insurer handling 140,000 claims per year
Your role	Senior Machine Learning Data Engineer
Core themes	Feature stores, training-serving consistency, entity resolution, explainability and fairness

The Scenario

Castlegate Insurance pays out a little over six hundred million pounds a year in motor claims, and its counter-fraud team estimates that somewhere between five and eight per cent of that value involves some element of fraud: exaggerated injuries, staged collisions, phantom passengers, inflated repair invoices, or organised networks that recycle the same vehicles, addresses and witnesses across dozens of claims. The industry has fought this for years with intelligence databases, specialist investigators and rules, and more recently with machine learning. Castlegate's current fraud scoring model was built three years ago, and its performance has been declining steadily. You have been brought in as senior machine learning data engineer to find out why and fix it.

It does not take long to find the first problem. The model was trained in a data science environment on a historical extract of claims, enriched with dozens of derived features: the number of previous claims by the policyholder, the time between policy inception and first claim, whether the claimant's address appears on other claims, the distance between the incident location and the claimant's home, the proportion of third-party claimants represented by particular solicitors, and so on. When the model went live, those features were re-implemented by the claims system integration team in a different language, reading from the operational claims database rather than the warehouse. When you compare the features computed in production for last month's claims with the same features recomputed from the warehouse, a third of them differ materially. Some differences are bugs. Others are more subtle: the training data used information that was only known weeks after first notification of loss, such as the final repair cost and whether a solicitor was eventually appointed, so the model learned patterns from data that is not available at the moment it is asked to score.

The second problem is entity resolution. Many of the most predictive fraud signals depend on recognising when the same person, vehicle, address or phone number appears across multiple claims. Castlegate's claims system stores claimant details as free text entered by call handlers. Names are misspelt, addresses are formatted inconsistently, and the same phone number might be recorded with or without an international prefix. The current features use exact matching and therefore miss many connections that investigators find manually. The counter-fraud team has licensed access to an industry-wide claims intelligence database, but its data arrives as periodic files, and matching against it is done by a separate process that nobody has integrated with the model.

The head of claims analytics, Dr Priya Nair, wants to build a feature store: a central platform where each feature is defined once, computed consistently for both training and live scoring, versioned, and served at low latency when a claim is notified. She argues that this is the only way to end the training-serving mismatch and that it will support future models for pricing, claims triage and customer retention. The IT architecture team, led by Graham Fletcher, is sceptical. They point out that Castlegate already has a data warehouse, a streaming platform used for policy events, and an operational data store, and that adding a feature store product creates another system to secure, operate and pay for. Graham would prefer the features to be built as well-governed views in the warehouse, with a caching layer for real-time scoring.

The head of counter-fraud, Mike Harrison, a former detective, is less interested in architecture than in outcomes. His investigators are overwhelmed with referrals from the current model, most of which turn out to be genuine. He also has a serious concern about fairness. Last year a customer complained to the Financial Ombudsman Service after their claim was delayed for a fraud investigation that found nothing, and the complaint revealed that one of the model's strongest features was the claimant's postcode area. The ombudsman did not find against Castlegate, but the case prompted the compliance team to ask whether postcode, vehicle age and other features might act as proxies for protected characteristics or for financial vulnerability. The Financial Conduct Authority's consumer duty requires insurers to demonstrate good outcomes for all groups of customers, including vulnerable ones, and Mike does not want his team accused of targeting particular communities.

There is also a question of speed. The business wants fraud scores within seconds of first notification of loss, so that suspicious claims can be routed to specialist handlers before any payment or courtesy car is arranged. Today, scoring happens in an overnight batch, and by the time a claim is flagged, the claimant may already have been assigned a hire car at considerable cost. Real-time scoring requires features that can be computed from data available at notification, which rules out many of the features that make the current model accurate in back-testing.

Your budget and time are finite. The chief data officer has approved six months of work with a team of four, including you. She has asked for three things: a model the investigators trust, features that are provably consistent between training and scoring, and a governance framework that satisfies compliance on fairness. She has not told you which architecture to use, but she has made clear that she will not approve an approach that Graham's architecture team refuses to support.

The claims handlers themselves have added a practical concern that you had not anticipated. When you visited the first notification of loss team, the call handlers explained that they often record claims in stages: a quick initial record while the customer is distressed at the roadside, then fuller details the next day when the customer calls back with the third party's information and photographs. A fraud score calculated at the moment of first notification would therefore be based on a skeleton record, while a score calculated a day later would be far more informed but would arrive after the hire car decision has been made. The team leader suggested that the business might accept a two-stage approach, with an immediate triage score on limited data and a fuller score once the claim is complete, but that would mean two models, two sets of features and two sets of monitoring. It would also mean explaining to investigators why the same claim received two different scores, a conversation Mike Harrison has already said he does not relish having with his team.

As you think about the plan, you realise that the architecture debate may be a distraction from a harder question: how to define, for every feature, what was knowable at the moment of scoring, and how to reconstruct that knowledge historically so the model is trained on the world as it actually looked at decision time. Whatever system you build, getting point-in-time correctness right will determine whether the next model works any better than the last one.

THE DECISION IN FRONT OF YOU

Do you adopt a dedicated feature store, build governed warehouse features with a serving cache, or pursue a hybrid, and how do you guarantee point-in-time correctness, integrate fuzzy entity resolution and prove to compliance that the resulting model treats customers fairly?

Data Sources

#	Source and link	How it is used in this project
1	Porto Seguro Safe Driver Prediction https://www.kaggle.com/c/porto-seguro-safe-driver-prediction	Anonymised motor insurance policy data used to prototype feature pipelines and claim likelihood modelling.
2	Vehicle Insurance Fraud Detection Dataset https://www.kaggle.com/datasets/shivamb/vehicle-claim-fraud-detection	Labelled motor claims fraud dataset used to develop and back-test fraud features and scoring.
3	STATS19 Road Safety Data (DfT) https://www.data.gov.uk/dataset/cb7ae6f0-4be6-4935-9277-47e5ce24a11f/road-safety-data	Official GB road collision data used to assess plausibility of reported incident locations and circumstances.
4	Feast Open Source Feature Store https://docs.feast.dev	Reference implementation for point-in-time joins, offline and online stores and feature definitions.
5	Splink Probabilistic Record Linkage https://moj-analytical-services.github.io/splink/	Open-source entity resolution library for fuzzy matching of people, addresses and phone numbers across claims.
6	Postcodes.io https://postcodes.io	Free UK postcode lookup and geocoding for normalising addresses and computing distances.
7	Insurance Fraud Bureau https://insurancefraudbureau.org	Industry intelligence and typology information informing feature design and network detection.
8	FCA Consumer Duty https://www.fca.org.uk/firms/consumer-duty	Regulatory requirements shaping the fairness and customer outcomes testing framework.

Deliverables (17)

- D1 Training-serving skew audit.** A feature-by-feature comparison of production and warehouse values with root causes classified and quantified.
- D2 Temporal leakage analysis.** Identification of every feature using information unavailable at notification time and its effect on model performance.
- D3 Feature architecture decision record.** A documented decision comparing feature store, warehouse and hybrid options endorsed by architecture.
- D4 Feature definition registry.** A single catalogue of feature logic, owners, versions, freshness and permitted use.
- D5 Point-in-time training dataset builder.** Tooling that reconstructs feature values as they were at each claim's notification time.

- D6 Entity resolution pipeline.** Probabilistic matching of claimants, vehicles, addresses and phone numbers with confidence scores and review workflow.
- D7 Claims network graph dataset.** A graph of linked entities across claims supporting ring and network features for investigators.
- D8 Industry intelligence integration.** Automated ingestion and matching of industry claims intelligence files into the feature pipeline.
- D9 Real-time scoring feature service.** Low-latency features served at first notification of loss with fallbacks for missing data.
- D10 Retrained fraud model.** A model trained on point-in-time correct features with performance compared against the current model.
- D11 Fairness and proxy assessment.** Analysis of features such as postcode for correlation with protected characteristics and vulnerability, with mitigations.
- D12 Explainability outputs for investigators.** Per-claim reason codes that tell investigators why a claim was referred.
- D13 Model monitoring framework.** Drift, stability and outcome monitoring comparing referral quality over time.
- D14 Investigator feedback loop.** A structured capture of investigation outcomes feeding back into labels for future training.
- D15 Data protection impact assessment.** A DPIA covering profiling, automated decision-making and data sharing with industry bodies.
- D16 Operational runbook.** Procedures for feature service failures, stale data and emergency model rollback.
- D17 Business impact report.** Measurement of referral precision, investigator workload, hire car cost avoided and customer complaints after launch.

CASE STUDY 08 · ADVANCED MANUFACTURING

Predictive Maintenance Data Pipeline for a Turbine Component Plant

Industry	Advanced Manufacturing
Organisation	Aldermoor Precision, a Midlands manufacturer of aerospace and power generation turbine components
Your role	Industrial Data Engineer, Operations Technology
Core themes	IT and OT convergence, edge versus cloud processing, time-series data management, labelling scarcity

The Scenario

The machining hall at Aldermoor Precision contains forty-two computer numerically controlled machines, eleven heat treatment furnaces, six coordinate measuring machines and a newly installed additive manufacturing cell. Together they produce turbine blades, discs and casings for aerospace engines and industrial gas turbines, parts where a single defect can scrap tens of thousands of pounds of nickel superalloy and where unplanned downtime on a bottleneck machine delays customer deliveries by weeks. Last year the plant lost just over nine hundred hours of production to unplanned breakdowns, and two late deliveries triggered contractual penalties from an engine manufacturer. The operations director, Ravi Chandran, has made predictive maintenance his top priority, and you have been hired as the plant's first industrial data engineer to make it happen.

On paper, the plant is rich in data. The newer machines have controllers that expose spindle load, vibration, temperature, feed rates and alarm codes through OPC UA, the industrial interoperability standard. The furnaces log temperature profiles to a supervisory system. A historian database, installed a decade ago, collects some signals at one-second intervals and stores them for ninety days before summarising them to hourly averages. Maintenance work orders are recorded in an enterprise asset management system. Quality inspection results, including dimensional measurements and scrap decisions, live in a manufacturing execution system. In practice, almost none of this has been joined together, and the reality on the shop floor is messier than the architecture diagrams suggest.

Only nineteen of the forty-two CNC machines have controllers capable of OPC UA. The rest are older machines with proprietary interfaces or none at all; for these, the only option is retrofitting external vibration and current sensors. The historian's ninety-day raw retention means that high-frequency signals from before most historical failures no longer exist. The maintenance work orders are written by technicians in free text, and a spindle bearing replacement might be described as "spindle noisy, replaced brg", "bearing change", or simply "fixed". Failure dates are recorded as the date the work order was closed, which is sometimes days after the actual failure. And the plant network is segmented for good reason: the operational technology network that connects machines is isolated from the corporate IT network by a firewall, following a ransomware incident at a sister company two years ago that halted production for eleven days.

The central architectural argument is about where processing should happen. The group IT function, led by Caroline Webb, has a cloud-first strategy and wants machine data streamed to the group's cloud data platform, where the data science team at head office can build models. Caroline argues that this centralises expertise, allows models to be reused across the group's three plants, and avoids building fragile local infrastructure. The plant's OT engineering manager, Stuart Bell, disagrees strongly. He points out that high-frequency vibration data from even twenty machines amounts to terabytes per month, that the plant's internet connection is shared and modest, and that sending machine data outside the plant creates a pathway that the cyber security team spent two years closing. Stuart wants edge computing: industrial PCs on the shop floor that process signals locally, compute features such as vibration spectra and alarm counts, and send only summarised data to the cloud. He also argues that any real-time protection, such as stopping a machine when a spindle shows signs of imminent failure, must work even if the network connection to the cloud is lost.

The data science team at head office, led by Dr Emma Walsh, has its own concern: labels. Predictive maintenance models need examples of failures, and the plant has recorded perhaps sixty genuine spindle and axis failures across all machines in five years, most without surviving high-frequency data. Emma believes the plant should start with anomaly detection rather than failure prediction, learning what normal looks like for each machine and flagging deviations, and should invest in carefully capturing the next

failures with full data so that supervised models become possible over time. She wants raw, high-frequency data retained for at least two years, which conflicts directly with Stuart's plan to summarise at the edge and discard.

The maintenance manager, Keith Dunn, has been through two previous "Industry 4.0" pilots that installed sensors, produced a dashboard, and were quietly abandoned. His technicians are sceptical and busy. He tells you that he will support the project only if it produces alerts his team can act on, with few false alarms, and if it does not require technicians to fill in more forms. At the same time, he acknowledges that without better work order data, no model will ever learn from what his team does.

Then there is the question of the quality data. The additive manufacturing cell and the furnaces produce parts whose defects are sometimes caused not by machine failure but by process drift: a furnace thermocouple slowly losing calibration, or powder quality varying between batches. The quality director, Anita Shah, sees the project as a chance to link process data to inspection results and predict scrap before it happens, which could save more money than predicting breakdowns. But it would require joining machine data to individual parts through the manufacturing execution system's traceability records, which currently identify parts by batch rather than by individual serial number for most product lines.

A further practical problem surfaced when you walked the shop floor with the night shift supervisor. Several of the older machines are run in a way that makes their signals hard to interpret: operators routinely override feed rates to meet shift targets, spindles are warmed up differently depending on who is on shift, and some machines switch between very different part programmes several times a day. A vibration level that is perfectly normal while roughing a disc forging would be alarming while finishing a blade root. Any anomaly detection that ignores the machine's current operating context will flood the maintenance team with false alarms, and the context itself, meaning which programme is running, which tool is loaded and which part is being cut, is held in the controller for the modern machines but only in paper job cards for the older ones. Capturing that context reliably may be as important as capturing the sensor data, and it touches the manufacturing execution system team, who have their own backlog and their own priorities for the year.

You have a budget for sensors, edge hardware and one year of your own time plus a contractor. Ravi wants a visible result within six months to justify further investment across the group. You have to choose a first use case, an architecture that the cyber security team will approve, a data retention approach that serves future modelling without breaking the network, and a way of getting better maintenance data from technicians who do not want more paperwork.

THE DECISION IN FRONT OF YOU

Do you centralise machine data in the group cloud, process it at the edge, or design a tiered architecture, and which first use case, anomaly detection on bottleneck machines or process-to-quality prediction, offers the best chance of a trusted six-month result?

Data Sources

#	Source and link	How it is used in this project
1	NASA C-MAPSS Turbofan Degradation Dataset https://data.nasa.gov/Aerospace/CMAPSS-Jet-Engine-Simulated-Data/ff5v-kuh6	Simulated run-to-failure sensor data used to prototype remaining useful life features and pipelines.
2	NASA Prognostics Center of Excellence Data Repository https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/	Bearing, milling and other run-to-failure datasets for developing vibration features.
3	AI4I 2020 Predictive Maintenance Dataset (UCI) https://archive.ics.uci.edu/dataset/601/ai4i+2020+predictive+maintenance+dataset	Synthetic machining dataset with failure modes used to design labelling and feature logic.
4	Case Western Reserve University Bearing Data https://engineering.case.edu/bearingdatacenter	High-frequency bearing vibration recordings used to build edge spectral feature extraction.
5	OPC Foundation (OPC UA) https://opcfoundation.org	Specifications for industrial data access from machine controllers.
6	Eclipse Sparkplug Specification https://sparkplug.eclipse.org	MQTT-based industrial payload and state management standard for edge-to-cloud messaging.
7	NCSC Operational Technology Guidance https://www.ncsc.gov.uk/collection/operational-technology	UK cyber security guidance shaping the network segmentation and data diode decisions.
8	Bosch Production Line Performance https://www.kaggle.com/c/bosch-production-line-performance	Large manufacturing process dataset used to prototype process-to-quality prediction.

Deliverables (17)

- D1 Asset and signal inventory.** A register of every machine, its controller capability, available signals, sampling rates and retrofit needs.
- D2 Use case prioritisation matrix.** A scored comparison of candidate use cases by value, data readiness, feasibility and time to benefit.
- D3 Tiered edge-to-cloud architecture.** A design specifying edge processing, plant storage, cloud transfer and control-loop independence.
- D4 OT security design and approval.** Network, identity and data flow controls approved by cyber security, including one-way transfer where required.
- D5 Retrofit sensor specification.** Sensor types, mounting positions, sampling rates and installation plan for legacy machines.
- D6 Edge feature extraction service.** Software computing vibration spectra, statistical features and alarm counts locally with buffering during outages.
- D7 Raw data retention strategy.** A policy retaining high-frequency data selectively, for example around anomalies and on bottleneck machines.
- D8 Time-series data model.** A consistent model for signals, machine states, work orders and quality results with asset hierarchy.
- D9 Maintenance work order structuring.** A lightweight coding scheme and text classification of historical work orders into failure modes.
- D10 Technician feedback capture.** A simple mobile or terminal-based confirmation of alert outcomes that avoids additional forms.
- D11 Anomaly detection baseline.** Per-machine normal behaviour models with alert thresholds tuned to limit false alarms.
- D12 Alerting and work order integration.** Automatic creation of maintenance notifications in the asset management system from confirmed anomalies.
- D13 Process-to-quality traceability study.** Assessment of linking furnace and additive process data to part inspection results.
- D14 Operational dashboards.** Machine health, alert history and downtime views designed with technicians and supervisors.
- D15 Data quality monitoring.** Checks on sensor dropouts, flatlining, clock skew and gaps across edge and cloud.
- D16 Six-month results report.** A measured account of downtime avoided, alert precision and lessons for group rollout.
- D17 Group scaling blueprint.** A reusable template for extending the architecture to the group's other plants.

CASE STUDY 09 • TELECOMMUNICATIONS

Call Detail Record Processing and Customer Churn Data Mart

Industry	Telecommunications
Organisation	Kestrel Mobile, a UK mobile network operator with 6.4 million subscribers
Your role	Principal Data Engineer, Customer Data
Core themes	High-volume batch processing, legacy Hadoop migration, customer identity, regulatory data retention

The Scenario

Kestrel Mobile's network generates roughly two and a half billion call detail records, data session records and messaging events every day. Every time a subscriber makes a call, sends a message, or opens a data session, the network elements produce a record describing who, when, where, for how long and how much. For more than a decade these records have flowed into an on-premises Hadoop cluster, where a set of batch jobs, some written in MapReduce and some in Hive, mediate them into billing-ready formats, aggregate them into usage summaries and feed a customer analytics data mart. That cluster is now the subject of the most contentious technology decision the company has faced in years, and you, as principal data engineer for customer data, are in the middle of it.

The immediate trigger is churn. Kestrel's monthly churn rate has crept up from 1.1 per cent to 1.6 per cent over eighteen months, which on a base of more than six million subscribers means tens of thousands of lost customers a month and a significant hit to revenue. Competitors are aggressive, and the rules on switching providers have made it easier than ever for customers to leave: a text message gets them the code they need to move within a working day. The retention team has an offer budget and a call centre ready to act, but they complain that the churn risk scores they receive are a month old by the time they arrive, because the data mart is rebuilt monthly from aggregates that take eleven days to compute. By the time the retention team calls a customer flagged as high-risk, many have already gone.

The Hadoop cluster itself is ageing. The hardware support contract ends in fourteen months, the distribution it runs is no longer supported by the vendor, and the two engineers who understood its configuration best left last year. Jobs fail regularly, and when they do, the recovery process

involves manually rerunning dependent steps in the right order, which only one current team member can do reliably. The infrastructure director, Andrew Pike, wants to migrate everything to a cloud data platform, and the group technology strategy supports that direction. He has received proposals from two cloud providers, both of which include substantial migration incentives.

The complications arrive quickly. The first is regulatory. Communications data is subject to strict legal retention and access requirements under UK investigatory powers legislation. Certain categories of data must be retained for defined periods and made available to authorised public bodies through tightly controlled processes, and the company's legal team, led by Rebecca Stone, is deeply uneasy about that data moving to a public cloud. She has not said it is impossible, but she has made clear that the legal and security assurance required would take many months and would need sign-off from government-facing compliance functions. The second is cost. The cloud providers' estimates assume that the workloads will be re-engineered to be efficient. A direct lift and shift of the existing Hive jobs, which scan full daily partitions repeatedly, would, by your own calculations, cost more per month than the entire current data centre budget.

The third complication is data quality and identity. Call detail records identify subscribers by their mobile number and the SIM identifier. The customer relationship management system identifies customers by account number. One account may have several numbers, numbers are recycled to new customers after a quarantine period, and customers who port their number in from another network appear under temporary identifiers for a period. The churn data mart attempts to resolve all of this through a set of mapping tables maintained by a nightly job, but you have found that for about four per cent of active numbers the mapping is either missing or points to the wrong account. Some high-value business customers are treated as low-value consumers, and some churn scores are calculated from a mixture of the old and new owners of a recycled number.

The retention director, Louise Grant, wants weekly, or better still daily, churn scores built on fresh usage data: drops in data consumption, increases in calls to competitors' customer service numbers, network quality problems in the subscriber's home area, and recent complaints. The data science team has shown that network experience features, such

as dropped calls and slow data sessions in the places where a customer spends most time, are among the strongest predictors of churn. Those features require joining call detail records to network performance data at cell level, something the current platform cannot do in reasonable time.

Andrew wants a single migration programme that moves everything to cloud in fourteen months. Rebecca wants regulated data to remain on-premises until legal assurance is complete. Louise wants faster churn scores within three months. Finance wants the new platform to cost less than the old one. And your own view is that the churn problem is solvable relatively quickly with a well-designed incremental aggregation layer, while the full migration is a much longer programme that needs careful separation of regulated and analytical data.

Your own team is stretched and anxious. Of the nine engineers, four have only ever worked on the Hadoop platform, and they are worried, not unreasonably, that a cloud migration will make their skills redundant. Two have already asked you privately whether the company plans to replace them with the cloud provider's professional services consultants, who feature prominently in both migration proposals. The remaining engineers are keen on cloud but have limited experience of running production workloads at this scale. Any plan you propose needs to include a credible path for the existing team, both because you need their knowledge of the undocumented job logic to migrate safely and because losing them mid-migration would be disastrous. At the same time, the day-to-day failures of the current cluster consume so much of their time that there is little capacity left for learning, design or migration work unless something else is deprioritised.

You have also noticed something that nobody has raised. The analytical uses of call detail records, including churn modelling, do not need the full records with precise times and locations of every call. They need aggregates: daily usage by type, counts of dropped calls, the main cells a customer uses. If the platform were redesigned so that detailed records remained in a secure regulated zone and only derived, pseudonymised features moved to the analytics environment, both the legal concerns and the cloud cost concerns might shrink dramatically. But such a design would need buy-in from legal, security and the infrastructure team, and would change the scope of Andrew's migration programme significantly.

THE DECISION IN FRONT OF YOU

Do you support a full cloud migration on the infrastructure timeline, a split architecture that keeps regulated records on-premises while moving derived analytics to cloud, or a faster tactical fix for churn first, and how do you resolve subscriber identity so that any churn score can be trusted?

Data Sources

#	Source and link	How it is used in this project
1	Telco Customer Churn Dataset (IBM) https://www.kaggle.com/datasets/blatchar/telco-customer-churn	Customer-level churn dataset used to design the churn feature and label model.
2	Telecom Italia Big Data Challenge (Harvard Dataverse) https://dataverse.harvard.edu/dataverse/bigdatachallenge	Real aggregated telecom activity data by grid cell for calls, SMS and internet used to prototype CDR aggregation.
3	OpenCellID Cell Tower Database https://opencellid.org	Open cell tower location data for mapping usage to geography and network experience.
4	Ookla Open Data Speed Tests https://github.com/teamookla/ookla-open-data	Aggregated mobile performance tiles used as external network quality features.
5	Ofcom Connected Nations and Mobile Coverage Data https://www.ofcom.org.uk/phones-and-broadband/coverage-and-speeds	UK coverage and performance data for benchmarking network experience by area.
6	Investigatory Powers Act 2016 https://www.legislation.gov.uk/ukpga/2016/25/contents	Legal framework governing retention and access of communications data that constrains architecture.
7	Apache Spark Structured Streaming Guide https://spark.apache.org/docs/latest/structured-streaming-programming-guide.html	Reference for incremental aggregation of usage records at scale.
8	Ofcom One Touch Switching https://www.ofcom.org.uk/phones-and-broadband/switching-provider	Rules on switching providers that frame churn timing and retention windows.

Deliverables (17)

- D1 Current platform assessment.** An inventory of Hadoop jobs, dependencies, data volumes, failure rates and knowledge risks.
- D2 Data classification framework.** A classification separating regulated communications data from derived analytical features.
- D3 Split architecture design.** A target design with a secure regulated zone and a cloud analytics zone connected by controlled feature export.
- D4 Legal and security assurance plan.** Workstreams, evidence and approvals required for any movement of communications-derived data.

- D5** **Subscriber identity resolution model.** A time-aware mapping of numbers, SIMs, ports and recycling to accounts with validity periods.
- D6** **Identity mapping quality report.** Measurement of missing and incorrect mappings with correction processes and ongoing monitoring.
- D7** **Incremental usage aggregation pipeline.** Daily aggregation of CDRs into subscriber usage features without rescanning full history.
- D8** **Network experience features.** Per-subscriber dropped call, data speed and cell congestion features derived from network data.
- D9** **Pseudonymisation service.** Consistent tokenisation of subscriber identifiers for analytical datasets with controlled re-identification.
- D10** **Weekly churn data mart.** A refreshed mart of churn features, scores and outcomes delivered weekly to retention.
- D11** **Retention campaign feedback loop.** Capture of retention contacts, offers and outcomes for model learning and campaign evaluation.
- D12** **Cloud cost model.** A workload-based cost projection comparing lift-and-shift, re-engineered and split approaches.
- D13** **Job orchestration modernisation.** Rebuilt scheduling with dependency management, automatic retries and backfill capabilities.
- D14** **Migration wave plan.** Sequenced workload migration with parallel running, reconciliation and decommissioning milestones.
- D15** **Data retention automation.** Enforcement of legally mandated retention and deletion periods across all zones.
- D16** **Operational runbooks.** Procedures for job failures, late network feeds, identity mapping issues and lawful access requests.
- D17** **Executive decision paper.** A concise paper setting out options, costs, legal risks and recommended sequencing for approval.

CASE STUDY 10 · AVIATION (AIRLINE OPERATIONS)

Operational Flight Disruption Data Hub

Industry	Aviation (Airline Operations)
Organisation	Skyhaven Airways, a UK short-haul airline operating 86 aircraft from five bases
Your role	Lead Data Engineer, Operations Control Data
Core themes	Real-time operational data integration, event correlation, passenger rights compliance, single source of truth

The Scenario

On the second Saturday of last July, a line of thunderstorms moved across northern France during the late afternoon, air traffic control imposed flow restrictions on routes to the Mediterranean, and one of Skyhaven Airways' aircraft went technical at Manchester with a hydraulic warning. By midnight, the airline had cancelled fourteen flights and delayed sixty-one more, stranding nearly four thousand passengers in six countries. The weeks that followed were worse than the night itself. The customer service team received eleven thousand compensation claims. The operations control centre and the finance team disagreed about which delays were caused by extraordinary circumstances outside the airline's control, which would exempt Skyhaven from paying fixed compensation under UK passenger rights law, and which were within its control. The Civil Aviation Authority opened an enquiry after complaints that passengers had not been offered rerouting and care as required. You were asked to lead the data review, and what you found is the reason you are now writing a proposal for an operational data hub.

The operations control centre runs on a patchwork of systems. Flight schedules and aircraft rotations live in the airline's operations control system. Crew rosters and duty times live in a crew management system from a different vendor. Aircraft technical status lives in the maintenance and engineering system. Air traffic flow restrictions arrive as messages from the European network manager and are read by controllers on a separate screen. Weather is viewed on a commercial aviation weather portal. Passenger bookings, connections and special assistance needs are held in the reservation system. Airport slot and stand information comes from the airports' own systems, often by email or phone. Each system records delays, but using its own codes and timestamps. The industry has standard delay codes, but controllers assign them quickly under pressure,

and on the night in July the root cause of many delays was coded as reactionary, meaning a knock-on effect from an earlier delay, without recording what the original cause was.

When the compensation claims arrived, the customer service team had to determine, for each disrupted flight, the chain of causes. Was the aircraft late because of the weather-related flow restriction, which would be extraordinary, or because the crew had run out of legal duty hours after an earlier technical delay, which would not? The data to answer this question existed in five systems but had never been correlated. Analysts spent weeks manually reconstructing timelines from logs, emails and controller notes. They got some of it wrong, the airline paid some claims it could arguably have rejected, rejected some it should have paid, and several rejected passengers escalated to the alternative dispute resolution body, which found against Skyhaven in most of the cases it heard.

The director of operations, Captain Mark Ellison, wants a real-time operational data hub that gives controllers a single, live view of every flight, aircraft, crew pairing and passenger connection, with automatic tracking of how delays propagate through the network. He believes that if controllers could see the knock-on effect of a decision, such as swapping an aircraft or cancelling a rotation, before making it, they would make better decisions and the airline would have an accurate record of causes. The head of customer operations, Lisa Morgan, has a narrower requirement: she needs a reliable, auditable disruption record for every flight, produced within twenty-four hours, that tells her team which passengers are entitled to what, and she needs it before next summer's peak. The chief financial officer, Paul Adeyemi, is focused on cost: the July disruption cost the airline several million pounds in compensation, care, rerouting and reputational damage, and he wants to know how much of that a data investment would recover.

The technical landscape makes the grand vision difficult. The operations control system vendor provides an event feed, but it is a proprietary message format with limited documentation and is licensed per consuming application. The crew management system can export duty data only in batch every thirty minutes. The maintenance system has no event feed at all; engineers update defect status when they get to a terminal, sometimes an hour after the event. Network manager messages are available through a standard interface, but only after the airline completes a technical and

security onboarding process that takes several months. Airport data varies wildly by airport.

The controllers themselves are wary. The senior duty controller, Janet Price, has worked in operations control for twenty-two years and has seen several tools introduced with fanfare and abandoned. She points out that, during disruption, controllers are making decisions every few seconds and have no time to update a new system. Any hub that relies on them entering more information will fail. She also warns that a live network view that is wrong, even occasionally, will be ignored completely, because controllers cannot afford to act on bad information.

The data protection officer has also raised a concern. Passenger data, including special assistance needs that may reveal health conditions, would need to be combined with operational data to calculate entitlements and identify vulnerable passengers who need priority rebooking. That is legitimate and arguably required by law, but it means the hub would hold sensitive personal data, which raises the bar for security, access control and retention.

Timestamps are a quieter but equally serious problem. The operations control system records times in coordinated universal time, the crew system in the local time of each crew base, and the reservation system in the local time of the departure airport. The maintenance system records the time an engineer saved a record, not the time the defect was found. When you tried to line up events for a single disrupted rotation from the July incident, two events that happened ten minutes apart appeared in the wrong order because one system had not adjusted for British Summer Time. For a disruption timeline that may be used as evidence in compensation disputes, getting the order of events right is not a cosmetic concern. Every event would need a normalised timestamp, a record of which clock it came from and, ideally, an indication of whether it reflects when something happened or merely when somebody recorded it.

You have a team of five engineers and a year. You also have an insight from your investigation: most of the value in the July case would have come not from real-time visibility but from correctly correlating events after the fact. A reliable, automatically built disruption timeline for each flight, showing the chain of causes from original event through reactionary delays, crew duty breaches and cancellations to passenger impact, would have saved weeks of analyst time, improved compensation decisions and provided

evidence for the regulator. Building that timeline requires the same data integration as the live hub, but it can tolerate latency of hours rather than seconds, and it can be built from batch extracts where event feeds are unavailable.

There is a risk, though. If you build the retrospective disruption record first, Mark may see it as a watered-down version of his vision and withdraw his support, and without the operations director behind the project, controllers will not engage. If you build the live hub first, Lisa will not get her compensation record by next summer, and the finance case will not materialise.

THE DECISION IN FRONT OF YOU

Do you build the live operational network view the operations director wants, the auditable disruption record customer operations needs, or a common event backbone that delivers the retrospective record first and evolves toward real-time, and how do you win over controllers who have seen tools fail before?

Data Sources

#	Source and link	How it is used in this project
1	OpenSky Network Historical Flight Data https://opensky-network.org/data	Real ADS-B aircraft position and flight data used to reconstruct actual flight movements and delays.
2	US BTS Airline On-Time Performance Data https://www.transtats.bts.gov	Large historical dataset of flight delays with cause categories used to model delay propagation logic.
3	EUROCONTROL Aviation Intelligence Portal https://ansperformance.eu	European air traffic flow management delay and punctuality data used to attribute network restrictions.
4	Aviation Weather Center Data API https://aviationweather.gov/data/api/	METAR and TAF weather observations and forecasts used to evidence weather-related causes.
5	OurAirports Open Data https://ourairports.com/data/	Airport and runway reference data for enriching schedules and routes.
6	UK CAA Passenger Rights Guidance https://www.caa.co.uk/passengers-and-public/your-rights/delays-and-cancellations/	Regulatory rules on compensation, care and rerouting that define entitlement logic.
7	UK CAA Punctuality Statistics https://www.caa.co.uk/data-and-analysis/uk-aviation-market/flight-punctuality/	Official UK airport punctuality data for benchmarking performance.
8	IATA Delay Codes Reference (AHM 730) https://www.iata.org/en/publications/manuals/airport-handling-manual/	Industry-standard delay coding used to normalise causes across systems.

Deliverables (17)

- D1** **Disruption data review report.** Findings from the July event on data gaps, coding practices and their effect on compensation decisions.
- D2** **Source system integration map.** A catalogue of operational systems, available interfaces, latency, formats and licensing constraints.
- D3** **Common operational event model.** A canonical schema for flight, aircraft, crew, delay, restriction and passenger events with consistent timestamps.
- D4** **Event backbone design.** An architecture capturing events from feeds and batch extracts into a shared, replayable log.
- D5** **Delay cause normalisation.** Mapping of system-specific delay codes into a standard hierarchy with root and reactionary causes separated.
- D6** **Delay propagation engine.** Logic linking reactionary delays to originating events through aircraft rotations and crew pairings.
- D7** **Automated disruption timeline.** A per-flight timeline showing the causal chain from first event to passenger impact produced within twenty-four hours.
- D8** **Passenger entitlement calculator.** Rules determining compensation, care and rerouting entitlements per passenger with evidence references.
- D9** **Extraordinary circumstances evidence pack.** Automatic assembly of weather, air traffic and technical evidence supporting claim decisions.
- D10** **Network manager integration plan.** Onboarding steps and timeline for air traffic flow management message access.
- D11** **Live network view prototype.** A controller-facing display of rotations and propagated delays tested during a controlled disruption exercise.
- D12** **Controller co-design sessions.** Documented workshops with duty controllers on information needs and zero-extra-input principles.
- D13** **Sensitive passenger data controls.** Access, masking and retention controls for assistance needs and other special category data.
- D14** **Data quality monitoring.** Checks on missing events, clock inconsistencies, orphan delays and uncoded disruptions.
- D15** **Claims outcome analytics.** Reporting on claim volumes, decision accuracy, dispute outcomes and cost per disruption.
- D16** **Regulator response support.** Structured data outputs supporting responses to CAA enquiries and audits.
- D17** **Business case and benefits tracker.** Quantified savings from compensation accuracy, analyst time and improved recovery, tracked after go-live.

CASE STUDY 11 · CAPITAL MARKETS (BROKERAGE)

Tick Data Lake and MiFID II Best Execution Reporting

Industry	Capital Markets (Brokerage)
Organisation	Ashdown Securities, a London agency broker executing equities and ETF orders for institutional clients
Your role	Head of Market Data Engineering
Core themes	High-frequency time-series storage, market data licensing, regulatory reporting, transaction cost analysis

The Scenario

Ashdown Securities executes around four hundred thousand equity and exchange-traded fund orders a month on behalf of pension funds, asset managers and hedge funds. As an agency broker, it does not trade on its own account; it earns commission by executing clients' orders well. What "well" means has been defined with increasing precision by regulation. Under the European and UK versions of the Markets in Financial Instruments Directive, brokers must take all sufficient steps to obtain the best possible result for clients, and must be able to demonstrate it. Clients, for their part, now expect detailed transaction cost analysis showing how each order performed against benchmarks such as the arrival price, the volume-weighted average price and the prevailing best bid and offer at the time of each fill. You lead market data engineering, and this morning you received an email from the Chief Compliance Officer, Victoria Hale, forwarding a note from a large pension fund client that has started a formal review of its brokers.

The client's note is polite but pointed. It asks Ashdown to provide, for a sample of three hundred orders over the past year, fill-level execution data alongside the consolidated best bid and offer at each fill timestamp, measured to the microsecond, together with evidence of which trading venues were considered and why particular venues were chosen. Two competing brokers have apparently already provided this. Ashdown can provide the fills. It cannot easily provide the rest.

The problem lies in how Ashdown stores market data. The firm subscribes to real-time feeds from several exchanges and multilateral trading facilities through a market data vendor, and its smart order router consumes these feeds in memory to make routing decisions. But the feeds

are not stored in full. A tick capture system records top-of-book quotes for the most actively traded instruments, sampled at one-second intervals, to a relational database that was designed for end-of-day reporting. Full depth of book and every quote update are discarded. The router's decision logs record which venue was chosen for each child order, but not what the router could see on each venue at the moment of decision. Reconstructing the market state at the time of any historical fill requires buying historical data from the vendor, at a cost per request that becomes significant at scale, and the vendor's timestamps come from a different clock source than Ashdown's own, with discrepancies of several milliseconds.

The Head of Electronic Trading, James Whitmore, wants to build a full tick data lake: capture every quote and trade message from every venue Ashdown connects to, store them in a columnar time-series format, and build transaction cost analysis and best execution evidence on top. He argues that this is now table stakes for a serious agency broker and that the same data would allow his quantitative team to improve the router, test new algorithms and win business. His estimate is that the lake would ingest around three to four billion messages a day across European venues.

The Chief Operating Officer, Sarah Kingsley, supports the goal but is alarmed by the costs. Storage is not the main issue; at compressed columnar rates, even several years of full tick data would be affordable. The issue is licensing. Exchanges and market data vendors charge separately for the right to store and use data for non-display purposes, such as automated analysis and algorithmic trading, and for redistributing data to clients in reports. Some exchanges' policies distinguish between data used internally and data derived for client reporting, and the fees for each category vary by venue. The market data administration team estimates that storing full depth of book from all venues and using it for client transaction cost analysis could add several hundred thousand pounds a year to licence costs, depending on how derived data is classified. Nobody is quite sure, because the licensing terms are complex and differently interpreted by each venue.

Victoria's priority is compliance. The regulator has fined firms for failing to monitor best execution effectively, and she wants Ashdown to be able to demonstrate, on demand, that its routing decisions were sound. She points out that best execution monitoring does not necessarily require full depth

of book; the consolidated best bid and offer at the relevant timestamps, with accurate clock synchronisation, may be sufficient for most regulatory purposes. She has also reminded everyone that regulation requires business clocks to be synchronised to a defined accuracy relative to coordinated universal time and that the firm's clock synchronisation evidence has gaps.

The quantitative team leader, Dr Chen Wei, has another perspective. He points out that best execution and transaction cost analysis depend not only on the quote at the moment of the fill but also on market impact, the extent to which Ashdown's own orders moved prices. Measuring impact properly requires trade and quote data for a window around each order, including periods before and after, and preferably depth of book to see liquidity being consumed. A top-of-book-only solution would satisfy compliance but would not help the business improve execution quality, which is what clients ultimately care about.

The infrastructure constraints are real as well. The capture must happen in Ashdown's co-location facility, where rack space and power are expensive and limited, and the capture process must not add any latency to the production trading path. Feed handlers for each venue use different protocols, and message formats change when venues upgrade their systems, which happens several times a year. The current tick capture system has silently lost data during at least two venue upgrades because nobody noticed the format change.

The pension fund's request has also exposed a gap in how orders are linked to their executions. When a client sends a large order, Ashdown's algorithms slice it into many child orders sent to different venues over minutes or hours, and each child order may receive several partial fills. The order management system, the algorithmic trading engine and the router each assign their own identifiers, and the links between them are held in logs that are rotated after thirty days. For older orders, reconstructing which fills belonged to which parent order requires matching on instrument, side, quantity and time, which is usually but not always unambiguous. Several of the three hundred sampled orders fall into the ambiguous category. Any long-term solution needs to preserve the full order lifecycle permanently, but the immediate client response will have to be honest about the handful of orders where the linkage is reconstructed rather than recorded.

You have been asked to present an options paper to the executive committee in four weeks. You need to address the immediate client request, the regulatory obligation, the business ambition and the licensing costs, and you need to propose something your team of five can actually build and run. You are acutely aware that underestimating the licensing implications could produce an unexpected bill that would end your credibility, while underbuilding the capability could cost Ashdown one of its largest clients.

THE DECISION IN FRONT OF YOU

Do you build a full-depth tick data lake, a compliance-focused top-of-book capture with synchronised clocks, or a tiered design that captures everything but licenses and exposes it selectively, and how do you answer the pension fund's request in the meantime?

Data Sources

#	Source and link	How it is used in this project
1	Nasdaq TotalView-ITCH Sample Files https://emi.nasdaq.com/ITCH/	Real full-depth market data sample files used to build and test the feed parsing and storage pipeline.
2	LOBSTER Limit Order Book Data https://lobsterdata.com	Reconstructed limit order book data used to prototype book reconstruction and impact analysis.
3	ESMA FIRDS Instrument Reference Data https://registers.esma.europa.eu	Official instrument reference data for mapping ISINs, venues and trading characteristics.
4	FCA Best Execution Guidance https://www.fca.org.uk/firms/best-execution	Regulatory expectations for best execution monitoring and evidence.
5	MiFID II Directive 2014/65/EU (EUR-Lex) https://eur-lex.europa.eu/eli/dir/2014/65/oj	Legal source for best execution and related obligations underpinning reporting requirements.
6	Commission Delegated Regulation (EU) 2017/574 (RTS 25) https://eur-lex.europa.eu/eli/reg_del/2017/574/oj	Clock synchronisation requirements that define timestamp accuracy obligations.
7	OpenFIGI API https://www.openfigi.com/api	Free instrument identifier mapping to reconcile vendor, venue and internal identifiers.
8	FIX Trading Community Standards https://www.fixtrading.org/standards/	FIX protocol standards for order and execution messages used to link orders, child orders and fills.

Deliverables (17)

- D1 Client response pack.** A defensible response to the pension fund with fills, reconstructed quotes and a clear statement of data limitations.

- D2 Options paper.** A comparison of full-depth, top-of-book and tiered capture on cost, licensing, compliance, value and delivery risk.
- D3 Market data licensing analysis.** A venue-by-venue summary of storage, non-display and client reporting fees under each option.
- D4 Tiered capture architecture.** A design capturing raw feeds passively in co-location with tiered storage and controlled access.
- D5 Feed handler framework.** Protocol-specific parsers with versioned schemas and automatic detection of venue format changes.
- D6 Clock synchronisation evidence.** Monitoring and records demonstrating business clock accuracy against regulatory requirements.
- D7 Tick data storage model.** Partitioned columnar storage by venue, date and instrument with efficient as-of joins.
- D8 Consolidated best bid and offer builder.** A process deriving a consolidated view across venues at any timestamp.
- D9 Order lifecycle linkage.** Joining parent orders, child orders, routing decisions and fills using FIX identifiers.
- D10 Router decision snapshotting.** Enhanced logging capturing the visible market state at each routing decision.
- D11 Transaction cost analysis engine.** Calculation of arrival price, VWAP and spread-based benchmarks per order and fill.
- D12 Market impact analytics.** Pre- and post-trade price movement analysis around orders for the quantitative team.
- D13 Best execution monitoring dashboard.** Ongoing surveillance of execution quality by venue, instrument and client with exception flags.
- D14 Data completeness controls.** Checks for message gaps, sequence breaks and venue outages with alerting.
- D15 Entitlement and access control.** Controls ensuring data use aligns with licences, separating internal, non-display and client-facing uses.
- D16 Operational runbook.** Procedures for feed changes, capture failures, backfills and regulatory data requests.
- D17 Executive committee presentation.** A concise briefing with recommendation, costs, risks and phased plan.

CASE STUDY 12 • PHARMACEUTICALS

Pharmacovigilance Adverse Event Signal Detection Pipeline

Industry	Pharmaceuticals
Organisation	Verantis Pharma, a mid-sized specialty pharmaceutical company with 34 marketed products in 40 countries
Your role	Senior Data Engineer, Drug Safety Informatics
Core themes	Regulated data pipelines, medical coding, validated systems, unstructured text ingestion

The Scenario

Pharmacovigilance is the part of the pharmaceutical industry that most people never think about and everyone depends on. Once a medicine is on the market, its manufacturer must collect, assess and report adverse events, meaning suspected side effects, from every possible source, and must continuously look for signals that a medicine may have a risk that was not identified in clinical trials. Verantis Pharma markets thirty-four products across forty countries, and its drug safety department processes roughly sixty thousand individual case safety reports a year. You are a senior data engineer in drug safety informatics, and you have been asked to help solve a problem that the Qualified Person for Pharmacovigilance, Dr Elizabeth Moreau, describes as the department's biggest risk.

The core safety database is a validated commercial system that holds every case report, coded using the MedDRA medical terminology for adverse events and a drug dictionary for products. Case processors enter reports received from healthcare professionals, patients, partners and literature, and the system generates regulatory submissions in the ICH E2B format to authorities including the MHRA, the European Medicines Agency and the US Food and Drug Administration. That part works. The problem is signal detection. Signals are currently detected by a small team of safety physicians who run periodic statistical analyses on data exported from the safety database, review cumulative case listings, and scan literature. They also check public regulatory databases, such as the FDA Adverse Event Reporting System, to see whether events reported for Verantis products, or for similar products from other companies, are showing disproportionate patterns.

These analyses are done in spreadsheets and a statistical package, using exports that are prepared manually each quarter. Last year, an inspection by a regulatory authority identified that one signal, a pattern of liver injury reports for one of Verantis's products, had been visible in public adverse event data for two quarters before the company's own review picked it up. The inspectors did not find that patients had been harmed as a result, but they classified the finding as major, and Verantis committed in its corrective action plan to implement "systematic, timely and documented signal detection incorporating internal and external data sources". The deadline for demonstrating that commitment is nine months away.

The obvious answer is a signal detection data pipeline: regular automated ingestion of internal case data and external sources, harmonisation of products and events, calculation of disproportionality statistics, and presentation of potential signals to safety physicians for assessment, with every step documented. Several commercial signal detection tools exist, and the IT director, Robert Hayes, favours buying one. He points out that commercial tools come with validation packages, that pharmacovigilance systems are subject to computerised system validation requirements, and that building in-house would mean validating custom code, which is expensive and slow.

Elizabeth is not opposed to buying a tool, but she has found that the commercial tools she has evaluated handle external data poorly. They work well with a company's own safety database, but ingesting and harmonising public data from the FDA, the European and UK regulators, and the medical literature requires substantial custom work regardless. The public FDA data, for example, is released quarterly, uses free-text drug names that vary enormously, contains duplicate reports from different reporters of the same event, and includes reports for Verantis products under brand names, generic names, misspellings and foreign trade names. The UK and European public data is available in different forms, with different levels of detail and different access conditions.

The head of the safety physicians, Dr Samuel Okonjo, has a further concern. He wants the pipeline to include not just structured case data but literature and, eventually, social media and patient forums, where early signals sometimes appear. He has seen a vendor demonstration of natural language processing that extracts adverse events from free text and is enthusiastic. Elizabeth is cautious: literature surveillance is a regulatory

obligation with well-defined processes, but social media monitoring raises questions about the volume of low-quality data, the obligation to follow up on any identifiable report, and the regulatory expectation that any adverse event the company becomes aware of must be processed. Opening a social media channel could create thousands of reportable cases that her team cannot handle.

Your own assessment is that the most important engineering work is not the statistics, which are well established, but the harmonisation. Mapping thousands of variant drug names in public data to Verantis products and their active ingredients, mapping reported events to MedDRA at the correct level, identifying and removing duplicates, and doing all of this reproducibly every quarter, is where the manual process has failed and where any tool, bought or built, will need help. You also recognise that everything you build will need to be validated in a way that satisfies inspectors: documented requirements, risk assessment, testing evidence, change control and audit trails.

Robert has proposed a middle way that he thinks might work: buy a signal detection tool for the core analytics and presentation, but build an external data harmonisation pipeline in-house, validated as a separate system, that feeds the tool. Elizabeth likes the idea but worries about the interface between the two becoming a point of failure. The quality assurance lead, Hannah Keane, has pointed out that splitting the solution means two validation efforts, two change control processes and two sets of documentation, and that the nine-month deadline is already tight for one.

A further difficulty sits inside Verantis's own portfolio. Several products are licensed to partners in particular territories, and those partners collect adverse event reports under safety data exchange agreements that specify what must be shared, in what format and within what timeframe. In practice, some partners send structured electronic reports, some send spreadsheets and some send scanned forms that case processors re-key. Timelines vary, and at least one partner has been consistently late. Signal detection that relies on the internal safety database will therefore see partner-territory cases later than Verantis-territory cases, and cumulative counts for some products will understate true reporting until partner data catches up. Any pipeline you build needs to account for this lag explicitly, rather than presenting counts that look complete but are not, and the

safety data exchange agreements may themselves need to be renegotiated to support timely signal detection.

There is also the question of what happens after a signal is detected. At present, the assessment of signals, the decisions made and the evidence considered are recorded in documents and emails. Inspectors want to see a traceable signal management process from detection through validation, assessment, decision and action. That is partly a workflow problem, but it depends on the data pipeline preserving exactly which data version produced each potential signal.

THE DECISION IN FRONT OF YOU

Do you recommend buying a commercial signal detection platform, building an in-house pipeline, or combining a purchased tool with a validated in-house harmonisation layer, and how do you scope external and unstructured sources so that the corrective action is met without overwhelming case processing?

Data Sources

#	Source and link	How it is used in this project
1	openFDA Drug Adverse Event API https://open.fda.gov/apis/drug/event/	Public FDA adverse event reports accessible via API for automated external signal detection.
2	FDA FAERS Quarterly Data Extract Files https://fis.fda.gov/extensions/FPD-QDE-FAERS/FPD-QDE-FAERS.html	Full quarterly FAERS data releases used for bulk ingestion, deduplication and disproportionality analysis.
3	MHRA Interactive Drug Analysis Profiles https://yellowcard.mhra.gov.uk/idaps	UK Yellow Card suspected adverse reaction data for external comparison.
4	EudraVigilance Public Reports (adrreports.eu) https://www.adrreports.eu	European public adverse reaction data by substance.
5	RxNorm (US National Library of Medicine) https://www.nlm.nih.gov/research/umls/rxnorm/	Normalised drug naming used to map variant product names to ingredients.
6	MedDRA https://www.meddra.org	The standard medical terminology for coding adverse events at the appropriate hierarchy level.
7	ICH E2B(R3) Guideline https://www.ich.org/page/efficacy-guidelines	International standard for individual case safety report data elements and transmission.
8	PubMed E-utilities API https://www.ncbi.nlm.nih.gov/books/NBK25501/	Programmatic literature search for automated literature surveillance feeds.

Deliverables (17)

- D1** **Corrective action traceability matrix.** A mapping of inspection findings and commitments to specific pipeline capabilities and evidence.
- D2** **Build, buy and hybrid appraisal.** A structured comparison covering capability, validation burden, cost, timeline and inspection risk.
- D3** **Validated system requirements specification.** User and functional requirements for signal detection and external data harmonisation.
- D4** **Computerised system validation plan.** A risk-based validation approach with test strategy, documentation and responsibilities.
- D5** **External data ingestion pipelines.** Automated, logged ingestion of FAERS, UK and EU public data with version tracking.
- D6** **Drug name harmonisation service.** Mapping of brand, generic, misspelt and foreign names to products and active ingredients with confidence scoring.
- D7** **Adverse event term normalisation.** Coding of reported terms to MedDRA preferred terms with review workflow for low-confidence mappings.
- D8** **Duplicate report detection.** Probabilistic identification of duplicate reports across and within sources.
- D9** **Disproportionality analysis engine.** Reproducible calculation of reporting odds ratios, proportional reporting ratios and Bayesian measures.
- D10** **Signal review workbench data feed.** Structured outputs of potential signals delivered to the chosen tool or dashboard for physician review.
- D11** **Signal management audit trail.** Recording of detection, validation, assessment and decisions linked to the exact data version used.
- D12** **Literature surveillance feed.** Automated weekly literature search results with deduplication and screening workflow.
- D13** **Social media monitoring scoping paper.** An assessment of volume, reportability obligations and resource impact before any implementation.
- D14** **Data lineage and version control.** Traceability from each statistic back to source files, mappings and code versions.
- D15** **Change control procedure.** A documented process for modifying mappings, code and configuration in the validated system.
- D16** **Inspection readiness pack.** Evidence and narratives demonstrating systematic, timely and documented signal detection.
- D17** **Operational monitoring and SOP updates.** Revised standard operating procedures and pipeline health monitoring for ongoing operation.

CASE STUDY 13 • MEDIA AND VIDEO STREAMING

Real-Time Viewing Event Pipeline for Personalised Recommendations

Industry	Media and Video Streaming
Organisation	Northlight Streaming, a subscription video service with 3.8 million subscribers in the UK and Ireland
Your role	Staff Data Engineer, Personalisation Platform
Core themes	Event streaming at scale, feature freshness, experimentation data, editorial versus algorithmic control

The Scenario

Northlight Streaming built its audience on British and Irish drama, documentaries and a well-curated library of international films. For its first four years, the home screen that subscribers saw was assembled mostly by an editorial team, with a simple algorithmic row of "because you watched" suggestions refreshed overnight. Growth was steady. Over the past year, however, subscriber growth has stalled and monthly cancellations have risen, particularly among subscribers who joined for a single high-profile series and left soon after finishing it. The chief executive, Fiona Gallagher, has set the company a clear target: reduce churn among new subscribers by a quarter within a year. The product team believes personalisation is the lever. You are the staff data engineer for the personalisation platform, and the data pipeline behind recommendations has become the bottleneck.

Every play, pause, seek, completion and abandonment on every device produces events. Subscribers watch on smart televisions, streaming sticks, games consoles, phones, tablets and web browsers, and each client platform was built by a different team or agency. The events arrive at an ingestion endpoint and are written to a data lake in hourly batches. From there, an overnight job computes viewing histories, completion rates and genre affinities, trains the recommendation model, and writes recommendations for each subscriber into a key-value store that the home screen reads. If a subscriber finishes a series at nine in the evening, they see recommendations influenced by it the following afternoon at the earliest. The data science team has shown in offline analysis that the moment immediately after finishing a series is exactly when subscribers are most at risk of drifting away, and that fresher recommendations at that moment could make a measurable difference.

The engineering challenge is not just speed. The events themselves are inconsistent. The smart television application, built by an agency that no longer works with Northlight, sends a heartbeat event every sixty seconds during playback but does not send a completion event. The mobile apps send completion events but define completion as reaching the start of the end credits, which is marked differently for each title. The web player sends duplicate events when the network is unreliable. Some older devices send timestamps from their local clock, which may be wrong by hours. When you tried to calculate a simple metric, the proportion of subscribers who finished the first episode of a new drama, the answer ranged from forty-one to fifty-eight per cent depending on which device's logic you trusted.

The head of data science, Dr Aisha Karim, wants a streaming feature pipeline that updates each subscriber's viewing profile within minutes and a recommendation service that can re-rank the home screen in real time. She has a clear model architecture in mind and argues that the business case depends on responsiveness. The platform engineering lead, Owen Barrett, is concerned about cost and reliability. Northlight's peak viewing, on Sunday evenings, produces roughly eighty thousand events per second, and a real-time pipeline must handle that peak without falling behind. He also points out that the recommendation service is on the critical path of the home screen: if it fails or slows down, subscribers see an empty or broken home page, which is far worse than slightly stale recommendations.

The editorial director, Siobhan Doyle, has concerns of a different kind. Northlight's brand is built on curation, and she has contractual commitments to rights holders and producers about prominence for certain titles. Several production partners have deals that guarantee a title will appear in specified positions on the home screen for a period after launch. Public service content obligations under a distribution agreement also require certain titles to be discoverable. An algorithmic home screen that overrides these commitments would create legal and commercial problems. Siobhan is not against personalisation, but she wants editorial rules to be first-class inputs to the system, not an afterthought, and she wants to be able to see and explain why any title appears where it does.

Then there is experimentation. The product team wants to run controlled experiments comparing recommendation approaches, and the churn

target will be judged on experimental evidence. But the current experimentation data is unreliable. Experiment assignments are logged by the home screen service, while outcomes are measured from viewing events, and joining the two depends on subscriber identifiers that differ between devices when subscribers are not logged in, such as on shared family televisions. Profiles within a household further complicate matters: a household might have four profiles, but many families use only one, so the viewing history of a single profile can reflect a parent's crime dramas and a child's cartoons at once. The data protection officer has also asked that any profiling of children's viewing be handled with particular care under the children's code for online services.

Your team has five engineers. You estimate that building a robust real-time pipeline, fixing event quality across all clients, integrating editorial rules and making experimentation reliable would take well over a year if done properly. Fiona wants to see a reduction in churn within twelve months, which means experiments must be running within six. You have to decide what to build first. You suspect that the event quality problems will undermine any recommendation approach, real-time or not, and that experimentation results will be meaningless if the underlying event data is inconsistent. But fixing event quality means persuading five client teams to change their instrumentation, which will take months and is not glamorous.

Content metadata is the other hidden weakness. Recommendations depend on knowing what each title is about, but Northlight's catalogue metadata comes from a mixture of distributor feeds, an editorial content management system and manual spreadsheets. Genres are inconsistent, some titles are tagged with dozens of labels and others with none, series and episodes are sometimes linked incorrectly, and availability windows, which determine whether a title can be shown at all, are held in a rights management system that updates nightly. Twice last quarter, the recommendation row promoted a film whose licence had expired the previous day, and subscribers who clicked on it were shown an error. Siobhan's team noticed before the data team did. Real-time viewing data is of limited value if the catalogue it is matched against is stale, inconsistent or occasionally wrong, and the rights data in particular would need to be at least as fresh as the recommendations themselves.

You also wonder whether real-time is really necessary for the highest-value moment. A near-real-time approach that detects series completion within a few minutes and triggers a targeted refresh for that subscriber alone might capture most of the value at a fraction of the cost and risk of rebuilding the whole recommendation pipeline as a stream.

THE DECISION IN FRONT OF YOU

Do you rebuild the pipeline for full real-time recommendations, pursue targeted near-real-time triggers at key moments, or prioritise event quality and experimentation foundations first, and how do you embed editorial and contractual rules into an algorithmic home screen?

Data Sources

#	Source and link	How it is used in this project
1	MovieLens Datasets (GroupLens) https://grouplens.org/datasets/movielens/	Large-scale ratings and tagging datasets used to prototype recommendation features and offline evaluation.
2	Netflix Prize Data https://www.kaggle.com/datasets/netflix-inc/netflix-prize-data	Historical user-title interaction data for testing collaborative filtering pipelines at scale.
3	Microsoft MIND News Recommendation Dataset https://msnews.github.io	Impression and click logs used to model impression-level experimentation and position effects.
4	IMDb Non-Commercial Datasets https://developer.imdb.com/non-commercial-datasets/	Title metadata, genres and credits used to enrich content features.
5	TMDB API https://developer.themoviedb.org/docs	Additional content metadata and imagery for catalogue enrichment.
6	ICO Age Appropriate Design Code https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/childrens-information/childrens-code-guidance-and-resources/	Requirements for processing children's data that govern profiling of child profiles.
7	Apache Kafka Streams Documentation https://kafka.apache.org/documentation/streams/	Stream processing reference for sessionisation and profile updates.
8	Snowplow Event Tracking Documentation https://docs.snowplow.io	Reference patterns for self-describing event schemas and validation across client platforms.

Deliverables (17)

- D1 Event quality audit.** A cross-device analysis of event types, definitions, duplication and clock errors with quantified impact.
- D2 Unified playback event specification.** A versioned schema and tracking plan defining play, heartbeat, completion and abandonment consistently.

- D3 Client instrumentation change plan.** Agreed changes and timelines for each client team with validation tests.
- D4 Server-side event normalisation layer.** A process correcting timestamps, deduplicating events and deriving completion where clients cannot.
- D5 Near-real-time completion trigger.** A streaming job detecting series and film completion within minutes and triggering targeted refreshes.
- D6 Subscriber profile feature pipeline.** Incremental computation of viewing history, affinities and recency features.
- D7 Household and profile disambiguation.** Methods to separate mixed viewing within shared profiles where permissible.
- D8 Editorial rules engine integration.** A mechanism applying contractual prominence and discoverability rules to ranked recommendations.
- D9 Recommendation explanation log.** Records of why each title appeared in each position, including model scores and editorial overrides.
- D10 Experimentation data foundation.** Reliable assignment logging, exposure tracking and outcome joins for controlled experiments.
- D11 Experiment analysis framework.** Standardised metrics, guardrails and statistical methods for recommendation experiments.
- D12 Churn outcome dataset.** A subscriber-level dataset linking recommendations, viewing and cancellation for impact measurement.
- D13 Children's data safeguards.** Controls limiting profiling for child profiles in line with the children's code.
- D14 Resilience and fallback design.** Degraded modes ensuring the home screen always renders if personalisation services fail.
- D15 Peak load test results.** Evidence that pipelines and services handle Sunday evening peak with headroom.
- D16 Cost model.** Infrastructure cost comparison of batch, near-real-time and full streaming options.
- D17 Twelve-month roadmap.** A phased plan tying engineering milestones to experiment start dates and churn targets.

CASE STUDY 14 · PUBLIC SECTOR (POLICING)

Police Force Crime Data Quality and Open Data Publication Platform

Industry	Public Sector (Policing)
Organisation	Wyedale Constabulary, a county police force serving 1.2 million residents
Your role	Head of Data Engineering, Information Management Department
Core themes	Public transparency, data quality under inspection, location anonymisation, operational versus statistical data

The Scenario

Every month, police forces in England and Wales publish street-level crime and outcome data that anyone can download or view on a map. Residents look up their own streets, journalists compare neighbourhoods, estate agents quietly check areas for clients, and academics use the data to study everything from the effects of street lighting to the geography of knife crime. For Wyedale Constabulary, that monthly publication has become a source of recurring embarrassment. You lead data engineering in the force's information management department, and last week the chief constable received a letter from the local police and crime commissioner asking why Wyedale's published figures for violent crime had dropped by thirty per cent in a single month when nothing on the ground had changed.

The answer, which your team pieced together over three stressful days, was a data pipeline failure. Wyedale migrated its crime recording system eight months ago to a new records management platform used by several forces. The monthly extract that feeds national publication was rebuilt for the new platform by the supplier, and it contains a filter that excludes crimes whose classification is still under review. Under the old system, crimes were classified at the point of recording and rarely changed. Under the new system, a larger share of violent crimes are recorded with a provisional classification and confirmed later by the force crime registrar's team, which is running a backlog. Those crimes were silently excluded. Nobody noticed until the commissioner's office, which tracks the published data independently, raised the alarm.

That incident sits on top of a longer-running problem. His Majesty's Inspectorate of Constabulary and Fire and Rescue Services inspects forces' crime data integrity, meaning whether crimes that are reported are recorded correctly and promptly in line with national crime recording

standards. Wyedale's last inspection graded it as requiring improvement, estimating that around one in eight reported crimes, and a higher proportion of violent and sexual offences, were not being recorded as they should have been. The inspectorate is due to return in nine months. The deputy chief constable, Helen Brightman, has made crime data integrity her personal priority and has asked for a data platform that allows the force to monitor its own compliance continuously, rather than finding out from inspectors.

You have been asked to design a platform that serves three purposes: reliable national publication, continuous crime data integrity auditing, and better operational analytics for the force's neighbourhood policing teams. Each purpose pulls the design in a different direction.

National publication has strict rules. Locations are anonymised by snapping each crime to the nearest of a set of predetermined map points, so that a published crime cannot be tied to a specific house. Certain offences, including sexual offences and some domestic abuse, are subject to additional protection. The data must be submitted monthly in a defined format, and once published, historical months are occasionally revised when outcomes change. The force's data protection officer, Graham Hollis, is adamant that nothing about the new platform can increase the risk of re-identifying victims. He points out that, in small villages, even snapped locations combined with offence type and month can sometimes point to a household, and that the force received a complaint two years ago from a domestic abuse victim who believed her neighbours had identified her from the published map.

Crime data integrity auditing needs something quite different: access to the full, detailed record of incidents reported to the force, including the free-text narratives of calls to the control room, so that auditors can identify incidents where a crime was described but not recorded. At present, a small team of auditors does this manually by sampling a few hundred incidents a month. The deputy chief wants the platform to flag likely under-recording automatically, for example incidents where the caller describes an assault but no crime record is linked. That means processing sensitive, unstructured text about victims, suspects and witnesses, and applying some form of text classification, which raises its own questions about accuracy, bias and oversight.

Neighbourhood policing teams want timely, mapped data on the crimes and incidents in their areas, with repeat locations, repeat victims and emerging patterns highlighted. The head of neighbourhood policing, Superintendent Marcus Obi, complains that the current reports arrive weeks after the events they describe and that his officers have given up using them. His teams want something closer to daily.

The technical environment is constrained. The new records management platform is hosted by the supplier and shared with other forces, and direct database access is not permitted. Data is available through a nightly extract in a supplier-defined format and through an application interface with limited capacity. The control room system is separate and older, with its own nightly export. The force's data warehouse is on-premises, and the national policing security framework imposes strict controls on where police data can be processed and by whom. Any cloud service would need to meet those controls, and vetting requirements apply to anyone, including contractors, who can access the data.

Your team is small, four engineers and two analysts, and one of the engineers is currently on secondment to a regional collaboration project. The supplier has offered to build additional extracts, but its change requests take months and are charged per request. Helen has hinted that funding for more staff might be available if you can show early results. Marcus has quietly asked whether his team could simply be given access to the raw extract so his own analyst can build what he needs, which would be faster but would create yet another unmanaged version of the truth.

There is also a cultural dimension that you have come to appreciate. Crime recording decisions are made by frontline officers and control room staff under considerable pressure, and many of them see the crime data integrity agenda as bureaucracy imposed from above, a set of rules that turn every neighbourly dispute into a crime statistic. The force crime registrar, Detective Inspector Joanne Price, has spent years trying to change that culture through training and feedback. She is supportive of automated auditing but has warned you that if it is perceived as a tool for catching out officers, it will generate resentment and possibly defensive recording practices that make matters worse. She would like the platform to produce feedback that helps supervisors coach their teams, showing patterns rather than naming individuals, and she wants to be involved in deciding how flags are presented and to whom.

You also know that national publication is the only one of the three purposes with a hard monthly deadline and external scrutiny every single month. If you prioritise the auditing capability or the neighbourhood analytics and the publication fails again, the force's credibility will suffer further. If you prioritise publication alone, you will fix the symptom that embarrassed the chief constable but leave the inspection problem, which is far more serious, largely unaddressed.

THE DECISION IN FRONT OF YOU

How do you design and sequence a platform that guarantees accurate, safely anonymised monthly publication, enables continuous and ethically governed crime data integrity auditing, and gives neighbourhood teams timely analytics, all within the security constraints of policing data?

Data Sources

#	Source and link	How it is used in this project
1	data.police.uk API and Bulk Downloads https://data.police.uk/docs/	Official street-level crime, outcome and stop and search data used to understand the publication format and validate outputs.
2	Home Office Crime Recording Rules for Frontline Officers and Staff https://www.gov.uk/government/publications/counting-rules-for-recorded-crime	National counting rules and crime recording standards defining correct recording.
3	HMICFRS Crime Data Integrity Inspections https://hmicfrs.justiceinspectorates.gov.uk/our-work/article/crime-data-integrity/	Inspection methodology and findings that define the audit capability's requirements.
4	ONS Crime in England and Wales https://www.ons.gov.uk/peoplepopulationandcommunity/crimeandjustice	National statistics and survey comparisons used to contextualise recorded crime trends.
5	Police Recorded Crime Open Data Tables https://www.gov.uk/government/statistics/police-recorded-crime-open-data-tables	Force-level recorded crime data for benchmarking against peer forces.
6	Chicago Crimes 2001 to Present https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2	Large open crime dataset used to prototype spatial analytics and repeat location detection without UK sensitive data.
7	ONS Geography Portal https://geoportal.statistics.gov.uk	Output areas, wards and boundaries for neighbourhood aggregation and small-area disclosure checks.
8	ICO Anonymisation Guidance https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/data-sharing/anonymisation/	Guidance informing the re-identification risk assessment for published data.

Deliverables (17)

- D1 Publication incident review.** A root cause analysis of the violent crime drop with contributing factors and immediate fixes.
- D2 Publication pipeline with controls.** A rebuilt monthly extract with completeness checks against expected volumes by crime type.
- D3 Month-on-month variance alerts.** Automated detection of implausible changes before submission, with sign-off workflow.
- D4 Provisional classification handling policy.** Agreed rules for including crimes under review in publication and later revisions.
- D5 Re-identification risk assessment.** An assessment of small-area disclosure risk, with suppression rules for sensitive offences and sparse areas.
- D6 Target platform architecture.** A design meeting national policing security requirements with zones for publication, auditing and analytics.
- D7 Integrated crime and incident data model.** A model linking control room incidents, crime records, outcomes, victims and locations.
- D8 Crime data integrity audit dataset.** Structured data supporting auditors with links between incidents and crime records.
- D9 Under-recording detection prototype.** Text classification flagging incidents likely to contain unrecorded crimes, with human review.
- D10 Model oversight and bias review.** Evaluation of classification accuracy, error types and differential performance across offence and victim groups.
- D11 Neighbourhood analytics mart.** Daily refreshed crime and incident data by neighbourhood with repeat location and victim indicators.
- D12 Access control and vetting model.** Role-based access aligned to vetting levels and need-to-know for sensitive records.
- D13 Supplier extract data contract.** A specification of extract content, change notification and quality expectations with the platform supplier.
- D14 Data quality dashboard.** Monitoring of recording timeliness, classification backlog and missing fields.
- D15 Inspection readiness report.** Evidence of continuous monitoring and improvement prepared ahead of the inspectorate visit.
- D16 Data protection impact assessment.** A DPIA covering text processing, profiling and publication risks.
- D17 Stakeholder governance forum.** Terms of reference for a data board including the deputy chief, DPO, registrar and neighbourhood leads.

CASE STUDY 15 · LOGISTICS AND PARCEL DELIVERY

Last-Mile Delivery Telemetry and Route Performance Platform

Industry	Logistics and Parcel Delivery
Organisation	Parcelwise, a UK parcel carrier delivering 1.4 million parcels a day with 9,000 drivers
Your role	Senior Data Engineer, Network Operations Analytics
Core themes	GPS telemetry processing, geospatial matching, workforce data ethics, operational versus planned data

The Scenario

Parcelwise delivers one parcel in every twelve in the United Kingdom. Its drivers, a mixture of employees and self-employed couriers, leave fifty-one depots every morning with vans loaded according to routes planned overnight by an optimisation system. Every van has a telematics unit, and every driver carries a handheld device that scans parcels, captures photographs and signatures, and records delivery outcomes. Together these produce around four hundred million GPS points and fifteen million scan events a day. You are a senior data engineer in network operations analytics, and you have been asked to turn that data into something the business has wanted for years: a reliable, detailed picture of what actually happens on the road compared with what was planned.

The motivation is financial. Last-mile delivery is the most expensive part of the parcel business, and Parcelwise's cost per parcel has risen for three years in a row. The route planning system assumes average driving speeds, stop durations and parking times, and the operations director, Gareth Llewellyn, is convinced that these assumptions are badly wrong in many areas. Routes in dense city centres routinely run over time, while some rural routes finish early. Drivers complain that the planned stop times ignore reality: blocks of flats with no parking, business parks where reception closes at lunchtime, and rural addresses that the mapping system places in the wrong field. If the planning assumptions could be calibrated from actual telemetry, Gareth believes routes could be rebalanced to reduce overtime, missed deliveries and failed first attempts.

The data, however, is messy in every way that GPS data can be messy. Telematics units report positions every thirty seconds while the van is moving and less often when stationary, but some older units report only every two minutes. Handheld devices record their own GPS positions at

each scan, which are often more accurate for delivery locations but are sometimes taken indoors or with a weak signal. Clocks on the two devices are not synchronised. In urban canyons, GPS positions jump between streets. Scan events are sometimes batched when the device loses signal and uploaded together later, with the scan time recorded but the position recorded as the upload location. And to calculate stop durations accurately, you need to determine when the van stopped, when the driver left it, when the parcel was delivered and when the van moved again, which requires fusing the two data streams and matching both to the road network.

The data science team has already tried a simple approach, using the handheld scan location as the delivery point and the time between consecutive scans as the stop duration, and found that it produces implausible results for a third of stops. You believe a proper solution requires map matching of the van tracks to the road network, stop detection from the telematics, association of scans with stops, and careful treatment of the edge cases. You have estimated that this is several months of work to do well.

The second complication is people. The drivers' union representative, Carol Mitchell, has written to the chief executive expressing concern that telemetry analysis will be used to monitor and discipline individual drivers. Many drivers already feel surveilled: the handheld devices track them all day, and there have been disputes in the past where GPS data was used in disciplinary cases. Some self-employed couriers have argued that detailed monitoring is inconsistent with their self-employed status, and a recent employment tribunal case involving another courier firm has made the legal team nervous. The head of people, Susan Blake, has asked for firm assurances about how the data will and will not be used. Gareth, for his part, has said openly that he does want to identify routes and depots where performance is poor, but he insists the aim is to fix the plans, not blame the drivers.

The third complication is the address data itself. The route planner geocodes each delivery address to a point, using a commercial address database. For most urban addresses this is accurate. For rural addresses, new-build estates and large buildings with multiple entrances, the geocoded point is often far from where the driver actually delivers. The telemetry could be used to learn better delivery points, the place where

drivers actually stop for each address, but storing learned delivery locations for residential addresses is itself a privacy question, and the address database licence may restrict creating derived data from it.

The chief technology officer, Priya Anand, wants the platform to be built on the company's cloud data platform and to reuse its streaming infrastructure. She also wants it delivered in a way that supports a future real-time use case: detecting during the day which routes are running late so that operations can intervene. Gareth wants calibrated planning parameters for the peak season, which starts in four months. The data protection officer wants a full impact assessment before any processing of telemetry for new purposes begins. The union wants to be consulted.

There is history here as well. Three years ago, a previous project attempted to calibrate stop times using an external consultancy. The consultants produced a report with recommended parameters for each depot, based on a two-week sample of data, and the planning team implemented them. Within a month, routes in several depots were running badly over time, because the sample had been taken in a quiet period and did not reflect seasonal variation, weather or the growth of parcels to apartment buildings. The parameters were quietly reverted, and the planning team has been reluctant to change assumptions ever since. The head of route planning, Neil Ashworth, told you plainly that he will only accept new parameters that come with evidence of how they vary by season, day of week and parcel mix, and with a way of monitoring whether they remain valid once deployed. That reasonable demand adds scope, but ignoring it would repeat the mistake that made the planners sceptical in the first place.

You recognise that the most valuable outputs for route planning are aggregated: typical stop durations by address type and area, realistic driving speeds by road and time of day, and accurate delivery points. None of these require identifying individual drivers in the final data. But the processing to produce them necessarily handles driver-level data, and it is tempting for managers, once the data exists, to want driver-level league tables. You also recognise that the peak season deadline is only achievable if you narrow the scope, perhaps to a subset of depots, and that the real-time use case, while strategically important, is a different problem with different latency and reliability requirements.

THE DECISION IN FRONT OF YOU

How do you scope and sequence a telemetry platform that delivers calibrated planning parameters before peak, builds toward in-day monitoring, handles noisy GPS fusion robustly, and embeds privacy and workforce safeguards that the union, legal team and drivers can trust?

Data Sources

#	Source and link	How it is used in this project
1	Amazon Last Mile Routing Research Challenge Data https://registry.opendata.aws/amazon-last-mile-challenges/	Real historical delivery routes, stops and timing data used to develop stop duration and route deviation analytics.
2	Cainiao LaDe Last-Mile Delivery Dataset https://huggingface.co/datasets/Cainiao-AI/LaDe	Large-scale courier pickup and delivery trajectories used to test GPS and event fusion methods.
3	OpenStreetMap Extracts (Geofabrik) https://download.geofabrik.de	Road network data for map matching and speed calibration.
4	OSRM Routing Engine http://project-osrm.org	Open-source routing and map-matching engine for snapping GPS traces to roads.
5	Ordnance Survey Open UPRN https://osdatahub.os.uk/downloads/open/OpenUPRN	Open unique property reference points for improving delivery point accuracy.
6	DfT Road Traffic Statistics https://roadtraffic.dft.gov.uk	Traffic counts and flows used to contextualise speeds and congestion by road.
7	H3 Geospatial Indexing https://h3geo.org	Hexagonal indexing for aggregating stop durations and speeds by area without exposing individual routes.
8	ICO Employment Practices: Monitoring Workers https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/employment/monitoring-workers/	Regulatory guidance governing worker monitoring and telemetry use.

Deliverables (17)

- D1 Telemetry data inventory.** A description of telematics and handheld data streams, sampling rates, clock behaviour and known defects.
- D2 Data protection impact assessment.** A DPIA covering telemetry processing purposes, driver data, derived delivery points and retention.
- D3 Worker data use charter.** A written commitment agreed with union and people teams on permitted and prohibited uses of telemetry.
- D4 Pilot depot selection.** A justified selection of representative depots for the pre-peak phase.

- D5** **GPS cleaning pipeline.** Processing that removes jumps, handles sparse sampling and corrects clock offsets.
- D6** **Map matching service.** Snapping of van tracks to the road network with confidence scoring.
- D7** **Stop detection algorithm.** Identification of stops, parking and dwell periods from telematics with validation against scans.
- D8** **Scan-to-stop association.** Linking handheld events to detected stops, handling batched uploads and indoor positions.
- D9** **Stop duration model.** Aggregated stop duration distributions by address type, building type, area and time of day.
- D10** **Road speed calibration.** Realistic speed profiles by road segment and time band for the route planner.
- D11** **Learned delivery points.** Improved delivery locations for problematic addresses with licensing and privacy review.
- D12** **Planner parameter export.** Calibrated parameters delivered in the format the route optimisation system consumes.
- D13** **Aggregation and anonymisation layer.** Outputs designed so managers see routes and areas, not individual driver rankings.
- D14** **Planned versus actual dashboard.** Route and depot level comparisons of planned and actual time with drill-down to causes.
- D15** **In-day monitoring design.** An architecture for real-time late-route detection as a later phase.
- D16** **Data quality monitoring.** Checks on device reporting gaps, stale units and implausible positions.
- D17** **Peak season impact review.** Measurement of overtime, failed deliveries and route completion after parameter changes.

CASE STUDY 16 · AGRICULTURE AND AGRI-FOOD

Satellite and Weather Data Platform for Crop Yield Forecasting

Industry	Agriculture and Agri-Food
Organisation	Fenland Grain Cooperative, a farmer-owned grain marketing cooperative representing 1,100 farms in eastern England
Your role	Data Engineering Lead, Market Intelligence
Core themes	Earth observation data processing, geospatial joins, data sharing with members, uncertainty communication

The Scenario

Fenland Grain Cooperative markets wheat, barley and oilseed rape on behalf of eleven hundred member farms. Its value to members depends on selling their grain at the best possible price, and that depends heavily on judging how much grain members will produce before harvest. If the cooperative commits to forward sales contracts with millers and exporters in spring and the harvest falls short, it must buy grain on the open market to fill the contracts, often at a loss. If it undersells and the harvest is strong, members' grain is sold later at whatever the market will pay. Last year's harvest was fourteen per cent below the cooperative's forecast after a wet autumn and a dry May, and the resulting shortfall cost the cooperative more than two million pounds. The board has asked for better forecasting, and you have been hired to lead data engineering in the market intelligence team.

Until now, forecasting has been done by the cooperative's agronomists, who visit a sample of farms, collect members' own estimates, and combine these with experience. Members report planted areas by crop in an annual survey, although response rates have fallen to around sixty per cent. The chief executive, Richard Coles, has been approached by several agritech companies offering satellite-based yield forecasting services, and one has offered a free pilot. The board is interested, but a board member who farms three thousand acres has argued that the cooperative should build its own capability using public satellite and weather data, because farm-level data is commercially sensitive and members will not want it held by a third party that might sell insights to grain traders.

You have spent a month exploring what building in-house would involve. Public satellite imagery is plentiful. The European Copernicus programme

provides optical imagery every few days at ten-metre resolution, free of charge, and radar imagery that can see through cloud, which matters in eastern England. Weather observations and reanalysis data are available from public sources, as are soil maps and historical yield statistics at regional level. The challenge is turning these into field-level crop monitoring. Optical images must be corrected, cloud and shadow masked, and converted into vegetation indices. Radar data needs different processing. All of it must be clipped to field boundaries, which means the cooperative needs accurate boundaries for its members' fields. Crop types must be known or inferred for each field. And the models relating satellite and weather signals to yield must be trained on actual yield data, which the cooperative only has at farm level from members' delivery records, not at field level.

Field boundaries are a problem in their own right. Members' fields are registered with government agencies for agricultural payment schemes, and boundary data exists, but the cooperative does not hold it and access for its purposes is not straightforward. Some members have provided field maps from their farm management software, in a variety of formats. Others have never been asked. The agronomy manager, Sarah Tunstall, suggests inferring boundaries from satellite imagery, which is technically feasible but introduces errors, particularly where hedges have been removed or fields are split between crops.

The members themselves are the most important consideration and the most varied. Some large, technologically sophisticated farms already use satellite services, yield-mapping combine harvesters and farm management platforms, and would willingly share detailed data in return for better services. Many smaller farms are wary of any data collection and suspicious of anything that might inform the cooperative's price negotiations with them. The cooperative's member relations manager, David Hurst, warns that if members believe their data could be used to judge their farming or be shared outside the cooperative, participation in the annual survey will fall further. He wants any new data platform to offer members something in return, such as field-level crop health monitoring they can see on their phones.

The trading team has a different perspective. The head of trading, Emma Ward, does not need field-level forecasts. She needs a reliable estimate of total cooperative production by crop, with a sensible range of uncertainty,

updated regularly from March to harvest. She is sceptical of satellite forecasts because she has seen commercial services confidently predict yields that turned out to be wrong, and she wants to understand how uncertain any forecast is, not just its central estimate. A forecast that says eight tonnes per hectare plus or minus half a tonne is useful; one that says eight tonnes with no indication of uncertainty is dangerous.

The engineering challenge is substantial for a team of two, yourself and a junior data engineer, supported by an agronomist who can code a little. Satellite data volumes are large: covering the cooperative's area at ten-metre resolution every few days generates terabytes over a season. Processing it requires either significant cloud compute or the use of one of the platforms that host the imagery and allow processing near the data. You also need to decide whether to process everything from raw imagery, or to use pre-computed products such as analysis-ready vegetation indices available through public or commercial platforms.

You have also looked closely at the terms of the free pilot. The agritech company would require members' field boundaries, crop types and historical yields to calibrate its models, and its standard agreement grants it a licence to use aggregated and anonymised data to improve its services generally. Its customers include several grain merchants and at least one large trading house. The company insists that no individual farm data would ever be disclosed, but an aggregate forecast of production across a region where the cooperative handles a large share of the crop is, in effect, a forecast of the cooperative's position. If that information reached traders on the other side of the cooperative's negotiations, even indirectly, it could weaken members' bargaining power. The company has indicated that it might agree to exclusivity terms, but only as part of a paid contract after the pilot, at a price that would consume most of the budget the board has in mind for forecasting.

Richard wants a recommendation on whether to accept the free pilot, build in-house, or do both. The board meets in six weeks. Planting of winter crops has already happened, and the first useful satellite signals for the coming harvest will appear in late winter. If you want a forecast system running for this season, you have roughly four months.

You are aware that accepting the pilot would produce results fastest but would involve sharing member data with a third party, and that building in-house would give the cooperative control but risks producing nothing

useful this season. You also suspect that the biggest single improvement to forecasting might not come from satellites at all, but from knowing accurately which crops are planted where, which the satellite data could help establish even if yield prediction itself remains uncertain.

THE DECISION IN FRONT OF YOU

Do you accept the commercial pilot, build an in-house Earth observation platform, or combine them, and how do you secure member trust and field boundary data while giving traders forecasts with honest uncertainty in time for this season?

Data Sources

#	Source and link	How it is used in this project
1	Copernicus Data Space Ecosystem https://dataspace.copernicus.eu	Free Sentinel-1 radar and Sentinel-2 optical imagery with processing APIs used for crop monitoring.
2	Copernicus Climate Data Store (ERA5) https://cds.climate.copernicus.eu	Reanalysis weather data such as temperature, rainfall and soil moisture used as yield model features.
3	NASA POWER Agroclimatology API https://power.larc.nasa.gov	Daily agro-meteorological data for field locations used as a lightweight weather feed.
4	Met Office MIDAS Open (CEDA Archive) https://catalogue.ceda.ac.uk/uuid/dbd451271eb04662beade68da43546e1	UK weather station observations for local calibration of weather features.
5	DEFRA Cereal and Oilseed Production Statistics https://www.gov.uk/government/statistics/cereal-and-oilseed-rape-production	Official regional yield and production data used to validate forecasts.
6	UKCEH Land Cover Maps https://www.ceh.ac.uk/data/ukceh-land-cover-maps	National land cover data used as a prior for crop and field identification.
7	SoilGrids https://soilgrids.org	Global gridded soil property data used as static yield model features.
8	FAOSTAT https://www.fao.org/faostat/en/	International crop production data for market context in trading reports.

Deliverables (17)

- D1 Build, buy and hybrid recommendation.** A board paper comparing pilot, in-house and combined approaches on cost, data control, timing and accuracy.
- D2 Member data principles.** A charter defining what member data is collected, how it is used, who sees it and what members receive in return.
- D3 Member consent and data sharing agreement.** Standard terms for members providing field boundaries and yield data.

- D4 Field boundary dataset.** A consolidated boundary layer from member uploads, public sources and imagery-based inference with quality flags.
- D5 Crop type mapping.** Classification of crop type per field from satellite time series, validated against survey returns.
- D6 Imagery ingestion pipeline.** Automated retrieval of Sentinel-1 and Sentinel-2 data for the cooperative's area with cloud masking.
- D7 Vegetation index time series.** Per-field time series of vegetation and radar indices updated as new imagery arrives.
- D8 Weather feature pipeline.** Daily field-level weather and soil moisture features from reanalysis and station data.
- D9 Historical training dataset.** A combined dataset linking past seasons' satellite, weather and farm-level delivery yields.
- D10 Yield forecasting model.** A model producing crop production estimates with prediction intervals, validated on held-out seasons.
- D11 Uncertainty communication design.** Charts and language that convey forecast ranges clearly to traders and the board.
- D12 Trading forecast report.** A regular automated report of cooperative production by crop with change explanations.
- D13 Member crop health app prototype.** A simple interface showing members their own fields' vegetation trends as a return benefit.
- D14 Data storage and cost plan.** A design for processing near the data and storing only derived products, with cost estimates.
- D15 Pilot evaluation framework.** A method for comparing the commercial pilot's forecasts with in-house results fairly.
- D16 Data quality checks.** Monitoring of missing imagery, cloud coverage, boundary errors and outlier fields.
- D17 Season retrospective.** A post-harvest review comparing forecasts with actual deliveries and planning improvements.

CASE STUDY 17 · REAL ESTATE AND MORTGAGE LENDING

Residential Property Valuation Data Warehouse

Industry	Real Estate and Mortgage Lending
Organisation	Marlow & Finch Building Society, a UK mutual lender with a £9 billion mortgage book
Your role	Senior Data Engineer, Credit Risk Data
Core themes	Reference data integration, address matching, model risk, stress testing and regulatory capital

The Scenario

Marlow & Finch Building Society has lent money to homebuyers for a hundred and thirty years. Its mortgage book is worth nine billion pounds, secured on around sixty thousand properties across England and Wales. The value of those properties matters enormously: it determines the loan-to-value ratios that drive the society's credit risk models, its provisions for expected losses, the capital it must hold, and its decisions about whom to lend to. For most of its history, the society has valued each property once, at origination, with a surveyor's report, and then updated values periodically by applying a regional house price index. You are a senior data engineer in the credit risk data team, and the Prudential Regulation Authority has just told the society that this approach is no longer good enough.

The regulator's feedback arrived after a routine review of the society's internal ratings-based models and its stress testing submissions. It noted that indexing each property's origination value by a broad regional index ignores the fact that different types of property in different places have performed very differently. Flats in some city centres have lost real value over the past decade, while houses in commuter towns have soared. Properties with poor energy efficiency ratings may face value pressure as minimum efficiency standards for rental properties tighten and buyers become more aware of energy costs. The regulator expects the society to use more granular valuation data and to understand the uncertainty in its property values, particularly for high loan-to-value loans where small valuation errors change risk estimates significantly.

The chief risk officer, Andrew Blackwood, has sponsored a programme to build a property valuation data warehouse that will support an automated valuation model, granular indexation and climate and energy risk analysis. The programme has a budget, a steering committee and a deadline: the

next stress testing submission, in ten months. You are responsible for the data.

The first challenge is the one that every property data project eventually faces: addresses. The society's mortgage system holds addresses as entered by brokers and staff over three decades. Some include flat numbers, some do not. Some use house names, some numbers, some both. Postcodes have occasionally changed. To link each mortgaged property to external data, including sales history from the Land Registry, energy performance certificates, flood risk assessments and local price indices, you need to match each address to a unique property reference number. In a trial on a thousand loans, a simple matching approach achieved a confident match for seventy-two per cent of properties. The rest required fuzzy matching or manual review, and for flats in converted buildings, even manual review was often inconclusive.

The second challenge is the external data itself. Land Registry price paid data records every residential sale in England and Wales, which is invaluable for building a valuation model, but it contains no information about the property's size or characteristics beyond broad type and whether it is new. Energy performance certificates contain floor area, construction age, heating and efficiency ratings, but a property may have several certificates over time, and not every property has one. Some certificate data is known to be inaccurate, with floor areas entered incorrectly. Matching sales to certificates to create a training dataset requires the same address matching that bedevils the mortgage book. Commercial valuation data providers offer ready-made automated valuations, but their models are proprietary, which makes it hard for the society to explain their outputs to the regulator, and their licences restrict how the data can be used and retained.

The head of credit risk modelling, Dr Jessica Tan, wants to build the society's own automated valuation model, trained on public sales and property characteristic data, so that the society understands and can validate every element of it. She believes this is what the regulator implicitly expects. The head of lending operations, Michael Crane, is sceptical. His team uses a commercial automated valuation service for remortgage and further advance applications, and it works well enough. He argues that building a competing model in-house is expensive, slow, and likely to be less accurate than a specialist provider with access to far

more data, including asking prices and surveyor valuations from many lenders. He would prefer to license a commercial valuation for every property in the book and use it for stress testing.

The finance director, Clare Hughes, is worried about cost and consistency. The society currently uses the commercial service for new lending decisions, a regional index for provisioning and capital, and surveyors' valuations for arrears cases. Three different values for the same property can coexist, and she has been asked by the external auditor which one the accounts rely on. She wants one authoritative value per property, with a documented method.

Model risk management adds its own requirements. Any valuation model, whether built or bought, that feeds capital calculations must be validated independently, with documented methodology, performance testing, and ongoing monitoring. For a commercial model, the society would need sufficient information from the provider to validate it, which not all providers will share. For an in-house model, the validation burden falls entirely on the society's small validation team, which is already stretched.

Then there is climate and energy risk. The regulator expects the society to assess how physical risks such as flooding and transition risks such as energy efficiency requirements might affect collateral values. That requires linking properties to flood risk data and energy certificates and developing scenarios for how values might change. The data engineering required overlaps heavily with the valuation work but adds geospatial processing and scenario management.

A practical difficulty lies in the society's own historical data. The origination valuations, the surveyors' figures on which every loan was originally based, are held in a document management system as scanned reports for loans written before about 2012, and as structured fields only for more recent loans. Several thousand older loans have no origination value recorded in the mortgage system at all, because the field was not migrated when the society changed its core platform. Indexation from origination value is impossible for those loans without first extracting values from the scanned reports, and the quality of those scans varies. Arrears cases have recent surveyor values, but they are biased towards properties that are harder to sell. Any valuation approach must fill these gaps, and the regulator will want to see how the society has treated loans where the underlying data is incomplete rather than assuming it away.

Your team comprises three engineers and a data analyst. You have ten months. You also know that the address matching problem underpins everything, whichever valuation approach is chosen, and that if the society ends up with poor matching, every downstream analysis, from valuation to flood risk, will be compromised.

THE DECISION IN FRONT OF YOU

Do you build an in-house automated valuation model on public data, license commercial valuations for the whole book, or combine them, and how do you establish a single authoritative property record and valuation method that survives model validation, audit and regulatory scrutiny?

Data Sources

#	Source and link	How it is used in this project
1	HM Land Registry Price Paid Data https://www.gov.uk/government/statistical-data-sets/price-paid-data-downloads	Every residential property sale in England and Wales, used for valuation training data and sales history.
2	UK House Price Index https://www.gov.uk/government/collections/uk-house-price-index-reports	Official local and property-type indices used for granular indexation and benchmarking.
3	Energy Performance of Buildings Data (EPC) https://epc.opendatacommunities.org	Property floor area, age, efficiency ratings and features used for valuation and transition risk.
4	Ordnance Survey Open UPRN https://osdatahub.os.uk/downloads/open/OpenUPRN	Open unique property references and coordinates to anchor address matching.
5	Environment Agency Risk of Flooding from Rivers and Sea https://www.data.gov.uk/dataset/bad20199-6d39-4aad-8564-26a46778fd94/risk-of-flooding-from-rivers-and-sea	Flood risk areas used to assess physical climate risk to collateral.
6	ONS Postcode Directory https://geoportal.statistics.gov.uk	Postcode to geography lookups for regional and small-area aggregation.
7	PRA Supervisory Statement on Climate-Related Risks https://www.bankofengland.co.uk/prudential-regulation/publication/2019/enhancing-banks-and-insurers-approaches-to-managing-the-financial-risks-from-climate-change-ss	Regulatory expectations for climate risk that shape scenario requirements.
8	Splink Record Linkage https://moj-analytical-services.github.io/splink/	Probabilistic address matching library for linking the mortgage book to external property data.

Deliverables (17)

- D1** **Valuation approach decision paper.** A comparison of in-house, commercial and hybrid valuation on accuracy, explainability, cost and validation effort.

- D2 Golden property record design.** A single master record per property with unique reference, attributes, sources and confidence.
- D3 Address matching pipeline.** Deterministic and probabilistic matching of loan addresses to property references with review queues.
- D4 Address match quality report.** Match rates, confidence distribution and manual review outcomes by property type and region.
- D5 External property data ingestion.** Scheduled ingestion of price paid, EPC, UPRN and flood data with version control.
- D6 Sales-to-characteristics training dataset.** A linked dataset of sales with property attributes for model development.
- D7 EPC data quality treatment.** Rules detecting implausible floor areas and selecting the most reliable certificate per property.
- D8 Automated valuation model prototype.** A transparent model producing property values with confidence intervals.
- D9 Granular indexation engine.** Indexation by local area and property type replacing the regional index.
- D10 Valuation reconciliation framework.** Comparison of in-house, commercial and surveyor values with rules for authoritative selection.
- D11 Model validation evidence pack.** Methodology, performance testing and limitations documentation for independent validation.
- D12 Flood risk collateral analysis.** Exposure of the mortgage book to flood risk bands with loan-to-value implications.
- D13 Energy transition scenario data.** Scenario inputs and outputs for value adjustments linked to energy efficiency ratings.
- D14 Stress testing data feed.** Property-level values and scenarios delivered to capital and stress testing models.
- D15 Data lineage documentation.** End-to-end lineage from source datasets to capital figures for audit and regulators.
- D16 Licensing compliance register.** A record of licence terms, permitted uses and retention limits for each external source.
- D17 Ongoing monitoring dashboard.** Tracking of valuation accuracy against new sales, match coverage and data freshness.

CASE STUDY 18 • WATER UTILITIES

Water Network Leakage Detection from Sensor and Smart Meter Data

Industry	Water Utilities
Organisation	Severnvalle Water, a regional water company supplying 2.9 million customers through 26,000 km of mains
Your role	Principal Data Engineer, Network Performance
Core themes	Sensor time-series integration, regulatory performance commitments, hydraulic model data, alarm fatigue

The Scenario

Severnvalle Water loses around one in five litres of the treated water it puts into its network before it reaches a customer's tap. That figure is roughly in line with the industry average, which is precisely the problem: the industry average is no longer acceptable. The economic regulator, Ofwat, has set challenging leakage reduction commitments for the current five-year period, with financial penalties for companies that miss them and rewards for those that outperform. The Environment Agency has classified the region as seriously water stressed. Hosepipe bans in two of the last four summers generated headlines about a company that lets water pour out of broken pipes while telling customers to save it. The board has made leakage reduction the company's top operational priority, and you, as principal data engineer for network performance, have been asked to make the data work harder.

Severnvalle's network is divided into around nine hundred district metered areas, each a zone of the network with metered inflows and outflows. The standard method for estimating leakage is to measure flow into each area in the middle of the night, when legitimate consumption is lowest, and subtract an estimate of night-time customer use. The remainder is assumed to be leakage. When the minimum night flow in an area rises, it suggests a new leak, and leakage technicians are sent with acoustic equipment to find it. This method has served the industry for decades, but it has well-known weaknesses. Customer night use is estimated rather than measured, and varies with shift workers, night-time irrigation and seasonal occupancy. Flow meters at area boundaries drift and fail. Boundary valves that should be closed are sometimes left open after maintenance, merging areas and corrupting the calculations. Around

fifteen per cent of Severnvalle's district metered areas are currently flagged as having unreliable data.

The data landscape has changed significantly in the last three years. Severnvalle is installing smart water meters at customer premises, with around four hundred thousand now in place and a target of a million within four years. These report hourly consumption, which in principle allows night use to be measured rather than estimated, and can reveal leaks on customers' own supply pipes, which account for a significant share of total leakage. The company has also installed thousands of acoustic loggers and pressure sensors across the network, which listen for leak noise and detect pressure transients that can damage pipes. And the telemetry system that monitors reservoirs, pumping stations and treatment works has been upgraded to report more frequently.

The problem is that all these data sources are managed separately. Smart meter data arrives from a meter data management platform run by the meter supplier. Acoustic logger alerts are viewed in a vendor web portal. Pressure data sits in the telemetry historian. District flow data is processed in a leakage analytics application that was built in-house fifteen years ago and runs on an ageing database server. The asset register, which describes every pipe, its material, age and diameter, is held in a geographic information system that is updated inconsistently after repairs. Work orders for repairs are in the field management system. Nobody can currently answer, for a given district, a simple question: what do all our data sources say about this area right now?

The director of water networks, Alison Pryce, wants an integrated leakage data platform that brings all these sources together, calculates leakage more accurately, and prioritises where technicians should go each day. She has been impressed by presentations from other water companies that claim significant leakage reductions from analytics. The head of leakage operations, Tony Wilkes, a former technician, is cautious. His teams already receive more alerts than they can investigate. The acoustic loggers generate thousands of alarms a week, most of which turn out to be background noise from traffic or customer usage. Smart meter continuous flow alerts, which suggest a leak on a customer's pipe, are sent to customers by letter, but many customers ignore them. Tony's view is that more data will simply mean more alarms unless the platform can combine signals intelligently and give his teams fewer, better leads.

Your data science colleague, Dr Nisha Kapoor, has proposed a model that combines night flow anomalies, acoustic alerts, pressure events, pipe characteristics and weather, since cold snaps and hot, dry spells both cause bursts, to rank district areas and specific pipe segments by likelihood of an active leak. The model requires all the data to be aligned in time and space, which requires the integration work you have been asked to design. It also requires labels, meaning records of where leaks were actually found, which depend on technicians recording their findings accurately in the field management system. At present, around a third of leak repair work orders have no precise location recorded, only a street name.

There are also regulatory reporting considerations. Leakage is reported annually to Ofwat using a defined methodology, and the company's reported figures are audited. Any change in how leakage is calculated internally must be carefully reconciled with the regulatory methodology, because the board is understandably nervous about any perception that the company is changing its sums to improve its reported performance. The regulatory reporting manager, Philip Hart, has asked that the new platform maintain a clear separation between operational analytics and the regulatory calculation, while allowing the two to be compared.

Data ownership is contested too. The smart meter supplier's contract gives Severnvalle access to consumption data, but the supplier also processes the data to generate its own alerts and insights, and it has proposed an analytics service that would compete with anything built in-house. The acoustic logger vendor provides only a limited export from its portal and charges for API access. The IT function wants all data consolidated on the corporate cloud platform, but the operational technology team is uneasy about sensor data flowing from the telemetry network to the cloud, citing security guidance for critical national infrastructure.

You have been given a budget and eighteen months, with an expectation that the first leakage benefits will be visible within a year, before the summer demand peak. You have also been told, informally, that the chief executive has promised the board a specific leakage reduction figure for the coming year, and that the data platform is expected to contribute.

As you plan, you are struck by how much of the value depends on things that are not data engineering at all: technicians recording repair locations, boundary valves being checked, customers acting on letters about supply

pipe leaks. You also notice that the smart meter data could transform the measurement of customer night use in the areas where meter penetration is highest, which might immediately improve leakage estimates for those districts without any complex modelling. The question is whether to start there, with a narrow, high-confidence improvement, or to build the full integrated platform that Alison wants and Nisha's model needs.

THE DECISION IN FRONT OF YOU

Do you start with a focused improvement using smart meter night-use data in high-penetration districts, build the full integrated leakage platform, or negotiate a phased path, and how do you reduce alarm fatigue while keeping operational analytics cleanly separated from audited regulatory reporting?

Data Sources

#	Source and link	How it is used in this project
1	BattLeDIM 2020 Leakage Detection Dataset (Zenodo) https://zenodo.org/records/4017659	Realistic simulated water network with pressure, flow and AMR data and known leak events, used to develop detection methods.
2	Environment Agency Hydrology Data API https://environment.data.gov.uk/hydrology/doc/reference	River flow and rainfall observations used as environmental context and demand drivers.
3	Met Office Weather DataHub https://www.metoffice.gov.uk/services/data/met-office-weather-datahub	Temperature and frost data used to model burst risk from weather conditions.
4	Ofwat Leakage and Performance Commitments https://www.ofwat.gov.uk/regulated-companies/company-obligations/outcomes/	Regulatory performance commitments that define targets and reporting expectations.
5	Discover Water https://www.discoverwater.co.uk	Industry comparative performance data for benchmarking leakage against other companies.
6	UK Water Industry Research (UKWIR) https://ukwir.org	Industry research on leakage methodology and consistent reporting.
7	NCSC Critical National Infrastructure Guidance https://www.ncsc.gov.uk/section/private-sector-cni/cni	Security guidance shaping the OT-to-cloud data flow design.
8	ONS Census Population Estimates https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates	Population and household data used to model expected night-time consumption by area.

Deliverables (17)

- D1 District data reliability assessment.** A scoring of every district metered area for meter health, boundary integrity and data completeness.
- D2 Integrated data architecture.** A design bringing flow, pressure, acoustic, smart meter, asset and work order data together securely.
- D3 OT data transfer security design.** Controls for moving telemetry data from operational networks to the analytics platform.
- D4 Smart meter night-use pipeline.** Calculation of measured night consumption for districts with sufficient meter penetration.
- D5 Improved night flow leakage estimates.** Revised operational estimates for high-penetration districts compared with existing assumptions.
- D6 Spatial-temporal alignment layer.** A common grid of time and network location joining all sensor and asset data.
- D7 Acoustic alert filtering.** Rules and models reducing noise alerts by correlating acoustic, pressure and flow signals.
- D8 Leak likelihood ranking model.** A prioritised daily list of districts and pipe segments with explanation of contributing signals.
- D9 Technician work queue integration.** Delivery of prioritised leads to field teams through the existing field management system.
- D10 Repair location capture improvement.** Changes to field workflows and validation ensuring precise leak find locations.
- D11 Customer supply pipe leak programme data.** Tracking of continuous flow alerts, customer contact and resolution outcomes.
- D12 Regulatory reconciliation report.** A comparison of operational estimates with the audited regulatory methodology, clearly separated.
- D13 Vendor data access agreements.** Revised terms with meter and logger vendors securing data access and API rights.
- D14 Data quality monitoring.** Automated detection of failed meters, open boundary valves and sensor dropouts.
- D15 Leakage performance dashboard.** District, zone and company views of estimated leakage, trends and interventions.
- D16 Benefits tracking framework.** Measurement of leakage reduction attributable to analytics-driven interventions.
- D17 Summer readiness plan.** A plan ensuring prioritisation and alerts perform under peak demand conditions.

CASE STUDY 19 • HIGHER EDUCATION

Learning Analytics Early Warning System for Student Retention

Industry	Higher Education
Organisation	University of Castlebridge, a UK university with 24,000 students, a third of whom study part-time or online
Your role	Lead Data Engineer, Planning and Insight
Core themes	Student data ethics, virtual learning environment data, predictive analytics governance, data minimisation

The Scenario

The University of Castlebridge has a proud record of widening participation. More than half its undergraduates are the first in their families to attend university, a third are over twenty-five, and a large and growing share study part-time or online while working. That profile is part of the university's identity, but it also brings a stubborn challenge. Continuation rates, the proportion of students who progress from first to second year, are several percentage points below the sector average for some courses, and the Office for Students, which regulates English higher education, has set minimum thresholds for continuation, completion and progression outcomes. Two of Castlebridge's online programmes are currently below the threshold. The pro vice-chancellor for education, Professor Ruth Adebayo, has asked for a learning analytics early warning system that identifies students at risk of withdrawing early enough for support to make a difference. You lead data engineering in the planning and insight team, and the task of building it has fallen to you.

The idea is not new, and it is not without controversy. Many universities have introduced learning analytics, using data from the virtual learning environment, attendance systems, library usage and assessment submissions to flag students whose engagement is falling. The evidence suggests that, done well, it can help personal tutors and student support teams reach struggling students earlier. Done badly, it can label students, reinforce disadvantage, overwhelm staff with alerts, and feel to students like surveillance. Castlebridge's students' union has already raised concerns after hearing about the project informally, and its president, Kemi Oladipo, has asked for a seat at the table.

The data available is extensive but fragmented. The virtual learning environment records every login, page view, resource download, forum post and quiz attempt, generating hundreds of millions of events a year. The student records system holds enrolment, demographics, entry qualifications, fee status, module registrations, marks and progression decisions. Lecture attendance is recorded by card swipes on campus and by joining online sessions, but the two systems are different, and many online students never attend live sessions by design. Library systems log borrowing and electronic resource access. The student support services case management system records contact with wellbeing, disability, financial and academic skills services, which is highly sensitive. Personal tutors record meetings, inconsistently, in a mixture of the student record and their own notes.

The first engineering challenge is simply joining these sources. Each uses its own identifiers: the student record uses a student number, the virtual learning environment uses a username that changed format four years ago, the attendance system uses a card number, and the library uses a borrower barcode. The mappings exist but are incomplete, particularly for students who have re-enrolled, changed programme or taken a break in studies.

The second challenge is defining engagement for very different kinds of student. A full-time campus student who stops attending lectures and logging into the virtual learning environment for two weeks is plausibly at risk. A part-time online student who works shifts and studies intensively in bursts, with long gaps in between, may look disengaged by the same measures while being perfectly on track. A model trained on historical data would learn patterns dominated by the majority full-time population and could systematically flag part-time and mature students, who already face stereotypes about commitment. The head of the online learning unit, Dr Paul Greaves, is worried about exactly this.

Ruth's preference is for a predictive model that produces a risk score for every student, updated weekly, drawing on all available data including demographics and entry qualifications, because these are among the strongest predictors of withdrawal. The university's data protection officer, Margaret Lowe, has serious reservations. Using demographic characteristics such as age, ethnicity, disability or socioeconomic background to predict risk may be statistically effective, but it treats

students differently on the basis of who they are rather than what they do, and it raises questions under both data protection and equality law. Margaret also questions whether data from student support services, which students disclose in confidence, should ever feed a predictive model. She would prefer an engagement-based approach that uses only behavioural signals and is transparent to students.

The personal tutors, represented by Dr Helen Marsh, have a practical concern. Tutors already have large tutee lists and limited time, and many feel that the university is asking them to become pastoral case workers without resources. An early warning system that sends each tutor a weekly list of at-risk students without any guidance on what to do, or any capacity to do it, will be ignored or resented. Helen wants any system to be designed around what tutors and support teams can actually do with the information.

Kemi, for the students' union, has set out conditions for the union's support: students must be told what data is used and how; they must be able to see their own engagement data; the system must not be used for disciplinary or attendance-monitoring purposes such as reporting international students to immigration authorities; and students should have some choice about participation.

There is also a technical debate. The IT department runs the virtual learning environment as a hosted service and can provide data through a standard learning analytics interface or through nightly database exports. The student record system is on-premises. The planning team's data warehouse was designed for regulatory returns and annual reporting, not for weekly operational analytics. You could extend it, or build a separate learning analytics store. The sector has produced open data standards and a shared learning analytics infrastructure that some universities use, which might save effort but would require the university to adopt particular data models.

There is an important lesson from the university's own recent history. Four years ago, a faculty introduced an attendance monitoring scheme in which students who missed three consecutive lectures received an automated email warning them that their enrolment might be reviewed. The intention was supportive, but the tone was not, and the emails went disproportionately to students with caring responsibilities and jobs. Several students complained that the message made them feel they were

being pushed out, and at least two withdrew shortly after receiving it, citing the email as the final straw. The scheme was abandoned, but the memory lingers among staff and students alike. Any early warning system will be judged against that experience, and the design of the first contact a student receives, who sends it, what it says and what support it offers, matters at least as much as the accuracy of the model behind it.

Ruth wants a pilot running for the start of the next academic year, seven months away, on the two online programmes below threshold and three campus programmes. The university's ethics committee must approve the project, and the data protection impact assessment must be completed before any processing begins.

THE DECISION IN FRONT OF YOU

Do you build the comprehensive predictive risk model the pro vice-chancellor wants, a transparent engagement-only approach favoured by the data protection officer, or something in between, and how do you design the data, the governance and the intervention workflow so that the system helps students rather than labels them?

Data Sources

#	Source and link	How it is used in this project
1	Open University Learning Analytics Dataset (OULAD) https://analyse.kmi.open.ac.uk/open_dataset	Anonymised VLE interaction, assessment and demographic data for 32,000 students used to prototype engagement features and models.
2	Office for Students B3 Student Outcomes Data https://www.officeforstudents.org.uk/data-and-analysis/student-outcomes-data-dashboard/	Regulatory continuation and completion data defining thresholds and benchmarks.
3	HESA Data and Analysis https://www.hesa.ac.uk/data-and-analysis	Sector-wide student data for benchmarking continuation by student characteristics.
4	Jisc Learning Analytics Resources https://www.jisc.ac.uk/learning-analytics	UK sector guidance, code of practice and architecture for learning analytics.
5	xAPI Specification (ADL) https://adlnet.gov/projects/xapi/	Standard for learning activity statements used to normalise VLE and other engagement events.
6	Moodle Developer Documentation https://moodledev.io	Reference for VLE log structures and data access.
7	ICO Guidance on AI and Data Protection https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/artificial-intelligence/guidance-on-ai-and-data-protection/	Guidance on fairness, transparency and lawful basis for predictive models about individuals.

#	Source and link	How it is used in this project
8	Equality Act 2010 https://www.legislation.gov.uk/ukpga/2010/15/contents	Legal framework for assessing indirect discrimination risks from model features.

Deliverables (17)

- D1 Ethics and governance proposal.** A submission to the ethics committee setting out purposes, data, safeguards and oversight.
- D2 Data protection impact assessment.** A DPIA addressing lawful basis, special category data, profiling and student rights.
- D3 Student transparency notice.** Plain-language information explaining what data is used, how and what students can see.
- D4 Identity resolution service.** Mapping of student numbers, usernames, card numbers and borrower IDs with history.
- D5 Learning event pipeline.** Daily ingestion and normalisation of VLE, attendance and library events into a common format.
- D6 Engagement feature definitions.** Documented, mode-aware engagement measures for full-time, part-time and online students.
- D7 Engagement baseline by study mode.** Typical engagement patterns per programme and mode to avoid misclassifying different study rhythms.
- D8 Risk model options analysis.** A comparison of behaviour-only and demographic-inclusive models on accuracy and fairness.
- D9 Fairness evaluation.** Testing of flag rates and error rates across age, mode, disability and background groups.
- D10 Tutor-facing alert design.** A co-designed view showing why a student was flagged and suggested actions.
- D11 Student-facing engagement view.** A dashboard allowing students to see their own engagement data.
- D12 Intervention recording workflow.** A simple way for tutors to record contact and outcomes for evaluation.
- D13 Access control model.** Role-based access limiting who sees flags and restricting support services data.
- D14 Purpose limitation controls.** Technical and policy controls preventing use for immigration monitoring or discipline.
- D15 Pilot evaluation plan.** A method, comparison groups and metrics for judging whether the pilot improves continuation.
- D16 Data warehouse extension design.** Changes needed to support weekly operational analytics alongside regulatory reporting.
- D17 Pilot operations runbook.** Procedures for weekly refresh, data issues, alert delivery and escalation.

CASE STUDY 20 · CYBERSECURITY (FINANCIAL SERVICES GROUP)

Security Log Lake Cost Crisis and SIEM Re-Architecture

Industry	Cybersecurity (Financial Services Group)
Organisation	Halden Group, a financial services group with 14,000 employees across banking, insurance and asset management
Your role	Principal Security Data Engineer
Core themes	Log volume economics, detection coverage, schema normalisation, retention obligations

The Scenario

Halden Group's security operations centre watches everything. Firewalls, endpoint detection agents, identity providers, cloud platforms, email gateways, web proxies, databases, mainframe audit logs and several hundred business applications all send their logs to the group's security information and event management platform, where detection rules look for signs of attack and analysts investigate alerts around the clock. The platform ingests about fourteen terabytes of log data a day. Two years ago it ingested five. The licence for the platform is priced by daily ingestion volume, and last month the chief information security officer, Daniel Harcourt, received a renewal quote that would increase the annual cost by just under two hundred per cent. The finance director has told him that the group cannot afford it. You are the principal security data engineer, and you have been asked to find a way out.

The growth in log volume has several causes. The group migrated most of its workloads to two public cloud providers, and cloud audit, network flow and service logs are voluminous. A new endpoint detection tool was rolled out to every laptop and server, and it produces far more telemetry than the one it replaced. Several business units onboarded their applications to the security platform during a regulatory review, sending everything they could rather than what was useful. And a well-intentioned engineer turned on verbose logging for the identity provider during an incident eighteen months ago and nobody turned it off. Nobody owns the question of what should be logged; application teams decide what they send, the security platform accepts whatever arrives, and the invoice is paid centrally, so no team has ever had a reason to send less.

When you analysed the ingested data, you found that around forty per cent of daily volume was never used by any detection rule, dashboard or investigation in the past ninety days. Some of it was genuinely useless, such as health check messages and debug output. Some of it was rarely queried but essential for investigations after an incident, when analysts need to reconstruct what an attacker did weeks earlier. And some was retained to meet regulatory and legal obligations; the group's financial services regulators expect firms to retain security-relevant logs for defined periods and to be able to produce them.

The head of the security operations centre, Rachel Ng, is anxious about any reduction. Her analysts rely on being able to search everything in one place during an investigation. She points out that the value of a log is often invisible until the day you need it, and that last year's investigation into a compromised privileged account depended on proxy logs that nobody had queried for months before the incident. She also notes that detection engineering is improving, and rules that do not exist today may need data that looks unused today.

The cloud platform team, led by Ian Forsyth, has proposed a security data lake: land all logs in cheap object storage in an open format, keep only the data needed for real-time detection in the expensive platform, and query the lake for investigations and hunting. Several vendors now sell security analytics tools that sit on top of such a lake, and the major cloud providers offer their own. Ian estimates that this could cut costs by more than half. Rachel's worry is that searching a data lake is slower and less convenient than searching the current platform, and that analysts under pressure during an incident will not tolerate query times of minutes instead of seconds.

The detection engineering lead, Oliver Mensah, has a different concern. Detection rules are written against the current platform's data model and query language. Moving data out of the platform, or changing its format, would break rules. Oliver is enthusiastic about adopting an open schema for security events, which would allow rules to be written once and run against any backend, but converting hundreds of rules and dozens of log sources to a new schema is a major undertaking. He also wants detection coverage to be measured systematically: which attack techniques the group can detect, on which systems, with which data. At present, nobody

can say with confidence which data sources are critical to which detections.

The compliance team has its own requirements. The group's retention policy, developed years ago, requires all security logs to be kept for seven years, searchable for the first year. Nobody can remember why seven years was chosen, and the legal team suspects that for some log types it is longer than necessary and may conflict with data minimisation principles for logs containing personal data, such as user browsing activity. For other log types, such as privileged access records, there may be specific regulatory requirements. Rewriting the retention policy requires legal input and approval from risk committees in three regulated businesses.

Daniel has given you eight weeks to produce a plan that he can take to the board. He wants it to reduce costs substantially, not reduce detection capability, satisfy retention obligations and be deliverable within the current contract term, which has fourteen months left. The vendor, sensing the risk of losing volume, has offered a discount if the group commits to a longer term at a slightly lower volume. Accepting it would reduce costs quickly but would lock the group into the current architecture for another five years.

There is an additional complication in the form of an open regulatory commitment. Following an operational resilience review, the group committed to its lead regulator that it would improve its ability to detect and investigate attacks on important business services, including payment processing and customer online banking. Any change that could be perceived as reducing monitoring, even if it merely moves data to a different tier, may need to be explained to the regulator, and the operational risk function has asked to review the plan before anything is switched off. The risk director, Claire Ellwood, has been clear that the group cannot present a cost-cutting story to the regulator; it must present a capability improvement story in which cost reduction is a consequence of better engineering. That framing is reasonable, but it raises the bar for evidence: you will need to demonstrate, not merely assert, that the redesigned architecture detects at least as much as the current one and supports investigations at least as well.

You are aware that the technically elegant answer, a data lake with an open schema and detection rules that run anywhere, is a multi-year programme, and that the cost problem is immediate. You also know that

simply filtering out unused logs, the quickest saving, risks removing data that a future investigation needs, and that if a serious incident occurs after you have cut something, you will be the person who cut it.

THE DECISION IN FRONT OF YOU

Do you accept the vendor's discounted long-term commitment, pursue a tiered architecture with a security data lake, or aggressively filter and optimise within the current platform first, and how do you prove that detection coverage and investigative capability are preserved throughout?

Data Sources

#	Source and link	How it is used in this project
1	CIC-IDS2017 Intrusion Detection Dataset https://www.unb.ca/cic/datasets/ids-2017.html	Labelled network traffic and attack scenarios used to test detection pipelines against lake and platform backends.
2	Loghub System Log Collection https://github.com/logpai/loghub	Real system log datasets from multiple platforms used to profile volume, parsing and compression.
3	MITRE ATT&CK Framework https://attack.mitre.org	Adversary technique knowledge base used to map detection coverage to data sources.
4	Open Cybersecurity Schema Framework (OCSF) https://schema.ocsf.io	Open schema for normalising security events across sources and backends.
5	Sigma Detection Rules https://github.com/SigmaHQ/sigma	Vendor-neutral detection rules used to test portability across query engines.
6	CISA Known Exploited Vulnerabilities Catalog https://www.cisa.gov/known-exploited-vulnerabilities-catalog	Threat intelligence used to prioritise detections and required data sources.
7	NCSC Logging and Protective Monitoring Guidance https://www.ncsc.gov.uk/collection/device-security-guidance/managing-deployed-devices/logging-and-protective-monitoring	UK guidance on which logs matter for security monitoring.
8	Apache Parquet Documentation https://parquet.apache.org/docs/	Columnar storage format for cost-efficient log retention in the data lake.

Deliverables (17)

- D1 Log volume and usage analysis.** A breakdown of ingestion by source, type and volume with usage by detections, dashboards and investigations.
- D2 Detection coverage map.** A mapping of detections to ATT&CK techniques and the data sources each requires.
- D3 Critical data source register.** A classification of sources as detection-critical, investigation-critical, compliance-only or low-value.

- D4 Retention policy review.** A legal and regulatory analysis proposing retention periods by log type.
- D5 Commercial options analysis.** A comparison of the vendor discount, tiered architecture and optimisation-first options over five years.
- D6 Tiered architecture design.** A design routing logs to hot detection, warm search and cold lake tiers by value.
- D7 Log pipeline and routing layer.** A collection and routing layer that filters, samples, enriches and forwards logs by policy.
- D8 Noise reduction rules.** Filters removing debug, health check and duplicate events with documented justification.
- D9 Schema normalisation to OCSF.** Mapping of priority log sources into a common schema with validation.
- D10 Security data lake implementation.** Object storage, open table format and query engine for long-term retention and hunting.
- D11 Investigation query performance tests.** Benchmarks of lake query times against analyst needs for common investigation scenarios.
- D12 Detection rule migration plan.** A sequenced plan for converting rules to portable formats with regression testing.
- D13 Coverage regression testing.** Automated attack simulation proving detections still fire after each change.
- D14 Personal data minimisation controls.** Masking or tokenisation of personal data in logs where not needed for security purposes.
- D15 Cost monitoring dashboard.** Daily ingestion and storage costs by source and tier with budget alerts.
- D16 Analyst training and playbooks.** Updated investigation procedures and training for querying across tiers.
- D17 Board briefing paper.** A concise paper on costs, risks, coverage assurance and the recommended path.

CASE STUDY 21 · TECHNOLOGY (SOFTWARE AS A SERVICE)

Multi-Cloud Cost Data Normalisation and FinOps Platform

Industry	Technology (Software as a Service)
Organisation	Corvid Systems, a B2B software company running workloads on three public clouds and spending £31 million a year on cloud
Your role	Staff Data Engineer, FinOps and Platform Economics
Core themes	Billing data normalisation, cost allocation, unit economics, organisational accountability

The Scenario

Corvid Systems sells workflow automation software to mid-sized and large enterprises. Its platform runs across three public clouds: the original product on one provider, a data analytics product acquired three years ago on a second, and a machine learning service launched last year on the third, chosen because it offered particular hardware for model training. The company spends thirty-one million pounds a year on cloud infrastructure, which is roughly a quarter of revenue. That share has crept up from eighteen per cent two years ago, and the chief financial officer, Laura Kinsella, has told the board that bringing it back below twenty per cent is essential if Corvid is to reach profitability and prepare for an eventual public listing. She has asked for a FinOps capability, meaning a discipline of financial accountability for cloud spending, and you have been recruited as staff data engineer to build its data foundations.

The problem begins with the data. Each cloud provider produces detailed billing data, running to hundreds of millions of line items a month across Corvid's accounts, but each uses its own format, terminology and structure. One provider's billing export describes usage in terms of products, usage types and operations; another in terms of services, meters and resource groups; the third in terms of SKUs and projects. Discounts, credits, reserved capacity and committed use agreements are represented differently. Some charges, such as support fees and marketplace purchases, appear as lump sums not attributable to any resource. Amortising upfront commitments across the months they cover requires understanding each provider's specific conventions. An open specification for normalised cloud billing data has emerged in the industry, and the

three providers now support it to varying degrees, but coverage of some fields is incomplete, and historical data predates it.

Allocation is harder still. To hold teams accountable, costs must be attributed to products, teams, environments and, ideally, customers. Corvid uses tags and labels on cloud resources for this, but tagging compliance is patchy. About sixty-five per cent of spend on the primary cloud carries a valid team tag, forty per cent on the analytics cloud, and almost none on the machine learning cloud, where the team moved quickly and never set up tagging. Shared resources, such as Kubernetes clusters that run services for many teams, network egress, logging and monitoring platforms, and shared databases, account for around thirty per cent of total spend and cannot be tagged to a single owner at all. Their costs must be split using some allocation method, and every allocation method creates winners and losers.

The engineering teams are wary. The vice president of engineering, Marcus Oyelaran, supports the goal but worries that FinOps will become a blame exercise that distracts engineers from building product. He has seen teams at previous companies spend more time arguing about allocation methodology than saving money. The platform engineering team, which runs the shared Kubernetes clusters, fears being made to look like the biggest cost centre in the company when it is actually providing infrastructure for everyone. Several product teams have already asked how their costs will be calculated before agreeing to any targets.

Laura wants something specific: unit economics. She wants to know the cloud cost of serving each customer, each product and each unit of usage, such as workflow executions or data processed, so that pricing decisions, contract negotiations and product investment can be informed by real margins. The head of sales has recently signed several large enterprise contracts at steep discounts, and Laura suspects that some of those customers cost more to serve than they pay. Answering that requires joining cloud costs with product telemetry about which customers use which resources, which in a multi-tenant architecture is far from straightforward.

The machine learning team, led by Dr Sofia Marin, is responsible for the fastest-growing slice of spend. Training runs on specialised hardware cost tens of thousands of pounds each, and inference for the new AI features runs continuously. Sofia is supportive of visibility but worries that cost

pressure will stifle experimentation at exactly the moment the company's AI features are its main competitive differentiator. She would like cost data that distinguishes between experimental and production workloads, and budgets that reflect the different nature of research spending.

There are also build versus buy questions. Several commercial FinOps platforms offer multi-cloud cost visibility, allocation and recommendations out of the box. They are quick to deploy but priced as a percentage of cloud spend, which at Corvid's scale would cost several hundred thousand pounds a year. They also struggle with customer-level unit economics, which requires internal telemetry that the tools cannot access. Building in-house on Corvid's own data warehouse would be cheaper to run and more flexible but would take longer, and the data engineering team is already committed to customer-facing analytics work.

Laura has asked for a credible cost visibility and allocation capability within three months, with unit economics within six. The board expects a plan for reducing cloud spend as a share of revenue at its next meeting. You have been given two engineers and a FinOps analyst recruited from a consultancy.

There is a previous attempt to learn from. Eighteen months ago, the finance team introduced a monthly cloud cost report that allocated all spend to teams using a simple rule: each team's share of shared costs was proportional to its share of directly tagged costs. The report was circulated to engineering managers without discussion. Within weeks, one team discovered that it was being charged for a large share of the logging platform despite sending very little data to it, and another found that it had been allocated costs for a database it had decommissioned months earlier, because the resource had been left running under the wrong account. The engineering managers stopped reading the report, and the finance team stopped producing it after three months. Several managers still cite it as evidence that finance does not understand engineering. Any new capability has to overcome that scepticism, which means the first reports must be demonstrably accurate, the allocation rules must be visible and open to challenge, and there must be a process for correcting errors quickly.

As you assess the situation, you realise that the most important outputs are not technical but social: an allocation methodology that engineering leaders accept as fair, targets that teams believe they can influence, and reporting that prompts action rather than argument. The data pipelines

are necessary but not sufficient. You also realise that the tagging problem will never be solved by reporting alone; it needs enforcement in the infrastructure provisioning process, which belongs to the platform team, who are the ones most worried about how they will be portrayed.

THE DECISION IN FRONT OF YOU

Do you buy a commercial FinOps platform, build a normalised cost data model in-house, or combine them, and how do you design allocation of shared costs, tagging enforcement and customer-level unit economics so that engineering teams accept the numbers and act on them?

Data Sources

#	Source and link	How it is used in this project
1	FinOps Open Cost and Usage Specification (FOCUS) https://focus.finops.org	Open specification for normalised cloud billing data used as the canonical cost schema.
2	AWS Cost and Usage Reports Documentation https://docs.aws.amazon.com/cur/latest/userguide/what-is-cur.html	Reference for detailed AWS billing exports, amortisation and discount representation.
3	Google Cloud Billing Export to BigQuery https://cloud.google.com/billing/docs/how-to/export-data-bigquery	Reference for GCP detailed billing exports and label handling.
4	Azure Cost Management Exports https://learn.microsoft.com/en-us/azure/cost-management-billing/costs/tutorial-export-acm-data	Reference for Azure cost and usage export formats.
5	FinOps Foundation Framework https://www.finops.org/framework/	Capabilities and maturity model for allocation, unit economics and accountability.
6	OpenCost https://www.opencost.io	Open-source Kubernetes cost allocation used to split shared cluster costs by namespace and workload.
7	Cloud Carbon Footprint https://www.cloudcarbonfootprint.org	Open-source tool estimating emissions from billing data for sustainability reporting alongside cost.
8	Google Cluster Workload Traces https://github.com/google/cluster-data	Real cluster usage traces used to prototype shared resource allocation methods.

Deliverables (17)

- D1 Build versus buy assessment.** A comparison of commercial FinOps platforms and in-house build on cost, time, flexibility and unit economics capability.
- D2 Canonical cost data model.** A FOCUS-aligned model normalising billing from all three providers with provider-specific extensions.

- D3 Billing ingestion pipelines.** Automated daily ingestion of detailed billing exports from each provider with reconciliation to invoices.
- D4 Amortisation and discount treatment.** Consistent logic for reserved capacity, committed use, credits and support fees.
- D5 Tagging standard.** A mandatory tag schema for team, product, environment and cost centre agreed across engineering.
- D6 Tagging enforcement in provisioning.** Policy controls in infrastructure-as-code pipelines rejecting untagged resources.
- D7 Tag coverage remediation.** Back-filling of ownership for untagged resources through account structure and inference.
- D8 Shared cost allocation methodology.** A documented, agreed method for splitting Kubernetes, network, logging and database costs.
- D9 Kubernetes cost allocation.** Namespace and workload-level cost attribution for shared clusters.
- D10 Customer usage telemetry join.** A mapping of multi-tenant resource usage to customers using product telemetry.
- D11 Unit economics model.** Cost per customer, per product and per usage unit with margin analysis.
- D12 Team cost dashboards.** Self-service views showing each team its costs, trends and drivers.
- D13 Anomaly detection and alerts.** Daily detection of unexpected cost spikes with notification to owners.
- D14 Experimental versus production spend tracking.** Classification and budgeting of machine learning research and production workloads.
- D15 Savings opportunity pipeline.** Identification of rightsizing, idle resources and commitment opportunities with estimated savings.
- D16 Governance and review cadence.** Monthly cost review meetings, roles and escalation agreed with engineering leadership.
- D17 Board reporting pack.** Cloud cost as a share of revenue, unit economics trends and savings delivered.

CASE STUDY 22 · HOSPITALITY

Hotel Revenue Management Data Platform and Demand Forecasting

Industry	Hospitality
Organisation	Harbourline Hotels, a group of 38 mid-market and boutique hotels in the UK, Ireland and the Netherlands
Your role	Lead Data Engineer, Commercial Systems
Core themes	Booking data integration, cancellation behaviour, third-party distribution data, pricing governance

The Scenario

Harbourline Hotels grew through acquisition. Over twelve years, the group bought independent hotels and small chains, rebranded some, left others with their local identities, and assembled a portfolio of thirty-eight properties ranging from a two-hundred-room conference hotel near Birmingham airport to a forty-room boutique hotel on an Amsterdam canal. Each acquisition brought its own property management system, its own channel manager for distributing rooms to online travel agents, and its own habits for setting prices. The commercial director, Alicia Monroe, has spent the past two years trying to turn that patchwork into a coherent revenue management operation, and she has now concluded that the missing piece is data. You were hired as lead data engineer for commercial systems to build it.

Revenue management in hotels is the art of selling the right room to the right guest at the right price at the right time. It depends on forecasting demand for each night at each hotel, understanding how demand responds to price, and managing the mix of channels, from direct bookings on the group's website to online travel agents that take commissions of fifteen to twenty-five per cent, corporate contracts, group bookings and wholesalers. At Harbourline, forecasting is done by a team of six revenue managers, each responsible for several hotels, using spreadsheets built from reports exported daily from each property management system. The revenue managers are experienced and good at their jobs, but they spend most of their mornings compiling data rather than analysing it, and each builds forecasts slightly differently.

The group's performance has been respectable but uneven. Some hotels consistently achieve revenue per available room well above their local

competitors; others lag. Alicia believes much of the difference comes down to data: the strong performers have revenue managers who track booking pace, cancellations and competitor prices obsessively, while the weaker ones react late. She wants a group-wide platform that consolidates bookings, forecasts demand consistently and gives every revenue manager the same quality of information. A commercial revenue management system vendor has pitched an integrated product that would do forecasting and price recommendations automatically, and the chief executive, Graham Pearce, is attracted by the idea of automation.

The data is harder to consolidate than it looks. Harbourline's thirty-eight hotels run five different property management systems. Each records reservations, but they disagree about fundamental concepts. One system treats a reservation for three nights as a single record with arrival and departure dates; another creates a record per night. Room types are coded locally, so a "superior double" in one hotel is a "deluxe king" in another, and the same physical room category may have been renamed after refurbishment. Rate codes, which identify the price plan and channel, number in the thousands across the group, many of them obsolete but still attached to historical bookings. Cancellations are recorded differently: some systems delete cancelled reservations after a period, others keep them with a status flag, and one overwrites the original booking date when a reservation is modified, which destroys the information needed to measure booking pace.

Cancellations deserve particular attention. Since the pandemic, guests have become used to flexible rates, and online travel agents promote free cancellation heavily. At some Harbourline hotels, more than forty per cent of bookings made through online agents are cancelled before arrival, and a fraction of those are speculative bookings made by guests holding several hotels at once. Any forecast based on reservations on the books must predict how many will cancel, and cancellation behaviour differs sharply by channel, rate type, lead time and nationality of the guest. The revenue managers compensate by overbooking, but inconsistently, and the Amsterdam hotel had to walk guests to other hotels eleven times last summer, generating complaints and compensation costs.

Competitor data adds another dimension. Revenue managers track competitor prices through a rate shopping service that scrapes prices from online travel agents daily. The data is useful but noisy: it captures the

lowest available price for a standard room, which may not be comparable, and it misses prices that are only available to loyalty members or through closed user groups. There are also questions about the terms of use of scraped data that the legal team has flagged but never resolved.

The vendor's integrated revenue management system would handle much of this: it connects to property management systems, forecasts demand, recommends prices and can push prices automatically to channels. But the vendor supports only two of Harbourline's five property management systems natively and would charge for custom integrations to the others. Its forecasting models are proprietary, and the revenue managers are uneasy about trusting recommendations they cannot interrogate. The head of revenue for the UK, Patrick Doherty, is openly hostile. He has used such systems before and says they perform well in stable markets but badly around unusual events, such as a concert, a conference or a transport strike, where local knowledge matters. He fears the group will pay a large fee for a system that the revenue managers override constantly.

Graham, however, notes that the group cannot scale its revenue management team as it grows, and that automation of routine pricing decisions would free the team to focus on the difficult ones. The finance director, Sunita Rao, has another requirement: she wants consistent group-wide reporting of occupancy, average daily rate and revenue per available room, calculated identically across hotels, because the current reports use different definitions for things like complimentary rooms and out-of-order rooms, making hotel comparisons unreliable.

There is also a history that shapes attitudes. Five years ago, before Alicia joined, the group attempted to consolidate hotel data into a central warehouse using a consultancy. The project ran over budget, delivered a set of reports that did not match the property management systems, and was quietly abandoned. The revenue managers remember it well. Patrick has said, only half jokingly, that he will believe in a group data platform when it tells him something he does not already know from his own spreadsheet.

Data protection also features, though less prominently than in some industries. Guest profiles include names, contact details, nationalities, stay histories and sometimes preferences and special requests that can reveal health or religious information. The Netherlands hotel falls under Dutch supervisory authority oversight, and the Irish hotels under the Irish

regulator, each with slightly different expectations. A group platform that centralises guest-level data would need a clear lawful basis and retention rules, and marketing would inevitably ask to use it for campaigns. Your working assumption is that revenue management needs bookings rather than people, and that guest identities can be pseudonymised in the analytical platform, but that assumption needs to be tested with the loyalty and marketing teams, who have their own plans.

Alicia has given you nine months and a small team. She wants to go into the next budget cycle with a clear recommendation on whether to buy the vendor system, and with the data foundations that any approach will need. You suspect that the most valuable thing you can build first is a clean, consolidated, history-preserving booking dataset with consistent definitions, because without it neither the vendor system nor in-house forecasting can succeed. But that is invisible work, and the pressure from the chief executive is for something visible and automated.

THE DECISION IN FRONT OF YOU

Do you recommend the vendor's automated revenue management system, build in-house forecasting on a consolidated data platform, or lay common data foundations and then decide, and how do you handle cancellations, room type mapping and competitor data so that revenue managers actually trust the output?

Data Sources

#	Source and link	How it is used in this project
1	Hotel Booking Demand Dataset https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand	Real booking records from two hotels with lead time, channel, cancellation and rate data used to prototype cancellation and pace models.
2	Inside Airbnb https://insideairbnb.com/get-the-data/	Short-term rental listings, prices and availability used as a competitive supply and pricing signal by city.
3	ONS Travel Trends and Tourism Statistics https://www.ons.gov.uk/peoplepopulationandcommunity/leisureandtourism	Official visitor flows used as macro demand features.
4	Eurostat Tourism Database https://ec.europa.eu/eurostat/web/tourism/database	European accommodation occupancy statistics for benchmarking the Irish and Dutch hotels.
5	VisitBritain Research and Insights https://www.visitbritain.org/research-insights	Inbound tourism data and forecasts informing demand context.

#	Source and link	How it is used in this project
6	HTNG Integration Specifications https://www.htng.org	Hospitality technology standards for reservation and rate message integration.
7	Ticketmaster Discovery API https://developer.ticketmaster.com/products-and-docs/apis/discovery-api/v2/	Local event data such as concerts and sports fixtures used as demand event features.
8	Open Exchange Rates API https://openexchangerates.org	Currency data for normalising revenue across sterling and euro properties.

Deliverables (17)

- D1 Recommendation paper on revenue management approach.** A comparison of vendor, in-house and staged approaches with costs, risks and success criteria.
- D2 Property management system integration layer.** Connectors extracting reservations from all five systems with change capture.
- D3 Canonical reservation model.** A stay-night grain model with consistent statuses, dates and history of modifications.
- D4 Booking history preservation.** Snapshotting of reservations on the books each day to measure pace, even where source systems overwrite data.
- D5 Room type harmonisation.** A group-wide room category hierarchy mapped from local codes, with refurbishment history.
- D6 Rate code rationalisation.** Classification of thousands of rate codes into channel, segment and restriction categories.
- D7 KPI definitions standard.** Agreed definitions of occupancy, ADR and RevPAR including treatment of complimentary and out-of-order rooms.
- D8 Cancellation prediction model.** Predicted cancellation probability by booking, used to adjust forecasts and overbooking.
- D9 Demand forecasting pipeline.** Consistent nightly forecasts by hotel, night and segment with accuracy tracking.
- D10 Local events calendar feed.** An automated calendar of concerts, conferences and holidays near each hotel.
- D11 Competitor rate data governance.** A review of rate shopping data quality, comparability and legal terms of use.
- D12 Overbooking guidance tool.** Recommended overbooking levels by hotel and night based on predicted cancellations and no-shows.
- D13 Revenue manager workbench.** Dashboards for pace, pickup, forecast versus actual and competitor positioning.
- D14 Forecast accuracy benchmark.** A comparison of in-house forecasts against revenue managers' spreadsheets and vendor trial outputs.
- D15 Data quality monitoring.** Checks on missing reservations, unmapped codes and system extract failures.

- D16 Group performance reporting.** Consistent hotel, region and group reporting for finance and executives.
- D17 Change management plan.** Engagement and training for revenue managers informed by lessons from the failed warehouse project.

CASE STUDY 23 · PROFESSIONAL SPORT

Match Event Data Platform for a Professional Football Club

Industry	Professional Sport
Organisation	Ashford Rovers FC, a Championship football club aiming for promotion to the Premier League
Your role	Head of Data Engineering, Performance and Recruitment Analytics
Core themes	Third-party data licensing, event and tracking data integration, player identity, analytics adoption

The Scenario

Ashford Rovers are a club with a long history, a passionate fan base and a budget that is modest by the standards of the league they want to join. The owner, a technology entrepreneur named Victor Hale, bought the club three years ago with a clear strategy: compete with wealthier rivals by making smarter decisions, particularly in player recruitment, where mistakes are expensive and hard to undo. He has invested in a data department of eight people, and you were hired eighteen months ago to lead its engineering. The department has had successes, including identifying a young winger from the Belgian second division who has since become the team's top scorer, but it has also made enemies, and the coming transfer window will decide whether the data department's influence grows or shrinks.

The club buys data from several providers. An event data provider supplies detailed records of every on-ball action in matches from dozens of leagues: passes, shots, tackles, dribbles, with locations and outcomes. A tracking data provider supplies player and ball positions at twenty-five frames per second for the club's own league matches. The club's own training ground collects GPS and heart rate data from wearable vests during training, as well as physical testing results. A scouting platform holds reports written by the club's network of scouts. Video is stored in yet another system. Each provider identifies players, teams and matches with its own identifiers, and the same player may appear under different names, spellings and birthdates across sources.

The most painful problem is player identity. When the recruitment team wants to evaluate a target, they need to combine his event data from the league he plays in, his physical data if tracking is available, scouting

reports and video, and his contract and market valuation information. Matching a player across sources is done by hand for each target, and errors happen. Last season, the recruitment analysts presented a data profile of a defender that turned out to combine statistics from two different players with the same name in the same country. The error was caught before any offer was made, but it was caught by the head of recruitment, who has been sceptical of the data team ever since.

The head of recruitment, Stuart Crane, is a former player and scout with thirty years in the game. He values data but believes it is a supplement to watching players, not a replacement. The first team manager, Luca Bernardi, is more receptive, particularly to tactical analysis of opponents and to physical load monitoring that helps him manage player fatigue. The head of performance science, Dr Ellie Mason, wants to link training load with match physical output and injury history to reduce soft tissue injuries, which cost the club dozens of player availability days last season. Victor wants the data department to give the club a genuine competitive advantage in the transfer market and has asked for a player valuation model that estimates what a player is worth, both on the pitch and in resale value.

The engineering challenges are significant. Event and tracking data arrive in different formats and coordinate systems; one provider measures the pitch in metres from a corner, another uses a normalised scale from zero to one hundred with the origin in the centre. Synchronising event data with tracking data requires aligning timestamps that differ by several seconds and sometimes drift during a match. Tracking data for a full season of the club's league amounts to several billion rows, and the analysts currently process it on their laptops, one match at a time. Training ground data is exported from the vendor's software as spreadsheets and stored on a shared drive.

Licensing is a further constraint. The event data contract permits use for internal analysis but prohibits sharing raw data with third parties, including agents and other clubs, and restricts retention after the contract ends. The tracking data contract is specific to the club's own league and does not cover other competitions. Several providers have changed their data models in recent seasons, breaking the club's analysis scripts. And the costs are rising: the event data provider has proposed a large price increase for next season, citing expanded league coverage the club did not

ask for. Victor has asked whether open or cheaper data sources could replace some of what the club buys. There are free event datasets for certain competitions, and a growing ecosystem of open-source tools, but coverage of the leagues where the club recruits is limited.

Within the department, there is tension between the analysts, who want flexibility to explore data in notebooks, and the need for a robust, shared platform. The lead recruitment analyst, Tom Whitaker, has built an impressive set of player metrics in his own code, but they depend on his personal data extracts and nobody else can reproduce them. The performance analysts have their own scripts for opponent analysis. Nothing is tested, and when a provider changes its format, everything breaks at once, usually on a Friday before a Saturday match.

Victor has also asked for a player valuation model to be ready for the summer window, four months away. You are wary. A valuation model depends on reliable player identity, consistent metrics across leagues of very different quality and a history of transfer fees, which are often undisclosed or reported inaccurately in the media. A model built hastily on shaky foundations could produce confident numbers that lead the club to overpay, and if it did, the data department's credibility would be damaged beyond repair.

Governance of player data is another consideration that the club has never formally addressed. Training ground data about the club's own players includes heart rate, sleep reports, body composition and injury records, which are health data under data protection law. Players have occasionally asked, through their agents, what the club holds about them, particularly during contract negotiations, where fitness data can influence terms. The players' union has published guidance on athletes' data rights, and at least one player at another club has challenged how his performance data was used in contract talks. Ellie wants to integrate physical and medical data for injury prevention, which is legitimate and valuable, but the platform will need strict access controls so that medical data is not casually visible to recruitment analysts or used in ways players have not been told about.

The window deadline concentrates minds. You have three engineers, a budget that can stretch to one new data subscription, and a recruitment department that will use data if it is trusted and ignore it if it is not. You need to decide whether to prioritise the player identity and data platform foundations that everything depends on, or the high-profile valuation

model the owner wants, and how to handle the event data contract renewal.

THE DECISION IN FRONT OF YOU

Do you prioritise a robust player identity and match data platform, the owner's player valuation model for the coming window, or a scoped combination, and how do you handle rising data licensing costs while winning the trust of a sceptical recruitment department?

Data Sources

#	Source and link	How it is used in this project
1	StatsBomb Open Data https://github.com/statsbomb/open-data	Free event data for selected competitions used to build the event schema, metrics and pipeline.
2	Metrica Sports Sample Tracking Data https://github.com/metrica-sports/sample-data	Anonymised tracking and event data for synchronisation and coordinate transformation work.
3	Football-Data.co.uk https://www.football-data.co.uk/data.php	Historical results and betting odds across many leagues used for team strength and league quality adjustment.
4	Kloppy Python Library https://kloppy.pysport.org	Open-source library standardising event and tracking data across providers and coordinate systems.
5	Transfermarkt (reference for market values and transfers) https://www.transfermarkt.co.uk	Public transfer histories and market value estimates used as reference data, subject to site terms.
6	FIFA Football Agent and Transfer Regulations https://inside.fifa.com/legal/football-regulatory	Regulatory context for transfer data and agent interactions.
7	Stats Perform (Opta) Data Services https://www.statsperform.com	Commercial event and tracking data provider used as a benchmark in the licensing review.
8	Wyscout Platform https://wyscout.com	Commercial scouting video and data platform used in the provider comparison.

Deliverables (17)

- D1 Data landscape and licensing review.** An inventory of providers, contract terms, costs, coverage and usage restrictions.
- D2 Player identity master.** A cross-provider player register with probabilistic matching, manual review and change history.
- D3 Team, competition and match reference data.** Harmonised identifiers for clubs, leagues, seasons and fixtures.
- D4 Unified event data model.** A provider-agnostic event schema with consistent coordinates and action definitions.

- D5 Event and tracking synchronisation.** Alignment of event timestamps with tracking frames with drift correction.
- D6 Tracking data processing pipeline.** Scalable processing of tracking data into physical and tactical metrics per player and match.
- D7 Training ground data integration.** Automated ingestion of GPS, heart rate and testing data linked to player identity.
- D8 Shared player metrics library.** Tested, documented implementations of recruitment metrics replacing personal scripts.
- D9 League strength adjustment.** A method for comparing player metrics across leagues of different quality.
- D10 Transfer history dataset.** A curated record of transfers, fees and contract information with source and confidence.
- D11 Player valuation model prototype.** A transparent model with uncertainty ranges and clear limitations.
- D12 Recruitment profile report.** A standard player profile combining data, scouting and video links for decision meetings.
- D13 Load and injury analytics.** Linking training and match load with injury records for performance science.
- D14 Provider change detection.** Automated schema validation that detects provider format changes before they break analysis.
- D15 Data access and contract compliance controls.** Controls enforcing licence restrictions on sharing and retention.
- D16 Event data contract negotiation brief.** An analysis of usage and alternatives to support renewal negotiations.
- D17 Transfer window readiness plan.** A timeline of capabilities available for the summer window with owners and fallbacks.

CASE STUDY 24 • VIDEO GAMES

Mobile Game Telemetry Pipeline and Experimentation Platform

Industry	Video Games
Organisation	Tidepool Games, a mobile game studio with three live titles and 4.5 million monthly active players
Your role	Senior Data Engineer, Live Operations
Core themes	High-volume event telemetry, schema evolution, A/B testing integrity, player privacy and monetisation ethics

The Scenario

Tidepool Games makes casual puzzle and simulation games for phones. Its flagship title, a match-three puzzle game with a gentle seaside theme, has been live for five years and still generates most of the studio's revenue through optional in-game purchases and advertising. Two newer titles are growing but not yet profitable. Like most free-to-play studios, Tidepool lives and dies by live operations: a constant cycle of new levels, events, offers and features, each tested and tuned based on how players respond. The data that informs those decisions comes from telemetry, the stream of events that the games send about every level started, completed or failed, every purchase, every advertisement viewed and every session begun or abandoned. You are a senior data engineer in the live operations team, and the telemetry pipeline has become a source of daily frustration.

The pipeline was built by the studio's first engineer on a managed analytics service that was cheap and simple when the flagship game had a few thousand players. It now processes around one and a half billion events a day. Events are sent from the game client in batches, stored by the analytics service and exported nightly to the studio's data warehouse. Analysts and game designers query the warehouse the next day. Each game sends events in its own format, because each was built by a different team and there was never a studio-wide tracking standard. The flagship game has more than four hundred event types, many of them created for experiments that ended years ago and never removed. Event parameters are stored as loosely typed key-value pairs, and the same concept, such as the level number, is called different things in different events.

The frustrations surface every week. When the game design team launches a new event, such as a limited-time tournament, they want to see results

within hours, not the next day. When a new game version ships with a tracking bug, as happened last month when a change caused purchase events to be sent twice on one Android device model, the error is not noticed for days, and by then revenue reports, experiment results and player segmentation are all contaminated. When analysts try to compare player behaviour across the three games, they spend days reconciling different event definitions.

Experimentation is where the stakes are highest. Tidepool runs dozens of A/B tests at any time: different level difficulty curves, offer prices, reward sizes, advertisement frequencies. Decisions based on these tests directly affect revenue and player experience. The head of product, Maya Lindgren, recently discovered that a test that had apparently increased revenue per player by eight per cent was invalid. The assignment of players to test groups was done in the game client, and a bug caused players who reinstalled the game to be reassigned, sometimes to a different group. Heavy spenders reinstall more often, because they play on multiple devices, and they were disproportionately moved into one group. The test result was an artefact. The change had already been rolled out to all players. Nobody knows how many other past decisions were based on similarly flawed tests.

The chief technology officer, Ben Ashworth, wants to replace the managed analytics service with an in-house streaming pipeline: events sent from the game to the studio's own collection endpoint, validated against schemas, streamed into a real-time processing layer and landed in the warehouse within minutes. He argues that this would give the studio control, speed and lower costs at current volumes, since the managed service is priced per event and costs have risen sharply. The data engineering lead, your manager Priya Shah, agrees in principle but worries about the operational burden: the managed service has never had a significant outage, whereas an in-house pipeline would need on-call support that the team of four engineers has never provided.

The game teams are ambivalent. They want faster data and better quality but are wary of any change that requires them to re-instrument their games, which takes development time away from content. The flagship game's lead developer, Callum Reid, has told you that his team has a packed roadmap and cannot spend weeks rewriting tracking code. He

would accept a new tracking library only if it were a near drop-in replacement.

Privacy and ethics add further complexity. The games are played by adults and, despite age gating, by some younger players. Advertising within the games uses advertising identifiers and involves sharing data with advertising networks, which requires consent under data protection law and platform rules. Apple's App Tracking Transparency framework means most iOS players have opted out of tracking, which has changed the economics of advertising and made attribution of player acquisition campaigns much harder. The studio's legal counsel, Nadia Farouk, has also raised concerns about monetisation analytics. Regulators and consumer groups have been examining whether free-to-play games use data to target vulnerable players, such as those who spend compulsively, with aggressive offers. Some of Tidepool's segmentation models do identify high-spending players and target them with personalised offers. Nadia wants any new platform to support responsible design, including the ability to detect and protect potentially vulnerable spenders, and she wants the studio to be able to demonstrate what data drives offer personalisation.

There is also cost pressure. The studio's finances depend heavily on one ageing title, and the chief financial officer has asked for infrastructure costs to be cut. Ben's in-house pipeline promises savings but requires upfront engineering investment. The managed service vendor, sensing the risk, has offered a discounted rate for a three-year commitment.

The game teams have also accumulated technical debt in how player identity is handled. A player might begin on a phone as an anonymous guest, later link a social account, then install the game on a tablet and restore progress through the social login. Each step can generate a new identifier, and the analytics service treats them as separate players until they are linked, if they ever are. Retention and lifetime value calculations are therefore distorted, because a single committed player can appear as several short-lived ones. The flagship game's player support team maintains its own manual process for merging accounts when players complain about lost progress, and those merges are not reflected in the analytics data at all. Any serious fix to experimentation or monetisation analytics depends on a reliable player identity graph, which does not currently exist.

You have been asked to recommend a path within a month. You believe the experimentation problem is the most urgent, because flawed tests lead directly to bad decisions, and that fixing it requires server-side assignment and exposure logging regardless of where events are processed. You also believe a studio-wide tracking standard is essential, but you know the game teams will resist it unless it saves them effort. And you are not sure the team can safely operate a real-time streaming pipeline, however attractive it looks on a whiteboard.

THE DECISION IN FRONT OF YOU

Do you build the in-house streaming telemetry pipeline, commit to the managed service with improved practices, or tackle experimentation integrity and tracking standards first, and how do you embed responsible monetisation safeguards into the data platform?

Data Sources

#	Source and link	How it is used in this project
1	Mobile Games A/B Testing (Cookie Cats) https://www.kaggle.com/datasets/yufengsui/mobile-games-ab-testing	Real A/B test data on player retention from a mobile puzzle game used to design experimentation analysis.
2	Snowplow Documentation https://docs.snowplow.io	Open-source event pipeline reference for schema validation, enrichment and self-describing events.
3	Apache Kafka Documentation https://kafka.apache.org/documentation/	Streaming backbone reference for the in-house telemetry option.
4	Apple App Tracking Transparency https://developer.apple.com/documentation/apptrackingtransparency	Platform rules on tracking consent that shape advertising data design.
5	Google Play Families Policy https://support.google.com/googleplay/android-developer/answer/9893335	Requirements for apps that may be used by children.
6	UK Department for Culture, Media and Sport Loot Box Response https://www.gov.uk/government/consultations/loot-boxes-in-video-games-call-for-evidence	Government position and industry principles on in-game monetisation.
7	GrowthBook Open Source Experimentation https://docs.growthbook.io	Reference implementation for feature flags, server-side assignment and experiment analysis.
8	Steam Web API https://steamcommunity.com/dev	Additional gaming telemetry and player statistics source for cross-platform benchmarking.

Deliverables (17)

- D1 Experiment audit report.** A review of recent experiments for assignment and exposure flaws with estimated decision impact.
- D2 Server-side experiment assignment.** Deterministic, persistent assignment keyed to player identity across reinstalls and devices.
- D3 Exposure logging standard.** Consistent logging of when players actually encounter experimental variants.
- D4 Experiment analysis framework.** Standard metrics, guardrails, sample ratio mismatch checks and statistical methods.
- D5 Studio-wide tracking plan.** A shared event taxonomy with typed parameters and naming conventions.
- D6 Tracking library.** A client library easing adoption of the tracking plan with minimal game code changes.
- D7 Event schema registry.** Versioned schemas with validation at ingestion and compatibility rules.
- D8 Pipeline options appraisal.** A comparison of in-house streaming, managed service and hybrid with cost and operational risk.
- D9 Event validation and quarantine.** Detection and isolation of malformed or duplicate events before they reach reporting.
- D10 Near-real-time live ops views.** Dashboards showing event performance within minutes of launch.
- D11 Legacy event rationalisation.** Retirement plan for obsolete event types with dependency checks.
- D12 Cross-game player data model.** Consistent player, session, economy and monetisation tables across titles.
- D13 Consent and advertising data controls.** Handling of consent states and advertising identifiers in line with platform rules and law.
- D14 Responsible spending indicators.** Detection of potentially harmful spending patterns with protective interventions.
- D15 Offer personalisation transparency record.** Documentation of data used in offer targeting for legal review.
- D16 On-call and reliability plan.** Monitoring, alerting and support rotas required for any in-house pipeline.
- D17 Cost model and recommendation.** A three-year cost comparison supporting the final decision.

CASE STUDY 25 · MARITIME AND PORT OPERATIONS

Vessel Tracking and Port Congestion Intelligence Platform

Industry	Maritime and Port Operations
Organisation	Estuary Port Authority, operator of a major UK container and bulk port handling 38 million tonnes a year
Your role	Principal Data Engineer, Port Digital Services
Core themes	AIS data processing, multi-party data sharing, geospatial streaming, commercial sensitivity

The Scenario

Estuary Port Authority runs one of the busiest ports on the east coast of Britain. Container ships, bulk carriers, tankers, ferries and roll-on roll-off vessels arrive around the clock, and every arrival depends on a choreography involving pilots, tugs, berths, cranes, lorries and trains. When the choreography works, ships are turned around quickly and cargo flows. When it fails, ships queue at anchor, berths sit idle waiting for delayed vessels, and hauliers wait hours at terminal gates. Last winter, a combination of storms, a crane failure at the container terminal and a surge of delayed vessels from disrupted Asian routes left twenty-three ships waiting at anchor at one point, and the port's reputation with shipping lines suffered. The chief executive, Martin Fairley, has asked for a port-wide intelligence platform that predicts arrivals and congestion, and you, as principal data engineer in port digital services, are responsible for building it.

The central data source is the Automatic Identification System. Every large vessel broadcasts its identity, position, speed, course and navigational status by radio every few seconds while moving, along with voyage information such as destination and estimated time of arrival, entered by the crew. The port receives these signals from its own coastal receivers, and commercial providers aggregate signals from terrestrial and satellite receivers worldwide. AIS data is invaluable but notoriously unreliable in its details. Destinations are typed free-form by crews and are often misspelt, abbreviated or left unchanged from the previous voyage. Estimated arrival times are optimistic or out of date. Some vessels switch off transmitters, and signal gaps over open ocean can last hours. Identifiers are occasionally duplicated or incorrect.

The port's current systems use AIS mainly for display on the vessel traffic services screens, where operators monitor vessel movements in the approaches. Arrival planning is done by the harbour master's team from shipping agents' notifications, which provide estimated arrival times usually twenty-four to seventy-two hours ahead and are updated by email and phone. The container terminal, which is operated by a separate company under a concession, has its own berth planning system and its own view of vessel arrivals. The pilotage service, the towage company and the port's rail operator each keep their own schedules. Nobody shares a single picture of what is coming and when.

Martin's vision is a shared platform, available to the port, terminal operators, pilots, towage, hauliers and rail operators, that predicts vessel arrivals from AIS data days in advance, shows anchorage and berth occupancy in real time, and forecasts congestion at terminal gates. Ports in the Netherlands and Belgium have built similar platforms under the banner of port community systems and data sharing initiatives, and he has seen them in action. He is keen to present the platform as part of the port's bid for government funding under a maritime decarbonisation programme, because better arrival coordination lets ships slow down at sea rather than racing to arrive and then waiting at anchor, which saves fuel and emissions.

The terminal operator's chief operating officer, Claire Jensen, is supportive in principle but cautious about data sharing. Her terminal competes with terminals at other ports, and its berth schedules, crane productivity and customer lists are commercially sensitive. She is willing to share some information with the port for planning purposes but not with hauliers or rival operators, and she wants contractual protections. The shipping lines, whose vessels generate the AIS data, have their own sensitivities. Their commercial teams do not like data about their vessel schedules and delays being visible to competitors, even though AIS is broadcast openly. The pilotage manager, Captain Ian Moss, wants anything that reduces short-notice changes to pilot schedules and is less concerned about commercial sensitivities.

The engineering challenges are substantial. Predicting arrival days in advance requires global AIS coverage, which means buying commercial satellite AIS data, since the port's own receivers only reach a few dozen miles offshore. Commercial providers price by area, vessel count and

update frequency, and satellite AIS updates for vessels in open ocean may be hours apart. The predictions themselves require knowing each vessel's likely route, its current speed, weather along the route, and behaviour such as waiting for tides or berths at previous ports. Voyage data must be cleaned: destinations standardised to port codes, port calls inferred from positions, and the port's own historical arrival records used to train and validate models. Real-time berth and anchorage occupancy requires geofencing and state tracking for every vessel in the port area. Gate congestion forecasts need data from the terminal's appointment system, which Claire has not yet agreed to share.

The IT director, Rebecca Lowe, is concerned about the platform's security and reliability. Port systems are considered critical national infrastructure, and a public-facing data sharing platform would expand the port's attack surface. She also notes that the vessel traffic services system that operators use for navigational safety must remain entirely separate and unaffected by any new platform.

There is also a question of ambition versus delivery. The funding bid deadline is in three months and requires a credible technical plan and evidence of stakeholder commitment. A full port community platform would take two to three years. But you believe a narrower initial product, such as a reliable arrival prediction service for the port's own planning, pilots and towage, could be built within six months using commercial AIS data and the port's own records, and would demonstrate value without requiring the terminal to share sensitive data.

The organisation itself is not set up for a data product. Port digital services is a team of eleven people who mainly support internal business systems, the port's website and the harbour dues billing system. Nobody has built or operated a real-time data service for external users before, and nobody owns the commercial relationships with the parties who would use it. The harbour master's team sees AIS as a safety tool and is nervous about operational staff being distracted by commercial dashboards. The commercial team, which manages relationships with shipping lines and terminal operators, sees the platform as a potential revenue stream and has already floated the idea of charging users for premium data, which would make the terminal operator even more cautious. Before any architecture is finalised, somebody needs to decide who owns the platform,

who funds it after any grant runs out and whether it is a public good for the port community or a commercial product.

You also know that the value of arrival predictions depends on how they are used. If ships are asked to slow down at sea to arrive just in time, the berth must actually be ready when they arrive; otherwise shipping lines will ignore the advice. That requires the terminal's berth plans, which brings you back to the data sharing problem.

THE DECISION IN FRONT OF YOU

Do you pursue the full multi-party port community data platform for the funding bid, a narrower arrival prediction service for the port's own operations first, or a staged approach with a data sharing framework, and how do you balance openness with the commercial sensitivities of terminal operators and shipping lines?

Data Sources

#	Source and link	How it is used in this project
1	MarineCadastre AIS Vessel Traffic Data https://marinecadastre.gov/ais/	Large historical AIS datasets for US waters used to develop port call inference and arrival prediction methods.
2	Global Fishing Watch APIs https://globalfishingwatch.org/our-apis/	Vessel identity, events and AIS-derived activity data used to study gaps and vessel behaviour.
3	Danish Maritime Authority AIS Data https://www.dma.dk/safety-at-sea/navigational-information/ais-data	Openly available historical AIS data for North Sea shipping lanes relevant to UK east coast traffic.
4	UN/LOCODE Code List https://unece.org/trade/cefact/unlocode-code-list-country-and-territory	Standard port codes for normalising free-text AIS destinations.
5	Copernicus Marine Service https://marine.copernicus.eu	Ocean currents, waves and weather along routes used as arrival model features.
6	UK Hydrographic Office Admiralty Tide Data https://admiralty.co.uk/access-data/tidal-data	Tidal predictions affecting port entry windows and arrival timing.
7	IMO Just-In-Time Arrival Guide https://greenvoyage2050.imo.org/just-in-time-arrival-guide/	Industry guidance on just-in-time arrival data exchange and emissions benefits.
8	UNCTADstat Maritime Statistics https://unctadstat.unctad.org	Port call and liner connectivity statistics for benchmarking port performance.

Deliverables (17)

- D1** **Stakeholder data needs and sensitivity map.** A matrix of what each party needs, can share and considers sensitive.

- D2 Data sharing framework.** Principles, legal agreements and access tiers governing multi-party data exchange.
- D3 Commercial AIS procurement specification.** Coverage, update frequency and licensing requirements for satellite AIS.
- D4 AIS ingestion and cleaning pipeline.** Streaming ingestion with deduplication, identity validation and gap handling.
- D5 Destination normalisation.** Mapping of free-text destinations to standard port codes with confidence scores.
- D6 Port call inference.** Detection of historical port calls, anchorage periods and berth stays from positions.
- D7 Arrival prediction model.** Predicted arrival times with uncertainty, validated against historical arrivals.
- D8 Anchorage and berth occupancy tracker.** Real-time geofenced state tracking of vessels within port limits.
- D9 Pilotage and towage planning feed.** Arrival predictions delivered to pilot and towage scheduling.
- D10 Terminal integration design.** A design for exchanging berth windows with the terminal under agreed protections.
- D11 Just-in-time arrival pilot.** A trial with willing shipping lines adjusting speed based on berth availability.
- D12 Emissions benefit estimation.** Calculation of fuel and emissions saved through reduced anchorage waiting.
- D13 Security architecture.** Isolation from vessel traffic services and controls appropriate for critical infrastructure.
- D14 Operational congestion dashboard.** Views of predicted arrivals, anchorage queue and berth utilisation.
- D15 Funding bid technical annex.** A credible plan, evidence of stakeholder commitment and phased milestones.
- D16 Data quality monitoring.** Checks on AIS coverage gaps, stale predictions and identity conflicts.
- D17 Roadmap to port community platform.** A multi-year plan from the initial service to a full shared platform.

CASE STUDY 26 · HUMANITARIAN AND INTERNATIONAL DEVELOPMENT

Humanitarian Needs Data Integration for a Refugee Response

Industry	Humanitarian and International Development
Organisation	Aid Alliance International, an NGO coordinating shelter, food and health assistance for displaced people in three countries
Your role	Senior Data Engineer, Information Management Unit
Core themes	Low-connectivity data collection, data protection for vulnerable people, interoperability between agencies, responsible data

The Scenario

Aid Alliance International works where things have gone badly wrong. Its teams provide shelter, food, cash assistance and primary healthcare to people displaced by conflict and disaster. For the past two years, its largest operation has been a response to a regional displacement crisis spanning three countries, where around four hundred thousand people have fled fighting and live in camps, informal settlements and host communities. The NGO is one of dozens of organisations working in the response, coordinated through clusters led by United Nations agencies. You are a senior data engineer in the information management unit at headquarters, and you have been asked to fix a problem that the country directors describe, with some understatement, as "not knowing who needs what, where".

The data flows in this response are a tangle that anyone who has worked in humanitarian information management will recognise. Field teams collect data using mobile data collection forms on tablets and phones, often offline for days before they can synchronise. Each programme, whether shelter, food, health or protection, designs its own forms, with its own questions, codes and definitions of a household. Needs assessments are carried out by multiple agencies using different methodologies. Distribution records are held in a beneficiary registration system that the NGO licenses, but health clinic records are kept in a separate system, and cash transfer records are held by a financial service provider. Coordination reporting to the clusters is done through spreadsheets submitted monthly, in formats defined by each cluster, which change from time to time.

The consequences are serious. Some households receive assistance from several organisations for the same need while others receive nothing,

because agencies cannot see each other's beneficiary lists. The NGO's own programmes cannot tell whether a family receiving food assistance also has a child enrolled in its malnutrition programme, because the records cannot be linked. Monthly reports to donors take weeks to compile and are frequently revised. When a cholera outbreak began in one camp last spring, the health team needed to know which shelters had inadequate water and sanitation access, and it took ten days to assemble the answer from shelter and water assessment data, by which time the outbreak had spread.

The regional director, Amara Diallo, wants an integrated data platform that brings the NGO's programme data together, links households across programmes, produces timely indicators and supports coordination with other agencies. A major institutional donor has signalled that funding for the next phase will depend on demonstrating improved targeting and reduced duplication, and has offered funds for information management.

The protection lead, Sofia Andersson, is deeply concerned about the risks. The people the NGO serves are among the most vulnerable in the world. Some have fled persecution by governments or armed groups that would like to know where they are. Biometric and personal data about refugees, if leaked or accessed by the wrong party, could lead to detention, deportation or violence. There have been real incidents in other responses where beneficiary data was exposed or shared inappropriately. Sofia argues that integrating data creates a larger, more valuable target, and that linking households across programmes, especially protection programmes that record sensitive information about gender-based violence, child protection and legal status, must be approached with extreme caution. She would prefer that protection data never enter any integrated system.

The country director for the largest operation, Hassan Mahmoud, has a practical view. His teams are overwhelmed, connectivity in the camps is poor and intermittent, and staff turnover is high. Any system must work offline, be simple to use and be maintainable by national staff after international advisers leave. He has watched several sophisticated systems introduced by headquarters fall into disuse because they were designed for conditions that do not exist in the field.

Interoperability with other agencies adds complexity. The United Nations refugee agency maintains registration data for refugees in two of the three

countries and has data sharing frameworks for partners. Humanitarian data exchange platforms publish aggregated data on needs, operational presence and population movements. The clusters use common standards for tagging data, which allow datasets from different organisations to be combined. Sharing beneficiary lists between agencies to reduce duplication has been proposed at coordination meetings, but agencies disagree on how to do it safely. Some advocate matching on shared identifiers under data sharing agreements; others propose privacy-preserving techniques that allow duplicates to be detected without exchanging names.

The donor's requirements create additional pressure. It wants quarterly results reported against a logical framework of indicators, disaggregated by sex, age and disability, and it wants evidence of reduced duplication within nine months. Its compliance team has also asked about counter-terrorism screening of beneficiaries, a request that humanitarian organisations generally resist because it conflicts with principles of impartiality and could endanger both staff and beneficiaries.

The technical environment is modest. The NGO uses an open-source mobile data collection platform, a commercial beneficiary registration system, spreadsheets, and a small cloud data warehouse at headquarters set up by a previous adviser. Your team consists of yourself, one data engineer at headquarters and three information management officers in the field, one per country. Internet connectivity at field offices ranges from reasonable to terrible.

Language and literacy shape everything in this response. The three countries use four main languages, and many of the displaced people speak minority languages that few staff understand. Enumerators translate questions on the fly, which means the same form question can be asked in subtly different ways in different places, and answers about sensitive topics such as disability or household composition vary depending on who asks and how. Names are transliterated from non-Latin scripts in inconsistent ways, so the same person may be recorded under several spellings across programmes. Dates of birth are often unknown and recorded as the first of January of an estimated year. Any attempt to link records about the same household using names and dates of birth will therefore be far less reliable than it would be in a European context, and

errors in linkage could lead to families being wrongly excluded from assistance as supposed duplicates.

There is also an honest question about what the platform is for. Amara wants coordination and targeting; the donor wants reporting; field teams want less paperwork; Sofia wants protection. You suspect that trying to satisfy all of them at once will produce a system that satisfies none, and that the first priority should be something tangible for field teams, such as harmonised forms and automated reporting, which would reduce their workload and build trust, before tackling cross-programme linkage.

At the same time, the cholera experience showed that the most important use of integrated data may be the ability to answer urgent operational questions quickly, and that does require some linkage, at least at the level of locations and facilities rather than individual households. You wonder whether a location-based integration, bringing together data by camp, block and shelter rather than by person, could deliver much of the operational value with far less protection risk.

THE DECISION IN FRONT OF YOU

Do you build household-level integration across programmes, a location-based integrated view that avoids linking individuals, or a phased approach starting with harmonised collection and reporting, and how do you handle inter-agency deduplication and donor demands without compromising the safety of the people you serve?

Data Sources

#	Source and link	How it is used in this project
1	Humanitarian Data Exchange (HDX) https://data.humdata.org	Shared humanitarian datasets on population, needs, operational presence and facilities used for location-based integration.
2	Humanitarian Exchange Language (HXL) https://hxlstandard.org	Hashtag standard for tagging humanitarian datasets to enable combination across organisations.
3	UNHCR Refugee Data Finder https://www.unhcr.org/refugee-statistics/	Official refugee population statistics for context and denominators.
4	ACLED Conflict Data https://acleddata.com	Conflict event data used to anticipate displacement and access constraints.
5	ReliefWeb API https://apidoc.reliefweb.int	Situation reports and humanitarian updates for contextual monitoring.

#	Source and link	How it is used in this project
6	OCHA Financial Tracking Service https://fts.unocha.org	Humanitarian funding flows for mapping who funds what where.
7	KoboToolbox Documentation https://support.kobotoolbox.org	Open-source mobile data collection platform supporting offline collection and APIs.
8	ICRC Handbook on Data Protection in Humanitarian Action https://www.icrc.org/en/data-protection-humanitarian-action-handbook	Authoritative guidance on protecting data about people affected by crises.

Deliverables (17)

- D1 Responsible data risk assessment.** An assessment of harms from data collection, integration and sharing for each data category.
- D2 Data protection impact assessment.** A DPIA with specific treatment of protection, biometric and health data.
- D3 Data classification and segregation policy.** Rules keeping protection case data outside integrated systems with strict access.
- D4 Harmonised form library.** Common questions, codes and household definitions across programmes with HXL tags.
- D5 Offline-first collection design.** Forms and sync processes suited to intermittent connectivity and field conditions.
- D6 Location master data.** A reference hierarchy of countries, camps, blocks, shelters and facilities with coordinates.
- D7 Location-based integrated dataset.** Programme indicators combined by location for operational decision-making.
- D8 Rapid operational query views.** Ready-made views for emergencies, such as water and sanitation gaps by shelter.
- D9 Automated donor reporting.** Indicator calculations with disaggregation feeding quarterly logical framework reports.
- D10 Cluster reporting automation.** Automatic production of cluster submissions in required formats.
- D11 Household linkage options paper.** An analysis of linking beneficiaries across programmes, including privacy-preserving methods.
- D12 Inter-agency deduplication protocol.** A proposed method and data sharing agreement for detecting duplicate assistance.
- D13 Donor screening position paper.** A principled response to counter-terrorism screening requests.
- D14 Access control and encryption design.** Role-based access, encryption at rest and in transit, and device security.

- D15 Data retention and deletion schedule.** Retention rules and deletion processes for beneficiary data.
- D16 Field capacity building plan.** Training and handover for national information management staff.
- D17 Data quality monitoring.** Checks on form completeness, duplicate submissions and inconsistent household records.

CASE STUDY 27 · FINANCIAL TECHNOLOGY (PAYMENTS)

Real-Time Card Payment Fraud Scoring Stream

Industry	Financial Technology (Payments)
Organisation	Payvine, a UK electronic money institution issuing debit cards to 1.8 million users and processing merchant payments
Your role	Staff Data Engineer, Risk Platform
Core themes	Sub-second streaming features, false positive cost, model governance, authorised push payment fraud

The Scenario

Payvine offers a mobile account and debit card aimed at younger customers and freelancers, with instant notifications, budgeting tools and cheap foreign exchange. It also runs a merchant acquiring business for small online shops. Both sides of the business process payments that must be authorised in real time, and both are targets for fraud. You are the staff data engineer on the risk platform team, and you have just come out of a meeting in which the chief risk officer, Hannah Doyle, set out the numbers that are now dominating board discussions.

Card fraud losses have doubled over the past year. The largest category is card-not-present fraud, where stolen card details are used online. A second, rapidly growing category is account takeover, where criminals gain access to a customer's app, often by tricking them into revealing one-time passcodes, and then spend or transfer their money. A third category, authorised push payment fraud, where customers are manipulated into sending money to fraudsters themselves, has become a regulatory priority: the Payment Systems Regulator's reimbursement rules now require payment firms to reimburse most victims and share the cost between sending and receiving firms. Payvine has found itself on both sides: its customers are victims, and fraudsters open Payvine accounts to receive stolen funds as money mules.

The fraud decision engine currently evaluates each card authorisation against a rules engine and a machine learning model. The model was trained offline on historical transactions and scores each authorisation in around forty milliseconds, well within the card scheme's time limits. But its features are limited. It sees the transaction amount, merchant category, country and a few customer attributes, plus aggregates such as the customer's spending over the past thirty days, which are computed nightly

in batch and cached. It cannot see what happened in the last few minutes: that the customer's app was logged into from a new device an hour ago, that the password was reset twenty minutes ago, that three small test transactions were made at different merchants in the last five minutes. Those signals are exactly the ones that distinguish account takeover from normal behaviour, and they exist in Payvine's systems, but in different places: login and device events in the app's backend, password resets in the identity service, card transactions in the processor's authorisation stream.

The head of fraud analytics, Dr Kwame Asante, wants real-time streaming features: rolling counts, amounts and velocity measures over windows from one minute to one day, joined across login, device, identity and transaction events, computed continuously and served to the model at authorisation time. He estimates that such features could cut account takeover losses by more than half. The platform engineering lead, Laura Kemp, supports the idea but has concerns about latency and reliability. The authorisation path has a strict time budget, and every millisecond spent fetching features reduces the time available for the model and rules. If the feature store is slow or unavailable, the system must fail safely, either approving with a fallback score or declining, and either choice has consequences.

False positives are the other side of the problem. When the system wrongly declines a genuine payment, the customer is embarrassed at a till or loses an online purchase, and many contact customer support or leave. Payvine's customer base is young and quick to switch providers. The head of customer experience, Olivia Grant, has data showing that customers who experience two false declines in a month are four times more likely to close their accounts. She has asked that any new model be judged not just on fraud caught but on false decline rates, and she wants customers to be able to confirm suspicious transactions in the app in real time rather than being declined outright.

The authorised push payment problem is different in shape. Outbound transfers through the faster payments scheme are also authorised in real time, but the fraud occurs with the customer's genuine consent, so signals of account takeover do not apply. Instead, the relevant signals are behavioural: an unusual payee, a first payment of a large amount, the customer being on a phone call during the transfer, the receiving account

being newly opened. For inbound payments, Payvine needs to identify mule accounts that receive stolen funds and quickly move them onward. The industry has introduced confirmation of payee and intelligence sharing between firms, and the scheme operator provides some network-level signals, but integrating these requires new data feeds.

Model governance adds constraints. Payvine's regulators expect firms to manage model risk, and the internal model risk policy requires validation of any new model and its features before deployment, including evidence that features are computed identically in training and production. With streaming features, that means the training data must be reconstructed from historical event logs with exactly the same windowing logic, which is harder than it sounds when events arrive late or out of order. Kwame's team has previously trained models on features that were subtly different from production, and the model's live performance was disappointing.

Data protection matters too. Device and behavioural data, such as typing patterns or whether the customer is on a call, are powerful signals but raise privacy concerns. The data protection officer has asked for a clear justification for each data type used and assurance that customers have been told.

Operations has also been raising a different issue. Fraud analysts review cases that the system flags for manual review, and they have access to a case management tool that shows the transaction and a customer summary. But the analysts told you that they make decisions using information the model never sees: the notes from previous customer calls, the tone of support chats, whether the customer recently complained about a lost phone. Some of those signals are structured and could become features; others are free text that would need careful processing. More importantly, the analysts' decisions are the source of the fraud labels the model learns from, and their confirmation process is inconsistent. A transaction the customer disputes weeks later is labelled fraud; one the customer never notices is labelled genuine. Label delay and label noise may be limiting the model as much as the missing features are.

The chief technology officer has approved a six-month programme with a team of five engineers. Hannah wants measurable loss reduction within that time. You have to choose between building a general streaming feature platform that serves card, transfer and mule detection, or delivering a focused set of account takeover features for card

authorisations first. You also need to decide how to reconstruct historical features for training, how to guarantee latency, and how to fail safely.

The temptation is to build the general platform, because the three fraud types share many underlying signals. But you also know that platforms take longer than expected, that the board wants results quickly, and that a focused account takeover capability could be live within three months and demonstrate value, at the risk of creating a special-purpose system that has to be rebuilt later.

THE DECISION IN FRONT OF YOU

Do you build a general-purpose real-time feature platform for all fraud types, deliver focused account takeover features for card authorisation first, or design a narrow first release on a general foundation, and how do you guarantee latency, safe failure, training-serving consistency and acceptable false decline rates?

Data Sources

#	Source and link	How it is used in this project
1	IEEE-CIS Fraud Detection Dataset https://www.kaggle.com/c/ieee-fraud-detection	Large real-world card-not-present transaction dataset with identity and device features used to prototype velocity features.
2	Credit Card Fraud Detection Dataset (ULB) https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud	Labelled card transactions used for model baseline and class imbalance testing.
3	PaySim Synthetic Mobile Money Dataset https://www.kaggle.com/datasets/ealaxi/paysim1	Synthetic transfers with fraud labels used to model push payment and mule behaviour.
4	Payment Systems Regulator APP Scams Reimbursement https://www.psr.org.uk/our-work/app-scams/	Regulatory requirements shaping authorised push payment detection and reimbursement data.
5	UK Finance Annual Fraud Report https://www.ukfinance.org.uk/policy-and-guidance/reports-and-publications/annual-fraud-report	Industry fraud trends and typologies informing feature priorities.
6	Apache Flink Documentation https://nightlies.apache.org/flink/flink-docs-stable/	Stateful stream processing reference for windowed features and event-time handling.
7	Feast Feature Store https://docs.feast.dev	Reference for online and offline feature serving and point-in-time training data.
8	Pay.UK Confirmation of Payee https://www.wearepay.uk/what-we-do/overlay-services/confirmation-of-payee/	Payee name checking service providing signals for push payment fraud detection.

Deliverables (17)

- D1 Fraud loss and false decline baseline.** A breakdown of losses by fraud type and false decline rates by segment.
- D2 Scoping decision record.** A documented choice between general platform and focused release with rationale.
- D3 Event source integration.** Streaming ingestion of login, device, identity, card and transfer events with consistent keys.
- D4 Streaming feature definitions.** Specifications for velocity, recency and cross-channel features with window semantics.
- D5 Real-time feature computation.** A stateful stream job producing features with late event handling.
- D6 Low-latency online feature store.** Feature serving within the authorisation latency budget with performance tests.
- D7 Latency budget and SLOs.** Documented time allocations for features, model and rules with alerting.
- D8 Safe failure design.** Fallback scoring and decision rules when feature services degrade or fail.
- D9 Historical feature reconstruction.** Replay of event logs to generate training features identical to production.
- D10 Training-serving parity tests.** Automated comparison of offline and online feature values.
- D11 Account takeover model.** A retrained model using streaming features with performance and false decline evaluation.
- D12 In-app confirmation integration.** A step-up flow asking customers to confirm suspicious transactions instead of declining.
- D13 Push payment risk signals.** Features for payee novelty, payment patterns and receiving account risk.
- D14 Mule account detection prototype.** Detection of inbound funds rapidly moved onward from new accounts.
- D15 Model validation evidence.** Documentation and results for the model risk committee.
- D16 Privacy justification register.** A record of each behavioural and device signal with purpose and customer notice.
- D17 Loss reduction tracking.** Ongoing measurement of fraud caught, losses, false declines and customer attrition.

CASE STUDY 28 • DIGITAL ASSETS AND CRYPTOCURRENCY

On-Chain Analytics and Crypto Compliance Data Platform

Industry	Digital Assets and Cryptocurrency
Organisation	Coinmere, a UK-registered cryptoasset exchange with 420,000 retail customers
Your role	Lead Data Engineer, Compliance Technology
Core themes	Blockchain data ingestion, graph analytics, travel rule compliance, vendor dependence

The Scenario

Coinmere is one of a small number of cryptoasset exchanges registered with the Financial Conduct Authority for anti-money laundering purposes. Registration was hard won: the regulator rejected or saw withdrawn a large majority of applications, and Coinmere's approval came with a clear expectation that its financial crime controls would be robust and continuously improved. Customers can buy, sell and hold a dozen cryptoassets and withdraw them to external wallets, or deposit assets from external wallets. It is that movement of assets on and off the platform, to and from addresses on public blockchains, that makes compliance in crypto both distinctive and difficult. You are the lead data engineer in compliance technology, and a recent incident has put your data platform under the microscope.

Two months ago, a customer deposited a significant quantity of a stablecoin that, it later emerged, had passed through several intermediate wallets from an address associated with a sanctioned entity. Coinmere's blockchain analytics vendor, which scores deposit and withdrawal addresses for risk, had assigned the deposit a medium risk score, below the threshold for automatic review, because the connection to the sanctioned address was several hops removed and the vendor's heuristics had not yet attributed one of the intermediate wallets. By the time the vendor updated its attribution a week later, the customer had traded the funds and withdrawn them to another exchange. Coinmere filed a suspicious activity report and notified the regulator, which asked pointed questions about how Coinmere monitors on-chain exposure beyond the vendor's point-in-time scores.

The money laundering reporting officer, Olivia Brennan, has asked for a review of the firm's on-chain monitoring. Coinmere relies entirely on the

vendor for blockchain intelligence: it sends each deposit and withdrawal address to the vendor's API and receives a risk score and exposure breakdown. It does not store the scores over time, it does not re-screen historical deposits when the vendor updates its attributions, and it does not have its own view of blockchain data. When the regulator asked whether any other deposits in the past year had similar indirect exposure, Coinmere could not answer without paying the vendor for a bulk re-screening.

The head of engineering, Raj Menon, wants to build an in-house on-chain analytics capability: run full nodes or use public blockchain datasets, index transactions for the chains Coinmere supports, build a graph of address relationships, and apply Coinmere's own analysis alongside the vendor's attributions. He argues that this would reduce dependence on a single vendor, allow retrospective analysis whenever attributions change, and enable more sophisticated detection. He has a talented team of engineers with blockchain experience who are keen to build it.

Olivia is sceptical. Blockchain analytics vendors have invested years in attributing addresses to entities, including exchanges, darknet markets, scams, mixers and sanctioned parties, using investigations, open-source intelligence and data sharing with law enforcement. Coinmere cannot replicate that. An in-house graph without attribution is just a very large network of anonymous addresses. She would rather strengthen how the firm uses the vendor: storing scores, re-screening periodically, lowering thresholds, and adding a second vendor for coverage.

The chief financial officer, Ben Harper, is concerned about cost. The vendor charges per screening and for bulk re-screening; adding a second vendor would roughly double costs; running full nodes and indexing multiple blockchains requires significant infrastructure and storage, since some chains are many terabytes and growing. On the other hand, public blockchain datasets hosted by cloud providers could reduce the infrastructure burden, although their coverage of newer chains and token standards varies.

A further regulatory obligation complicates matters. The travel rule, which requires cryptoasset businesses to collect and share originator and beneficiary information for transfers, came into force in the UK recently. Coinmere must now obtain and transmit information about the sender and recipient for qualifying transfers to and from other cryptoasset businesses,

and must decide what to do when a counterparty business does not comply. This requires identifying whether an external address belongs to a customer of another regulated business, which again relies on attribution data, and it requires integrating with one of several competing travel rule messaging protocols. Counterparty exchanges in different jurisdictions have adopted different protocols, and some have adopted none, so a transfer can stall for days while compliance staff exchange emails to establish who the recipient is. Implementation has been patchy, and Coinmere currently handles many transfers manually.

The data engineering challenges are considerable whatever path is chosen. Deposit and withdrawal events live in Coinmere's transaction ledger; customer due diligence data lives in the onboarding system; vendor scores are fetched on demand and discarded; travel rule data is held in a separate tool. To answer the regulator's retrospective question, you need a historical record of every on-chain movement associated with every customer, linked to risk assessments as they stood at the time and as they stand now. That requires, at a minimum, persistent storage of vendor responses and a way of re-screening historical addresses efficiently.

The engineering team's enthusiasm for the in-house graph is itself a factor. Several of Raj's best engineers joined Coinmere because they wanted to work on blockchain data, and they have been maintaining trading infrastructure instead. Raj worries about retention if the answer is simply to call a vendor's API more often. Olivia worries that engineering enthusiasm is driving a compliance decision.

There is a further technical wrinkle that makes the vendor-only approach more fragile than it appears. Coinmere recently added support for two newer blockchains and several tokens on existing chains, driven by customer demand and competitive pressure. The primary vendor's coverage of one of these chains is limited, returning scores only for addresses it has already attributed and a default unknown category for everything else, which in practice means most addresses on that chain. The product team was aware of this at launch but judged that volumes would be small. Volumes have since grown faster than expected, and a meaningful share of deposits now arrive on chains where the firm effectively has no on-chain risk visibility at all. Whatever the long-term decision, the regulator is likely to ask how Coinmere decided to list assets

it could not monitor, and future listing decisions will need a compliance data readiness check as a gate.

You have been asked to produce a proposal within a month that the MLRO, the head of engineering and the CFO can support, and that gives the regulator confidence. You suspect the right answer involves using vendor attribution as the foundation while building in-house data capabilities for retention, re-screening and graph analysis on top, but the balance of investment is the crux, and you need to be honest about what an in-house graph can and cannot do.

THE DECISION IN FRONT OF YOU

Do you invest in an in-house blockchain indexing and graph analytics platform, strengthen use of vendor intelligence with persistent storage, re-screening and a second provider, or build a layered capability, and how do you meet travel rule obligations and answer retrospective regulatory questions reliably?

Data Sources

#	Source and link	How it is used in this project
1	Elliptic Bitcoin Transaction Dataset https://www.kaggle.com/datasets/ellipticco/elliptic-data-set	Labelled Bitcoin transaction graph with illicit and licit classes used to prototype graph features and models.
2	Google BigQuery Public Crypto Datasets https://cloud.google.com/blog/products/data-analytics/introducing-six-new-cryptocurrencies-in-bigquery-public-datasets	Hosted blockchain datasets for Bitcoin, Ethereum and others enabling indexing without running full nodes.
3	Etherscan API https://docs.etherscan.io	Ethereum transaction, token transfer and contract data for address history lookups.
4	OFAC Sanctions List Search https://sanctionssearch.ofac.treas.gov	Sanctions designations including digital currency addresses used for direct screening.
5	FCA Cryptoasset AML Registration https://www.fca.org.uk/firms/cryptoassets-aml-ctf-regime	Regulatory expectations for registered cryptoasset businesses.
6	FATF Guidance on Virtual Assets https://www.fatf-gafi.org/en/publications/Fatfrecommendations/Guidance-rba-virtual-assets-2021.html	International guidance on virtual asset risks and the travel rule.
7	InterVASP Messaging Standard (IVMS101) https://www.intervasp.org	Data model for travel rule originator and beneficiary information.
8	Neo4j Graph Data Science Documentation https://neo4j.com/docs/graph-data-science/current/	Graph algorithms reference for exposure tracing and community detection.

Deliverables (17)

- D1 Incident retrospective.** An analysis of the sanctioned-exposure deposit covering detection gaps and control failures.
- D2 Options paper.** A comparison of vendor-enhanced, in-house and layered approaches on capability, cost and regulatory confidence.
- D3 Vendor response persistence.** Storage of every screening result with timestamps, attribution versions and exposure details.
- D4 Historical address register.** A complete record of all customer deposit and withdrawal addresses and transactions.
- D5 Periodic re-screening pipeline.** Scheduled re-screening of historical addresses with change detection and alerting.
- D6 Retrospective exposure report.** An answer to the regulator's question on indirect sanctioned exposure over the past year.
- D7 Blockchain data ingestion.** Indexing of supported chains from hosted datasets or nodes for customer-relevant transactions.
- D8 Address graph model.** A graph of address interactions around customer activity for multi-hop exposure tracing.
- D9 Direct sanctions address screening.** Screening against published sanctioned addresses independent of vendor scores.
- D10 Second vendor evaluation.** A structured trial comparing coverage and attribution quality across providers.
- D11 Travel rule data integration.** Collection, storage and exchange of originator and beneficiary information via a chosen protocol.
- D12 Counterparty VASP identification.** Determination of whether external addresses belong to regulated businesses.
- D13 Risk threshold calibration.** Revised alert thresholds informed by incident analysis and alert capacity.
- D14 Compliance case linkage.** Linking on-chain alerts to customer due diligence and case management records.
- D15 Cost model.** Infrastructure, vendor and staffing costs across options over three years.
- D16 Regulator briefing pack.** A narrative and evidence demonstrating strengthened on-chain monitoring.
- D17 Operational runbooks.** Procedures for re-screening alerts, vendor outages and travel rule exceptions.

CASE STUDY 29 · FOOD RETAIL AND SUPPLY CHAIN

Store-Level Demand Forecasting Data Platform for Fresh Food

Industry	Food Retail and Supply Chain
Organisation	Greenacre Markets, a regional supermarket chain with 186 stores and a central distribution network
Your role	Principal Data Engineer, Supply Chain Analytics
Core themes	Forecasting data pipelines, promotional data quality, waste versus availability, model operations at scale

The Scenario

Greenacre Markets built its reputation on fresh food: bakery baked in store, local produce, a butcher and fishmonger in larger branches, and prepared meals made in its own kitchens. Fresh food is also where the chain loses the most money. Products with a shelf life of one to five days must be ordered accurately every day for every store. Order too much and the surplus is marked down or thrown away; order too little and shelves are empty in the evening and customers go elsewhere. Last year, the chain wasted around four per cent of its fresh food sales value and suffered availability gaps on key lines that the customer insight team links to declining visit frequency. The supply chain director, Robert Ellison, has set targets to halve waste and improve availability at the same time, and he believes the answer is better forecasting. You lead data engineering for supply chain analytics.

Store orders for fresh products are currently generated by an automatic replenishment system that forecasts demand using a simple moving average of recent sales, adjusted by store managers who override orders based on their judgement. The system was installed fifteen years ago and handles ambient products reasonably well, but it struggles with fresh food, where demand swings with weather, day of the week, holidays, local events and promotions. Store managers override around thirty per cent of fresh orders, some with great skill and some with little, and there is no systematic record of why.

The data science team, led by Dr Lucy Harrington, has developed a new forecasting model using gradient boosted trees that incorporates weather forecasts, promotions, holidays, price changes and store characteristics. In an offline back-test over six months, it reduced forecast error for fresh

lines by roughly a quarter compared with the current system. Robert wants it in production for all 186 stores and around four thousand fresh products within six months. That means producing forecasts for about seven hundred thousand store-product combinations every night, feeding them into the replenishment system in time for orders to be placed by five in the morning, and doing so reliably every day of the year.

As you examine what production would require, several data problems emerge. Sales data is recorded at the till, but it measures demand only when products are available. When a product sells out at two in the afternoon, the sales figure understates true demand, and a model trained on sales learns to under-order. Correcting for this requires knowing when products were out of stock, which the chain does not record directly; it can be inferred from gaps in sales and from stock records, but stock records for fresh products are notoriously inaccurate because waste is often not recorded properly. Promotional data is held in a promotions planning system, but the records of what was actually executed in store, such as which products were on end-of-aisle displays, are incomplete. Prices are held in a pricing system that records planned prices, while markdowns applied in the evening to clear short-dated stock are recorded only at the till. Weather forecasts must be bought from a provider and mapped to stores. Local events are not recorded anywhere.

Lucy's back-test used a carefully cleaned historical dataset that took her team two months to assemble, including manual corrections. Producing the same quality of input data every night, automatically, is a different challenge. You estimate that around a fifth of the model's apparent improvement depends on features that are currently unreliable in daily operation.

The IT director, Andrew Kerr, is wary of the model running outside the replenishment system. The replenishment vendor offers its own advanced forecasting module, at considerable cost, which would integrate seamlessly. Andrew would prefer to buy that module than to run a custom model with a nightly pipeline that the data team must support around the clock. Robert dismisses the vendor module, having seen its results at another retailer, but acknowledges that operational reliability is essential: if forecasts fail to arrive one night, the fallback must be safe.

The store operations director, Karen Mills, raises the human dimension. Store managers are proud of their ordering skills, and many believe no

model can understand their stores as well as they do. If the new forecasts are imposed without explanation, managers will continue to override them, and the benefits will disappear. Karen wants managers to see why the forecast is what it is, and she wants override decisions and reasons recorded so the team can learn from the best managers and improve the model. She also points out that some stores have particularities, such as a hospital nearby or a university that empties in summer, that no national dataset captures.

The finance director has asked how benefits will be measured. Waste and availability are affected by many factors, including supplier performance, store execution and customer trends, and she does not want the data team claiming savings that would have happened anyway. She suggests a controlled rollout, with some stores using the new model and matched stores continuing with the old process, but Robert is impatient and wants all stores switched quickly.

There is also a history of fragile data processes in supply chain analytics. The weekly availability report, which the executive team reviews every Monday, is produced by a set of spreadsheet macros that pull from the warehouse and the replenishment system. It failed three times last quarter, each time because a column changed in a source extract, and each time the failure was noticed only when an executive asked why the report was late. The data science team's back-test itself was assembled in notebooks with manual steps that are not documented. Your own team is proud of the core warehouse pipelines but acknowledges that testing and monitoring have never been a priority. A nightly forecasting pipeline that drives ordering for every fresh product in every store demands a level of operational discipline that the function has not yet demonstrated, and you know that the first high-profile failure would give Andrew all the evidence he needs to insist on the vendor module.

You have a team of five engineers. The data science team has three people. The replenishment system can accept forecasts through a nightly file interface. You need to decide how to productionise the model, how to address the data quality issues, how to monitor forecasts and model drift, how to fail safely, how to integrate managers' knowledge, and how to measure benefits honestly.

The honest assessment is that the model is good, but the data pipeline to feed it reliably does not yet exist, and building it properly will take longer

than six months. You could launch on the data that is reliable today, accepting smaller gains, while improving stock-out and promotion data in parallel. Or you could delay launch until the data is right, which Robert would find hard to accept.

THE DECISION IN FRONT OF YOU

Do you productionise the in-house model on currently reliable data, delay for better input data, or adopt the vendor module, and how do you handle censored demand, promotional data gaps, store manager overrides and honest benefit measurement?

Data Sources

#	Source and link	How it is used in this project
1	M5 Forecasting Competition Data https://www.kaggle.com/c/m5-forecasting-accuracy	Hierarchical store-item sales with prices and events used to build and benchmark the forecasting pipeline.
2	Corporación Favorita Grocery Sales Forecasting https://www.kaggle.com/c/favorita-grocery-sales-forecasting	Large grocery sales dataset with promotions, perishables and store metadata for fresh demand modelling.
3	Rossmann Store Sales https://www.kaggle.com/c/rossmann-store-sales	Store sales with promotions, holidays and competition used to test promotional feature logic.
4	Met Office Weather DataHub https://www.metoffice.gov.uk/services/data/met-office-weather-datahub	Weather forecasts mapped to stores as daily demand features.
5	UK Bank Holidays (GOV.UK API) https://www.gov.uk/bank-holidays.json	Official bank holiday calendar for England, Wales, Scotland and Northern Ireland.
6	WRAP Food Waste Data https://wrap.org.uk/resources/report/food-surplus-and-waste-uk-key-facts	Industry food waste benchmarks for target setting and reporting.
7	Nixtla StatsForecast and MLForecast https://nixtlaverse.nixtla.io	Open-source forecasting libraries for baseline and scalable hierarchical models.
8	ONS Retail Sales Index https://www.ons.gov.uk/businessindustryandtrade/retailindustry	Market trend context for demand planning.

Deliverables (17)

- D1 Production readiness assessment.** A review of each model feature's availability, reliability and latency in daily operation.
- D2 Launch options paper.** A comparison of launching on reliable data, delaying, and buying the vendor module.
- D3 Nightly forecasting pipeline.** An orchestrated pipeline producing 700,000 forecasts before the replenishment cut-off.

- D4 Censored demand correction.** Inference of stock-outs and adjustment of training demand for lost sales.
- D5 Stock-out detection method.** A method identifying out-of-stock periods from sales gaps and stock records.
- D6 Promotion execution data capture.** Improved recording of in-store promotional execution with store feedback.
- D7 Markdown and price feature pipeline.** Integration of planned prices and till-level markdowns into model features.
- D8 Weather-to-store mapping.** Assignment of weather forecasts to stores with fallback locations.
- D9 Local events and store attributes register.** A curated record of store-specific demand drivers maintained with store managers.
- D10 Forecast fallback mechanism.** Safe default forecasts delivered automatically if the pipeline fails.
- D11 Model monitoring framework.** Tracking of forecast error, bias and drift by category, store and product.
- D12 Forecast explanation for managers.** Simple explanations of key drivers shown alongside suggested orders.
- D13 Override capture and analysis.** Recording of manager overrides and reasons with analysis of override value.
- D14 Controlled rollout design.** A test-and-control store design with matched comparison stores.
- D15 Benefits measurement methodology.** An agreed approach to attributing waste and availability changes to forecasting.
- D16 Replenishment interface specification.** A validated file contract with the replenishment system.
- D17 Operational support model.** On-call arrangements, runbooks and escalation for the nightly pipeline.

CASE STUDY 30 · HUMAN RESOURCES (PROFESSIONAL SERVICES FIRM)

People Analytics Platform and Statutory Pay Gap Reporting

Industry	Human Resources (Professional Services Firm)
Organisation	Brackenridge Partners, a UK consultancy and accountancy firm with 7,200 employees across 22 offices
Your role	Senior Data Engineer, People Data and Insights
Core themes	HR data integration, statutory reporting accuracy, sensitive employee data, ethical analytics

The Scenario

Every April, employers in Great Britain with two hundred and fifty or more employees must publish their gender pay gap: the difference between average hourly pay for men and women, the difference in bonus pay, the proportion of each gender receiving a bonus, and the gender mix in each pay quartile. The figures go on a government website where anyone can compare them, and journalists reliably write stories about the firms with the largest gaps. Brackenridge Partners has published a gap of around twenty-two per cent for several years, typical for its sector but not something its leadership is proud of. This year, something went wrong. Two weeks after publication, an employee in the finance team noticed that the published bonus gap was inconsistent with figures shared internally, and further checking revealed that around three hundred partners' bonus payments had been excluded from the calculation because they were paid through a different payroll run. The corrected bonus gap was significantly larger. The firm had to republish, and the story made the trade press.

You are the senior data engineer in the people data and insights team, and the chief people officer, Sarah Whitfield, has asked you to make sure nothing like this happens again. But she has also asked for more. The firm has made public commitments on diversity, including targets for women and ethnic minority representation at senior levels, and the executive board wants to understand what is driving the pay gap and whether its initiatives are working. It also wants to publish an ethnicity pay gap voluntarily, as several peer firms now do, and a disability pay gap is under discussion. Beyond pay gaps, Sarah wants a people analytics platform that helps the firm understand attrition, promotion fairness, workload and wellbeing.

The underlying data is scattered. Employee records live in a human resources information system implemented five years ago, but historical records before that are in an old system that was only partially migrated. Payroll is run by an outsourced provider using its own system, with separate payroll runs for employees, partners and some overseas secondees. Bonuses are calculated in spreadsheets by each business unit and fed to payroll. Performance ratings live in a talent management system. Timesheets, which record hours worked against client projects, live in a practice management system and are the only source of information about actual hours worked, which matters because the pay gap calculation uses hourly rates. Diversity data, including ethnicity, disability, sexual orientation and social background, is collected through a voluntary self-declaration form in the HR system, with completion rates between forty and seventy per cent depending on the characteristic.

The statutory gender pay gap calculation is precisely defined in regulations, but applying the definitions requires judgement. Who counts as a relevant employee? Partners are generally not employees and are excluded from statutory figures, but some senior staff are salaried partners with employment contracts, and the firm has been inconsistent about how they are treated. Which pay elements count as ordinary pay and which as bonus? How should the hourly rate be calculated for employees whose contracted hours differ from their actual hours? The error this year arose partly because nobody had written down the rules, and the calculation was performed by one analyst in a spreadsheet built from several extracts.

The ethnicity pay gap raises additional issues. There is no statutory method, although government guidance has been published. Self-declaration rates are too low in some offices to produce reliable figures, and categories must be aggregated in ways that are statistically sound without hiding differences between groups. The head of diversity and inclusion, Priya Raman, wants detailed breakdowns to understand where the gaps are largest. The data protection officer, Martin Lee, is cautious: ethnicity and disability are special category data, and analysis at small group sizes risks identifying individuals. He insists that any analysis must have a clear lawful basis and appropriate safeguards, and that results must not be published or shared internally at levels where individuals could be identified.

The executive board's appetite for people analytics goes further than some employees are comfortable with. The chief operating officer, Daniel Price, has suggested using timesheet, email and calendar metadata to identify burnout risk and predict which high performers are likely to leave. The employee forum, which represents staff, has heard about this and objected strongly, saying it would feel like surveillance and damage trust. Sarah is sympathetic to both the business interest and the employees' concerns. She wants a platform that supports genuine insight without crossing lines that would undermine the firm's culture.

Your team is small: two data engineers, including you, and three analysts. The HR technology team, which owns the HR system, is separate and has its own priorities. The outsourced payroll provider charges for custom data extracts and responds to requests slowly.

The most urgent requirement is clear: next year's statutory gender pay gap figures must be correct, reproducible and auditable. The snapshot date for the next calculation is five months away, and publication must happen within a year of that date, but Sarah wants the figures ready well in advance so they can be reviewed. The ethnicity pay gap is a strategic commitment. The broader people analytics platform is an ambition. The burnout prediction idea is a controversy.

You have been thinking about the relationship between these goals. A trusted, well-governed people data model, with clear definitions of employees, pay elements, hours and characteristics, underpins all of them. If you build the statutory calculation on that foundation, you fix the immediate risk and create the basis for everything else. But the foundation takes time, and there is a temptation to simply rebuild the gender pay gap spreadsheet more carefully and move on. There is also the question of how far the platform should go: whether to deliberately exclude certain data sources, such as email and calendar metadata, from the design, which would make a principled statement but might be overruled later by an executive who wants the analysis.

There is an additional complication in the firm's partnership structure. Equity partners share profits rather than receiving salaries and bonuses, so they fall outside the statutory calculation, yet they are the most senior and highest-paid people in the firm, and they are overwhelmingly men. Several peer firms now publish an additional voluntary figure that includes partners, which produces a much larger gap, and journalists increasingly

ask why firms exclude them. The senior partner has asked privately whether Brackenridge should do the same. Doing so would require partner profit share data, which is held by a small partnership finance team and treated as highly confidential. Whatever the decision, the data model must be able to support both the statutory and voluntary figures from the same foundations, with the differences between them clearly explained.

You also recognise that the error this year was not purely technical. It happened because nobody owned the definitions, nobody reconciled the inputs to payroll totals, and nobody reviewed the calculation independently. Any solution needs governance: a documented methodology, reconciliation controls and sign-off by accountable people, as well as better data pipelines.

THE DECISION IN FRONT OF YOU

Do you rebuild the statutory calculation as a focused, controlled process first, or invest in a comprehensive people data model that serves pay gap reporting and broader analytics, and where do you draw the line on employee monitoring when the chief operating officer wants burnout prediction and staff are opposed?

Data Sources

#	Source and link	How it is used in this project
1	UK Gender Pay Gap Service https://gender-pay-gap.service.gov.uk	Official published pay gap data for all reporting employers used for peer benchmarking and format validation.
2	Gender Pay Gap Reporting Guidance (GOV.UK) https://www.gov.uk/guidance/gender-pay-gap-reporting-overview	Official guidance on relevant employees, pay definitions and calculation methods.
3	Ethnicity Pay Reporting Guidance for Employers https://www.gov.uk/government/publications/ethnicity-pay-reporting-guidance-for-employers	Government guidance on voluntary ethnicity pay gap methods and aggregation.
4	ONS Annual Survey of Hours and Earnings https://www.ons.gov.uk/employmentandlabourmarket/peopleinwork/earningsandworkinghours	National earnings data for context and sector comparison.
5	IBM HR Analytics Attrition Dataset https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset	Synthetic employee data used to prototype attrition analytics without real employee data.
6	ICO Employment Practices and Data Protection Guidance https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/employment/	Regulatory guidance on processing employee data and monitoring.

#	Source and link	How it is used in this project
7	Equality and Human Rights Commission Guidance https://www.equalityhumanrights.com/guidance	Guidance on equal pay and fair treatment informing promotion and pay analyses.
8	ONS Harmonised Ethnicity Standards https://analysisfunction.civilservice.gov.uk/policy-store/ethnicity-harmonised-standard/	Standard ethnicity categories for consistent collection and aggregation.

Deliverables (17)

- D1 Pay gap incident review.** A root cause analysis of the bonus exclusion error covering data, process and governance failures.
- D2 Statutory methodology document.** Written rules for relevant employees, pay elements, bonus periods and hourly rate calculation.
- D3 Payroll reconciliation controls.** Reconciliation of all payroll runs and bonus payments to calculation inputs before sign-off.
- D4 Controlled gender pay gap pipeline.** A reproducible, version-controlled calculation replacing the spreadsheet.
- D5 Independent review and sign-off process.** A defined review by finance and HR with documented approval.
- D6 People data model.** A governed model of employees, contracts, pay, hours, roles and characteristics over time.
- D7 Historical HR data reconciliation.** Integration of pre-migration records with gaps and assumptions documented.
- D8 Diversity data quality assessment.** Declaration rates by characteristic and office with implications for reliable analysis.
- D9 Ethnicity pay gap methodology.** A method with aggregation rules and minimum group sizes aligned to government guidance.
- D10 Disclosure control rules.** Small-number suppression and aggregation standards for internal and external outputs.
- D11 Data protection impact assessment.** A DPIA covering special category data and people analytics purposes.
- D12 Analytics ethics principles.** A published set of principles, co-developed with the employee forum, including prohibited uses.
- D13 Promotion and pay fairness analysis.** Analysis of promotion rates and pay progression by characteristic with appropriate controls.
- D14 Attrition insight dashboard.** Aggregate attrition trends by level, office and practice without individual risk scoring.
- D15 Access control model.** Role-based access to people data with audit logging.

- D16 Burnout analysis options paper.** An evaluation of aggregate, consent-based alternatives to individual monitoring.
- D17 Annual reporting calendar.** A timeline of snapshot, calculation, review and publication with owners.

CASE STUDY 31 • LEGAL SERVICES

Legal Document Ingestion and Case Law Intelligence Pipeline

Industry	Legal Services
Organisation	Whitlock Sterne LLP, a UK commercial law firm with 900 lawyers and a litigation practice handling large disputes
Your role	Lead Data Engineer, Legal Technology and Innovation
Core themes	Unstructured document pipelines, client confidentiality, retrieval-augmented generation, provenance and accuracy

The Scenario

Whitlock Sterne is a firm that has always prided itself on being careful. Its litigation practice handles some of the largest commercial disputes in the English courts, where a single case can involve millions of documents, and its advisory practices draft and negotiate contracts worth billions of pounds. Like every large law firm, it is under intense pressure from clients to use technology, including generative artificial intelligence, to reduce costs and speed up work. The managing partner, Helena Marsh, announced a firm-wide AI programme six months ago, and you were hired as lead data engineer in legal technology and innovation to build the data foundations that the programme needs. You have discovered that those foundations are thinner than anyone realised.

The firm's knowledge is spread across several systems. A document management system holds more than forty million documents, organised by client and matter, going back twenty years: contracts, advice notes, court filings, correspondence and drafts in many versions. A knowledge management team maintains a curated library of precedents, practice notes and model clauses. Litigation matters use a separate electronic disclosure platform, where documents from clients and opponents are processed for review. The firm subscribes to commercial legal research databases that contain case law, legislation and commentary, accessed through their own interfaces under licences that restrict bulk downloading. And increasingly, public sources such as the National Archives' service for judgments and the official legislation website offer free, structured access to court decisions and statutes.

The AI programme's flagship idea is an internal legal assistant: a system that allows lawyers to ask questions in plain English and receive answers

grounded in the firm's own precedents and advice, relevant case law and legislation, with citations. The innovation partner, James Hollis, has seen demonstrations from vendors and wants a pilot running within four months. Several vendors offer products that connect to document management systems and use retrieval-augmented generation, where relevant documents are retrieved and passed to a language model to generate an answer. James is keen to build rather than buy, because he believes the firm's precedents are its competitive advantage and he does not want them processed by a vendor that also serves competitors.

The general counsel and risk partner, Catherine Ashby, has concerns that go to the heart of the firm's obligations. Client documents are confidential and often subject to legal professional privilege. Many client engagement letters restrict how client information may be used, and some clients, particularly financial institutions and government bodies, have explicitly prohibited use of their data to train or improve AI systems. Documents from one client's matter must never influence advice to another client, especially where clients are competitors or where the firm has information barriers in place for conflict reasons. The document management system enforces access controls by matter, and ethical walls restrict certain lawyers from seeing certain matters. Any AI system that indexes documents must respect those controls exactly, at the moment of retrieval, for every user.

Accuracy is the other concern. Lawyers in other firms, and litigants in person, have been criticised by judges for submitting documents that cited non-existent cases generated by AI tools. The firm's professional indemnity insurers have asked what controls are in place. Catherine wants any system to show its sources, to distinguish clearly between what a source says and what the system infers, and to be tested rigorously before lawyers rely on it.

As you investigate the data, practical problems accumulate. Around a third of documents in the document management system are scanned images without text layers, requiring optical character recognition. Many documents exist in dozens of versions, with only some marked as final. Metadata is inconsistent: document types are chosen by secretaries from a long list, and many are classified simply as "other". The precedent library is well curated but small and partly out of date; some model clauses predate significant legislative changes. The commercial research databases'

licences prohibit ingesting their content into internal systems, so the assistant could not use them directly. Public judgments and legislation are available and well structured, but coverage of older and lower-court judgments varies.

The head of knowledge management, Dr Rachel Obi, has a different emphasis. She believes the most valuable and safest starting point is the curated precedent library and public law, not the entire document management system. Curated content has been reviewed by senior lawyers, carries no client confidentiality issues, and is exactly what junior lawyers need when drafting. She worries that indexing forty million unreviewed documents would surface outdated, incorrect or client-specific content, and that lawyers would not be able to tell the difference.

James counters that the precedent library covers only a fraction of the firm's expertise, and that the real value lies in the tacit knowledge embedded in past advice and negotiated documents. He points out that competitors are building exactly this and that clients are asking about it in pitches.

The litigation practice has its own request. The head of litigation, Andrew Rhys, wants tools for the disclosure platform that classify documents, identify key issues and build chronologies from very large document sets. That work involves client documents within a single matter, which is easier from a confidentiality perspective, but it involves opponents' documents too, and the court rules on disclosure impose their own obligations about how review is conducted and explained.

The firm's own history with knowledge tools is instructive. Ten years ago, it invested heavily in an enterprise search product that indexed the document management system. Within months, a trainee found a draft settlement agreement from a matter behind an ethical wall because the search index had been built with a service account that could see everything, and permission filtering was applied only at display time, with a caching bug. The incident was contained, but it led to the product being withdrawn and to a firm-wide rule that no system may index client documents without the risk partner's approval. Catherine was the partner who handled the incident. Her caution is therefore not abstract; she has personally seen what happens when a retrieval system's permissions lag behind the source system's. She will want to know precisely how

permissions are evaluated, how quickly changes to ethical walls propagate, and how failures are detected.

You have a team of four engineers and a budget for cloud infrastructure and model access. The firm's IT security team insists that any language model used must run in a private deployment with contractual guarantees that data is not retained or used for training. You need to recommend what to build first, what data to include, and how to enforce permissions and provenance.

THE DECISION IN FRONT OF YOU

Do you index the full document management system to capture the firm's tacit knowledge, start with curated precedents and public law, or focus on single-matter litigation tools, and how do you guarantee that access controls, information barriers, client restrictions and citation accuracy are enforced by design?

Data Sources

#	Source and link	How it is used in this project
1	The National Archives Find Case Law https://caselaw.nationalarchives.gov.uk	Free, structured judgments from UK courts and tribunals with an API, used as the authoritative case law corpus.
2	legislation.gov.uk API https://www.legislation.gov.uk/developer	Official UK legislation in structured formats with point-in-time versions for grounding answers.
3	CourtListener API https://www.courtlistener.com/help/api/	Large open US case law and court filings dataset used to test ingestion and citation extraction at scale.
4	EUR-Lex https://eur-lex.europa.eu	EU legislation and case law relevant to retained and comparative law questions.
5	CUAD Contract Understanding Dataset https://www.atticusprojectai.org/cuad	Expert-annotated commercial contracts used to evaluate clause extraction and classification.
6	Solicitors Regulation Authority Guidance https://www.sra.org.uk/solicitors/guidance/	Regulatory expectations on confidentiality, competence and use of technology.
7	Apache Tika https://tika.apache.org	Document parsing and text extraction across file formats in the ingestion pipeline.
8	OCRmyPDF https://ocrmypdf.readthedocs.io	Open-source OCR for adding text layers to scanned documents.

Deliverables (17)

- D1 Knowledge source inventory.** A catalogue of document repositories, volumes, formats, quality and restrictions.
- D2 Client restriction register.** A structured record of engagement letter terms restricting data use, linked to matters.
- D3 Scope recommendation paper.** A phased recommendation on which corpora to include first, with risk analysis.
- D4 Permission-aware retrieval design.** An architecture enforcing matter-level permissions and ethical walls at query time.
- D5 Document ingestion pipeline.** Extraction, OCR, version selection and metadata normalisation with lineage.
- D6 Public law ingestion.** Automated ingestion of judgments and legislation with point-in-time versioning.
- D7 Precedent library enrichment.** Tagging, currency checks and structuring of curated precedents for retrieval.
- D8 Chunking and embedding strategy.** Methods for splitting legal documents that preserve clause and section context.
- D9 Citation verification service.** Automatic checking that every cited case or statute exists and supports the statement.
- D10 Evaluation dataset and benchmarks.** Lawyer-written questions with reference answers for measuring accuracy.
- D11 Private model deployment design.** A secure deployment with no data retention, meeting IT security requirements.
- D12 Answer provenance display.** An interface showing sources, excerpts and confidence for every answer.
- D13 Litigation chronology prototype.** A single-matter tool extracting dated events from disclosure documents.
- D14 Audit logging.** Logging of queries, retrieved documents and outputs for supervision and compliance.
- D15 Insurer and client assurance pack.** Documentation of controls for professional indemnity insurers and client questionnaires.
- D16 Lawyer training and usage policy.** Guidance on appropriate use, verification duties and limitations.
- D17 Pilot evaluation report.** Measured accuracy, adoption and time savings from the pilot group.

CASE STUDY 32 · ENVIRONMENTAL MONITORING (LOCAL GOVERNMENT)

City-Wide Air Quality Sensor Network Data Platform

Industry	Environmental Monitoring (Local Government)
Organisation	Greater Hallam Combined Authority, a city region of 1.9 million people with a new clean air zone
Your role	Data Engineering Lead, Environmental Intelligence Unit
Core themes	Low-cost sensor calibration, open data publication, public health communication, policy evaluation

The Scenario

The Greater Hallam Combined Authority introduced a clean air zone in its city centre eighteen months ago. Older, more polluting commercial vehicles are charged to enter, and the policy was introduced after legal action forced the government to require the region to reduce nitrogen dioxide concentrations to legal limits in the shortest possible time. The policy was controversial. Taxi drivers, small businesses and some residents' groups campaigned against it; health charities and parents' groups campaigned for it. The political leadership has promised to evaluate its effectiveness transparently. You lead data engineering in the authority's environmental intelligence unit, and the evaluation depends on air quality data that nobody fully trusts.

The region's official air quality monitoring consists of eleven reference-grade monitoring stations operated to national standards, measuring nitrogen dioxide and particulate matter continuously, plus several hundred diffusion tubes, simple passive samplers that are collected monthly and analysed in a laboratory. Reference stations are accurate but expensive and sparse: they tell you a great deal about eleven locations and nothing about the streets in between. Diffusion tubes are cheap but only give monthly averages, with results arriving six weeks after collection. To fill the gaps, the authority installed two hundred and forty low-cost sensors on lamp posts across the region, funded by a government grant, each reporting readings every five minutes over a mobile network.

The low-cost sensors are the source of both excitement and trouble. They produce vastly more data than the official network, at street level, in near real time. But their readings are affected by temperature, humidity and interference from other gases, they drift over time, and individual units vary. Several sensors near busy junctions report nitrogen dioxide levels

that would, if accurate, be among the highest in the country; others at quiet suburban locations report implausibly low or negative values. Some go offline for weeks when their mobile connection fails. The vendor provides a cloud dashboard with calibrated readings, but the calibration method is proprietary, and the vendor changed it six months ago without warning, causing a step change in the data that looked like a sudden improvement in air quality.

Last month, a local newspaper obtained sensor data through a freedom of information request, found a location near a primary school with readings well above legal limits, and ran a front-page story. The school's parents are furious, and councillors are demanding action. The environmental scientists in your unit are not sure the readings are right. The nearest reference station is two kilometres away and shows much lower levels, but that does not prove the sensor is wrong; it could be a genuine hotspot.

The director of public health, Dr Amanda Clarke, wants the platform to support public health messaging: real-time information that helps vulnerable people, such as those with asthma, avoid exposure. She is also concerned about the effect of alarming but possibly inaccurate data on public anxiety. The transport policy team, led by Mark Stevens, wants the data to evaluate the clean air zone rigorously, which requires a long, consistent time series and careful statistical methods to separate the effect of the zone from weather, traffic trends and other factors such as the gradual replacement of old vehicles nationally. The political leadership wants a public dashboard that shows progress. A campaign group has asked for all raw data to be published openly under the authority's open data commitments.

The engineering challenges begin with calibration. The scientific literature on low-cost sensors recommends co-locating some sensors with reference stations and using the paired data to build calibration models that correct for temperature, humidity and drift, then transferring those models to sensors elsewhere, periodically rotating sensors through the reference sites. This requires a disciplined data pipeline: ingesting raw sensor readings, reference data and weather observations, training and applying calibration models, tracking which calibration version was applied to which reading, and flagging readings with high uncertainty. It also requires the raw data, which the vendor's contract provides only through a limited export.

Then there is integration of the different data types. Reference stations report hourly data through the national air quality archive, with ratified data released months later after quality assurance. Diffusion tube results arrive monthly from the laboratory in spreadsheets. Traffic counts, needed for the evaluation, come from automatic counters and the clean air zone's own number plate recognition cameras, which record every vehicle entering the zone. Weather data comes from the national meteorological service. Each source has its own time resolution, quality flags and revision practices.

Open data raises questions of its own. The authority is committed to publishing data openly, and the campaign group is right that the public has a right to know. But publishing uncalibrated low-cost sensor readings without context could mislead, and publishing number plate recognition data is out of the question for privacy reasons, although aggregated traffic counts could be shared. The data protection officer has also asked whether highly localised readings near homes could be considered personal data in combination with other information, which seems unlikely but needs a documented answer.

There is also a difference of professional culture inside your own unit. The environmental scientists come from a regulatory monitoring background, where every measurement is subject to rigorous quality assurance, instruments are serviced and audited, and data is ratified before anyone draws conclusions. They are deeply uncomfortable with publishing low-cost sensor data at all and would prefer it to be used only internally until calibration is proven. The engineers, several of whom came from technology companies, see the sensors as a data product that should be open, iterative and improved in public. Both views have merit, and the tension between them has already caused friction over the vendor dashboard, which the engineers wanted to link from the authority's website and the scientists wanted to keep private. Any plan you propose needs to reconcile these instincts, perhaps by tiering data products by quality level and being explicit about which are suitable for which purposes.

Your unit has three engineers and two environmental scientists. The vendor contract for the sensors is up for renewal in eight months, and the vendor has proposed a new contract with an analytics platform bundled in. The chief executive wants a response to the school hotspot story within

weeks, and the formal evaluation of the clean air zone is due to be published in ten months.

You believe the core problem is that nobody can currently say how accurate any given sensor reading is, and that until that problem is solved, every use of the data, from public messaging to policy evaluation, is on shaky ground. But solving it properly takes months of co-location and modelling, and the political pressure is immediate.

THE DECISION IN FRONT OF YOU

Do you build an in-house calibration and data platform, accept the vendor's bundled analytics, or run both during a transition, and how do you respond to the school hotspot story, publish open data responsibly and produce a defensible clean air zone evaluation?

Data Sources

#	Source and link	How it is used in this project
1	UK-AIR (DEFRA Air Information Resource) https://uk-air.defra.gov.uk	Official reference monitoring data and ratified measurements used for calibration and evaluation.
2	OpenAQ https://openaq.org	Open global air quality data platform and API used for comparison and open data publication patterns.
3	openair R Package https://openair-project.github.io/book/	Widely used air quality analysis toolkit for trend analysis and meteorological normalisation.
4	Met Office MIDAS Open (CEDA Archive) https://catalogue.ceda.ac.uk/uuid/dbd451271eb04662beade68da43546e1	Weather observations used for sensor calibration and weather normalisation.
5	DfT Road Traffic Statistics https://roadtraffic.dft.gov.uk	Traffic count data used as evaluation covariates.
6	Clean Air Zone Framework (GOV.UK) https://www.gov.uk/government/publications/clean-air-zone-framework-for-england	National framework for clean air zones informing evaluation requirements.
7	US EPA Air Sensor Performance Targets https://www.epa.gov/air-sensor-toolbox	Guidance on evaluating and calibrating low-cost sensors.
8	DEFRA Local Air Quality Management Technical Guidance https://laqm.defra.gov.uk	Technical guidance on monitoring, diffusion tubes and bias adjustment.

Deliverables (17)

- D1 Hotspot rapid assessment.** An urgent co-location of a reference-grade instrument or mobile monitor near the school with an interim public statement.

- D2** **Sensor data quality audit.** An assessment of each sensor's completeness, drift, anomalies and plausibility.
- D3** **Raw data access agreement.** Revised vendor terms securing raw readings, calibration disclosure and change notification.
- D4** **Unified ingestion pipeline.** Ingestion of sensor, reference, diffusion tube, weather and traffic data with source metadata.
- D5** **Co-location programme design.** A rotation schedule placing sensors alongside reference stations for calibration data.
- D6** **Calibration model pipeline.** Models correcting sensor readings for temperature, humidity and drift with version tracking.
- D7** **Uncertainty flags.** Per-reading uncertainty estimates and quality labels for downstream use.
- D8** **Data revision handling.** Treatment of provisional and ratified reference data and restatement of derived results.
- D9** **Public health messaging feed.** A cautious, calibrated real-time index for vulnerable groups with clear caveats.
- D10** **Open data publication.** Published datasets with raw and calibrated values, quality flags and methodology notes.
- D11** **Traffic aggregation from zone cameras.** Anonymised, aggregated traffic counts derived without exposing number plates.
- D12** **Meteorological normalisation.** Weather-adjusted pollution trends to isolate policy effects.
- D13** **Clean air zone evaluation dataset.** A consistent long time series combining all relevant sources with documented choices.
- D14** **Evaluation statistical analysis.** A counterfactual analysis estimating the zone's effect with uncertainty.
- D15** **Public dashboard.** An accessible dashboard explaining data sources, limitations and trends.
- D16** **Vendor renewal assessment.** An evaluation of the bundled analytics offer against in-house capabilities.
- D17** **Communications guidance.** Guidance for officers and councillors on interpreting and discussing sensor data.

CASE STUDY 33 · AUTOMOTIVE MANUFACTURING

Connected Vehicle Telemetry and Early Warranty Defect Detection

Industry	Automotive Manufacturing
Organisation	Arden Motors, a British manufacturer of electric vans and light commercial vehicles selling 28,000 vehicles a year
Your role	Principal Data Engineer, Vehicle Data Platform
Core themes	Vehicle telemetry at scale, warranty and recall analytics, data ownership and consent, safety-critical decisions

The Scenario

Arden Motors builds electric vans for delivery companies, tradespeople and public sector fleets. It launched its current model four years ago, and around sixty thousand are now on the road in the UK and Europe. Every van contains a telematics control unit that connects to the company's cloud platform, sending battery state, charging events, motor temperatures, fault codes, location and dozens of other signals. The data was originally collected to support fleet management services that Arden sells to large customers, but it has become something more important. Last year, a fault in a batch of battery cooling pumps caused several vans to lose power on motorways. Nobody was hurt, but Arden had to issue a recall affecting nine thousand vehicles, and an investigation by the vehicle safety regulator found that warning signs had been visible in telemetry data for months before the first roadside failure. You are the principal data engineer for the vehicle data platform, and the chief engineer, Dr Fiona Walsh, has asked you to make sure the next problem is spotted early.

The telemetry is voluminous and heterogeneous. Each van sends summary data every few minutes when driving and a burst of detailed data when a fault occurs. Over a year, the fleet generates around forty billion signal readings. The signals were defined by the vehicle engineering teams for their own purposes, and each software release has added, renamed or changed signals. Three generations of telematics hardware are in the field, with different capabilities. Vans that are out of mobile coverage buffer data and send it later, sometimes days later. Fault codes, the diagnostic messages that vehicle systems raise when they detect a problem, number in the thousands and are documented inconsistently across suppliers.

The warranty data is separate. When a van is repaired under warranty, the dealer or service partner files a claim describing the fault, the parts replaced and the labour time. Claims arrive weeks after the repair and are coded by dealers using a parts catalogue and a fault code list that many dealers use carelessly. Warranty analysis has traditionally been the main way manufacturers detect emerging defects: an unusual increase in claims for a particular part triggers an investigation. But claims lag the problem significantly, and the cooling pump failures had been generating fault codes and temperature anomalies in telemetry long before they generated claims.

Fiona wants an early warning system that combines telemetry and warranty data to detect emerging defects, prioritise investigations and estimate how many vehicles might be affected. The quality director, Ian Morrison, supports the goal but is wary of false alarms. Every investigation consumes engineering time, and a system that raises dozens of spurious alerts will quickly be ignored. He also points out that a signal visible in hindsight is not necessarily detectable in advance: the cooling pump anomalies were subtle and mixed with many similar anomalies that turned out to be harmless.

Legal and commercial questions surround the data. In the UK and European Union, data generated by connected vehicles is increasingly regulated. The European data act gives users of connected products rights to access the data their products generate and to share it with third parties such as independent repairers. Location data is personal data when the van can be linked to a driver, which is the case for sole traders and for fleet drivers whose employers track them. Arden's customer terms allow it to use vehicle data for product improvement and safety, but some large fleet customers have negotiated contracts that restrict Arden's use of their data, and one major customer, a national delivery company, has asked whether Arden is using its vehicles' data to develop services it will sell to that customer's competitors. The general counsel, Rebecca Hughes, wants a clear data use framework before any new analytics are built.

The platform itself needs work. Telemetry currently lands in a time-series database designed for the fleet management service, which retains detailed data for only thirty days and aggregates after that. For defect detection, engineers want long-term retention of detailed data, particularly around faults, so they can study patterns over months. Storing

all detailed telemetry indefinitely would be expensive; storing none of it means the evidence is gone when a problem is discovered. The fleet services team, which owns the current platform and generates revenue from it, is reluctant to change a system that customers depend on.

Then there is the regulatory dimension of recalls. When a defect affecting safety is identified, the manufacturer must notify the regulator and determine which vehicles are affected. Last year, determining the affected population took weeks because it required matching vehicle identification numbers to battery pack serial numbers, supplier batch records and production dates, held in different systems, some of them at suppliers. The regulator was critical of the delay. Fiona wants the platform to support rapid determination of affected vehicles, which means integrating manufacturing traceability data, the record of which parts from which batches went into which vehicle.

Your team has six engineers. The vehicle engineering organisation has its own data scientists who have built analyses on extracts. Suppliers hold some of the most important data, such as battery cell test results, and share it reluctantly. The chief executive has approved funding for a year. Fiona wants early warning capability for the next model year, which starts production in nine months, and for the existing fleet as soon as possible.

The organisation also has to face an uncomfortable fact about the recall itself. The internal review found that an engineer in the thermal systems team had noticed unusual pump temperature readings in a fleet data extract five months before the first roadside failure, raised it informally with a colleague and moved on when the readings seemed to settle. There was no mechanism for logging such observations, no owner for cross-cutting signal anomalies and no link between the engineering extracts and the warranty team's monitoring. Nobody was at fault in the individual sense, but the organisation had no process for turning an engineer's hunch into an investigation. Fiona is determined that the new platform will include a way of capturing, triaging and tracking such observations, but she also knows that engineers will only use it if doing so is quick and if they see their observations acted on.

You have come to believe that three foundations matter more than any detection algorithm: a consistent, versioned catalogue of signals and fault codes across hardware and software versions; long-term retention of detailed data around fault events; and a complete vehicle genealogy

linking each vehicle to its parts, software and production history. Without these, detection models will be fragile and recall scoping will remain slow. But these foundations are not what Fiona asked for, and they touch systems owned by other teams.

THE DECISION IN FRONT OF YOU

Do you build defect detection analytics on existing data, invest first in signal cataloguing, fault-centred retention and vehicle genealogy, or pursue both in parallel, and how do you respect customer data rights and contractual restrictions while meeting safety and recall obligations?

Data Sources

#	Source and link	How it is used in this project
1	NHTSA Datasets and APIs https://www.nhtsa.gov/nhtsa-datasets-and-apis	US vehicle complaints, recalls and investigations data used to study defect signal patterns and recall scoping.
2	DVSA Vehicle Recalls Search https://www.check-vehicle-recalls.service.gov.uk	UK recall records for benchmarking recall frequency and response.
3	UK Vehicle Safety Defects Code of Practice https://www.gov.uk/government/publications/code-of-practice-on-vehicle-safety-defects-and-recalls	Obligations for manufacturers on defect reporting and recall management.
4	EU Data Act https://digital-strategy.ec.europa.eu/en/policies/data-act	Rules on user access to connected product data that shape data use and sharing design.
5	NASA Li-ion Battery Aging Datasets https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/	Battery degradation data for developing battery health features and anomaly detection.
6	Vehicle Energy Dataset (VED) https://github.com/gsoh/VED	Real vehicle telemetry from electric and hybrid vehicles for prototyping ingestion and features.
7	COVESA Vehicle Signal Specification https://covesa.github.io/vehicle_signal_specification/	Open standard taxonomy for vehicle signals used to build the signal catalogue.
8	SAE J1939 and OBD Diagnostic Standards https://www.sae.org/standards	Diagnostic trouble code standards informing fault code harmonisation.

Deliverables (17)

- D1 Data use framework.** Legal and policy rules on permitted uses of telemetry by purpose, customer contract and data type.
- D2 Customer contract restrictions register.** Structured capture of fleet customer data restrictions enforced in the platform.

- D3** **Signal and fault code catalogue.** A versioned catalogue mapping signals and codes across hardware and software generations.
- D4** **Telemetry ingestion upgrade.** Ingestion handling late buffered data, multiple hardware versions and schema evolution.
- D5** **Fault-centred retention policy.** Long-term retention of detailed data windows around fault events with cost analysis.
- D6** **Vehicle genealogy model.** Linkage of each vehicle to parts, batches, suppliers, software versions and production records.
- D7** **Rapid affected-population query.** A capability to determine vehicles affected by a part or batch within hours.
- D8** **Warranty claim normalisation.** Cleaning and coding of dealer claims with part and fault harmonisation.
- D9** **Telemetry-warranty linkage.** Joining claims to pre-failure telemetry for each repaired vehicle.
- D10** **Early warning detection models.** Detection of emerging anomalies by component, batch and software version with alert thresholds.
- D11** **Retrospective validation on the cooling pump case.** Testing whether detection would have flagged last year's defect and how early.
- D12** **Investigation triage workflow.** An alert review process for quality engineers with outcome recording.
- D13** **Supplier data sharing agreements.** Terms for obtaining batch test and traceability data from key suppliers.
- D14** **Customer data access service.** A mechanism allowing customers to access their vehicle data as required by regulation.
- D15** **Platform separation design.** Isolation of fleet services from engineering analytics to protect customer-facing reliability.
- D16** **Cost and storage model.** Projected storage and compute costs for retention and analytics options.
- D17** **Regulator readiness pack.** Documentation of detection, recall scoping and data governance for the vehicle safety regulator.

CASE STUDY 34 • SEMICONDUCTOR MANUFACTURING

Semiconductor Wafer Yield Analytics and Process Data Integration

Industry	Semiconductor Manufacturing
Organisation	Caldera Semiconductors, a specialty chip foundry in South Wales producing power and compound semiconductor devices
Your role	Senior Data Engineer, Yield Engineering Systems
Core themes	High-dimensional process data, genealogy and traceability, root cause analysis, customer data sharing

The Scenario

Caldera Semiconductors makes chips that most people will never see but increasingly depend on: silicon carbide and gallium nitride power devices that switch electricity efficiently inside electric vehicle inverters, fast chargers, solar inverters and data centre power supplies. Demand has grown so quickly that Caldera is running its fab close to capacity, and every wafer that fails to yield good devices is revenue lost. Over the past two quarters, yield on its flagship silicon carbide product has fallen from eighty-one per cent to seventy-three per cent, with no obvious cause. The yield engineering team has spent weeks investigating, and the chief operating officer, Dr Gareth Morgan, has asked you, as senior data engineer in yield engineering systems, why a factory that generates terabytes of data cannot find the problem.

A wafer at Caldera passes through several hundred process steps over eight to twelve weeks: cleaning, oxidation, lithography, etching, implantation, annealing, deposition, metallisation and many measurements in between. At each step, the processing tool records dozens or hundreds of parameters, such as temperatures, pressures, gas flows and plasma power, sampled many times per second. Metrology tools measure film thicknesses, critical dimensions and overlay on sample wafers. At the end of the line, every device on every wafer is electrically tested, producing parametric measurements and a pass or fail result for each die, and wafer maps show the spatial pattern of failures. Some patterns are signatures: a ring of failures near the edge suggests a problem with an etch or deposition tool; a cluster in one location suggests a particle or handling defect; a gradient suggests a temperature non-uniformity.

The data exists in many systems. The manufacturing execution system records which lot of wafers went through which tool and which chamber at which time, using which recipe. Tool sensor data is collected by a fault detection and classification system that monitors each tool in real time and alarms when parameters drift outside limits, but it keeps detailed traces for only sixty days. Metrology data lives in a statistical process control system. Electrical test data lives in a test data system, stored in formats defined by the test equipment vendors. Defect inspection images and classifications live in yet another system. Each system uses its own identifiers for lots and wafers, and wafers are sometimes split into sub-lots, merged or reworked, which complicates tracing any wafer's full history.

When yield engineers investigate a problem, they pull extracts from each system into spreadsheets or statistical tools and try to correlate yield with the process history. This works for simple cases, such as a single tool that went out of calibration, but the current yield decline seems to involve interactions. Preliminary analysis suggests that wafers processed on one of three implantation tools and then annealed in one particular furnace show lower yields, but only for lots started in certain weeks, and only for devices in certain positions. The engineers suspect a subtle interaction between implant dose uniformity and anneal temperature profiles, but proving it requires joining detailed tool traces with die-level test results for thousands of wafers, and the tool traces from the relevant period have been deleted under the sixty-day retention policy.

The head of yield engineering, Dr Anjali Rao, wants a yield data platform: a unified store linking every die's test results to its wafer's complete genealogy and process history, including tool sensor summaries, metrology and inspection data, with long-term retention and tools for engineers to explore correlations. She has seen such platforms at larger fabs where she previously worked and believes Caldera is years behind. The IT manager, Huw Davies, points out that commercial yield management systems exist and are mature, but they are expensive, typically priced for large fabs, and require integration work with Caldera's specific tools and systems. He also notes that the fab's IT infrastructure is mostly on-premises, for security and latency reasons, and that storage for long-term retention of tool traces would be substantial: each tool generates gigabytes per day.

There is a data sharing dimension too. Caldera's largest customers, automotive suppliers, require extensive quality data under industry standards for automotive components. After a field failure, a customer may ask for the complete manufacturing history of a specific device, and Caldera must be able to trace it within days. Two customers have recently asked for ongoing access to process and test data for their products to support their own reliability analyses, and one has proposed a data sharing platform. The commercial director, Emma Pritchard, is keen to accommodate them because they are strategic accounts, but the head of process engineering warns that detailed process data reveals Caldera's proprietary recipes and know-how, and that sharing it could benefit competitors if it leaked.

The engineering team is small. You have two data engineers, plus the support of a process engineer who writes Python. The fab's systems vendors are responsive but charge for interface development. The yield decline is costing the company significant revenue every week, and Gareth wants progress within a month, not a platform in a year.

Culture within the fab complicates any change. Process engineers are organised by module, such as lithography, etch or implant, and each module owns its tools and its data. When yield falls, the natural instinct of each module is to demonstrate that its own processes were within specification, which they usually were, since the problem lies in interactions between modules that nobody owns. Investigations can become defensive exercises in which each team produces charts showing its parameters were in control. A shared platform that links data across modules would make interactions visible, which Anjali sees as its greatest value, but some module leads see it as a way for yield engineering to second-guess their work. Their cooperation matters, because they understand what their parameters mean, and a platform populated with thousands of unexplained sensor channels is of little use without them.

There is also a question of analytical approach. Some engineers want to throw machine learning at the problem: feed thousands of process parameters into a model and let it find the variables most associated with yield loss. Anjali is cautious. In her experience, such analyses frequently find spurious correlations, especially when process choices are confounded, for example when certain products always run on certain tools. She prefers a disciplined approach grounded in process physics,

using data to test hypotheses rather than generate them blindly, although she acknowledges that data-driven screening can help prioritise.

The current crisis and the long-term need point in different directions. To solve the current yield problem, the fastest route may be a targeted, one-off data integration of the relevant lots, tools and test data, combined with a designed experiment splitting wafers across the suspect tools and furnace settings. To prevent future crises, the fab needs a permanent platform with longer retention and automated genealogy. Doing the first quickly could delay the second; doing the second first leaves the current problem unsolved for months.

You also recognise that retention is the single decision you can make immediately that will matter most in future. Every day that passes, more tool trace data is deleted. Extending retention, even crudely by archiving raw traces to cheap storage without integrating them, would preserve evidence for future investigations, but it requires storage capacity and a decision about which tools and parameters matter, which nobody has formally agreed.

THE DECISION IN FRONT OF YOU

Do you focus on a targeted investigation to resolve the current yield decline, begin building a permanent yield data platform, or buy a commercial yield management system, and how do you handle tool trace retention now and customer demands for data without exposing proprietary process know-how?

Data Sources

#	Source and link	How it is used in this project
1	SECOM Manufacturing Dataset (UCI) https://archive.ics.uci.edu/dataset/179/secom	Real semiconductor process sensor data with pass and fail labels used to prototype feature screening and yield correlation.
2	WM-811K Wafer Map Dataset https://www.kaggle.com/datasets/qingyi/wm811k-wafer-map	Over 800,000 real wafer maps with failure pattern labels used to build spatial signature classification.
3	Bosch Production Line Performance https://www.kaggle.com/c/bosch-production-line-performance	Large manufacturing traceability dataset for genealogy and station-level analysis.
4	SEMI Standards (E10, E164, EDA) https://www.semi.org/en/products-services/standards	Industry standards for equipment data acquisition and interfaces.

#	Source and link	How it is used in this project
5	AEC-Q101 Automotive Qualification Standard https://www.aecouncil.com/AECDocuments.html	Automotive qualification requirements that drive traceability and data retention.
6	IATF 16949 Overview https://www.iatfglobaloversight.org	Automotive quality management requirements for traceability and customer data.
7	Apache Parquet https://parquet.apache.org/docs/	Columnar format for efficient long-term tool trace archiving.
8	NIST Engineering Statistics Handbook https://www.itl.nist.gov/div898/handbook/	Reference for designed experiments and statistical process control methods.

Deliverables (17)

- D1 Yield decline investigation plan.** A hypothesis-driven plan combining targeted data integration and a designed split-lot experiment.
- D2 Targeted investigation dataset.** A one-off joined dataset of affected lots, tool history, metrology and die-level test results.
- D3 Designed experiment specification.** A split-lot experiment across implanters and anneal settings with analysis plan.
- D4 Interim trace retention extension.** Immediate archiving of raw tool traces for priority tools to low-cost storage.
- D5 Retention policy by tool and parameter.** An agreed classification of which traces to retain, at what resolution and for how long.
- D6 Wafer genealogy service.** Tracking of every wafer through splits, merges and rework with consistent identifiers.
- D7 Die-level test data pipeline.** Parsing of vendor test formats into a unified die and wafer result model.
- D8 Tool sensor summary features.** Per-wafer, per-step summary statistics derived from tool traces.
- D9 Wafer map signature classification.** Automated classification of spatial failure patterns.
- D10 Yield correlation workbench.** Tools for engineers to link yield to process history with confounding checks.
- D11 Commercial yield system evaluation.** A structured assessment of vendor systems against requirements and cost.
- D12 Customer traceability response process.** A capability to produce complete device histories within customer deadlines.
- D13 Customer data sharing policy.** Rules defining what process and test data can be shared, at what aggregation, under what terms.

- D14 Proprietary data protection controls.** Controls preventing recipe and process know-how exposure in shared datasets.
- D15 Platform architecture roadmap.** A phased design for a permanent yield data platform on-premises.
- D16 Data quality checks.** Validation of identifier consistency, missing test data and genealogy breaks.
- D17 Yield recovery report.** Documentation of the root cause, corrective action and yield improvement achieved.

CASE STUDY 35 · FOOD AND BEVERAGE MANUFACTURING

Product Traceability and Rapid Recall Data Platform for a Food Manufacturer

Industry	Food and Beverage Manufacturing
Organisation	Halewood Foods, a UK manufacturer of chilled ready meals, sandwiches and salads supplying major retailers
Your role	Lead Data Engineer, Operations and Quality Systems
Core themes	Farm-to-fork traceability, allergen data, supplier data integration, regulatory response time

The Scenario

Halewood Foods makes chilled food that appears under supermarket own labels: ready meals, sandwiches, wraps, salads and snack pots, produced in four factories and delivered daily to retailers' distribution centres. Most of what it makes has a shelf life of three to seven days, which means it is eaten within days of being produced. When something goes wrong, it goes wrong fast. Last spring, a supplier of spinach notified Halewood that a batch might be contaminated with listeria. Halewood needed to determine, urgently, which of its products contained spinach from that batch, which production runs they came from, which retailers had received them and which were still on sale. It took thirty-one hours to produce a definitive answer. During that time, Halewood and its retail customers withdrew a far larger range of products than necessary, as a precaution, at a cost of more than a million pounds. The Food Standards Agency, which oversees food safety incidents, asked why it took so long. You were brought in as lead data engineer for operations and quality systems to make sure the next answer takes minutes.

The traceability chain at Halewood runs through several systems and a surprising amount of paper. Goods-in records, capturing each delivery of ingredients with the supplier's batch codes, are entered into the enterprise resource planning system by warehouse staff, but some suppliers' batch codes are recorded incompletely or not at all, especially for fresh produce, where a single delivery may include product from several farms. Ingredients are transferred to production in quantities recorded on paper sheets that are later keyed into the system, sometimes days later. A production run may use ingredients from several batches, and batches may be split across several runs and several factories. Recipes are held in a

product lifecycle management system, but factory recipes are sometimes adjusted on the line, for example substituting one supplier's spinach for another's when supplies run short, and these substitutions are recorded on paper. Finished products are labelled with a production code and use-by date, and dispatch records show which pallets went to which retailer depot.

To answer the question "which products contain this batch of spinach?", someone must trace forward from the goods-in record through the paper transfer sheets and production records to the finished products and dispatches. During the incident, the quality team did this manually, calling factory managers at home, searching filing cabinets and cross-referencing spreadsheets. Several transfer sheets were illegible. One factory had used spinach from the suspect batch in a product that was not on any recipe card for spinach, because of an on-line substitution.

The operations director, Philip Carter, wants a traceability platform that records every ingredient movement digitally and can trace forward and backward instantly. He has been looking at systems that use barcode scanning of every ingredient container at each transfer, linked to production orders. The finance director, Laura Evans, supports this but is concerned about cost and disruption: scanning at every transfer requires hardware, network coverage in cold and wet production areas, process changes and training for hundreds of operatives, many of whom work through agencies and turn over frequently.

The technical director, Dr Neil Barker, who is responsible for food safety and quality, raises a different point. Traceability is not only about recalls. Allergens are the most common cause of food recalls in the UK, usually because a product contains an allergen that is not declared on the label, often due to a recipe change, an ingredient substitution or cross-contact on a shared production line. Accurate, real-time knowledge of what went into each product is essential for allergen control as well. Neil wants the platform to check, at the point of production, that the ingredients being used match the recipe and label declarations, and to alert if a substitution introduces an undeclared allergen. That would change the platform from a record-keeping system into a control system.

The retailers add another layer. Each of Halewood's major retail customers has its own technical standards and increasingly its own supplier portals, where suppliers must upload product specifications, allergen declarations,

and in some cases traceability data. One retailer has launched a programme requiring suppliers to share batch-level traceability data electronically using global standards, so that in an incident the retailer can trace products across its entire supply base. Halewood's commercial director, Sophie Grant, is keen to comply, because the retailer is Halewood's largest customer and compliance may influence future contracts. But the standards require consistent product and location identifiers and event data formats that Halewood's systems do not currently produce.

Upstream, suppliers vary enormously. Large ingredient suppliers provide batch codes, certificates of analysis and electronic delivery notes. Small farms and growers provide handwritten delivery notes. Some ingredients come through wholesalers who consolidate produce from many sources and cannot always say which farm a particular crate came from. Halewood can only trace as well as its suppliers allow.

There is a human side to the factory floor that any design has to respect. Production operatives work long shifts in cold rooms at two to four degrees Celsius, wearing gloves, hairnets and protective clothing, and handling heavy crates in a hurry to keep lines running. Many are agency workers who may not speak English as a first language and who may be on a different line each week. The factory managers are measured on output and waste, and anything that slows a line is resisted. The paper transfer sheets survive precisely because they are quick and require no training. Any digital capture must be at least as fast as writing a number on a sheet, must work with gloved hands and in wet conditions, and must tolerate network outages in parts of the building where wireless coverage is poor. Philip's scanning vision is sound in principle, but a pilot that slows the lines will be abandoned within a week.

You have a year, a budget that the board has approved after the recall, and a team of three engineers plus the support of the quality team. The Food Standards Agency expects to see improvements, and the retailer wants its data standard implemented within nine months.

You see several possible starting points. The most visible is scanning hardware on the factory floor, which Philip wants. The most urgent from a safety perspective is allergen control, which Neil wants. The most commercially pressing is the retailer's data standard, which Sophie wants. The most fundamental, in your view, is a clean, consistent data model of

ingredients, batches, recipes, production runs and dispatches, with identifiers that link them, because without it, neither scanning nor allergen control nor retailer data sharing will work reliably. But a data model does not look like progress to anybody except you.

You also recognise that the paper-based steps will not disappear overnight, and that the platform must work with partial digital data for some time, handling gaps and uncertainty explicitly rather than pretending to precision it does not have. A trace that returns "these products definitely contain the batch, and these products might" would have been far more useful last spring than thirty-one hours of silence.

THE DECISION IN FRONT OF YOU

Do you start with factory-floor scanning, allergen control at the line, the retailer's traceability data standard or the underlying traceability data model, and how do you design the platform to give fast, honest answers while parts of the chain remain on paper?

Data Sources

#	Source and link	How it is used in this project
1	GS1 EPCIS and Core Business Vocabulary https://www.gs1.org/standards/epcis	Global standard for supply chain event data used to model traceability events and share with retailers.
2	GS1 UK Standards and GTIN Guidance https://www.gs1uk.org	Product, batch and location identification standards underpinning traceability.
3	Food Standards Agency Food Alerts API https://data.food.gov.uk/food-alerts/ui/reference	Official UK product recalls and allergy alerts used to analyse recall causes and test alert ingestion.
4	openFDA Food Enforcement Reports https://open.fda.gov/apis/food/enforcement/	US food recall records used to study recall scope and root causes.
5	Open Food Facts https://world.openfoodfacts.org/data	Open product ingredient and allergen data used to prototype label and allergen consistency checks.
6	FSA Allergen Guidance for Food Businesses https://www.food.gov.uk/business-guidance/allergen-guidance-for-food-businesses	Regulatory requirements for allergen labelling and control.
7	BRCGS Food Safety Standard https://www.brcgs.com/our-standards/food-safety/	Certification standard requirements for traceability testing and timings.

#	Source and link	How it is used in this project
8	FDA Food Traceability Rule (FSMA 204) https://www.fda.gov/food/food-safety-modernization-act-fsma/fsma-final-rule-requirements-additional-traceability-records-certain-foods	International reference model for key data elements and critical tracking events.

Deliverables (17)

- D1 Recall incident data review.** A reconstruction of the spinach incident identifying every data gap and delay.
- D2 Traceability data model.** A model of suppliers, ingredient batches, transfers, recipes, production runs, products and dispatches.
- D3 Identifier standardisation.** Adoption of GS1 product, batch and location identifiers across systems.
- D4 Goods-in capture improvement.** Mandatory batch code capture with validation and supplier data requirements.
- D5 Supplier traceability requirements.** Revised supplier specifications for batch coding and electronic delivery data.
- D6 Transfer recording design.** A phased approach from digitised paper forms to scanning at critical transfers.
- D7 Line-side scanning pilot.** A scanning pilot in one factory with hardware, process and training evaluation.
- D8 Recipe and substitution control.** Digital recording of on-line substitutions linked to production runs.
- D9 Allergen consistency checks.** Automated checks that ingredients used match recipe and label allergen declarations.
- D10 Forward and backward trace service.** Queries returning definite and possible affected products within minutes.
- D11 Uncertainty-aware trace outputs.** Explicit distinction between confirmed and possible links where data is incomplete.
- D12 Retailer EPCIS data feed.** Traceability events shared in the retailer's required standard format.
- D13 Mock recall testing programme.** Regular timed trace exercises with performance tracking.
- D14 Recall response dashboard.** An incident view showing affected batches, products, customers and stock locations.
- D15 Data quality monitoring.** Alerts for missing batch codes, late paper entry and broken links in the chain.
- D16 Regulator and retailer assurance pack.** Evidence of improved traceability for the FSA and retail customers.

- D17 Implementation roadmap.** A sequenced plan balancing data model, scanning, allergen and retailer priorities.

CASE STUDY 36 • PUBLIC HEALTH

Integrated Infectious Disease Surveillance Data Pipeline

Industry	Public Health
Organisation	The Regional Health Protection Agency, responsible for surveillance and outbreak response for 5.6 million people
Your role	Principal Data Engineer, Surveillance Systems
Core themes	Multi-source surveillance integration, timeliness versus completeness, record linkage, emergency scalability

The Scenario

The Regional Health Protection Agency exists to spot threats to the public's health early and respond to them. Its epidemiologists track notifiable diseases, investigate outbreaks of food poisoning, measles and tuberculosis, monitor seasonal influenza and respiratory viruses, and advise hospitals, schools, care homes and local authorities. The agency learned hard lessons during the pandemic, when surveillance systems designed for routine reporting were overwhelmed by volume and by the need for daily, even hourly, data. A national inquiry found that data flows were fragile, fragmented and slow, and made recommendations about building resilient surveillance infrastructure for future emergencies. You are the principal data engineer for surveillance systems, and you have been tasked with implementing the agency's response.

Surveillance data arrives from many directions. Laboratories report positive test results for notifiable organisms electronically, but through several different routes: hospital laboratories through one national system, private laboratories through another, and some specialised reference laboratories by secure email. Clinicians notify suspected cases of certain diseases by phone, online form or letter. Hospitals report admissions for severe respiratory illness through a separate collection. General practice data provides syndromic signals, such as consultations for influenza-like illness. Wastewater monitoring, introduced during the pandemic, measures viral material in sewage. Care homes report outbreaks. Schools report absences. Each source has its own format, timeliness and identifiers. Some sources report individual patients, others report counts by area or setting, and a few, such as wastewater, report measurements that cannot be tied to any individual at all, so the platform must

accommodate fundamentally different kinds of data rather than a single case-based model.

The core problem is linkage. A single patient may appear in a laboratory report, a clinician notification and a hospital admission record, each with slightly different identifying information: a correct national health number in one, a missing number and a misspelt name in another. To count cases accurately, understand severity, and investigate outbreaks, the agency needs to link these records to individuals, deduplicate them and build a single case record. Currently, linkage is done by a nightly batch process using deterministic matching on health number, followed by manual review of unmatched records by a team of administrators. The manual review backlog grows rapidly whenever case numbers rise.

The chief epidemiologist, Dr Rebecca Adeyemi, wants a surveillance data platform that integrates all sources in near real time, links records probabilistically, and supports both routine surveillance and emergency surges. She points out that during the pandemic, the agency's systems buckled at a few thousand cases a day, and that the next emergency might be a novel influenza strain producing tens of thousands of cases a day. The head of data and analytics, Colin Marsh, agrees but warns that previous attempts at a new surveillance system foundered because they tried to replace everything at once. The current system, though creaking, is trusted by epidemiologists and connected to national reporting. He favours incremental improvement: modernising linkage, adding sources one by one and building surge capacity, while keeping the existing case management system in place.

There are governance considerations. Surveillance data about identifiable individuals is processed under specific public health legal powers that allow processing without consent for the purposes of protecting public health. Those powers have limits, and the agency's Caldicott guardian, Dr Helen Price, is responsible for ensuring that identifiable data is used appropriately. During the pandemic, emergency legal notices expanded data sharing temporarily, and some data flows that began under those notices have continued without clear legal basis since they expired. Helen wants every data flow reviewed and documented. She is also concerned about proposals to share linked surveillance data with other public bodies, such as local authorities, which want neighbourhood-level information to target interventions.

Timeliness and completeness pull against each other. Laboratory reports arrive quickly but often lack clinical and demographic detail. Clinician notifications contain more detail but arrive later. Hospital data is complete but lags by days. Epidemiologists want early signals, but they also need to avoid false alarms that trigger unnecessary interventions, and they know that recent data is always incomplete because of reporting delays. Statistical methods for adjusting recent counts for expected delays exist, but they depend on well-understood, stable reporting patterns, which break down in emergencies.

The technical environment is mixed. The agency has a case management system used by health protection teams, a data warehouse used by epidemiologists, and a variety of scripts and databases built during the pandemic by staff and contractors who have since left. Some pandemic-era pipelines run on a cloud platform procured quickly under emergency arrangements; others run on the agency's on-premises servers. There is little documentation. The national public health body provides some shared infrastructure and data, and there is an expectation of alignment with national standards and systems.

Your team consists of four engineers, one of whom is new. The agency has secured funding for a three-year programme, with the first year focused on resilience. Rebecca wants a platform that could have coped with the pandemic's peak; Colin wants nothing that jeopardises routine surveillance; Helen wants every flow legally sound. The winter respiratory season starts in five months, and last winter's surge nearly broke the manual linkage process.

The people side is as fragile as the technology. The manual linkage team consists of eight administrators, several of whom joined during the pandemic on temporary contracts that have since been made permanent. They have developed deep, informal expertise in resolving tricky matches: they know which laboratories habitually send dates of birth in the wrong format, which hospitals use temporary numbers for emergency admissions and how to spot twins. That expertise is undocumented and would be invaluable in designing and tuning an automated linkage process, but the team is understandably anxious that automation will eliminate their roles. Colin has assured them that their work will shift towards resolving the genuinely hard cases and improving data quality at source, but trust is limited. Meanwhile, two of the four engineers on your team were involved

in building the pandemic-era pipelines under intense pressure and are sensitive about any suggestion that their work was poor, even though the lack of documentation was a consequence of circumstances rather than carelessness.

You see the linkage problem as the heart of it. If record linkage were fast, accurate and automated, with manual review reserved for genuinely ambiguous cases, much of the downstream pain would ease. But linkage depends on data quality at source, and the agency cannot directly control what laboratories and clinicians send. You also see an immediate risk in the undocumented pandemic-era pipelines: nobody knows exactly what they do, whether they still work correctly or whether their legal basis is sound, yet some of them feed weekly reports that the government relies on.

THE DECISION IN FRONT OF YOU

Do you pursue a comprehensive new surveillance platform, incremental modernisation around the existing system, or a winter-first resilience plan focused on linkage and surge capacity, and how do you regularise pandemic-era data flows and balance timeliness against completeness?

Data Sources

#	Source and link	How it is used in this project
1	UKHSA Data Dashboard https://ukhsa-dashboard.data.gov.uk	Published infectious disease surveillance data used for benchmarking outputs and format design.
2	WHO Global Health Observatory https://www.who.int/data/gho	International surveillance indicators for context and comparison.
3	CDC Data Catalog https://data.cdc.gov	US public health surveillance datasets such as respiratory virus and wastewater data used for prototyping.
4	CDC National Wastewater Surveillance System https://www.cdc.gov/nwss/	Wastewater surveillance methods and data used to design the wastewater ingestion component.
5	Notifiable Diseases and Causative Organisms (GOV.UK) https://www.gov.uk/guidance/notifiable-diseases-and-causative-organisms-how-to-report	Legal reporting requirements defining data sources and obligations.
6	NHS Data Dictionary https://www.datadictionary.nhs.uk	Standard definitions for health data items used in linkage and integration.
7	Splink Record Linkage https://moj-analytical-services.github.io/splink/	Probabilistic linkage library for automating patient record matching at scale.

#	Source and link	How it is used in this project
8	Health Service (Control of Patient Information) Regulations 2002 https://www.legislation.gov.uk/uksi/2002/1438/contents	Legal basis for processing confidential patient information for public health.

Deliverables (17)

- D1 Surveillance data flow inventory.** A catalogue of every source, route, format, timeliness and legal basis.
- D2 Pandemic-era pipeline audit.** A review of undocumented pipelines covering function, correctness, dependencies and legal status.
- D3 Legal basis regularisation plan.** Actions to confirm, document or cease each data flow with Caldicott guardian approval.
- D4 Winter resilience plan.** Priority actions to protect surveillance through the coming respiratory season.
- D5 Probabilistic linkage service.** Automated record linkage with match weights, thresholds and clerical review queue.
- D6 Linkage quality evaluation.** Measured precision and recall of linkage against a manually reviewed sample.
- D7 Canonical case record model.** A single case representation combining laboratory, notification and hospital data.
- D8 Source ingestion adapters.** Standardised ingestion for each laboratory and notification route with validation.
- D9 Surge capacity testing.** Load tests simulating tens of thousands of daily cases across the pipeline.
- D10 Reporting delay adjustment.** Nowcasting methods estimating true recent counts from incomplete data.
- D11 Timeliness and completeness monitoring.** Metrics per source showing reporting lag and data completeness.
- D12 Wastewater data integration.** Ingestion and normalisation of wastewater viral measurements alongside case data.
- D13 Outbreak investigation workspace.** Linked data views supporting epidemiologists during outbreak investigations.
- D14 Local authority data sharing framework.** Aggregation and disclosure rules for sharing neighbourhood-level information.
- D15 Platform architecture and hosting decision.** A consolidated architecture resolving cloud and on-premises fragmentation.
- D16 Documentation and knowledge base.** Documentation of pipelines, definitions and runbooks to reduce key-person risk.

- D17 Three-year roadmap.** A phased programme aligned with national systems and inquiry recommendations.

CASE STUDY 37 · URBAN MOBILITY (MICROMOBILITY)

Bike Share Demand Forecasting and Fleet Rebalancing Data Platform

Industry	Urban Mobility (Micromobility)
Organisation	Cyclo, operator of a docked and e-bike share scheme with 9,500 bikes and 780 stations in a UK city
Your role	Senior Data Engineer, Operations Intelligence
Core themes	Real-time station state, demand forecasting, operational optimisation, public contract reporting

The Scenario

Cyclo runs the bike share scheme in a large UK city under a contract with the city council and the regional transport authority. The scheme has grown steadily since it launched, and the arrival of electric bikes two years ago transformed its usage: e-bikes now account for half the fleet and more than two thirds of trips, because they make hills and longer commutes easy. Growth has brought an operational problem that every bike share scheme knows well. In the morning, bikes flow from residential areas into the city centre and business districts; in the evening, they flow back. Stations in the centre fill up in the morning and empty in the evening; stations in residential areas do the opposite. Customers who find an empty station when they want a bike, or a full station when they want to return one, are frustrated, and the contract includes service level targets for station availability, with financial penalties when they are missed. You are the senior data engineer in Cyclo's operations intelligence team, and the rebalancing problem has become yours.

Rebalancing is done by a fleet of vans that move bikes between stations, directed by a team of dispatchers who watch a live map of station fill levels and decide where to send the vans. The dispatchers are experienced and know the city's patterns intimately, but they react to what they see rather than anticipating it. By the time a central station is full, the morning rush is often over. The e-bikes add complexity: they need charging, which happens either at charging stations or by swapping batteries, and a flat e-bike is effectively unavailable. Customers strongly prefer e-bikes, so a station full of classic bikes can be as frustrating as an empty one.

The general manager, Nadia Ferreira, wants predictive rebalancing: forecasts of demand at each station over the next few hours, combined

with an optimisation that recommends where the vans should go and which bikes to move. She has seen published research and vendor products claiming large improvements in availability. The contract renewal is eighteen months away, and the council has made clear that availability performance will weigh heavily in its decision.

The data is plentiful in principle. Every docking and undocking event is recorded by the stations. Every e-bike reports its location, battery level and status over a mobile connection. Trips are recorded with start and end stations and times, and customer accounts show usage patterns. Van locations are tracked by telematics. External factors matter too: weather strongly affects demand, as do events such as football matches and concerts, school holidays, public transport disruption and even the time of sunset.

In practice, the data has problems. Station status feeds occasionally freeze, reporting stale fill levels for minutes or hours when a station loses connectivity. Docking events are sometimes duplicated or missed. E-bike location reports are sparse when bikes are docked, and battery levels are reported inconsistently between two generations of e-bike. Trips that end with a bike being returned to a full station and left nearby, which is permitted for e-bikes at some locations, create orphan records. The historical data used to train any demand model reflects not true demand but constrained usage: when a station is empty, nobody can take a bike from it, so observed trips understate demand at exactly the times and places where availability is worst.

The operations manager, Tom Brennan, who leads the dispatchers, is sceptical. He has tried two vendor tools before. Both produced recommendations that the dispatchers found unrealistic: sending vans across the city during rush hour, ignoring road closures, or moving bikes to stations that the dispatchers knew would empty naturally within the hour. Tom believes the dispatchers' judgement is the scheme's greatest asset and that any tool should support rather than replace it. He also points out that vans, drivers and battery-swapping staff are the real constraint: there are only fourteen vans, and no forecast can make them move bikes faster. Traffic in the city centre during rush hour can turn a two-kilometre move into a forty-minute journey, so a recommendation that looks optimal on a map may be useless on the road.

The council's transport data team has its own interest. It wants trip data from the scheme to inform cycling infrastructure planning, such as where to build segregated cycle lanes. The contract requires Cyclo to share anonymised trip data, and the council wants to publish it as open data, as some other cities do. Cyclo's data protection lead, Chloe Watson, has concerns: trip data with precise start and end locations and times can, in combination with other information, reveal where individuals live and work, particularly for regular commuters. She wants aggregation or other protections before publication.

There is also a question of what to optimise. The contract's availability metric counts the percentage of time stations are neither empty nor full, weighted equally across all stations. Operationally, however, a busy central station matters far more to customers than a quiet suburban one. Optimising for the contract metric might mean spending van time keeping quiet stations tidy rather than serving busy ones. Nadia wants to maximise the renewal chances; Tom wants to serve customers; the finance director wants to minimise van hours.

Your team is three engineers and a data scientist. The platform currently consists of the station management system's database, a data warehouse updated nightly, and some dashboards. There is no real-time processing capability beyond the live map in the dispatch software, which is a vendor product that can accept external recommendations only through a limited interface.

There is a strategic uncertainty hanging over all of this. The council is considering expanding the scheme into two neighbouring districts and adding a dockless e-bike zone in the city centre, where bikes can be left anywhere within designated parking bays rather than at stations. Dockless operation changes the rebalancing problem fundamentally: instead of moving bikes between a few hundred fixed stations, vans and staff would need to collect and redistribute bikes scattered across many bays, and availability would be measured by the density of bikes within walking distance rather than by station fill levels. A platform designed narrowly around stations might need significant rework within two years. Nadia has asked that anything you build should be able to accommodate dockless operation, but she cannot yet say whether or when it will happen, or what the service levels would be.

You believe that the first step is making the real-time data trustworthy and building an honest demand model that accounts for censored demand, before attempting optimisation. But Nadia wants visible improvements in availability this summer, and summer is when demand peaks and availability suffers most.

THE DECISION IN FRONT OF YOU

Do you buy a vendor rebalancing optimisation product, build in-house forecasting and dispatcher decision support, or first stabilise real-time data and demand modelling, and how do you reconcile the contract availability metric, customer needs and van capacity while sharing trip data safely with the council?

Data Sources

#	Source and link	How it is used in this project
1	Citi Bike System Data https://citibikenyc.com/system-data	Large public trip history and station data used to prototype demand forecasting and station pattern analysis.
2	TfL Cycling Open Data https://cycling.data.tfl.gov.uk	London bike share usage data used for UK-specific demand patterns and seasonality.
3	General Bikeshare Feed Specification (GBFS) https://github.com/MobilityData/gbfs	Open standard for real-time station and vehicle status feeds used for ingestion design.
4	Met Office Weather DataHub https://www.metoffice.gov.uk/services/data/met-office-weather-datahub	Weather forecasts as key demand features.
5	OpenStreetMap Extracts (Geofabrik) https://download.geofabrik.de	Road and cycle network data for van routing and station context.
6	Google OR-Tools https://developers.google.com/optimization	Open-source optimisation library for vehicle routing and rebalancing prototypes.
7	ONS Census Travel to Work Data https://www.ons.gov.uk/census	Commuting patterns by area used as static demand features.
8	DfT Walking and Cycling Statistics https://www.gov.uk/government/collections/walking-and-cycling-statistics	National cycling data for context and council reporting.

Deliverables (17)

- D1 Real-time data quality assessment.** An analysis of station feed freezes, duplicate events and e-bike telemetry gaps.
- D2 Station state reconstruction service.** A reliable real-time station state combining docking events and e-bike telemetry.

- D3 Stale feed detection.** Automatic detection and flagging of frozen or inconsistent station feeds.
- D4 Censored demand estimation.** A method estimating true demand when stations are empty or full.
- D5 Station demand forecasting model.** Short-term forecasts of pick-ups and returns per station by bike type.
- D6 External feature pipeline.** Weather, events, holidays and transport disruption features.
- D7 E-bike battery availability model.** Prediction of usable e-bike availability accounting for charge levels.
- D8 Dispatcher decision support prototype.** Ranked rebalancing suggestions designed with dispatchers and respecting van constraints.
- D9 Recommendation acceptance tracking.** Recording of dispatcher decisions and outcomes for learning and evaluation.
- D10 Service metric analysis.** A comparison of contract availability with customer-weighted availability measures.
- D11 Van capacity and productivity analysis.** An analysis of van hours, moves and effectiveness by time and area.
- D12 Vendor product evaluation.** A structured trial of vendor optimisation against in-house decision support.
- D13 Trip data anonymisation for open data.** Aggregation and privacy protections before sharing trip data with the council.
- D14 Council data sharing agreement.** Terms for sharing and publishing trip data with defined protections.
- D15 Summer operations plan.** Measures deliverable before peak season, with measurement of availability impact.
- D16 Operations dashboards.** Live and historical views of availability, forecasts and rebalancing activity.
- D17 Contract renewal evidence pack.** Performance data and improvement evidence to support contract renewal.

CASE STUDY 38 · ADVERTISING TECHNOLOGY

Programmatic Advertising Bid Log Pipeline and Supply Path Transparency

Industry	Advertising Technology
Organisation	Adwell, an independent demand-side platform buying digital advertising for 300 agency and brand clients
Your role	Principal Data Engineer, Bidding Data Infrastructure
Core themes	Extreme event volumes, sampling versus completeness, log reconciliation, privacy regulation in advertising

The Scenario

Adwell is a demand-side platform, a piece of advertising technology that buys ad impressions on behalf of advertisers in real-time auctions. Every time someone opens a web page or app that sells advertising programmatically, an auction takes place in about a tenth of a second. Supply-side platforms and exchanges send bid requests describing the impression to many buyers, the buyers decide whether to bid and how much, and the highest bidder's ad is shown. Adwell receives around two million bid requests per second at peak, bids on a small fraction, and wins a small fraction of those. Each request, bid, win, impression, click and conversion generates log data. You are the principal data engineer responsible for bidding data infrastructure, and the platform that stores and processes those logs has become the company's most expensive and most contested asset.

The logs matter for three reasons. First, they train the bidding models that decide how much each impression is worth to each advertiser, which is the core of Adwell's value. Second, they drive billing and reporting: advertisers pay for impressions and clicks, and Adwell pays exchanges for impressions won, so every number must reconcile. Third, they underpin a new and increasingly urgent demand from advertisers for supply path transparency. Several large advertisers commissioned an industry study that found a significant share of their programmatic spending could not be traced from advertiser to publisher, lost somewhere in the chain of intermediaries. Adwell's largest client, a global consumer goods company, has asked for impression-level logs showing exactly what it paid, what each intermediary took and what the publisher received.

The current architecture stores full logs for wins, impressions and clicks, which are relatively low volume, but samples bid requests at one per cent, because storing every request would cost several million pounds a year. The bidding models are trained on sampled data, which works reasonably well for common impressions but poorly for rare, valuable ones, such as a particular audience on a particular premium site. Loss data, the bids Adwell made but did not win, including the winning price when exchanges share it, is kept for seven days. The data science team, led by Dr Marcus Lee, argues that the models would improve significantly with full bid request data and longer loss data retention, particularly for bid shading, the practice of bidding just enough to win in first-price auctions. He estimates that better models could improve advertiser performance by several per cent, which in a competitive market could be decisive.

The finance director, Joanna Price, has a different problem. The monthly reconciliation of impressions and costs between Adwell's logs, the exchanges' reports and the advertisers' invoices is a nightmare. Discrepancies of one to three per cent are normal in the industry, because of timing differences, filtered invalid traffic, currency conversion and differences in how each party counts. But at Adwell's volumes, a two per cent discrepancy is a large sum, and the finance team spends two weeks every month investigating and negotiating. Joanna wants a reconciliation pipeline that identifies discrepancies automatically, explains them by cause and supports disputes with evidence.

The supply path transparency request raises its own engineering challenges. To show what each intermediary took, Adwell needs to trace each impression through the chain of exchanges and resellers that passed the request to it. Industry standards exist for this: a supply chain object in the bid request lists every intermediary, and files published on publishers' and sellers' websites declare authorised sellers. But adoption is inconsistent, some intermediaries misreport, and matching Adwell's data with publisher-side data requires the publishers' cooperation. The client's request also raises data sharing questions: impression-level logs contain information about users, and privacy regulations restrict sharing it.

Privacy is the fourth pressure. The advertising industry's data practices have been under sustained regulatory scrutiny, and the UK data protection regulator has published findings that real-time bidding involves extensive processing of personal data, often without adequate legal basis. Bid

requests can contain device identifiers, IP addresses, location data and inferred interests. Adwell stores these in its logs. The company's general counsel, Sarah Okonkwo, has asked for a review of what personal data is retained, for how long and on what legal basis. She suspects that retaining full bid request logs for model training would be difficult to justify, and that longer retention of any identifiers would increase regulatory risk. She is also watching industry initiatives to replace third-party cookies and device identifiers with privacy-preserving alternatives, which would change the data available in bid requests.

Your infrastructure lead, Ravi Sharma, points out that the platform is already stretched. The streaming pipeline that processes logs runs on a large cluster of machines that is expensive and difficult to scale. Log formats change whenever exchanges update their integrations, and parsing errors silently drop data. Storage costs have grown faster than revenue. Any expansion of retention or sampling rates must come with efficiency gains elsewhere.

The chief executive, Oliver Grant, sees all four demands as strategic and wants a plan that addresses them together. Budget is available, but not unlimited, and the board expects infrastructure costs as a share of revenue to fall, not rise.

You see trade-offs everywhere. Full bid request retention would help the models but raise costs and privacy risk. Longer retention of identifiers would help reconciliation and transparency but would face legal pushback. Aggregation and anonymisation would reduce privacy risk but might limit model improvements. Smarter sampling, such as keeping all requests for rare and valuable segments while sampling common ones heavily, could offer much of the modelling benefit at a fraction of the cost, but it would complicate every downstream analysis, because sampled data must be weighted correctly.

Invalid traffic adds another dimension. A proportion of bid requests come from bots, fraudulent apps and websites that fabricate impressions, and Adwell uses a third-party verification vendor to screen them. The verification vendor's results arrive hours after the impression, in a separate feed, and are joined to Adwell's logs using an impression identifier that some exchanges populate inconsistently. When the join fails, impressions that should have been filtered are billed to advertisers, and impressions that were legitimately served are sometimes disputed by

exchanges. The data science team also suspects that some of the training data for bidding models is contaminated by invalid traffic that was never identified, which would cause models to value fraudulent inventory. Any redesign of the logging platform needs to treat verification data as a first-class input rather than an afterthought joined at month end.

There is also a question of data architecture. The current logs are stored as compressed text files partitioned by hour. Moving to a columnar format with schema enforcement would reduce storage and speed up queries dramatically, and would catch parsing errors, but migrating petabytes of history and dozens of downstream jobs is a substantial project in its own right. And it would do nothing on its own to answer the client's transparency request, which has a deadline of three months before the client's annual agency review.

THE DECISION IN FRONT OF YOU

How do you balance the data science team's demand for fuller bid data, finance's need for automated reconciliation, a major client's supply path transparency request and legal's privacy concerns, within a mandate to reduce infrastructure cost as a share of revenue?

Data Sources

#	Source and link	How it is used in this project
1	Criteo 1TB Click Logs and Attribution Datasets https://ailab.criteo.com/ressources/	Large-scale real advertising logs used to prototype sampling strategies, storage formats and model training pipelines.
2	Avazu Click-Through Rate Prediction Data https://www.kaggle.com/c/avazu-ctr-prediction	Mobile ad impression and click logs for testing feature pipelines and sampling effects.
3	IAB Tech Lab OpenRTB Specification https://iabtechlab.com/standards/openrtb/	Bid request and response schema that defines log structure and supply chain fields.
4	IAB Tech Lab ads.txt, sellers.json and SupplyChain Object https://iabtechlab.com/ads-txt/	Standards for authorised sellers and supply path tracing used in transparency analysis.
5	ICO Update Report into Adtech and Real Time Bidding https://ico.org.uk/about-the-ico/research-reports-impact-and-evaluation/research-and-reports/technology-and-innovation/update-report-into-adtech-and-real-time-bidding/	Regulatory findings on personal data in RTB informing retention and minimisation.
6	ISBA Programmatic Supply Chain Transparency Study https://www.isba.org.uk/knowledge/programmatic-supply-chain-transparency-study	Industry study on unattributable spend that frames the client's transparency request.

#	Source and link	How it is used in this project
7	Apache Iceberg https://iceberg.apache.org/docs/latest/	Open table format for columnar log storage with schema evolution and partitioning.
8	Privacy Sandbox Documentation https://privacysandbox.google.com	Emerging privacy-preserving advertising APIs that will change bid request data.

Deliverables (17)

- D1 Log data inventory and cost analysis.** Volumes, retention, storage costs and usage for every log type.
- D2 Personal data retention review.** A mapping of personal data fields in logs to purposes, legal bases and retention periods.
- D3 Stratified sampling design.** A sampling strategy retaining rare and valuable segments fully with correct weighting.
- D4 Model impact study.** An experiment quantifying model improvement from fuller data versus smarter sampling.
- D5 Columnar log storage migration.** Migration from text files to a schema-enforced columnar table format with partitioning.
- D6 Parser validation and schema registry.** Detection of exchange format changes and parsing failures before data loss.
- D7 Identifier minimisation layer.** Hashing, truncation or removal of identifiers not needed for each downstream purpose.
- D8 Loss and clearing price dataset.** Extended retention of bid outcomes without unnecessary personal data for bid shading models.
- D9 Automated reconciliation pipeline.** Daily matching of Adwell, exchange and advertiser records with discrepancy categorisation.
- D10 Discrepancy evidence packs.** Structured evidence supporting disputes with exchanges and clients.
- D11 Supply path tracing.** Analysis of supply chain objects and sellers.json to map intermediaries per impression.
- D12 Client transparency report.** A privacy-safe, impression-level or aggregated fee breakdown for the major client.
- D13 Supply path optimisation insights.** Identification of inefficient or unauthorised paths for bidding strategy.
- D14 Streaming pipeline efficiency plan.** Changes reducing cluster cost while maintaining throughput and latency.
- D15 Privacy change readiness assessment.** Analysis of how cookie and identifier deprecation will affect data and models.
- D16 Cost governance dashboard.** Infrastructure cost per billion requests and as a share of revenue.

- D17 Integrated roadmap.** A sequenced plan across modelling, reconciliation, transparency and privacy workstreams.

CASE STUDY 39 · CENTRAL GOVERNMENT

Public Procurement Spend Transparency and Supplier Risk Data Platform

Industry	Central Government
Organisation	The Department for Infrastructure Delivery, a central government department spending £14 billion a year through contracts
Your role	Head of Data Engineering, Commercial Directorate
Core themes	Open contracting data standards, supplier entity resolution, transparency obligations, fraud and conflict detection

The Scenario

The Department for Infrastructure Delivery spends around fourteen billion pounds a year buying goods, works and services: road and rail maintenance, construction projects, consultancy, IT systems and much more. Every contract above certain thresholds must be advertised and awarded under public procurement law, and the department must publish information about contracts and spending so that citizens, journalists and suppliers can see how public money is used. A new procurement regime has recently come into force, replacing the rules that applied for decades. It introduces a central digital platform on which contracting authorities must publish a series of notices across the life of every procurement, from pipeline planning through award to contract performance and termination, and it expects data to be published in a standardised, machine-readable form. You lead data engineering in the department's commercial directorate, and you have inherited the job of making the department compliant, while an uncomfortable story unfolds in the press.

The story began when a data journalist, analysing the department's published spending data, identified a small consultancy that had received contracts worth more than thirty million pounds over four years through a series of direct awards and call-offs from frameworks, each below the threshold that would have required open competition. The journalist also found that one of the consultancy's directors had previously worked in the department. The department's internal audit team is investigating whether rules were broken; so far, there is no evidence of wrongdoing, but the permanent secretary, Dame Caroline Hughes, was embarrassed that the department could not answer the journalist's questions quickly. Nobody

could produce, within a day, a complete list of all contracts and payments to the consultancy across the department's agencies and business units.

The problem is data fragmentation. The department and its four executive agencies use three different finance systems, two e-procurement tools and a contract management system that covers only some contracts. Supplier records are duplicated across systems, with the same company appearing under different names, addresses and supplier numbers. Contracts awarded through frameworks run by other government bodies are recorded inconsistently. Monthly spending over twenty-five thousand pounds is published as spreadsheets, as required by transparency policy, but the spreadsheets are produced manually from each finance system, use different formats and are sometimes late. Contract award notices are published by procurement teams individually, and their data does not reliably link to the spending data.

The commercial director, Andrew Sinclair, wants a single commercial data platform that consolidates procurement, contract and spend data across the department and its agencies, resolves supplier identities, publishes notices to the central platform in the required format, and provides analytics to spot risks. He is particularly interested in using data to detect patterns such as contract splitting to avoid thresholds, repeated direct awards to the same suppliers, concentration of spending, supplier financial distress, and possible conflicts of interest.

The chief digital and information officer, Dr Fatima Rahman, supports the goal but is wary of scope. The department is also in the middle of replacing one of its finance systems, and she does not want a commercial data platform built on data structures that will change within a year. She also points out that central government provides shared services and platforms, and that the department should reuse what exists rather than building bespoke tools.

The head of counter-fraud, Robert Mills, is enthusiastic about analytics for fraud detection but warns about the risks of accusing suppliers or staff based on data patterns. Many patterns that look suspicious, such as frequent awards to one supplier, have innocent explanations: a specialist supplier with unique capabilities, or a framework designed for repeated call-offs. Conflict of interest detection is particularly sensitive: linking suppliers' directors to departmental staff involves processing personal data about employees and company officers, which requires a clear legal

basis and proportionality. The department's data protection officer, Janet Moore, has asked for an impact assessment before any such linkage is attempted.

Supplier entity resolution is the technical crux. To answer the journalist's question, you need to know that "ABC Consulting Ltd", "A.B.C. Consulting Limited" and supplier number 400123 are the same company, and ideally that they are part of the same corporate group as "ABC Advisory Services Ltd", which received further contracts. Company registration numbers would solve much of this, but they are missing from around forty per cent of supplier records. Public registers of companies provide authoritative data on company names, numbers, directors and group structures, but matching to them requires careful probabilistic linkage, and some suppliers are partnerships, sole traders, charities, overseas companies or public bodies not on the companies register.

The central digital platform for the new regime has its own requirements. Notices must be published at defined stages with specified data fields, including unique identifiers for procurements and suppliers. The platform provides interfaces for submitting notices, but the department must supply the data. Some agencies want to use the central platform's own interface manually; others want automated integration from their e-procurement tools. The first performance notices, which require publishing key performance indicators for large contracts, are due within the year.

The agencies themselves present different levels of readiness. Two agencies have mature commercial teams with well-maintained e-procurement data; one has a small commercial function that relies on spreadsheets and email; the fourth has recently merged two predecessor bodies and still runs both of their legacy finance systems. The agency chief executives are accountable for their own spending and are wary of a departmental platform that might be used to scrutinise their decisions. One has already asked whether the department intends to publish league tables of agency procurement practice. Building consolidated data requires their cooperation, their data and, in some cases, changes to their processes, yet you have no authority over them. The permanent secretary's support gives you leverage, but you know that data programmes imposed on reluctant organisations tend to receive compliance with the letter rather than the spirit, producing data that is technically complete and practically unreliable.

You have a team of five engineers and two analysts. The department's cloud platform is available. The permanent secretary wants to be able to answer journalists' questions within hours, not weeks. Andrew wants risk analytics. Fatima wants alignment with the finance system replacement and shared services. Robert wants fraud detection but with safeguards. Janet wants lawful processing. And the new regime's compliance deadlines are fixed.

You suspect that the single most valuable capability, serving all of these goals, is a reliable supplier master with resolved identities linked to authoritative registers, together with a consolidated view of contracts and spend keyed to it. But building it well takes months, the finance system replacement will change supplier records, and there is pressure to deliver visible analytics quickly.

THE DECISION IN FRONT OF YOU

Do you prioritise compliance with the new procurement regime's publication requirements, a consolidated supplier and spend data foundation, or risk and fraud analytics, and how do you design supplier entity resolution and conflict-of-interest analysis so that they are accurate, lawful and fair to suppliers and staff?

Data Sources

#	Source and link	How it is used in this project
1	Find a Tender Service https://www.find-tender.service.gov.uk	UK public procurement notices used to study notice data and published contract awards.
2	Contracts Finder https://www.contractsfinder.service.gov.uk	Published UK public sector contract opportunities and awards for historical analysis.
3	Open Contracting Data Standard https://standard.open-contracting.org	Standard for publishing procurement data across the contracting lifecycle, used as the canonical model.
4	Companies House API https://developer.company-information.service.gov.uk	Authoritative company names, numbers, officers and filings for supplier entity resolution.
5	Companies House Persons with Significant Control Data https://download.companieshouse.gov.uk/en_pscdata.html	Beneficial ownership data used to identify corporate groups and controlling persons.
6	Procurement Act 2023 https://www.legislation.gov.uk/ukpga/2023/54/contents	Legal framework defining notice and transparency obligations.
7	USAspending API https://api.usaspending.gov	Large open federal spending dataset used to prototype spend analytics and anomaly detection at scale.

#	Source and link	How it is used in this project
8	Charity Commission Register https://register-of-charities.charitycommission.gov.uk	Register of charities for resolving non-company suppliers.

Deliverables (17)

- D1 Rapid response capability.** A temporary consolidated query of supplier payments and contracts to answer urgent questions within hours.
- D2 Source system inventory.** A catalogue of finance, e-procurement and contract systems across the department and agencies.
- D3 Canonical commercial data model.** An OCDS-aligned model of procurements, awards, contracts, suppliers and payments.
- D4 Supplier master and entity resolution.** Deduplicated supplier records linked to company, charity and public body registers with confidence scores.
- D5 Corporate group mapping.** Linkage of suppliers to parent companies and group structures.
- D6 Notice publication integration.** Automated submission of required notices to the central digital platform.
- D7 Contract performance KPI publication.** A pipeline producing performance notices for large contracts.
- D8 Spend transparency automation.** Automated, consistent monthly publication of spend over the threshold.
- D9 Contract-to-spend linkage.** Matching of payments to contracts to show spend against awarded values.
- D10 Threshold avoidance detection.** Analytics identifying patterns of low-value awards that may indicate contract splitting.
- D11 Supplier concentration and dependency analysis.** Measurement of spend concentration and reliance on single suppliers.
- D12 Supplier financial risk monitoring.** Integration of filings and credit indicators to flag financial distress.
- D13 Conflict of interest analysis design.** A proportionate, lawful approach for comparing declared interests with supplier officers.
- D14 Data protection impact assessment.** A DPIA covering supplier officer and staff data processing.
- D15 Analytics safeguards and review process.** Human review, context gathering and fair treatment before any risk flag leads to action.
- D16 Finance system transition alignment.** A plan ensuring the platform adapts to the finance system replacement.
- D17 Executive and ministerial briefing dashboard.** Summary views of spend, suppliers and risk for leadership.

CASE STUDY 40 • CHARITY AND NON-PROFIT

Donor Data Platform and Supporter Journey Analytics for a National Charity

Industry	Charity and Non-Profit
Organisation	Brightpath, a UK children's charity raising £96 million a year from 600,000 supporters
Your role	Senior Data Engineer, Supporter Insight
Core themes	Fundraising regulation, consent management, data minimisation, legacy CRM migration

The Scenario

Brightpath is one of the country's best-known children's charities. It runs services for vulnerable children and families, campaigns on child poverty and relies on the generosity of around six hundred thousand supporters who give through regular direct debits, one-off donations, events such as sponsored runs, raffles and lotteries, legacies in wills, and corporate partnerships. Fundraising is the lifeblood of its work, and data is the lifeblood of fundraising. You are a senior data engineer in the supporter insight team, and you have joined at a moment when the charity's approach to supporter data is under strain from every direction.

The supporter database is a customer relationship management system that has been in use for sixteen years. It has been customised so heavily that upgrades are no longer possible, and it holds records not only for supporters but for event participants, campaign signatories, volunteers, service enquirers and anyone who has ever bought something from the charity's online shop. Data arrives from dozens of channels: online donation forms, a telephone fundraising agency, face-to-face fundraising agencies that sign up donors on the street, events platforms, a lottery provider, social media fundraising tools, and third-party giving platforms. Each channel has its own data format and its own approach to capturing consent. The database contains around two and a half million records, of which the insight team estimates about a fifth are duplicates.

Consent is the heart of the problem. Charities in the UK have been under close scrutiny since a series of scandals a decade ago revealed aggressive fundraising practices, including repeated contact with elderly and vulnerable people and the sharing of supporter data between charities. The data protection regulator fined several major charities for practices such as wealth screening, where supporters' data was analysed to identify

wealthy individuals for targeted approaches, without their knowledge. The fundraising regulator introduced a preference service allowing people to stop contact from specific charities. Brightpath was not fined, but its trustees remain acutely conscious of reputational risk. The charity's data protection officer, Anita Shah, has recently found that consent records in the database are inconsistent: some supporters have opted in to email marketing on one form and opted out on another, and the system records only the latest status without history, making it impossible to demonstrate when and how consent was given.

The director of fundraising, Jonathan Reeves, has an ambitious plan. Income from regular giving has plateaued, and the cost of acquiring new regular donors through face-to-face fundraising has doubled in five years as attrition has risen. Jonathan wants a modern supporter data platform that creates a single view of each supporter across all channels, enables personalised supporter journeys, predicts which supporters are likely to cancel their direct debits so the charity can thank and retain them, and identifies supporters with potential to give larger gifts or leave legacies. He has been talking to vendors of charity-specific CRM platforms and marketing automation tools, and he wants a decision on replacing the CRM within three months.

Anita's position is not opposed to modernisation, but she insists that the charity must be able to demonstrate lawful processing for everything it does. Predictive modelling of attrition may be acceptable under legitimate interests if supporters are told about it, but modelling legacy potential or wealth is precisely what attracted regulatory fines in the past, and she wants a formal assessment and clear transparency before anything is built. She also points out that the charity holds sensitive data: some supporters have told the charity about personal experiences, such as their own childhood abuse or a child's illness, when explaining why they support the cause, and this information is scattered in free-text notes fields. It is special category data and should probably not be there at all.

The head of supporter care, Linda Moore, has a practical concern. Her team handles thousands of supporter calls and emails each month, including complaints about unwanted contact, requests to update details and subject access requests under data protection law. A subject access request currently takes her team several days, because they must search the CRM, the email marketing platform, the events platform and the

agencies' records. Complaints about contact after supporters have opted out are rising, and she has traced several to agencies uploading old data that overwrote updated preferences.

The finance team has its own requirement: accurate reconciliation of income across channels, particularly Gift Aid, the tax relief scheme that allows charities to reclaim basic rate tax on donations from UK taxpayers. Gift Aid claims depend on having valid declarations from donors, and the tax authority audits them. The charity's Gift Aid declarations are held in the CRM but have not been properly linked to donations from some channels, and the finance director estimates that the charity is failing to claim several hundred thousand pounds a year it is entitled to, while also possibly claiming on some donations where declarations are invalid.

The charity's culture adds a further layer. Brightpath's staff are deeply committed to its mission, and many in the services directorate are uncomfortable with commercial marketing techniques being applied to supporters. The director of children's services, Dr Grace Oduya, has argued at executive meetings that the charity should never treat supporters as customers to be optimised, and that its fundraising should reflect the same values of respect and safeguarding it applies to children and families. Jonathan, for his part, points out that every pound not raised is a pound not spent on services, and that supporters expect a modern, responsive experience from the charities they support. Both perspectives are sincerely held, and the trustees will want any data strategy to show how the charity's values are reflected in how it uses supporters' data, not merely that it complies with the law.

You have a team of two engineers, yourself included, and one analyst. The chief operating officer has approved a budget for a CRM replacement programme but has asked for a data strategy first. You realise that the questions of which CRM to buy and which analytics to build are secondary to more fundamental issues: consent history, deduplication, data minimisation and agency data quality. Any new CRM migrated from dirty, inconsistent data will inherit the same problems. But Jonathan's income targets are pressing, and the trustees want to see growth.

You also recognise that the charity's supporters are its greatest asset and that their trust is fragile. A data-driven fundraising programme that feels intrusive could damage Brightpath's reputation far more than any short-term income gain would justify. Yet a charity that cannot use data well will

lose supporters to rivals that do, and ultimately fund fewer services for children.

THE DECISION IN FRONT OF YOU

Do you prioritise CRM replacement, a data foundation of consent history, deduplication and minimisation, or predictive fundraising analytics, and where do you draw the line on modelling supporters' wealth and legacy potential given the sector's regulatory history?

Data Sources

#	Source and link	How it is used in this project
1	Charity Commission Register of Charities https://register-of-charities.charitycommission.gov.uk	Official charity data for benchmarking income sources and sector context.
2	360Giving GrantNav https://grantnav.threesixtygiving.org	Open grants data used to analyse institutional funding and corporate giving patterns.
3	Fundraising Regulator Code of Fundraising Practice https://www.fundraisingregulator.org.uk/code	Standards governing fundraising conduct and use of supporter data.
4	Fundraising Preference Service https://www.fundraisingregulator.org.uk/fps	Suppression requests that must be ingested and honoured.
5	ICO Guidance for the Charity Sector https://ico.org.uk/for-organisations/charity/	Data protection guidance on consent, legitimate interests and wealth screening.
6	HMRC Gift Aid Guidance https://www.gov.uk/claim-gift-aid	Rules on declarations and claims that define Gift Aid data requirements.
7	Royal Mail Postcode Address File (PAF) https://www.poweredbypaf.com	Address validation reference used in deduplication and data quality.
8	Telephone Preference Service https://www.tpsonline.org.uk	Suppression list for telephone fundraising compliance.

Deliverables (17)

- D1 Supporter data strategy.** A strategy setting priorities across consent, quality, CRM replacement and analytics.
- D2 Data landscape assessment.** An inventory of sources, agencies, record volumes, duplication and data flows.
- D3 Consent history model.** A design recording every consent event with channel, wording, date and source.
- D4 Consent reconstruction.** Back-filling of consent history from available records with confidence levels.

- D5 Deduplication and golden record.** Matching and merging of duplicate supporter records with survivorship rules.
- D6 Agency data contract.** Standard formats, validation and preference-protection rules for agency uploads.
- D7 Preference protection controls.** Prevention of opted-out preferences being overwritten by stale uploads.
- D8 Special category data remediation.** Discovery and removal or restriction of sensitive information in free-text notes.
- D9 Data minimisation and retention schedule.** Retention periods by record type with automated deletion of lapsed contacts.
- D10 Subject access request automation.** A consolidated search across systems to fulfil requests within statutory time limits.
- D11 Suppression list integration.** Automated application of fundraising, telephone and mail preference services.
- D12 Gift Aid reconciliation.** Linking of declarations to donations across channels with identification of missed and invalid claims.
- D13 Attrition prediction with transparency.** A regular-giving attrition model with privacy notice updates and legitimate interests assessment.
- D14 Wealth and legacy modelling position paper.** A documented decision on whether and how such modelling may be done.
- D15 CRM replacement data requirements.** Data model and migration requirements for vendor selection.
- D16 Migration readiness plan.** Cleansing, mapping and testing steps before any CRM migration.
- D17 Trustee governance report.** A report giving trustees assurance on supporter data practices and risks.

CASE STUDY 41 · BIOTECHNOLOGY (CLINICAL RESEARCH)

Clinical Trial Data Standardisation and Submission Pipeline

Industry	Biotechnology (Clinical Research)
Organisation	Novaris Therapeutics, a biotech company running three Phase II and one Phase III trial in rare diseases
Your role	Lead Clinical Data Engineer
Core themes	CDISC standards, regulated data transformations, multi-vendor integration, submission deadlines

The Scenario

Novaris Therapeutics is a biotechnology company developing treatments for rare genetic diseases. It has no marketed products yet; its value depends entirely on its clinical trials. Four trials are currently running: three Phase II trials testing whether its lead compounds are safe and show signs of efficacy, and one pivotal Phase III trial of its most advanced compound, which, if successful, will form the basis of applications for marketing authorisation to the US Food and Drug Administration, the European Medicines Agency and the UK's Medicines and Healthcare products Regulatory Agency. The Phase III trial's database lock, the point at which the data is frozen for final analysis, is scheduled for eight months from now. You are the lead clinical data engineer, and you have just discovered that the data pipeline for that trial is not in a state that will survive a regulatory submission.

Regulators require clinical trial data to be submitted in standardised formats defined by the Clinical Data Interchange Standards Consortium. Raw data collected during the trial must be transformed into the Study Data Tabulation Model, which organises data into standard domains such as demographics, adverse events, laboratory results, vital signs and exposure to study drug. From those tabulations, analysis datasets are derived in the Analysis Data Model, which supports the statistical analyses in the clinical study report. Every transformation must be traceable, reproducible and documented, and the submission must include machine-readable metadata describing every dataset and variable. The FDA's technical conformance guide and validation rules check submissions automatically, and non-conformant submissions can be refused.

Novartis outsources much of its clinical operations to contract research organisations. The Phase III trial's electronic data capture system, where investigators at sixty hospital sites across twelve countries enter data, is managed by one contract research organisation. Central laboratory results come from a specialist laboratory vendor. Pharmacokinetic samples are analysed by a bioanalytical laboratory. Patient-reported outcomes are collected on tablets through an electronic clinical outcome assessment vendor. Imaging is read centrally by an imaging core laboratory. Genetic testing results, essential for confirming eligibility in a rare genetic disease, come from yet another laboratory. Randomisation and drug supply are managed through an interactive response technology vendor. Each vendor delivers data in its own format, on its own schedule, under data transfer agreements that were negotiated at different times by different people.

Until recently, the contract research organisation was responsible for producing the standardised datasets. Six months ago, Novartis decided to bring this work in-house, to gain control and reduce costs, and hired you to lead it. What you have found since is sobering. The contract research organisation's transformation programs were written in a statistical programming language, poorly documented and dependent on manual steps. Several vendor data transfers have changed format during the trial without formal notification. The laboratory vendor's reference ranges were updated mid-trial, and it is unclear which ranges apply to which results. Adverse event coding uses one version of the medical dictionary for most of the trial but a newer version for recent entries. Subject identifiers differ between vendors: the imaging laboratory uses screening numbers, while other vendors use randomisation numbers, and the mapping between them is held in a spreadsheet.

The head of biostatistics, Dr Laura Chen, is anxious. Her team needs clean analysis datasets to finalise the statistical analysis plan's programming and to run dry runs of the final analysis well before database lock. Every week of delay compresses the timeline for the clinical study report and the submission. She would prefer to pay the contract research organisation to finish what it started, even though it is expensive and the quality is questionable, because switching midstream creates risk.

The vice president of clinical operations, Mark Sullivan, made the decision to bring the work in-house and is reluctant to reverse it. He points out that Novartis will run many more trials if this one succeeds, and that building an

in-house capability now will pay off. He wants a modern, automated pipeline that can be reused across trials, with standardised mappings, automated conformance checks and version-controlled code.

The quality assurance head, Dr Priya Nandakumar, reminds everyone that the pipeline is a regulated computerised system. Under good clinical practice and regulatory expectations on data integrity, every change must be controlled, every program validated and every dataset traceable back to the source. An inspection by a regulatory authority before approval could examine exactly how the submission datasets were produced. She is uneasy about building a new system under deadline pressure, because validation shortcuts are exactly what inspectors find.

Your team consists of two clinical programmers and one data engineer. The company has a cloud environment qualified for regulated work. Several open-source and commercial tools exist for standardising clinical data, generating metadata and running conformance checks, and the industry has been moving towards open-source statistical programming for submissions, with regulators accepting submissions produced using modern tools if properly validated.

There is also a scientific urgency that the data team cannot ignore. Because the trial is in a rare disease, every patient's data is precious; the trial has enrolled a little over two hundred patients worldwide, and losing or mishandling even a few records could affect the statistical power of the final analysis. Several patients have withdrawn and some have transferred between sites after moving home, which has created duplicate or fragmented records across vendors. The independent data monitoring committee, which reviews unblinded safety data periodically, has already had to delay one of its meetings because the safety datasets it received contained inconsistencies between adverse events and laboratory results. The committee's chair wrote to Novartis expressing concern, and the letter has been circulated to the board. The pressure on data quality is therefore not merely procedural; it bears directly on patient safety oversight and on whether the trial can answer its scientific question.

You face a set of uncomfortable choices. Completing the Phase III pipeline in-house with a modern, reusable design is attractive but risky under the deadline. Handing it back to the contract research organisation is safer in the short term but abandons the strategy and pays for work of questionable quality. A hybrid, in which the contract research organisation

produces the Phase III datasets under closer oversight while you build the reusable pipeline on the Phase II trials, might balance risk and strategy, but it doubles effort and management overhead.

You also know that whatever path you choose, the vendor data problems must be fixed: identifier mapping, reference ranges, dictionary versions and undocumented format changes will cause conformance failures regardless of who writes the transformation programs.

THE DECISION IN FRONT OF YOU

Do you complete the Phase III submission pipeline in-house on a new reusable platform, return the work to the contract research organisation under tighter oversight, or split the work between Phase III and Phase II trials, and how do you resolve vendor data defects and meet validation expectations before database lock?

Data Sources

#	Source and link	How it is used in this project
1	CDISC SDTM and ADaM Standards https://www.cdisc.org/standards	The standard tabulation and analysis data models required for regulatory submission.
2	FDA Study Data Technical Conformance Guide https://www.fda.gov/regulatory-information/search-fda-guidance-documents/study-data-technical-conformance-guide-technical-specifications-document	Technical requirements for submitted study data and metadata.
3	CDISC Pilot Study Data (PHUSE) https://github.com/phuse-org/phuse-scripts	Public example SDTM and ADaM data and scripts for building and testing transformation pipelines.
4	ClinicalTrials.gov API https://clinicaltrials.gov/data-api/api	Public registry data for trial design references and results reporting obligations.
5	pharmaverse (admiral and sdtm.oak) https://pharmaverse.org	Open-source R packages for producing SDTM and ADaM datasets in regulated settings.
6	Pinnacle 21 Community Validator https://www.certara.com/software/pinnacle-21-community/	Conformance checking tool for SDTM and ADaM datasets against regulatory rules.
7	MedDRA https://www.meddra.org	Medical dictionary for adverse event coding with version management.
8	ICH E6 Good Clinical Practice https://www.ich.org/page/efficacy-guidelines	Regulatory expectations for data integrity and computerised systems in trials.

Deliverables (17)

- D1 Pipeline state assessment.** An evaluation of inherited programs, manual steps, documentation gaps and data defects.
- D2 Options and risk appraisal.** A comparison of in-house, CRO and split approaches against the database lock timeline.
- D3 Vendor data transfer specifications.** Updated, agreed specifications for every vendor with change notification obligations.
- D4 Subject identifier mapping service.** A controlled, validated mapping of screening, randomisation and vendor identifiers.
- D5 Laboratory reference range reconciliation.** Determination of applicable ranges per result with documentation of changes.
- D6 Dictionary version harmonisation.** Recoding of adverse events and medications to a single agreed dictionary version.
- D7 SDTM mapping specifications.** Variable-level mapping documents for every domain.
- D8 SDTM transformation programs.** Version-controlled, tested programs producing tabulation datasets.
- D9 ADaM derivation programs.** Analysis dataset programs aligned with the statistical analysis plan.
- D10 Define-XML metadata.** Machine-readable metadata for all submission datasets.
- D11 Automated conformance checks.** Integration of regulatory validation rules into every pipeline run.
- D12 Traceability documentation.** Source-to-submission traceability for each variable.
- D13 Computerised system validation package.** Validation plan, risk assessment, test evidence and change control.
- D14 Dry run schedule and results.** Iterative dry runs of datasets and analyses before database lock.
- D15 Reviewer's guides.** Study data reviewer's guides explaining data and conformance issues.
- D16 Reusable pipeline framework.** A standardised, parameterised framework for future trials.
- D17 Inspection readiness plan.** Preparation for regulatory inspection of data handling and submission production.

CASE STUDY 42 • GENOMICS AND DIAGNOSTICS

Genomic Variant Processing Pipeline for a Clinical Diagnostics Laboratory

Industry	Genomics and Diagnostics
Organisation	Helix Clinical Genomics, an accredited laboratory providing rare disease and cancer genomic testing to NHS and private clients
Your role	Principal Bioinformatics Data Engineer
Core themes	Petabyte-scale sequencing data, reproducible pipelines, clinical accreditation, variant knowledge management

The Scenario

Helix Clinical Genomics sequences DNA from patients with suspected rare genetic diseases and from tumour samples of patients with cancer, and reports clinically significant genetic variants back to the doctors treating them. A report from Helix might confirm a diagnosis that a family has been seeking for years, identify a variant that makes a tumour susceptible to a particular targeted therapy, or reveal an inherited cancer risk that affects a patient's relatives. The stakes of getting it wrong are obvious. The laboratory is accredited to the international standard for medical laboratories, inspected regularly, and processes around fifteen thousand samples a year, a number that has tripled in three years as genomic testing has moved into routine care. You are the principal bioinformatics data engineer, and the pipeline that turns raw sequencing data into reportable variants is straining under that growth.

The process begins in the wet laboratory, where DNA is extracted, prepared into libraries and loaded onto sequencing instruments. Each whole-genome sequencing run produces hundreds of gigabytes of raw data; exome and targeted panel runs produce less. The bioinformatics pipeline aligns sequence reads to the human reference genome, calls variants, annotates them with information about genes, predicted effects, population frequencies and known clinical significance, and filters them according to the clinical question. Clinical scientists then review the filtered variants, interpret them using established guidelines for variant classification, and write reports. Every step must be reproducible: if a patient's result is questioned years later, the laboratory must be able to show exactly which software versions, reference data and parameters produced it.

The current pipeline runs on an on-premises high-performance computing cluster purchased five years ago. It was sized for a third of today's volume. Samples now queue for days waiting for compute, and turnaround times for urgent cases, such as critically ill newborns in intensive care, where rapid genomic diagnosis can change treatment within days, have been missed several times. Storage is nearly full: the laboratory retains raw data and aligned reads for every sample, as required by professional guidelines and by contracts, and the total now exceeds three petabytes. The cluster's support contract ends in a year.

The laboratory director, Dr Sarah Whitmore, wants to move the pipeline to a cloud platform. She has seen other genomics laboratories do so and believes the elasticity of cloud computing would solve the queueing problem, allowing urgent samples to be processed immediately. She has also been approached by a cloud provider offering credits and a genomics-specific managed service. The finance director, Paul Hendry, is concerned about cost: cloud storage for petabytes of data, retained for years, and data egress charges if data ever needs to be moved, could be expensive. He wants a detailed cost model, not a vendor's estimate.

The quality manager, Dr Chioma Eze, raises the accreditation issue. The pipeline is a validated clinical procedure. Any change to its components, whether software versions, reference genomes, annotation databases or computing environment, requires revalidation, demonstrating that the changed pipeline produces results at least as accurate as the old one, using reference samples with known variants. Moving to a new computing platform, even with identical software, would require revalidation, which takes months and consumes scientist time. Chioma also notes that the laboratory processes NHS patient data under specific data protection and information governance requirements, and that any cloud hosting must meet NHS data security standards, with data stored in approved locations.

The head of clinical science, Dr Imran Qureshi, has a different priority. He is less concerned with where the pipeline runs than with the knowledge base his scientists rely on. When a clinical scientist classifies a variant as pathogenic, likely pathogenic, uncertain or benign, that classification is recorded in a laboratory database. But classifications change as evidence accumulates: a variant of uncertain significance today may be reclassified as pathogenic next year when more families are found with it. The laboratory has thousands of previously reported variants of uncertain

significance, and professional guidelines increasingly expect laboratories to reanalyse and recontact patients when classifications change. The current database was built in-house years ago, has no structured evidence capture, and cannot easily identify which previously reported patients are affected by a reclassification. Imran wants a variant knowledge management system integrated with the pipeline, with links to public variant databases and structured recording of evidence.

The research team adds another demand. With patients' consent, the laboratory contributes anonymised data to research programmes and national databases, and several academic collaborators want access to aggregated variant frequency data from the laboratory's population. Data sharing requires careful de-identification: genomic data is inherently identifying, and even aggregated frequencies of rare variants can reveal individuals in a small population.

Your bioinformatics team consists of four engineers and three bioinformaticians. The pipeline is a collection of shell scripts and a workflow manager configured years ago, with some components pinned to software versions that are no longer maintained. Modern workflow languages and community-maintained pipelines now exist, designed for portability between on-premises and cloud environments and with built-in reproducibility features.

You see several options. You could modernise the pipeline using a portable workflow framework, containerise every component and validate it once, running on-premises now and in the cloud later, or both at once in a hybrid arrangement where urgent samples burst to the cloud. You could buy the managed genomics service and validate it. You could replace the cluster with a larger on-premises cluster, the least disruptive option but one that postpones the problem. Each has different implications for cost, turnaround, validation effort and data governance. Meanwhile, the variant knowledge problem is arguably more important clinically, but it competes for the same people.

The laboratory's relationship with the NHS genomic medicine service adds a further consideration. Helix performs some tests under contract to a regional genomic laboratory hub, which expects results in specified formats and within specified turnaround times, and which is itself moving towards national standards for genomic data exchange and a national genomic data platform. There is a possibility that in the next few years,

some of the pipeline work Helix currently does in-house will be expected to run on shared national infrastructure, or at least to produce outputs compatible with it. Investing heavily in a bespoke cloud architecture that later has to be adapted to national systems would be wasteful; ignoring the direction of travel would leave Helix exposed when contracts are renewed. Sarah has asked for any design to be portable and standards-based, but the standards themselves are still evolving.

There is also the storage question, which may dominate costs whatever you choose. Retaining raw data indefinitely for every sample is expensive. Guidelines specify minimum retention periods, but the laboratory has kept everything because nobody decided otherwise. Compressed formats could reduce storage significantly, and tiered storage could move old data to cheaper archives, but any data transformation must itself be validated to ensure nothing clinically relevant is lost.

THE DECISION IN FRONT OF YOU

Do you move the pipeline to the cloud, modernise it for portability and hybrid operation, or expand on-premises capacity, and how do you balance revalidation effort, storage costs and NHS data requirements against the clinically urgent need for variant reclassification and recontact?

Data Sources

#	Source and link	How it is used in this project
1	Genome in a Bottle Reference Samples (NIST) https://www.nist.gov/programs-projects/genome-bottle	Benchmark genomes with high-confidence variant calls used to validate pipeline accuracy.
2	1000 Genomes Project Data https://www.internationalgenome.org	Public whole-genome data for testing pipeline throughput and scaling.
3	gnomAD https://gnomad.broadinstitute.org	Population allele frequencies used for variant annotation and filtering.
4	ClinVar https://www.ncbi.nlm.nih.gov/clinvar/	Public archive of variant clinical significance used in the knowledge base and reclassification monitoring.
5	nf-core Community Pipelines https://nf-co.re	Community-maintained Nextflow pipelines such as sarek for portable, reproducible variant calling.
6	GA4GH Standards https://www.ga4gh.org	Global genomics data standards for file formats, workflow execution and data sharing.

#	Source and link	How it is used in this project
7	ACGS Best Practice Guidelines for Variant Classification https://www.acgs.uk.com/quality/best-practice-guidelines/	UK guidelines for variant interpretation and reclassification.
8	NHS Data Security and Protection Toolkit https://www.dsptoolkit.nhs.uk	Security standards required for processing NHS patient data, including hosting.

Deliverables (17)

- D1 Capacity and turnaround analysis.** Current queue times, compute utilisation and projected demand by test type.
- D2 Platform options appraisal.** A comparison of cloud, hybrid and on-premises options on cost, turnaround, validation and governance.
- D3 Five-year cost model.** Compute, storage, egress, staffing and validation costs for each option.
- D4 Portable workflow implementation.** Re-implementation of the pipeline in a portable workflow framework with containers.
- D5 Reproducibility controls.** Pinned software, reference and database versions recorded for every sample.
- D6 Validation plan and evidence.** Revalidation using reference samples with accuracy metrics compared to the current pipeline.
- D7 Urgent sample fast path.** A prioritised processing route meeting rapid turnaround targets.
- D8 Storage tiering and compression.** A validated scheme for compressing and tiering data by age and retention requirement.
- D9 Retention policy.** Documented retention periods by data type aligned with guidelines and contracts.
- D10 Information governance assessment.** Evidence that hosting meets NHS data security and residency requirements.
- D11 Variant knowledge base.** A structured store of classifications, evidence and history per variant.
- D12 Reclassification monitoring.** Automated detection of classification changes in public databases affecting reported variants.
- D13 Patient recontact workflow data.** Identification of previously reported patients affected by reclassification.
- D14 Research data sharing framework.** De-identification and aggregation rules for sharing variant frequencies.
- D15 Pipeline monitoring and alerting.** Tracking of run status, failures, turnaround and resource usage.

- D16 Accreditation documentation updates.** Updated procedures and records for the accreditation body.
- D17 Migration and cutover plan.** A plan transitioning to the new platform without interrupting clinical service.

CASE STUDY 43 • BANKING (CLIMATE RISK)

Physical Climate Risk Data Platform for a Commercial Lending Portfolio

Industry	Banking (Climate Risk)
Organisation	Thamesmead Commercial Bank, lending £22 billion to businesses, real estate and agriculture in the UK and Europe
Your role	Lead Data Engineer, Climate and Sustainability Risk
Core themes	Geospatial hazard data, scenario modelling, asset location data quality, regulatory disclosure

The Scenario

Thamesmead Commercial Bank lends to businesses: property developers, manufacturers, retailers, farmers, logistics companies and many more. Its twenty-two billion pound loan book is secured on factories, warehouses, offices, farmland, shops and homes built for sale. Every one of those assets sits somewhere on a map, and the map is changing. Rivers flood more often, coastal areas face rising seas and storm surges, summers bring heatwaves and droughts that damage crops and buildings, and subsidence from shrinking clay soils cracks foundations. Supervisors now expect banks to understand how physical climate risks could affect the value of collateral and the ability of borrowers to repay, and to report on it. You lead data engineering for climate and sustainability risk, and the bank's first serious attempt to quantify its physical risk exposure has hit a wall of data problems.

The Prudential Regulation Authority expects banks to integrate climate risk into their risk management and to conduct scenario analysis. Last year, the bank participated in an exploratory exercise in which it had to estimate how its loan book would perform under different climate scenarios over several decades. The results were produced by consultants using a vendor's climate risk model, and they were, in the words of the chief risk officer, Dr Eleanor Bright, "a black box built on sand". The model produced numbers, but nobody inside the bank could explain them, and when the risk team examined the inputs, they found that only about half of the bank's collateral had usable location data. The rest had been assigned to the centroid of a postcode, a local authority or, in some cases, the whole of the United Kingdom.

Location data is the foundation of physical risk analysis, and it is weak. Collateral records in the bank's lending systems contain addresses, but these were captured for legal documentation, not geospatial analysis. Commercial properties often have multiple buildings, and the address may refer to a head office rather than the site. Agricultural loans are secured on land described by title deeds that cover many fields. Development finance is secured on sites that did not have postal addresses when the loan was made. Some loans are secured on portfolios of dozens or hundreds of properties, with the individual properties listed in schedules attached to loan agreements as scanned documents. And unsecured corporate loans have no collateral, but the borrower's operations, such as a factory or a distribution centre, are exposed to physical risks that affect its ability to repay.

Hazard data is the second foundation. Public sources provide flood maps for rivers, seas and surface water, projections of sea level rise, historical weather data, climate model projections under different emissions pathways, and datasets on subsidence risk. Commercial vendors provide more granular, forward-looking hazard models, often with property-level risk scores. The public data is free but fragmented, varies in resolution and methodology between England, Scotland, Wales and European countries, and generally describes current rather than future risk. Vendor data is convenient and forward-looking but expensive and opaque, and different vendors produce surprisingly different scores for the same property.

The chief risk officer wants a transparent, in-house physical risk capability that the bank understands and can defend to the regulator. The head of credit risk, Martin Doyle, is more pragmatic. He wants climate risk incorporated into credit decisions for new lending, so that a relationship manager approving a loan secured on a warehouse in a floodplain knows it. He does not need a perfect model; he needs a reasonable, consistent flag at origination. The head of sustainability, Aisha Rahman, has another requirement: the bank must disclose climate-related risks in its annual report under the disclosure framework that has become mandatory for large UK companies, and it has made public commitments on sustainable finance. She wants disclosures that are credible and consistent year to year, and she is worried that changing methodologies will produce numbers that jump around.

The regulator's expectations are evolving and somewhat ambiguous. It expects proportionate, improving capabilities rather than perfection, and it has acknowledged that data gaps are an industry-wide problem. But it also expects banks to be honest about limitations and to have credible plans to close gaps. Eleanor wants the bank to present a data improvement plan at the next supervisory meeting in six months.

The data engineering challenges are substantial. Geocoding collateral requires cleaning addresses, matching them to authoritative address and property datasets, and in many cases extracting site information from scanned documents. Assigning hazards requires spatial joins between property locations and hazard layers, which differ in format, resolution and projection. Scenario analysis requires combining hazard projections under different climate pathways with damage functions that translate hazard intensity into financial impact, and then with credit models that translate impact into default probability and loss. Each step involves assumptions that must be documented. And the whole thing must be reproducible, because the bank will run it repeatedly as data and methods improve, and will need to explain changes.

Your team consists of three data engineers and a geospatial analyst. The bank's data platform supports spatial data, but nobody has used that capability at scale. The lending systems are old, and adding new fields, such as validated coordinates, requires change requests that take months.

You see a tension between speed and foundations. The credit team wants a simple flag at origination soon; the risk team wants a transparent model; sustainability wants stable disclosures; the regulator wants a credible plan. The fastest route to all of them is to buy a vendor's property-level risk scores, but that repeats the black box problem. Building everything in-house would take years. And none of it works without better location data, which requires changes to lending processes that relationship managers will resist as extra paperwork.

Internally, climate risk sits awkwardly across organisational boundaries. The sustainability team reports to the chief executive and owns the public commitments; the risk team reports to the chief risk officer and owns the regulatory relationship; credit sits within the business lines; and data sits within the chief operating officer's organisation. Each has commissioned its own climate data work at different times: sustainability bought one vendor's scores for its disclosure, risk used another vendor through the

consultants, and a business line has been piloting a third vendor's tool for agricultural lending. The same loan can therefore have three different climate risk ratings, and nobody has reconciled them. Eleanor wants one authoritative capability, but each team is attached to its own vendor and its own numbers, and changing any of them will change figures that have already been published or shared with the regulator.

There is also a question of where to start geographically and by portfolio. Real estate lending is the largest and most exposed portfolio, with generally better address data. Agriculture is highly exposed to drought and flooding but has poor location data. Corporate lending is the hardest because exposure depends on operations rather than collateral. Starting with real estate would produce credible results quickest but would leave the agricultural and corporate books unexamined.

THE DECISION IN FRONT OF YOU

Do you buy vendor property-level climate risk scores, build a transparent in-house hazard and scenario platform on public data, or combine them, and how do you fix collateral location data, prioritise portfolios and keep disclosures stable while methods improve?

Data Sources

#	Source and link	How it is used in this project
1	Copernicus Climate Data Store https://cds.climate.copernicus.eu	Climate projections and reanalysis data used for forward-looking hazard indicators under different scenarios.
2	NGFS Scenarios Portal https://www.ngfs.net/ngfs-scenarios-portal/	Central bank climate scenarios used as the basis for scenario analysis.
3	Environment Agency Risk of Flooding from Rivers and Sea https://www.data.gov.uk/dataset/bad20199-6d39-4aad-8564-26a46778fd94/risk-of-flooding-from-rivers-and-sea	Flood risk data for England used in spatial joins with collateral.
4	UK Climate Projections (UKCP18) https://www.metoffice.gov.uk/research/approach/collaboration/ukcp	UK-specific climate projections including sea level rise and temperature.
5	British Geological Survey GeoSure (shrink-swell) https://www.bgs.ac.uk/datasets/geosure-shrink-swell/	Subsidence hazard data for clay soils affecting property risk.
6	Ordnance Survey Open UPRN https://osdatahub.os.uk/downloads/open/OpenUPRN	Open property references and coordinates for geocoding collateral.

#	Source and link	How it is used in this project
7	HM Land Registry INSPIRE Index Polygons https://use-land-property-data.service.gov.uk/datasets/inspire	Freehold title boundaries used to locate land and site-based collateral.
8	PRA Climate Change Adaptation Report and Expectations https://www.bankofengland.co.uk/prudential-regulation/publication/2023/report-on-climate-related-risks-and-the-regulatory-capital-frameworks	Supervisory expectations informing the capability and data plan.

Deliverables (17)

- D1 Collateral location data assessment.** Quantification of location quality by portfolio, collateral type and value.
- D2 Geocoding and address matching pipeline.** Matching of collateral addresses to property references and coordinates with confidence levels.
- D3 Title and site extraction.** Extraction of site locations from title data and scanned schedules for land and portfolio collateral.
- D4 Location quality tiers.** A classification of each exposure's location precision used to caveat results.
- D5 Hazard data catalogue.** A documented inventory of public and vendor hazard layers with resolution, method and coverage.
- D6 Spatial hazard join service.** Reproducible joins of exposure locations to hazard layers with version control.
- D7 Vendor score comparison.** A comparison of vendor property-level scores with in-house indicators to understand divergence.
- D8 Damage function library.** Documented functions translating hazard intensity into asset value or cash flow impact.
- D9 Scenario analysis pipeline.** A reproducible end-to-end pipeline from scenarios to credit impact.
- D10 Origination risk flag.** A physical risk indicator available to relationship managers at loan approval.
- D11 Lending process data capture change.** Requirements for capturing validated coordinates and site details at origination.
- D12 Portfolio prioritisation plan.** A sequence for extending analysis from real estate to agriculture and corporate lending.
- D13 Methodology documentation.** Full documentation of assumptions, limitations and changes for regulators and auditors.
- D14 Disclosure data pack.** Stable, reconciled metrics for annual climate-related disclosures with change explanations.
- D15 Data improvement plan for the regulator.** A credible, time-bound plan to close location and hazard data gaps.

- D16 Governance and model risk review.** Submission of the climate risk model to model risk governance.
- D17 Results dashboard.** Exposure to hazards by portfolio, region, scenario and horizon.

CASE STUDY 44 · CONSUMER GOODS (SUSTAINABILITY REPORTING)

Scope 3 Supply Chain Emissions Data Platform for a Consumer Goods Company

Industry	Consumer Goods (Sustainability Reporting)
Organisation	Pemberton Brands, a UK-headquartered household and personal care products company with 1,400 suppliers
Your role	Senior Data Engineer, Sustainability Data
Core themes	Emissions accounting data, supplier data collection, estimation versus measurement, assurance readiness

The Scenario

Pemberton Brands makes shampoo, washing-up liquid, laundry detergent, deodorant and dozens of other everyday products sold in supermarkets across Europe. Like many consumer goods companies, it has committed publicly to reaching net zero emissions, with an interim target to reduce emissions across its value chain by forty-two per cent within the decade. Its own factories, offices and vehicles, the direct emissions known as scope 1 and the purchased electricity known as scope 2, account for less than five per cent of its total footprint. The rest, over ninety-five per cent, is scope 3: emissions from the raw materials it buys, the packaging, the transport, the energy consumers use when they wash clothes in warm water with its detergent, and the disposal of its products and packaging. You are a senior data engineer in the sustainability data team, and the scope 3 numbers are the problem.

Pemberton's current scope 3 footprint is calculated once a year by a consultancy, mostly using spend-based methods: the amount spent with each category of supplier is multiplied by an average emissions factor for that sector. This approach is quick but crude. It cannot reflect a supplier's actual performance, so if a key supplier of surfactants switches to renewable electricity and cuts its emissions, Pemberton's footprint does not change. Worse, spend-based estimates move with prices: when raw material prices rose sharply, Pemberton's calculated emissions rose with them, even though it bought less material. The chief sustainability officer, Dr Helen Carter, has told the board that progress towards the reduction target cannot be demonstrated credibly with spend-based numbers, and the company's climate target validation body expects better data over time.

Regulatory pressure is increasing too. European sustainability reporting rules will require Pemberton's European subsidiaries to report detailed emissions data, subject to external assurance. UK sustainability disclosure standards are being introduced. Investors ask detailed questions through disclosure platforms. Retail customers, including the largest supermarket chains, have their own supplier emissions programmes and are asking Pemberton for product-level carbon footprints. Assurance is the key word: auditors will need to see how numbers were calculated, from which data, with which factors, and with what controls.

Helen's plan is to move from spend-based to activity-based and supplier-specific data. Activity-based means using physical quantities, such as tonnes of each raw material and packaging type, multiplied by emission factors for those materials. Supplier-specific means obtaining actual emissions data from suppliers for the products they sell to Pemberton. The procurement system holds purchase orders, but in monetary terms and often with vague descriptions; physical quantities are recorded in the enterprise resource planning system for raw materials used in factories, but not for indirect purchases such as marketing services or IT. Bills of materials describe which raw materials and packaging go into each product, but they are maintained by research and development teams in a product lifecycle management system that does not talk to procurement.

Supplier data is the hardest part. Pemberton has fourteen hundred suppliers, of which around two hundred account for eighty per cent of spend on materials and packaging. Some large chemical suppliers already calculate product carbon footprints and are willing to share them. Many smaller suppliers have never calculated their emissions and do not know how. Suppliers receive requests from dozens of customers, each asking for slightly different data in slightly different formats, and they are suffering from survey fatigue. Several industry initiatives aim to standardise how suppliers share product carbon footprints, including a data exchange specification and platforms that allow suppliers to publish data once for many customers.

The procurement director, Graham Walsh, is wary. His team is already stretched, and he does not want sustainability data requests to damage supplier relationships or to become a burden that increases prices. He suggests focusing on a small number of strategic suppliers first. The head of research and development, Dr Priya Menon, sees an opportunity: better

emissions data at material level would help her teams design lower-carbon products, for example by choosing surfactants with lower footprints or reducing packaging weight. She wants data at a level of granularity that informs design decisions, not just annual totals. The finance director, David Burke, is focused on assurance: he wants sustainability data held to the same standard of controls as financial data, with reconciliation, audit trails and sign-off, because errors in reported emissions could expose the company to accusations of greenwashing.

The consumer use phase adds a complication. A large share of Pemberton's footprint comes from the energy used to heat water when consumers use its products, especially laundry detergents. These emissions depend on consumer behaviour, washing temperatures and national electricity grid carbon intensity, and are estimated using models and survey data. They are highly uncertain but material. Some stakeholders argue they should be reported but excluded from targets; others argue that encouraging cold washing is one of the company's biggest levers.

There is also a tension between precision and progress. The sustainability team knows that the first year of activity-based and supplier-specific data will almost certainly produce a different baseline from the spend-based figures published previously, and that the difference could go in either direction. If the new method shows higher emissions, the company will need to explain why its footprint appears to have grown despite years of claimed progress. If it shows lower emissions, critics may suspect that the method was changed to flatter the numbers. Either way, the restatement will need careful communication with investors, customers and the target validation body. Helen is prepared for this, but she wants the data team to be able to run old and new methods side by side for at least one reporting cycle, so that the effect of the methodology change can be isolated and explained transparently.

Your team comprises two data engineers, an analyst and a sustainability specialist. The company has a cloud data platform used for commercial analytics. There are commercial carbon accounting software products that promise to do all of this, and the sustainability team has been approached by several vendors.

You realise that the problem is less about calculation, which is arithmetic, than about data linkage and provenance. Emissions must be traced from purchases to materials to products, with the right factor for each, with

supplier-specific data where available and estimates where not, and with a clear record of which is which. Each year's figures must be reproducible, and changes in method must be distinguishable from real changes in emissions. That requires a data model connecting procurement, bills of materials, supplier data and emission factor libraries, all of which are currently disconnected. Each of these sources has its own owner, update cycle and identifiers, and none was designed with emissions accounting in mind.

THE DECISION IN FRONT OF YOU

Do you buy a carbon accounting platform, build an in-house emissions data model linking procurement, bills of materials and supplier data, or combine them, and how do you prioritise supplier engagement and design for assurance so that reported reductions reflect real change rather than methodology?

Data Sources

#	Source and link	How it is used in this project
1	GHG Protocol Scope 3 Standard and Guidance https://ghgprotocol.org/standards/scope-3-standard	The accounting standard defining scope 3 categories and calculation methods.
2	UK Government GHG Conversion Factors https://www.gov.uk/government/collections/government-conversion-factors-for-company-reporting	Official emission factors for energy, transport and materials.
3	EXIOBASE Multi-Regional Input-Output Database https://www.exiobase.eu	Environmentally extended input-output data for spend-based estimation by sector and region.
4	PACT Pathfinder Framework and Technical Specifications (WBCSD) https://www.carbon-transparency.org	Standard for exchanging product carbon footprint data between suppliers and customers.
5	US EPA Supply Chain GHG Emission Factors https://www.epa.gov/climateleadership/ghg-emission-factors-hub	Supply chain emission factors for spend-based comparison.
6	Science Based Targets initiative https://sciencebasedtargets.org	Target-setting criteria that shape data requirements for progress tracking.
7	European Sustainability Reporting Standards (EFRAG) https://www.efrag.org/en/sustainability-reporting	Reporting requirements for EU subsidiaries including emissions disclosure and assurance.
8	Carbon Intensity API (NESO) https://carbonintensity.org.uk	UK grid carbon intensity used in use-phase modelling.

Deliverables (17)

- D1** **Current footprint methodology review.** An assessment of the spend-based approach, its limitations and price sensitivity.
- D2** **Build, buy and hybrid evaluation.** A comparison of carbon accounting platforms and in-house build against requirements.
- D3** **Emissions data model.** A model linking purchases, materials, bills of materials, suppliers, factors and products.
- D4** **Procurement data classification.** Mapping of purchase lines to scope 3 categories and material types.
- D5** **Physical quantity extraction.** Derivation of material and packaging quantities from ERP and procurement records.
- D6** **Bill of materials integration.** Linkage of product formulations and packaging specifications to purchased materials.
- D7** **Emission factor library.** A versioned library of factors with source, year, geography and applicability.
- D8** **Supplier prioritisation.** Selection of strategic suppliers by emissions contribution and engagement readiness.
- D9** **Supplier data collection process.** A streamlined request aligned with industry exchange standards.
- D10** **Supplier data quality scoring.** Assessment of supplier-provided data for completeness, method and verification.
- D11** **Hybrid calculation engine.** Calculations combining supplier-specific, activity-based and spend-based methods with labelling.
- D12** **Product carbon footprints.** Product-level footprints for retail customer requests and design decisions.
- D13** **Use-phase modelling.** A documented model for consumer use emissions with uncertainty.
- D14** **Baseline recalculation policy.** Rules for restating baselines when methods or data change.
- D15** **Assurance controls framework.** Reconciliations, audit trails, reviews and sign-offs aligned with financial controls.
- D16** **Reporting outputs.** Data feeds for annual reports, regulatory disclosures and investor questionnaires.
- D17** **Progress tracking dashboard.** Tracking of emissions against targets distinguishing real reductions from method changes.

CASE STUDY 45 · MANUFACTURING SUPPLY CHAIN (ELECTRONICS)

Supplier Risk and Shipping Disruption Early Warning Platform

Industry	Manufacturing Supply Chain (Electronics)
Organisation	Orbitek, a UK designer and manufacturer of industrial control electronics with 320 component suppliers in 18 countries
Your role	Lead Data Engineer, Supply Chain Intelligence
Core themes	External data integration, multi-tier supplier visibility, risk scoring, alert relevance

The Scenario

Orbitek designs and assembles control systems used in factories, water treatment plants and energy infrastructure. Its products contain hundreds of electronic components, from microcontrollers and power semiconductors to connectors, capacitors and printed circuit boards, sourced from three hundred and twenty direct suppliers and, through them, from many more manufacturers around the world. The company survived the global chip shortage, but only just: for eighteen months, it spent enormous effort chasing parts, paid brokers inflated prices for components of uncertain provenance, redesigned products to use available parts, and missed deliveries to customers who then turned to competitors. The chief executive, Julia Harrington, has told the board that Orbittek will never again be caught unprepared, and has sponsored a supply chain intelligence programme. You lead its data engineering.

The ambition is an early warning platform that monitors risks affecting Orbittek's supply chain and alerts the right people in time to act. Risks come in many forms. A supplier might be in financial difficulty. A factory might be hit by an earthquake, flood, fire or power outage. A port might be congested or closed. A trade restriction might block exports of a component category from a country. A component might be announced as end of life by its manufacturer. Shipping rates might spike. Geopolitical tensions might threaten supplies from a region. Each of these can be monitored, in principle, using external data: news feeds, financial filings, natural hazard alerts, port and shipping data, trade and sanctions announcements, and component lifecycle databases.

The first challenge is knowing what to monitor. Orbittek knows its direct suppliers, their addresses and what it buys from them. It knows much less about where those suppliers make things or source their own inputs. A

distributor in Germany may supply a microcontroller made in Taiwan, packaged in Malaysia and tested in the Philippines. When an earthquake hits Taiwan, it matters to Orbitek, but nothing in its supplier master data connects that distributor to that island. Mapping the multi-tier supply chain requires information from suppliers, which many are reluctant to share, and from component manufacturers' published data, which is incomplete.

The procurement director, Stephen Ward, wants the platform to focus on direct suppliers first: their financial health, their delivery performance and their exposure to obvious risks. He is sceptical of multi-tier mapping, which he has seen attempted before at a previous employer, where it consumed a large budget and produced a map that was out of date within months. He wants something his buyers will actually use.

The head of engineering, Dr Neil Foster, has a different perspective. The components that caused the most pain during the shortage were not bought from risky suppliers; they were bought from large, stable distributors. The problem was that the components themselves were single-sourced from one manufacturer, with no approved alternatives, and demand exceeded global supply. Neil wants the platform to analyse Orbitek's bills of materials to identify components with single sources, long lead times, end-of-life risk or concentrated manufacturing geography, and to drive engineering work on alternatives. To him, the risk lies in the design, not the supplier.

The logistics manager, Karen Liu, is focused on shipping. Orbitek imports components by air and sea and exports finished products globally. She wants visibility of shipments in transit, early warning of port congestion and freight rate movements, and the ability to reroute when disruptions occur. She currently tracks shipments through the freight forwarder's portal and spreadsheets.

The data landscape is fragmented. The enterprise resource planning system holds purchase orders, supplier master data and inventory. The product lifecycle management system holds bills of materials and approved manufacturer part lists. Supplier performance data, such as on-time delivery, is calculated in spreadsheets. External data would come from many sources, some free, such as hazard alerts and trade data, and some commercial, such as supplier financial risk scores, component lifecycle and availability data, and shipping visibility services. Several

vendors sell integrated supply chain risk platforms that combine these, and Julia has received pitches from three of them.

The fundamental difficulty with risk monitoring is relevance. A platform that monitors news and hazards globally will generate thousands of events a day. Almost none of them will matter to Orbitek. The value lies in linking events to specific suppliers, sites, components and products, estimating impact, and alerting only when action is warranted. Vendors claim to do this, but their accuracy depends on the quality of the customer's own data, particularly supplier locations and component mapping, which is exactly what Orbitek lacks. In one vendor's trial on a sample of Orbitek's data, eighty per cent of the alerts generated in a week related to supplier head offices or sales branches rather than the sites where anything was actually made, and the buyers who reviewed them concluded, fairly, that the tool was crying wolf.

Your team is two engineers and a supply chain analyst. The budget can cover either a vendor platform subscription or a year of in-house development with some commercial data feeds, but not both at full scale. Julia wants a demonstration of value within four months, ahead of the board's strategy review.

You believe the foundations are internal: a clean supplier master with accurate site locations, a bill of materials risk analysis linking components to manufacturers and sources, and a clear mapping between components, suppliers, products and customers so that impact can be estimated. External data feeds are relatively easy to obtain; making them relevant is the hard part. But foundations are not what the board wants to see in four months, and a vendor platform could produce an impressive dashboard quickly, even if its alerts are mostly noise.

The company's recent experience with brokers adds urgency to a quieter risk. During the shortage, Orbitek bought components through independent brokers when authorised distributors had none, and a quality investigation later found that a batch of voltage regulators bought this way included counterfeit parts, remarked from lower-grade devices. The parts were caught at incoming inspection, but only because an experienced technician noticed inconsistent markings. Customers in critical infrastructure sectors have since asked Orbitek to demonstrate traceability to the original manufacturer for components in safety-related products. Any supply chain intelligence platform will therefore need to

hold provenance information about where each batch of components came from, and to flag purchases from non-franchised sources, which extends its scope from risk anticipation into quality and compliance.

There is also a question of ownership and action. Risk alerts are only useful if someone acts on them. At present, nobody at Orbitek is responsible for supply chain risk as a discipline. Buyers are measured on price and availability; engineers on product features; logistics on cost. Any platform will need a process and owners, not just data.

THE DECISION IN FRONT OF YOU

Do you buy an integrated supply chain risk platform, build in-house foundations with selected data feeds, or combine them, and do you prioritise supplier risk, bill of materials component risk or shipping visibility for the four-month demonstration?

Data Sources

#	Source and link	How it is used in this project
1	UN Comtrade Database https://comtradeplus.un.org	International trade flows by product and country used to assess geographic concentration of component categories.
2	Freightos Baltic Index https://fbx.freightos.com	Container freight rate indices used as shipping cost risk signals.
3	New York Fed Global Supply Chain Pressure Index https://www.newyorkfed.org/research/policy/gscpi	Macro supply chain pressure indicator for context and trend alerts.
4	GDACS Global Disaster Alert and Coordination System https://www.gdacs.org	Real-time natural hazard alerts for earthquakes, floods and cyclones linked to supplier sites.
5	USGS Earthquake Catalog API https://earthquake.usgs.gov/fdsnws/event/1/	Earthquake events for geospatial matching to manufacturing locations.
6	GDELT Project https://www.gdeltproject.org	Global news event data used to detect supplier and regional disruption signals.
7	DataCo Smart Supply Chain Dataset https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis	Supply chain order and shipping data used to prototype delivery risk and delay analytics.
8	UK Export Control and Sanctions Notices https://www.gov.uk/government/collections/notices-to-exporters	Trade restriction updates relevant to component flows.

Deliverables (17)

- D1 Risk use case prioritisation.** A scored assessment of supplier, component and shipping risk use cases for the demonstration.
- D2 Vendor platform evaluation.** A structured trial of risk platforms using Orbitek's own data to measure alert relevance.
- D3 Supplier master remediation.** Cleaned supplier records with verified manufacturing and shipping site locations.
- D4 Supplier site geocoding.** Coordinates for supplier and manufacturer sites enabling hazard matching.
- D5 Bill of materials risk analysis.** Identification of single-sourced, long lead time and end-of-life components.
- D6 Component-to-product-to-customer mapping.** Linkage enabling estimation of revenue at risk from a component disruption.
- D7 Multi-tier mapping for critical components.** Targeted mapping of manufacturers and sites for the highest-risk components.
- D8 External event ingestion.** Pipelines for hazard, news, trade and freight data with geocoding and entity tagging.
- D9 Event-to-exposure matching.** Logic linking external events to affected suppliers, sites and components.
- D10 Impact estimation model.** Estimated revenue and production at risk for each matched event.
- D11 Alert relevance tuning.** Thresholds and filters that keep alerts actionable, with feedback capture.
- D12 Supplier financial health monitoring.** Integration of financial risk indicators for key suppliers.
- D13 Shipment visibility feed.** Consolidated in-transit shipment status from forwarders with delay alerts.
- D14 Risk ownership and response process.** Defined owners, playbooks and escalation for each risk type.
- D15 Risk dashboard.** A view of active risks, exposure and mitigation status for leadership.
- D16 Four-month demonstration report.** Evidence of detected risks, actions taken and value for the board.
- D17 Long-term roadmap.** A plan for extending coverage, depth and automation over two years.

CASE STUDY 46 · NEWS MEDIA AND PUBLISHING

News Archive Digitisation and Content Enrichment Pipeline

Industry	News Media and Publishing
Organisation	The Northern Chronicle Group, publisher of a regional daily newspaper, six weekly titles and a news website with 9 million monthly visitors
Your role	Lead Data Engineer, Digital Products
Core themes	Unstructured content processing, metadata and taxonomy, licensing of archives to AI developers, editorial standards

The Scenario

The Northern Chronicle has been published every weekday since 1868. Its archive, stored as bound volumes, microfilm and, since the late 1990s, digital files in the editorial system, is a record of a century and a half of life in the north of England: industrial disputes, local elections, football triumphs, floods, royal visits, crimes, births, marriages and deaths. The archive has always been valued by historians and family researchers, and the paper sells access to scanned pages through a genealogy partner. In the past year, it has suddenly acquired a different kind of value. Three companies developing artificial intelligence models have approached the group's chief executive, Richard Holloway, offering substantial sums to license the archive for training, and a fourth has asked about licensing the paper's daily news output for use in AI-powered search products. You are lead data engineer for digital products, and you have been asked to make sense of what the archive actually is and what it could become.

The archive is not one thing. Content from 1868 to 1995 exists as scanned page images from microfilm, digitised a decade ago by the genealogy partner under a contract that gave the partner rights to host and sell access, with optical character recognition text of variable quality. Older pages, printed in small type on poor paper, produce text that is often barely readable. Content from 1996 to 2012 sits in an old editorial system's database export, with articles as text and some metadata, but without reliable links to photographs. Content from 2012 onwards is in the current content management system, with structured articles, images, tags and author information, but tagging has been inconsistent, and the tag taxonomy has been changed several times. Photographs from all periods are held separately, with rights that vary: staff photographers' work

belongs to the group, but agency photographs and pictures supplied by readers do not.

The editor-in-chief, Margaret Doyle, has strong views. She sees the archive as a public trust and a journalistic resource, not merely an asset to sell. She wants it made searchable for her journalists, who currently rely on memory and an unreliable internal search tool, and for readers, who could discover historical coverage connected to today's stories. She is uneasy about licensing it to AI companies: the paper's journalism is its competitive advantage, and she worries that AI products trained on or summarising Chronicle content would reduce the reasons for readers to visit the Chronicle itself. She also points out that the archive contains material that would not be published today, including coverage of crimes where the accused were later acquitted, names of victims, and content that reflects the prejudices of its time.

The commercial director, Simon Park, sees the licensing offers as a lifeline. Regional newspapers face relentless decline in print revenue, and digital advertising has not replaced it. The sums offered would fund the newsroom for years. But he acknowledges that the offers vary greatly depending on what is licensed: clean, structured, well-tagged text commands far more than raw optical character recognition output, and some buyers want guarantees about rights, such as confirmation that the group owns the content and that personal data has been handled lawfully.

The legal counsel, Joanna Fraser, has identified several problems. The group's rights to its own content are not as clear as everyone assumed. Freelance contributors in past decades often retained copyright, and contracts from that era are incomplete or lost. Syndicated content from news agencies was licensed for publication, not for onward licensing. The genealogy partner's contract may restrict what the group can do with the scanned pages. Personal data is pervasive: the archive names millions of people, including private individuals in court reports, and data protection law applies to archives, with some exemptions for journalism and archiving in the public interest that may not extend to commercial licensing for AI training. Joanna wants a rights and data protection assessment before anything is licensed.

The engineering task has several layers. Improving the text of the scanned archive would require reprocessing page images with modern optical character recognition and layout analysis, separating articles, headlines,

advertisements and captions, which the original digitisation did not do. Enriching the content would involve extracting people, places, organisations and events, linking them to reference datasets, and classifying articles by topic, so that journalists and readers can search meaningfully. Consolidating the three eras into one consistent content store would require a common schema. Tracking rights at article and image level, so that only content the group controls is licensed, would require linking articles to contributors and contracts. And any licensing would require packaging and delivery mechanisms that preserve provenance and exclude restricted content.

Your team is three engineers and a data scientist, supplemented by the archive librarian, Brenda Morris, who has worked at the paper for thirty years and knows the archive intimately. The group has limited cash for technology, although a licensing deal could change that. One AI company has offered to do the optical character recognition reprocessing and enrichment itself, for free, as part of a licensing agreement, which is tempting because it would save the group significant cost, but it would mean handing over the raw archive before rights are resolved and before the group understands the value of what it holds.

The newsroom has its own reasons for caring about the archive. Journalists at the Chronicle regularly need to know what the paper has previously reported about a person, a company or a planning dispute, and they rely on a search tool that indexes only the post-2012 content management system and returns results by keyword with no understanding of context. Last year, the paper published a profile of a local businessman that omitted his conviction for fraud twenty years earlier, which the paper had itself reported at the time; a rival title pointed it out within hours. On another occasion, a reporter repeated a claim from a 1990s article that had been the subject of a correction and apology, which nobody could find because corrections were never linked to the original stories. Margaret sees a well-structured, searchable archive as a matter of journalistic integrity as much as convenience. She would like corrections, legal notices and apologies linked to the articles they concern, and she wants reporters to see, when they open an archived story, whether it was later corrected, challenged or superseded. That requirement touches the same enrichment pipeline the licensing buyers want, but with very different priorities: completeness and accuracy of linkage matter more to her than volume, and a missed

correction is a far more serious defect than an imperfectly tagged sports report.

You see the risk of two opposite mistakes. One is to sell the archive quickly, cheaply and in raw form, before the group understands its value or its rights, and to regret it. The other is to spend years on a perfect digitisation project that never produces value, while the licensing window closes as AI developers find other sources.

THE DECISION IN FRONT OF YOU

Do you accept the AI company's offer to process the archive as part of a licence, invest in in-house reprocessing and enrichment first, or prioritise internal journalistic and reader use, and how do you resolve rights, personal data and historical sensitivities before any licensing?

Data Sources

#	Source and link	How it is used in this project
1	Chronicling America (Library of Congress) https://chroniclingamerica.loc.gov/about/api/	Large digitised historical newspaper collection with OCR text and API, used to prototype page segmentation and enrichment.
2	British Newspaper Archive https://www.britishnewspaperarchive.co.uk	Reference for UK historical newspaper digitisation practices and commercial partnership models.
3	GDELT Project https://www.gdeltproject.org	Global news event and entity extraction data used to benchmark enrichment outputs.
4	Tesseract OCR https://github.com/tesseract-ocr/tesseract	Open-source OCR engine for reprocessing historical page images.
5	IPTC Media Topics and NewsML-G2 https://iptc.org/standards/media-topics/	News industry taxonomy and metadata standards for consistent classification.
6	Wikidata Query Service https://query.wikidata.org	Knowledge base used to link extracted people, places and organisations to identifiers.
7	ICO Guidance on Journalism and Data Protection https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/data-protection-and-journalism/	Code of practice on the journalism exemption and archival processing.
8	UK IPO Copyright and AI Consultation https://www.gov.uk/government/consultations/copyright-and-artificial-intelligence	Policy context on text and data mining and licensing for AI training.

Deliverables (17)

- D1** **Archive inventory.** A catalogue of content by era, format, volume, quality and storage location.
- D2** **Rights assessment.** An analysis of ownership by content type, contributor category, agency material and partner contracts.
- D3** **Personal data and sensitivity assessment.** Identification of personal data categories and historical sensitivities with legal treatment.
- D4** **Licensing options paper.** A comparison of licensing models, prices, protections and strategic risks.
- D5** **OCR quality benchmark.** Measurement of current text quality and improvement from reprocessing on sample pages.
- D6** **Page segmentation pipeline.** Separation of articles, headlines, captions and advertisements in scanned pages.
- D7** **Unified content schema.** A common model for articles, images, contributors, dates and rights across all eras.
- D8** **Content consolidation pipeline.** Migration of all three archive eras into the unified store.
- D9** **Entity extraction and linking.** Identification of people, places and organisations linked to reference identifiers.
- D10** **Topic classification.** Assignment of standard news topics to articles across eras.
- D11** **Rights tagging at article level.** Linkage of each article and image to its rights status and licensability.
- D12** **Sensitive content flags.** Identification of acquittals, victims, minors and outdated language for handling decisions.
- D13** **Journalist search tool.** An internal search interface with filters by entity, topic and period.
- D14** **Reader discovery features.** Links from current stories to relevant archive coverage.
- D15** **Licensing delivery package.** A controlled export mechanism with provenance, exclusions and usage terms.
- D16** **Editorial governance policy.** Rules agreed with the editor on archive publication and licensing boundaries.
- D17** **Value and investment case.** An estimate of archive value under different processing levels to inform negotiations.

CASE STUDY 47 · SOCIAL MEDIA AND ONLINE COMMUNITIES

Trust and Safety Content Moderation Data Platform

Industry	Social Media and Online Communities
Organisation	Hearthside, a community discussion platform with 14 million monthly users and 60,000 active communities
Your role	Staff Data Engineer, Trust and Safety
Core themes	Online safety regulation, harmful content data handling, moderator wellbeing, transparency reporting

The Scenario

Hearthside is where people talk about things they care about: gardening, parenting, video games, local news, chronic illness, football, knitting, politics and tens of thousands of other subjects. Communities are created and run by volunteer moderators, with the company's own trust and safety team handling serious issues and platform-wide rules. For most of its history, Hearthside's approach to harmful content was reactive: users reported posts, volunteer and staff moderators reviewed them, and automated filters caught obvious spam. The Online Safety Act has changed that. Hearthside, as a user-to-user service with a significant UK user base, now has legal duties to assess the risks of illegal content and content harmful to children appearing on its platform, to take proportionate steps to mitigate those risks, and to report to the regulator, Ofcom. You are the staff data engineer in trust and safety, and the data needed to meet those duties does not exist in usable form.

Moderation data is scattered. User reports are stored in one database, with free-text reasons and a category chosen by the reporter from a list that has changed several times. Moderation decisions by volunteer moderators are recorded in community logs, which vary in detail and are not consistently categorised. Decisions by staff moderators are recorded in a case management tool that the trust and safety team adopted two years ago. Automated filter actions, such as removing spam or blocking certain keywords, are recorded in application logs. Appeals are handled by email. Law enforcement requests and reports to child protection organisations are handled by a small specialist team using a separate secure system. Nobody can currently answer basic questions such as how much illegal content is reported each month, how quickly it is removed, or how often removal decisions are reversed on appeal.

The regulator's codes of practice set out measures that services are expected to take, including effective content moderation systems, accessible reporting and complaints processes, and, for services likely to be accessed by children, measures to protect them. Hearthside must complete a risk assessment, grounded in evidence about its own service, and keep it updated. It must be able to demonstrate that its measures are working. Transparency reporting requirements will oblige certain services to publish information about how they deal with harmful content. The head of trust and safety, Rachel Adeyemi, needs data to support all of this, and she needs it before the regulator's deadlines.

The chief product officer, Leo Martens, has concerns about the approach. Hearthside's culture is built on community self-governance, and volunteer moderators are fiercely protective of their autonomy. Many are sceptical of the company imposing centralised rules and monitoring their decisions. If the new data platform is seen as a way of auditing volunteers, some of the platform's most important communities could revolt. Leo wants any data collection from community moderation to be lightweight and framed as support rather than surveillance.

The machine learning team, led by Dr Aarav Singh, wants to build classifiers that detect harmful content proactively, rather than waiting for reports. Training such classifiers requires labelled examples of harmful content, which means storing and handling the very content the company is trying to remove, including some of the worst material on the internet. For certain categories, such as child sexual abuse material, it is illegal to possess the content except in narrowly defined circumstances, and industry practice relies on hash matching against databases maintained by specialist organisations rather than training in-house classifiers. For other categories, such as terrorist content, self-harm and suicide content, hate speech and harassment, labelled data is needed but must be handled with great care. Access must be tightly controlled, retention limited, and people who view the content protected from its psychological effects.

Moderator wellbeing is a serious issue. Staff moderators who review harmful content are exposed to distressing material daily, and research on content moderation has documented high rates of psychological harm. The trust and safety team's wellbeing lead, Dr Hannah Clarke, wants the data platform designed to minimise unnecessary exposure: blurring images by default, grouping similar content so that moderators do not repeatedly

view the same material, and prioritising queues so that the most harmful content is reviewed by specialists with appropriate support.

Privacy and freedom of expression add further constraints. The data platform will hold information about users who post and report content, including sensitive information such as health conditions discussed in support communities. The Act also places duties on services to have regard to users' rights to freedom of expression and privacy, and to protect content of democratic importance and journalistic content. Over-removal is itself a harm, and data on appeals and reversals is essential for demonstrating balance.

Your team is four engineers. The company's data platform is a cloud data warehouse used mainly for product analytics. Trust and safety data has never been brought into it, partly because of sensitivity concerns. The regulator's timelines are fixed, and Rachel wants a risk assessment grounded in data within four months and a sustainable reporting capability within a year.

There is also the matter of what the company already knows but has never measured. The trust and safety team has anecdotal evidence that certain types of community carry disproportionate risk: communities for teenagers discussing mental health, communities that trade in second-hand goods where scams proliferate, and a handful of political communities where harassment of public figures is common. Staff moderators know which communities generate the most escalations, but this knowledge is held in people's heads and in informal channels, not in data. The regulator's risk assessment guidance expects services to consider risk factors such as user base, functionality and business model, and to evidence their conclusions. Rachel suspects that a rigorous analysis will reveal risks the company has not fully confronted, including features such as private messaging between users who met in public communities, which several child safety organisations have identified as a grooming vector on other platforms. She is prepared for uncomfortable findings, but she needs the leadership to understand that the purpose of the data platform is to surface risks honestly, including those that implicate product design choices the company is proud of. Leo has agreed in principle, but you sense that his agreement will be tested the first time the data points at a feature on his roadmap.

You see a need for careful separation. Aggregated metrics about reports, decisions, timeliness and appeals can live in the analytics environment with appropriate controls. The content itself, particularly the most harmful categories, must not. Personal data about users involved in moderation should be minimised. Volunteer moderation data should be collected in a way that respects community autonomy. And the most sensitive categories should be handled through specialist, external mechanisms rather than in-house data stores.

THE DECISION IN FRONT OF YOU

How do you design a trust and safety data platform that provides the evidence Hearthside needs for its risk assessments and transparency reporting, supports proactive detection responsibly, protects moderators and respects volunteer communities, without creating new stores of harmful content or new privacy risks?

Data Sources

#	Source and link	How it is used in this project
1	Ofcom Online Safety Codes and Guidance https://www.ofcom.org.uk/online-safety	Regulatory codes of practice, risk assessment guidance and transparency requirements.
2	Online Safety Act 2023 https://www.legislation.gov.uk/ukpga/2023/50/contents	Legal duties that define the data and evidence the platform must support.
3	Jigsaw Toxic Comment Classification Data https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge	Labelled comments used to prototype classification pipelines without platform content.
4	Perspective API https://perspectiveapi.com	External toxicity scoring service for comparison with in-house classifiers.
5	Internet Watch Foundation Services https://www.iwf.org.uk/our-technology/	Hash lists and services for detecting child sexual abuse material without in-house storage.
6	Tech Against Terrorism and GIFCT Hash-Sharing https://gifct.org/hsdb/	Industry hash-sharing database for terrorist content detection.
7	Trust and Safety Professional Association Resources https://www.tsipa.org/curriculum/ts-fundamentals/	Industry practice on moderation operations, metrics and wellbeing.
8	Samaritans Online Harms Guidance https://www.samaritans.org/about-samaritans/research-policy/internet-suicide/guidelines-tech-industry/	Guidance on handling self-harm and suicide content safely.

Deliverables (17)

- D1** **Moderation data inventory.** A map of report, decision, appeal and enforcement data across systems.
- D2** **Harm taxonomy.** A consistent classification of content categories aligned with legal definitions and platform rules.
- D3** **Data zoning architecture.** Separation of aggregated metrics, restricted case data and prohibited content handling.
- D4** **Unified moderation event model.** A schema for reports, actions, appeals and outcomes across staff, volunteer and automated moderation.
- D5** **Volunteer moderation data design.** Lightweight, community-respecting capture of volunteer decisions co-designed with moderators.
- D6** **Risk assessment evidence base.** Metrics on prevalence, reporting, response times and outcomes by harm category.
- D7** **Timeliness and outcome metrics.** Measurement of time to action and reversal rates on appeal.
- D8** **Transparency reporting pipeline.** Automated production of transparency report figures with documented definitions.
- D9** **Hash matching integration.** Integration with specialist hash databases for the most severe categories.
- D10** **Classifier training data governance.** Rules on storage, access, retention and labelling of harmful content samples.
- D11** **Proactive detection pipeline.** Deployment of classifiers with human review and measured precision.
- D12** **Moderator wellbeing features.** Blurring, deduplication, exposure tracking and specialist routing in review queues.
- D13** **Freedom of expression safeguards.** Tracking of over-removal, appeals and protected content categories.
- D14** **User data minimisation controls.** Pseudonymisation and retention limits for users involved in moderation.
- D15** **Access control and audit.** Strict role-based access with logging for restricted data.
- D16** **Regulator engagement pack.** Documentation demonstrating measures, evidence and governance to Ofcom.
- D17** **Implementation roadmap.** A plan aligned with regulatory deadlines and community consultation.

CASE STUDY 48 • MUSIC PUBLISHING AND RIGHTS MANAGEMENT

Streaming Royalty Calculation and Rights Data Pipeline

Industry	Music Publishing and Rights Management
Organisation	Songline Music Rights, an independent music publisher administering 180,000 compositions for 4,000 songwriters
Your role	Head of Data Engineering, Royalty Operations
Core themes	Rights metadata matching, high-volume usage reports, financial accuracy, creator trust

The Scenario

Songline Music Rights represents songwriters. When a song is streamed, downloaded, broadcast, performed live or used in a film, the people who wrote it are entitled to royalties, and publishers like Songline collect those royalties on their behalf and pay them out, taking a commission. Songline administers a catalogue of a hundred and eighty thousand compositions for four thousand songwriters, from established hitmakers to emerging artists in the independent scene. Its reputation depends on a simple promise: that it will find every penny its writers are owed and pay it accurately and on time. You lead data engineering for royalty operations, and that promise is under strain.

Royalties from streaming now make up two thirds of Songline's income. Every quarter, and increasingly every month, the company receives usage reports from digital service providers and collecting societies, listing streams of recordings with the number of plays, the territory, the revenue generated and the royalty due. The volumes are enormous: a single quarter's reports contain several billion lines. The challenge is that streaming services report usage of recordings, identified by recording codes, artist names and track titles, whereas Songline owns rights in compositions, the underlying songs. One composition may have hundreds of recordings: the original, remixes, live versions, cover versions by other artists, karaoke versions. To claim royalties, Songline must match each reported recording to the composition it embodies and to Songline's share of that composition, which may be split between several writers and publishers.

Matching is imperfect everywhere in the industry. Recording codes are often missing or incorrect in reports. Composition codes, the identifiers for

songs, are frequently absent from recording metadata. Titles vary in spelling, punctuation and language. Songwriter names are credited inconsistently. Songline's own catalogue data, built up over decades through acquisitions of smaller publishers, contains duplicates, missing identifiers and conflicting ownership shares. The industry estimates that a significant share of streaming royalties goes unmatched and ends up in pools of unclaimed money that are eventually distributed by market share, benefiting the largest publishers. Songline's current matching rate is around eighty-two per cent by value, which its chief executive, Amanda Price, believes is well below what is achievable.

The processing system was built eight years ago by a contractor using a relational database and a set of batch scripts. Each quarter, the operations team loads reports, runs matching, reviews exceptions in spreadsheets and produces royalty statements for writers. The process takes eleven weeks from receipt of reports to payment. Writers complain about delays and about statements that are hard to understand. Several prominent writers have audited Songline under the audit rights in their contracts, and one audit found underpayments caused by a matching error that had misallocated royalties for a popular song to a different writer with a similar title for three years. Songline paid the arrears and interest, but the writer left, and word spread in the songwriting community.

Amanda wants a new royalty data platform that improves matching, shortens the payment cycle and gives writers transparent, detailed statements through an online portal. The finance director, Graham Kelly, is focused on accuracy and controls: royalty payments are the company's largest expense, and errors either cost money through overpayment or cost reputation and legal risk through underpayment. He wants a system with the rigour of a financial ledger, where every penny received can be traced to every penny paid.

The head of copyright administration, Nadia Osei, highlights the catalogue problem. Her team maintains ownership records for every composition: who wrote it, what share each writer and publisher owns, in which territories, for what period and under which agreements. Ownership changes when catalogues are bought and sold, when agreements expire and when disputes are resolved. Many of these records exist only in contracts and correspondence. She believes that matching improvements

will have limited impact unless catalogue data is cleaned first, and that her team needs better tools for maintaining it.

Industry initiatives offer some help. Standard message formats for sales and usage reporting and for work registrations exist and are increasingly adopted by digital services and societies. Public databases of recordings and compositions, maintained by industry bodies and community projects, can help fill gaps in identifiers. Collective initiatives in some countries allow publishers to access shared matching infrastructure. Songline has never joined any of them, partly because of cost and partly because the previous chief executive distrusted sharing catalogue data with larger rivals. But adoption is uneven, and many reports still arrive in proprietary formats.

Your team comprises four engineers and two analysts. The operations team of fifteen people handles exceptions and writer queries. The contractor who built the current system is no longer available, and the matching rules it applies are embedded in stored procedures that nobody on the current team has fully read, let alone tested. Amanda wants the next quarterly cycle to show improvement and a new platform within eighteen months.

The songwriters themselves have changed. A generation ago, most of Songline's writers were represented by managers and lawyers who read statements once a year and trusted the publisher. Today, many independent writers track their own streams on dashboards provided by streaming services and distributors, compare them with what their publisher reports, and share their findings on social media when the numbers do not add up. The comparisons are often misleading, because streams on a dashboard do not translate simply into publishing royalties, which pass through several intermediaries, currencies and deductions before they arrive. But the perception of opacity is damaging, and the writer who left after the misallocation has spoken publicly about it on podcasts. The head of writer relations, Tom Ashby, receives dozens of queries a week from writers asking why a song with millions of streams produced only a few hundred pounds. He believes that a large share of these queries could be answered, and trust rebuilt, if statements explained the journey from stream to payment in plain terms, with each deduction and intermediary visible. That is as much a data design problem as a

communication one, because it requires the platform to retain and expose the full chain of calculation for every line, rather than just the final amount.

You see three possible starting points. You could focus on matching, applying modern fuzzy matching and machine learning to recover unmatched income quickly. You could focus on the catalogue, cleaning and enriching ownership data as the foundation for everything else. Or you could focus on the platform, rebuilding processing to handle volumes efficiently and shorten the cycle. Each helps, but each depends on the others. Better matching against a dirty catalogue risks new misallocations like the one that lost a major writer. A clean catalogue without better matching leaves money unclaimed. A faster platform without better data simply produces wrong statements more quickly.

THE DECISION IN FRONT OF YOU

Do you prioritise recovering unmatched income through improved matching, cleaning the rights catalogue, or rebuilding the processing platform, and how do you ensure financial accuracy and transparency so that writers trust Songline with their royalties?

Data Sources

#	Source and link	How it is used in this project
1	MusicBrainz Database https://musicbrainz.org/doc/MusicBrainz_Database	Open music metadata linking recordings, works and artists, used for identifier enrichment and matching.
2	DDEX Standards (DSR and CWR) https://ddex.net/standards/	Industry standards for digital sales reporting and work registration messages.
3	ISWC International Standard Musical Work Code https://www.iswc.org	Standard identifier for compositions used to link recordings to works.
4	ISRC Handbook (IFPI) https://isrc.ifpi.org	Standard identifier for recordings and guidance on usage.
5	Spotify Web API https://developer.spotify.com/documentation/web-api	Recording metadata and identifiers for enrichment of usage reports.
6	Million Song Dataset http://millionsongdataset.com	Large dataset of track metadata used to test matching algorithms at scale.
7	The MLC Public Work Search https://www.themlc.com/dataandreports	Public data on works and unmatched usage from the US mechanical licensing collective.
8	PRS for Music https://www.prsformusic.com	UK collecting society data, registration and reporting practices.

Deliverables (17)

- D1 Unmatched income analysis.** A breakdown of unmatched royalties by source, territory, cause and value.
- D2 Catalogue data quality assessment.** Duplicate works, missing identifiers and conflicting ownership shares quantified.
- D3 Catalogue remediation programme.** Prioritised cleaning of high-value works with ownership verification.
- D4 Rights ownership data model.** A temporal model of shares by writer, publisher, territory, right type and agreement.
- D5 Identifier enrichment pipeline.** Enrichment of works and recordings with ISWC and ISRC from external sources.
- D6 Usage report ingestion.** Scalable ingestion of standard and proprietary report formats with validation.
- D7 Matching engine.** Tiered matching using identifiers, fuzzy text and machine learning with confidence scores.
- D8 Match review workflow.** Human review of low-confidence matches with decisions feeding back into rules.
- D9 Misallocation safeguards.** Checks preventing matches to similar-titled works without corroborating evidence.
- D10 Royalty calculation ledger.** A double-entry ledger tracing income received to royalties paid per writer.
- D11 Reconciliation controls.** Reconciliation of reports, receipts, calculations and payments each cycle.
- D12 Writer statement redesign.** Clear, detailed statements explaining sources, uses and calculations.
- D13 Writer portal data services.** Data feeds powering an online portal with statement history and catalogue views.
- D14 Cycle time reduction plan.** Process and platform changes shortening the report-to-payment cycle.
- D15 Audit readiness pack.** Evidence and traceability supporting writer audits.
- D16 Unclaimed income recovery report.** Quantified income recovered through improved matching each cycle.
- D17 Platform roadmap.** An eighteen-month plan sequencing catalogue, matching and platform work.

CASE STUDY 49 · US HEALTHCARE (HEALTH INSURANCE)

Hospital Price Transparency Data Ingestion and Analytics

Industry	US Healthcare (Health Insurance)
Organisation	Cascade Health Partners, a regional health insurer covering 850,000 members in the Pacific Northwest
Your role	Senior Data Engineer, Network Analytics
Core themes	Massive semi-structured public files, schema inconsistency, provider identity, negotiation analytics

The Scenario

For decades, the prices that American hospitals and health insurers negotiated with each other were closely guarded secrets. That changed with federal price transparency rules. Hospitals must now publish machine-readable files listing their standard charges, including the rates they have negotiated with each insurer for each service. Health insurers must publish their own machine-readable files listing the rates negotiated with every in-network provider for every covered item and service, along with historical allowed amounts for out-of-network services. The goal was to empower consumers and employers and to increase competition. In practice, it has produced one of the largest and messiest public datasets in the world. You are a senior data engineer in the network analytics team at Cascade Health Partners, and your job is to turn that mess into competitive intelligence.

Cascade negotiates contracts with hospitals, physician groups and other providers across Washington, Oregon and Idaho. Its network management team, led by vice president Diane Morales, has always negotiated knowing only its own rates and, informally, what providers claimed other insurers paid. Now, in principle, it can see the rates competitors have negotiated with the same hospitals for the same services, and the rates hospitals report for every insurer. Diane wants to know, for each major hospital system in Cascade's market, whether Cascade is paying more or less than its competitors for common procedures, where the biggest gaps are, and how to use that information in upcoming contract renewals with the three largest hospital systems, the first of which begins in five months.

The data is enormous and chaotic. Insurer files, published monthly, run to terabytes for the largest national insurers, with individual files sometimes

hundreds of gigabytes in compressed form. They use a standard schema defined by the federal government, but insurers have interpreted it in different ways, and the schema itself has evolved. Rates are listed by provider identifiers and billing codes, with negotiated arrangements that may be fee-for-service amounts, percentages of billed charges, per diem rates or bundled payments, each requiring different interpretation. Many files contain rates for services that providers never actually perform, because insurers publish entire fee schedules for every provider in a network, producing billions of rows of theoretical prices. Hospital files are smaller but even less consistent: early versions used dozens of different formats, and while a standard template now exists, compliance and data quality vary widely. Some hospitals list rates by insurer plan names that are ambiguous or out of date.

Provider identity is a central challenge. Insurer files identify providers by national provider identifiers and tax identification numbers, while hospital files identify themselves by name and location. A hospital system may operate many facilities under different identifiers, and physicians may bill under group or individual identifiers. To compare rates for the same hospital across insurers, you must resolve all of these to a common entity, using public provider registries that are themselves imperfect.

The chief analytics officer, Dr Robert Chen, wants a robust, repeatable platform: automated monthly ingestion of competitors' insurer files and all relevant hospitals' files, normalisation to a common model, entity resolution, and a curated dataset of comparable rates. He knows that several commercial vendors now sell exactly this, having invested heavily in processing the files, and that their products are expensive but ready. He is uncertain whether building in-house is sensible. The chief financial officer, Karen Liu, would prefer to buy, on the grounds that price transparency processing is not Cascade's core competency. Diane, however, is suspicious of vendor data: she has seen vendor reports with obvious errors, such as rates that are clearly wrong by orders of magnitude, and she wants to understand exactly how any number she uses in a negotiation was derived, because hospital negotiators will challenge it.

The legal team, led by general counsel Michael Brooks, has raised questions about how the data may be used. The files are public, and using public competitor pricing in negotiations is lawful, but antitrust law restricts sharing of competitively sensitive information between

competitors, and Cascade must be careful that its use of the data does not amount to coordination. Michael has also asked that the platform not be used to disclose Cascade's own rates beyond what regulation requires, and that analyses shared with providers during negotiations be reviewed first.

The data engineering challenges are formidable. Downloading and processing terabytes of files monthly requires scalable compute and storage and careful handling of compressed, nested formats. Files frequently fail to parse because of malformed content. Deduplicating rates across the many files in which the same rate appears is essential. Filtering to relevant providers, services and plans early is key to making the problem tractable. Interpreting arrangements such as percentage-of-charges rates requires joining to hospital charge masters. And validating the results against Cascade's own claims data, which shows what it actually paid, is the only way to test whether the published rates correspond to reality.

There is an internal political dimension as well. Cascade's own rates are now visible to its competitors and to the hospitals, which means the network management team is operating in a world where every counterpart has the same information and where hospitals may use insurer files to argue that Cascade pays less than its rivals. Diane is concerned that hospital negotiators already have analytics from vendors, and that Cascade would enter negotiations at a disadvantage without its own. At the same time, Cascade's leadership has been cautious about technology investments since a claims system replacement ran over budget last year, and there is little appetite for another large platform project. Robert has been told that any new platform must demonstrate measurable financial return within a year, which in this case means savings achieved in contract negotiations. That creates a subtle risk: the pressure to show savings could push the team to present comparisons more confidently than the data warrants. You know from early exploration that for many services, comparable rates across insurers are genuinely ambiguous because of different arrangement types and plan structures, and that honest analysis will often conclude that the difference is uncertain. Presenting that uncertainty clearly, without undermining the negotiators' confidence, will be as important as producing the numbers.

Your team is three engineers and an analyst. You have five months until the first negotiation, and the hospital system involved is the largest in the

region, accounting for nearly a fifth of Cascade's total medical spend. A vendor trial could be set up in weeks. A robust in-house platform would take at least six months, but a targeted in-house analysis of the three hospital systems and their key competitors could be ready in time.

THE DECISION IN FRONT OF YOU

Do you buy a commercial price transparency data product, build an in-house ingestion and normalisation platform, or deliver a targeted in-house analysis for the upcoming negotiations while evaluating vendors, and how do you ensure every number used at the negotiating table is traceable, validated and legally safe?

Data Sources

#	Source and link	How it is used in this project
1	CMS Transparency in Coverage Schema https://github.com/CMSgov/price-transparency-guide	Official schema and documentation for insurer machine-readable files.
2	CMS Hospital Price Transparency Resources https://www.cms.gov/hospital-price-transparency	Hospital standard charges requirements, templates and data dictionary.
3	CMS Hospital Price Transparency Template Repository https://github.com/CMSgov/hospital-price-transparency	Standard CSV and JSON templates and validators for hospital files.
4	NPPES NPI Registry API https://npiregistry.cms.hhs.gov/api-page	National provider identifiers, names and addresses for entity resolution.
5	CMS Provider of Services File https://data.cms.gov/provider-characteristics/hospitals-and-other-facilities/provider-of-services-file-hospital-non-hospital-facilities	Facility characteristics and identifiers for hospitals and other providers.
6	CMS Medicare Physician Fee Schedule https://www.cms.gov/medicare/payment/fee-schedules/physician	Medicare reference prices used to express commercial rates as percentages of Medicare.
7	Turquoise Health Resources https://turquoise.health	Commercial price transparency data provider used as a benchmark in vendor evaluation.
8	DuckDB https://duckdb.org	Embedded analytical engine for processing large JSON and CSV files efficiently.

Deliverables (17)

- D1** **Negotiation analysis scope.** Defined hospitals, competitors, services and plans for the first negotiation.

- D2 Vendor evaluation.** A trial of commercial products with accuracy checks against Cascade's own claims.
- D3 File discovery and download pipeline.** Automated retrieval of relevant insurer and hospital files each month.
- D4 Streaming parser.** Memory-efficient parsing of large nested JSON and CSV files with error handling.
- D5 Early filtering layer.** Filtering to relevant providers, codes and plans during ingestion to control volume.
- D6 Canonical rate model.** A normalised model of rates, arrangement types, providers, services and plans.
- D7 Provider entity resolution.** Mapping of NPIs, tax identifiers and facility names to hospital systems and facilities.
- D8 Arrangement interpretation.** Conversion of percentage, per diem and bundled arrangements into comparable amounts.
- D9 Theoretical rate filtering.** Removal of rates for services providers do not perform, using claims and volume data.
- D10 Claims-based validation.** Comparison of published rates with Cascade's actual paid amounts.
- D11 Rate comparison dataset.** Validated comparisons of Cascade and competitor rates by hospital and service.
- D12 Negotiation briefing packs.** Hospital-specific analyses with provenance for every figure.
- D13 Data lineage and provenance store.** Traceability from each comparison back to source files and transformations.
- D14 Legal review process.** Antitrust and disclosure review of analyses before external use.
- D15 Data quality monitoring.** Detection of parse failures, implausible rates and missing files each month.
- D16 Cost and scalability plan.** Compute and storage estimates for expanding to full-market coverage.
- D17 Build versus buy recommendation.** A final recommendation informed by the targeted analysis and vendor trial.

CASE STUDY 50 · HEALTH INSURANCE AND REAL-WORLD EVIDENCE

Converting Health Insurance Claims to the OMOP Common Data Model for Outcomes Research

Industry	Health Insurance and Real-World Evidence
Organisation	Meridian Health Plan, a US health insurer with 2.1 million members and a real-world evidence partnership programme
Your role	Lead Data Engineer, Health Data Science
Core themes	Common data models, vocabulary mapping, de-identification, research data governance

The Scenario

Meridian Health Plan insures around two million people across several US states through employer plans, individual marketplace coverage and Medicare Advantage. Every time a member sees a doctor, fills a prescription, has a laboratory test or is admitted to hospital, Meridian receives a claim, coded with diagnoses, procedures, drugs and costs. Over fifteen years, those claims have accumulated into a rich longitudinal record of the health and healthcare of millions of people. Pharmaceutical companies, academic researchers and public health agencies increasingly want to use such data to study how treatments perform in the real world, beyond the controlled conditions of clinical trials. Meridian's leadership sees an opportunity: a real-world evidence programme that partners with researchers and generates revenue while contributing to medical knowledge. You lead data engineering in health data science, and the programme depends on converting Meridian's claims into a form researchers can use.

The research community has converged on a small number of common data models, standard structures into which healthcare data from different sources can be transformed so that the same analysis code can run across many databases. The most widely used for observational research is the OMOP common data model, maintained by an international open-science collaborative. It defines standard tables for people, visits, conditions, drugs, procedures, measurements and observations, and, crucially, a standardised vocabulary that maps source codes, such as diagnosis codes, procedure codes and national drug codes, to standard concepts. Hundreds of databases worldwide have been converted, and a rich ecosystem of open-source tools supports analyses across them. Pharmaceutical partners

have told Meridian that an OMOP-converted database would be far more attractive than raw claims, because their analysts can use existing tools and network studies.

The conversion is harder than it sounds. Meridian's claims live in a data warehouse designed for financial and operational reporting. Claims are submitted, adjusted, reversed and resubmitted; a single encounter may generate several claim versions, and the warehouse holds all of them. Institutional claims from hospitals and professional claims from doctors for the same hospital stay must be combined into visits. Pharmacy claims use national drug codes, which change frequently and must be mapped to ingredients and strengths. Diagnosis codes changed from one coding system to another in 2015, so the history spans two systems. Enrollment periods determine when a member's data is complete; a member who leaves and rejoins has gaps that researchers must know about. Laboratory results are available only for some members through a partnership with a laboratory company, with values and units that need standardisation.

The chief data officer, Dr Linda Park, wants the conversion completed within six months, because a major pharmaceutical partner is negotiating a multi-year agreement contingent on OMOP availability. The head of research partnerships, Kevin Brooks, has already promised the partner a preview dataset within three months.

The privacy officer, Sandra Mitchell, has a different set of priorities. Claims data is protected health information under federal health privacy law. Research use requires either member authorisation, which is impractical at scale, a waiver from an institutional review board, or de-identification to the law's standard. The programme intends to provide de-identified data, which requires either removing specified identifiers under the safe harbour method or obtaining an expert determination that the risk of re-identification is very small. Safe harbour removes dates more precise than year and geographic detail below the first three digits of postal codes, which would severely limit research value because many analyses depend on the timing of events. Expert determination allows shifting dates consistently rather than removing them, but requires a statistical expert's assessment and ongoing controls. Sandra also notes that some state laws impose additional restrictions, particularly on data about substance use treatment, reproductive health and mental health.

The medical director, Dr Ahmed Hassan, raises data quality and interpretation concerns. Claims capture what was billed, not necessarily what was wrong with the patient: a diagnosis code on a claim for a test may reflect a condition being ruled out rather than confirmed. Researchers unfamiliar with claims data routinely misinterpret it. He wants the converted database to come with clear documentation of its limitations and with data quality assessments that researchers can review.

The IT infrastructure team points out that the data warehouse is heavily used for operations and cannot support large research extracts during business hours. Any conversion will need its own environment, and research partners will need secure access, either through data delivered to their own environments or through a controlled analytics environment where they run analyses without data leaving Meridian.

Your team is three data engineers and a clinical informaticist. The open-source community provides extract, transform and load frameworks, vocabulary files and data quality tools for OMOP conversion, and several consulting firms offer conversion services. You have to decide whether to build the conversion yourselves with community tools or engage a consultancy, how to handle de-identification, how to balance speed against quality, and how to provide partner access.

You are acutely aware that a poorly converted database, delivered fast to meet a commercial deadline, could produce misleading research findings. Observational research using claims data is already controversial, and high-profile studies that later turned out to be wrong have damaged trust in real-world evidence. If Meridian's data contributes to a published study that is later retracted because of conversion errors, the reputational damage would far exceed any partnership revenue. But you also know that conversions are never perfect, that the community's approach accepts iterative improvement with transparent quality reporting, and that waiting for perfection would mean never delivering.

Meridian's members are another constituency whose views have not yet been sought. The health plan's member advisory council, a group of members who advise on service changes, learned about the real-world evidence programme from a trade press article and asked why members had not been told. Several council members said they would be comfortable with their data contributing to medical research, but not with it being sold to pharmaceutical companies for marketing purposes. Sandra

believes that the programme is lawful under a de-identification approach, but she also recognises that lawful and acceptable are not the same thing, and that a public backlash could damage member trust and employer relationships. She would like the programme to publish a clear statement of how data is used, what is excluded and how research partners are vetted, and to restrict partner agreements to research uses with prohibitions on re-identification and marketing.

There is also the question of what the three-month preview should contain. A small, carefully converted sample might satisfy the partner's evaluation while the full conversion continues, but it could create expectations about completeness and quality that the full database might not meet. Alternatively, a full but unvalidated conversion would give a true picture of scale but risk the partner discovering errors before Meridian does.

THE DECISION IN FRONT OF YOU

Do you build the OMOP conversion in-house with community tools, engage a consultancy to meet the partner timeline, or deliver a limited validated preview while the full conversion proceeds, and how do you choose a de-identification approach that preserves research value while meeting privacy law and member expectations?

Data Sources

#	Source and link	How it is used in this project
1	OMOP Common Data Model Specification https://ohdsi.github.io/CommonDataModel/	Table and field specifications for the target data model.
2	OHDSI Athena Standardised Vocabularies https://athena.ohdsi.org	Vocabulary downloads mapping ICD, CPT, HCPCS, NDC and other source codes to standard concepts.
3	OHDSI Book of OHDSI https://ohdsi.github.io/TheBookOfOhdsi/	Guidance on ETL design, data quality and research methods.
4	OHDSI Data Quality Dashboard https://github.com/OHDSI/DataQualityDashboard	Automated data quality checks for converted OMOP databases.
5	CMS Medicare Claims Synthetic Public Use Files (SynPUF) https://www.cms.gov/data-research/statistics-trends-and-reports/medicare-claims-synthetic-public-use-files	Synthetic Medicare claims used to build and test the ETL without real member data.
6	Synthea Synthetic Patient Generator https://synthea.mitre.org	Synthetic longitudinal patient data for testing edge cases.

#	Source and link	How it is used in this project
7	HHS Guidance on De-identification under HIPAA https://www.hhs.gov/hipaa/for-professionals/special-topics/de-identification/index.html	Official guidance on safe harbour and expert determination methods.
8	42 CFR Part 2 (Substance Use Records) https://www.ecfr.gov/current/title-42/chapter-I/subchapter-A/part-2	Federal rules restricting use of substance use disorder treatment records.

Deliverables (17)

- D1 Conversion approach decision.** A comparison of in-house, consultancy and hybrid conversion with timeline and quality risks.
- D2 Source data profiling report.** Profiling of claims, enrollment, pharmacy and laboratory data with known issues.
- D3 ETL design specification.** Detailed mapping from source tables to OMOP tables with business rules.
- D4 Claim versioning resolution.** Logic selecting final adjudicated claims and handling reversals and resubmissions.
- D5 Visit construction logic.** Rules combining institutional and professional claims into visits.
- D6 Vocabulary mapping pipeline.** Mapping of source codes to standard concepts with unmapped code tracking.
- D7 Drug code normalisation.** Mapping of national drug codes to ingredients and strengths across code changes.
- D8 Observation period derivation.** Enrollment-based observation periods identifying complete data windows.
- D9 Laboratory measurement standardisation.** Harmonisation of laboratory codes, values and units.
- D10 Sensitive data exclusion rules.** Identification and handling of substance use and other restricted records.
- D11 De-identification approach.** A chosen method with expert determination or safe harbour justification and date shifting design.
- D12 Data quality dashboard results.** Automated quality check results published with the database.
- D13 Characterisation reports.** Descriptive statistics on the population, conditions and treatments for partners.
- D14 Limitations documentation.** Clear guidance on claims data interpretation and known gaps.
- D15 Partner access environment.** A secure analytics environment or controlled delivery mechanism with auditing.

- D16 Validated preview dataset.** A carefully validated subset delivered for partner evaluation.
- D17 Refresh and versioning process.** A repeatable process for periodic database refreshes with change documentation.

CASE STUDY 51 · BANKING (CREDIT RISK AND FINANCE)

IFRS 9 Expected Credit Loss Data Lineage and Model Input Remediation

Industry	Banking (Credit Risk and Finance)
Organisation	Cardinal Bank, a UK retail and SME bank with a £31 billion loan book
Your role	Principal Data Engineer, Finance and Risk Data
Core themes	Regulatory data lineage, model input controls, audit findings, finance and risk data reconciliation

The Scenario

Every quarter, Cardinal Bank calculates the provision it must hold against expected credit losses on its loans, under the international accounting standard known as IFRS 9. The provision is one of the largest and most judgemental numbers in the bank's financial statements. It depends on models that estimate, for each loan, the probability that the borrower will default, the likely loss if they do, and the exposure at the time of default, all adjusted for forward-looking economic scenarios. It also depends on classifying each loan into stages according to whether its credit risk has increased significantly since it was originated, with stage two and three loans requiring provisions for lifetime losses rather than just twelve months. Last year's external audit identified significant deficiencies in the controls over data feeding these models, and the audit committee has demanded that they be fixed before the year-end audit, nine months away. You are the principal data engineer for finance and risk data, and the remediation is yours to lead.

The audit findings were uncomfortable reading. Auditors could not trace several model inputs back to source systems without manual intervention. The origination credit scores used to assess significant increase in credit risk were missing for about eight per cent of mortgages written before a system migration, and the model filled the gaps with a default value that was not documented. Days-past-due information, a key indicator of credit deterioration, came from two systems that disagreed for a small but material number of loans. Forbearance flags, which identify borrowers who have received concessions because of financial difficulty, were maintained in a spreadsheet by the collections team and uploaded manually each month. Macroeconomic scenario data, such as forecasts of unemployment and house prices, was copied from economists'

spreadsheets into the model environment by hand. And the reconciliation between loan balances in the risk data and balances in the general ledger showed unexplained differences in the small business portfolio.

The chief financial officer, Elizabeth Hart, and the chief risk officer, Dr Samuel Grant, jointly own the remediation. They agree on the goal but approach it differently. Elizabeth's priority is the audit: she wants every audit finding closed with evidence, and she wants the year-end provision process to be demonstrably controlled. Samuel's priority is the models: the Prudential Regulation Authority has also been reviewing banks' IFRS 9 models and data, and he wants a sustainable data foundation that supports model improvements, not just a set of controls bolted onto the existing mess.

The data architecture is the product of history. Cardinal was formed through the merger of a building society and a small business bank a decade ago. Mortgages sit on one core banking system, personal loans and credit cards on another, and business lending on a third. A risk data warehouse extracts data from each system nightly, applies transformations and feeds the credit models. The general ledger receives balances from each system through a separate finance data flow. The transformations in the risk warehouse are implemented in thousands of lines of SQL written by many people over many years, with limited documentation. Some of the most important staging logic lives in a single stored procedure more than two thousand lines long, which has been modified by at least nine different people and has no automated tests. The bank's data governance function has a data catalogue, but lineage is documented only at a high level.

The head of credit risk modelling, Dr Anita Rao, is anxious about the effect of data fixes on model outputs. Correcting the missing origination scores, for example, will change which loans are classified as stage two, which will change provisions. Changing the days-past-due source will reclassify some loans. Each change is a correction, but together they could move the provision by a material amount, which would need to be explained to the audit committee, the auditors and possibly investors. Anita wants every data change assessed for its impact on staging and provisions before implementation, and she wants the changes timed to avoid a disorderly movement at year end.

The collections team, led by Mark Davies, resists replacing the forbearance spreadsheet. The spreadsheet captures nuanced information about arrangements with customers that the core banking systems cannot record, and his team trusts it. He agrees it should be controlled but fears a system change will lose information. The economics team is similarly attached to its spreadsheets, which contain complex adjustments that analysts understand intimately.

Your team comprises four data engineers, a data analyst and a business analyst. The bank's data platform is being modernised by a separate programme that is migrating the risk data warehouse to a cloud platform over the next two years. The modernisation programme's director, Paul Evans, wants the IFRS 9 remediation to be built on the new platform, so that controls are not implemented twice. But the new platform will not be ready for production credit risk workloads before year end.

You see a set of competing paths. You could implement controls on the existing warehouse quickly: automated reconciliations, lineage documentation, data quality checks, and system-controlled uploads for forbearance and scenarios. This would address the audit findings in time but would need to be redone on the new platform. You could accelerate the migration of IFRS 9 data flows to the new platform and build controls there, which would be more sustainable but riskier given the timeline. Or you could implement minimal tactical controls now and plan strategic remediation on the new platform next year, accepting that auditors may view tactical controls as insufficient.

The people involved add their own constraints. Two of the analysts who understand the staging logic best are in the credit risk team rather than your team, and both are needed for the quarterly provision run, which takes most of their time for three weeks of every quarter. Model changes, data changes and the migration programme all compete for them. The external auditors, for their part, have indicated that they will test the remediated controls in the fourth quarter and that controls implemented later than that may not give them enough evidence to rely on. That means the controls must not only exist but must have operated effectively for at least one quarter before year end, which effectively brings the real deadline forward by three months and shortens the time available for design and build to about six months.

There is also the question of lineage. The audit committee has asked for end-to-end lineage from source systems to the provision. Documenting lineage manually for thousands of lines of SQL would take months and would be out of date as soon as it was written. Automated lineage tools can parse SQL and capture lineage from pipelines, but their coverage of the old warehouse's stored procedures is uncertain.

THE DECISION IN FRONT OF YOU

Do you implement tactical controls on the existing warehouse to meet the audit deadline, accelerate migration to the new platform and build controls there, or combine them, and how do you manage the provision impact of data corrections and replace trusted spreadsheets without losing information?

Data Sources

#	Source and link	How it is used in this project
1	IFRS 9 Financial Instruments (IFRS Foundation) https://www.ifrs.org/issued-standards/list-of-standards/ifrs-9-financial-instruments/	The accounting standard defining expected credit loss requirements.
2	BCBS 239 Principles for Risk Data Aggregation https://www.bis.org/publ/bcbs239.htm	Principles for risk data accuracy, completeness, timeliness and lineage.
3	PRA Dear CFO Letters on IFRS 9 https://www.bankofengland.co.uk/prudential-regulation/letter/2024/thematic-feedback-on-ifrs-9	Supervisory feedback on IFRS 9 practices including data and model controls.
4	Home Credit Default Risk Dataset https://www.kaggle.com/c/home-credit-default-risk	Loan-level data used to prototype PD model input pipelines and data quality checks.
5	Lending Club Loan Data https://www.kaggle.com/datasets/wordsofthewise/lending-club	Loan performance data used to test staging and days-past-due logic.
6	Bank of England Database https://www.bankofengland.co.uk/boeapps/database/	Macroeconomic and lending statistics used for scenario inputs.
7	ONS Labour Market and House Price Data https://www.ons.gov.uk/economy	Economic series such as unemployment and house prices for scenario variables.
8	OpenLineage https://openlineage.io	Open standard for capturing lineage from pipelines to support automated lineage evidence.

Deliverables (17)

- D1** **Audit finding remediation plan.** A plan mapping each finding to actions, owners, evidence and target dates.

- D2 Remediation approach decision.** A documented choice between tactical, strategic and combined paths.
- D3 Critical data element inventory.** Identification of all model inputs with owners, sources and quality thresholds.
- D4 Origination score back-fill.** Recovery or justified estimation of missing origination scores with documentation.
- D5 Days-past-due golden source.** Agreed authoritative source and reconciliation between conflicting systems.
- D6 Forbearance data capture solution.** A controlled replacement for the spreadsheet preserving arrangement detail.
- D7 Scenario data ingestion control.** System-controlled loading of macroeconomic scenarios with approval workflow.
- D8 Risk to ledger reconciliation.** Automated reconciliation of loan balances with explanation of breaks.
- D9 Data quality rules and thresholds.** Automated checks on completeness, validity and consistency of model inputs.
- D10 Automated lineage capture.** Lineage from source to provision captured through tooling and supplemented manually.
- D11 Provision impact assessment.** Quantified staging and provision effects of each data correction.
- D12 Change sequencing plan.** Timing of data corrections to manage provision movements and disclosure.
- D13 Control documentation.** Documented controls with evidence suitable for external audit testing.
- D14 Migration alignment design.** A design ensuring controls carry over to the new data platform.
- D15 Management information dashboard.** Data quality and control status reporting to the audit committee.
- D16 Year-end dry run.** A full rehearsal of the controlled provision process ahead of year end.
- D17 Lessons learned and target state.** A post-remediation review defining the sustainable target architecture.

CASE STUDY 52 · CONSUMER LENDING (FINTECH)

Open Banking Transaction Categorisation and Affordability Data Service

Industry	Consumer Lending (Fintech)
Organisation	Kindred Credit, an online lender offering personal loans and credit-builder products to 400,000 customers
Your role	Senior Data Engineer, Decisioning Data
Core themes	Third-party financial data, merchant categorisation, consumer duty, model explainability

The Scenario

Kindred Credit lends to people that traditional banks often turn away: young adults with thin credit files, self-employed workers with irregular incomes, and people rebuilding credit after past difficulties. Its proposition relies on understanding applicants better than a credit score alone can. Since the arrival of open banking, which allows customers to share their bank account transaction data securely with authorised third parties, Kindred has asked applicants to connect their current accounts during the application process. The transactions show income, essential spending, existing debt repayments, gambling, overdraft use and much more. Kindred uses them to verify income, assess affordability and inform credit decisions. You are a senior data engineer in the decisioning data team, and the transaction data pipeline has become a source of both competitive advantage and regulatory worry.

The pipeline works like this. When an applicant consents, an open banking aggregator retrieves up to twelve months of transactions from each connected account and passes them to Kindred. Each transaction has a date, amount, description and sometimes a merchant category code or the aggregator's own category. Kindred's categorisation engine assigns each transaction to categories such as salary, benefits, rent, utilities, groceries, gambling, loan repayments and buy-now-pay-later instalments. From these categories, the affordability engine calculates monthly income, essential expenditure, debt commitments and disposable income. The credit decision combines these with credit bureau data and application information.

Categorisation accuracy is the crux, and it is less accurate than anyone would like. Transaction descriptions vary wildly between banks: the same

supermarket might appear as a clean name at one bank and as a cryptic string of letters, numbers and location codes at another. Salary payments might be described with the employer's name, a payroll bureau's name or simply "BGC". Transfers between the applicant's own accounts look like income at one end and spending at the other unless both accounts are connected. Cash withdrawals hide spending entirely. Payments to peer-to-peer apps could be rent to a flatmate, repayment of a loan to a friend, or money for gambling. The aggregator's categories are general and sometimes wrong. Kindred's own rules-based engine was built three years ago and has been patched repeatedly; it now contains around four thousand rules, some contradicting others.

A recent internal review, prompted by the Financial Conduct Authority's consumer duty, found worrying patterns. For a sample of declined applicants, affordability had been calculated as negative because regular transfers to savings accounts at another bank were categorised as expenditure. For a sample of approved applicants who later fell into arrears, the engine had missed significant buy-now-pay-later commitments because those providers' payments were not recognised. Gambling transactions through certain payment intermediaries were not being detected. The chief risk officer, Dr Laura Benson, is concerned that the categorisation errors are causing both unfair declines and harmful approvals, and that the firm cannot currently measure how often either happens.

The head of product, Jamie Collins, wants to move to a machine learning categorisation model trained on labelled transactions, which he believes will generalise better than rules. He has seen vendors offering categorisation as a service with claimed accuracy above ninety-five per cent. The compliance director, Ruth Adebayo, is cautious. Under the consumer duty and consumer credit rules, Kindred must be able to explain its decisions and demonstrate that its affordability assessments are reasonable and proportionate. A black-box categorisation model feeding a lending decision raises explainability questions, particularly when a customer complains or the Financial Ombudsman Service investigates a declined or unaffordable loan. Ruth also notes that gambling and health-related spending can reveal information about vulnerability, which the firm must handle sensitively.

The data science lead, Dr Omar Farouk, raises a practical point about labels. To train or evaluate any categorisation model, Kindred needs transactions labelled with their true categories. Labelling at scale requires human reviewers, clear definitions and quality checks. Some categories are ambiguous even to humans. And the distribution of transactions changes over time as new merchants, payment methods and fintech providers appear. A categorisation model that is accurate today may degrade within months as new buy-now-pay-later providers, gambling sites and payment apps enter the market, so any approach needs continuous monitoring rather than a one-off evaluation. Omar estimates that a useful ground truth would need several tens of thousands of labelled transactions spread across banks, customer types and time periods, which would take a small team months to produce.

There is also the question of data retention and purpose. Kindred retains transaction data from applications for model development and to defend decisions. The data protection officer, Priya Shah, has questioned whether retaining twelve months of detailed transactions for every applicant, including those who were declined and never became customers, is proportionate. Open banking consent is also time-limited, and the use of data for purposes beyond the application needs a clear lawful basis.

Your team comprises three data engineers and works closely with the data science and risk teams. The aggregator contract is up for renewal, and a competing aggregator offers enriched data with better categorisation and merchant identification, at a higher price. Laura wants a remediation plan within six weeks and measurable improvements in categorisation accuracy within six months.

The customers' perspective is not abstract either. Kindred's complaints data shows that declined applicants frequently contact the firm to say that the affordability figures quoted in their decline letters are wrong: that a large outgoing payment was a one-off car purchase funded from savings, that the income figure ignores regular cash-in-hand earnings, or that a gambling transaction was a single birthday bet. Some of these challenges are valid and some are not, but the firm currently has no efficient way for a customer to see and correct the categorisation of their own transactions. Several competitors have introduced interactive affordability reviews in which applicants confirm their income and major commitments before a decision is made, which both improves accuracy and gives customers a

sense of fairness. Jamie is keen on this idea; Laura worries that it invites applicants to game the process by misrepresenting their spending. Designing a review process that captures genuine corrections without undermining the independence of the assessment is part of the data problem.

You see that the categorisation problem has several layers: data quality from banks and aggregators, entity recognition to identify merchants and counterparties, category assignment, and aggregation into affordability measures. Improving any layer helps, but measuring improvement requires a ground truth that does not yet exist. You also see that switching aggregators, adopting a vendor model or building a machine learning model are each significant changes to a credit decision process that must remain explainable and fair throughout.

THE DECISION IN FRONT OF YOU

Do you refine the rules-based engine, adopt a vendor categorisation service or enriched aggregator data, or build an in-house machine learning model, and how do you build the ground truth, explainability and retention controls needed to show that affordability decisions are accurate and fair?

Data Sources

#	Source and link	How it is used in this project
1	Open Banking Standards (UK) https://standards.openbanking.org.uk	Account and transaction API standards defining the data fields available from banks.
2	Open Banking Limited Read/Write API Specifications https://openbankinguk.github.io/read-write-api-site3/	Detailed specifications for transaction data structures and permissions.
3	Financial Dataset (CTU Relational Repository, Berka) https://relational.fel.cvut.cz/dataset/Financial	Real anonymised bank transaction data used to prototype categorisation and income detection.
4	FCA Consumer Credit Sourcebook (CONC) Creditworthiness Rules https://www.handbook.fca.org.uk/handbook/CONC/5/	Rules on affordability and creditworthiness assessment.
5	FCA Consumer Duty https://www.fca.org.uk/firms/consumer-duty	Outcome requirements informing fairness monitoring and explainability.
6	Financial Ombudsman Service Decisions https://www.financial-ombudsman.org.uk/decisions-case-studies/ombudsman-decisions	Published decisions on irresponsible lending informing categorisation risk areas.

#	Source and link	How it is used in this project
7	ONS Family Spending https://www.ons.gov.uk/peoplepopulationandcommunity/personalandhouseholdfinances/expenditure	National expenditure benchmarks for plausibility checks on essential spending.
8	Plaid Transactions Documentation https://plaid.com/docs/transactions/	Reference for commercial transaction enrichment categories and merchant data.

Deliverables (17)

- D1** **Categorisation error analysis.** A review of declined and defaulted samples quantifying errors by category and impact.
- D2** **Category taxonomy.** A documented taxonomy with definitions and decision rules for ambiguous cases.
- D3** **Labelled ground truth dataset.** A representative, human-labelled transaction set with quality checks.
- D4** **Labelling operations process.** A sustainable process for ongoing labelling of new and ambiguous transactions.
- D5** **Accuracy measurement framework.** Precision and recall by category and bank, tracked over time.
- D6** **Merchant and counterparty recognition.** Normalisation of descriptions to identified merchants and payment intermediaries.
- D7** **Own-account transfer detection.** Logic identifying transfers between an applicant's accounts.
- D8** **Buy-now-pay-later and credit commitment detection.** Recognition of all credit repayment patterns.
- D9** **Gambling detection enhancement.** Detection including gambling through payment intermediaries.
- D10** **Rules engine rationalisation.** Removal of conflicting and obsolete rules with regression testing.
- D11** **Vendor and aggregator evaluation.** A comparison of enriched data and categorisation services against ground truth.
- D12** **Model options assessment.** Evaluation of machine learning categorisation with explainability analysis.
- D13** **Affordability calculation transparency.** Customer and complaint-ready explanations of income and expenditure figures.
- D14** **Fairness and outcome monitoring.** Monitoring of declines and arrears linked to categorisation errors by customer group.
- D15** **Vulnerability data handling policy.** Rules for using gambling and health-related spending sensitively.

- D16 Data retention and purpose review.** Revised retention periods for applicant transaction data.
- D17 Remediation plan and roadmap.** A six-week plan and six-month improvement roadmap for the chief risk officer.

CASE STUDY 53 • ENERGY TRADING AND RENEWABLES

Renewable Generation Forecasting Data Feeds for a Grid-Scale Energy Trader

Industry	Energy Trading and Renewables
Organisation	Westerly Power, an independent renewable generator and trader with 1.6 GW of wind and solar assets in Great Britain
Your role	Lead Data Engineer, Trading Analytics
Core themes	Weather model data at scale, real-time market data, forecast accuracy economics, operational resilience

The Scenario

Westerly Power owns and operates wind farms and solar parks across Great Britain with a combined capacity of about one point six gigawatts, and it sells the electricity they generate through a trading desk that operates around the clock. In the British electricity market, generators contract to deliver specific volumes of power for each half-hour period. If a wind farm produces less than contracted, the generator must pay the imbalance price for the shortfall, which can be extremely high when the system is tight. If it produces more, it receives the imbalance price for the surplus, which can be low or even negative when the system is long. Accurate forecasting of wind and solar output is therefore directly worth money: every percentage point of forecast error costs Westerly substantial sums each year. You lead data engineering in trading analytics, and the forecasting data platform is in trouble.

Westerly's forecasts are produced by combining numerical weather predictions with models of each site's power output. Weather predictions come from several sources: a commercial weather vendor providing site-specific forecasts, global and regional weather models from national meteorological services, and ensemble forecasts that give a range of possible outcomes. Each is updated several times a day, with forecasts extending from hours to days ahead. Site power models translate forecast wind speed, direction, temperature and solar irradiance into expected output, accounting for turbine characteristics, wake effects, curtailment and availability. Actual output is measured by each site's metering and supervisory control systems and reported in near real time.

The current platform was built piecemeal by quants on the trading desk. Weather data is downloaded by scripts running on a server under a desk,

stored in files and read by models written in several languages. Site availability, meaning whether turbines are running or out for maintenance, is recorded by the asset management team in a separate system and copied into a spreadsheet that traders update manually. When the server crashed during a storm last winter, forecasts stopped updating for nine hours, and the desk traded blind during a period of extreme price volatility. The loss that day ran to several hundred thousand pounds. The head of trading, Simon Burke, has made reliability his first priority.

The head of quantitative analysis, Dr Maria Novak, wants more than reliability. She believes forecast accuracy could improve significantly by using higher-resolution weather models, more ensemble members, machine learning models trained on years of site data, and real-time adjustments based on recent actual output. Short-term forecasting, from minutes to a few hours ahead, benefits enormously from real-time data: if a wind farm is producing less than forecast right now, the next hour is likely to be similar. Maria's team has built promising prototypes, but they need a platform that can ingest large weather model outputs, store long histories of forecasts and actuals, and run models on schedule.

The volumes are larger than they look. Global weather models are published as gridded files covering the whole planet at many vertical levels and many forecast steps. Downloading only the relevant region, variables and levels reduces volumes but requires careful handling of formats designed for meteorologists rather than software engineers. Ensemble forecasts multiply volumes by the number of members. Storing a history of every forecast issued, not just the latest, is essential for training and evaluating models, because a model must be trained on forecasts as they looked at the time, not on later revisions. Over several years, this adds up to tens of terabytes.

Market data adds another dimension. System prices, imbalance volumes, national demand and generation by fuel type, and forecasts from the system operator are published by industry data services at half-hourly or finer resolution. Traders use them to anticipate prices and to decide how much forecast risk to hedge. Integrating market data with generation forecasts would let the desk quantify the financial value of each forecast and optimise trading decisions, but the market data services have their own interfaces and occasional outages.

The asset management director, Rachel Evans, has a concern about data ownership. Site metering and supervisory data belongs to her team, which uses it for performance monitoring and warranty claims with turbine manufacturers. Some turbine manufacturers' service agreements restrict how high-resolution operational data can be used or shared. She wants to be consulted on any new use of site data and wants her team's systems protected from additional load.

The chief financial officer, Andrew Stone, wants to know the return on investment. He has seen a business case for a new forecasting platform that promises large savings, but he wants a clear baseline of current forecast accuracy and imbalance costs so that improvements can be measured.

Your team is three data engineers. Maria's team has four quants. The company uses a cloud platform for some corporate systems but not for trading. Simon wants the reliability problem solved before the next winter, six months away, and he has made clear that he would rather have a simple forecast that always arrives than a sophisticated one that sometimes does not. Maria wants her models in production within a year. Commercial forecasting vendors offer complete forecasting services, which some competitors use, and Andrew has asked whether outsourcing forecasting would be cheaper and more reliable than building a platform.

There is a market change on the horizon that affects the design. Reform of the wholesale electricity market has been under consultation for several years, and although the government has decided against the most radical options, changes to settlement periods, balancing arrangements and the treatment of renewable generation could all affect how forecasts translate into money. Westerly's regulatory affairs manager has warned that the imbalance pricing methodology may change again within a few years, and that the company is also considering adding battery storage at several sites, which would turn forecasting from a question of predicting output into a question of optimising when to store and when to export. A platform designed narrowly around today's wind and solar forecasting might need substantial rework if batteries arrive. Simon wants resilience now; Maria wants a research platform; the chief executive wants whatever is built to be ready for storage.

You see a distinction between the reliability problem, which is urgent and largely an engineering discipline issue, and the accuracy opportunity,

which is a longer-term research and platform question. You also see that both depend on the same foundations: robust ingestion of weather and site data, a store of historical forecasts and actuals, and scheduled, monitored model execution with fallbacks. The question is whether to build those foundations quickly and minimally for resilience, or properly for the research ambitions, or to buy forecasts and focus on integration.

THE DECISION IN FRONT OF YOU

Do you outsource forecasting to a commercial vendor, build a resilient in-house forecasting data platform, or first stabilise the existing pipeline for winter while designing the long-term platform, and how do you establish an accuracy and imbalance cost baseline that proves the value of whichever path you choose?

Data Sources

#	Source and link	How it is used in this project
1	Elexon Insights Solution API (BMRS) https://developer.data.elexon.co.uk	System prices, imbalance volumes, generation by fuel type and other GB market data.
2	NESO Data Portal https://www.neso.energy/data-portal	System operator forecasts, demand and embedded generation data.
3	Sheffield Solar PV Live https://www.solar.sheffield.ac.uk/pvlive/	National and regional solar generation estimates for benchmarking solar forecasts.
4	ECMWF Open Data https://www.ecmwf.int/en/forecasts/datasets/open-data	Open global weather forecast data including ensembles in GRIB format.
5	Met Office Weather DataHub https://www.metoffice.gov.uk/services/data/met-office-weather-datahub	UK high-resolution weather model data.
6	Copernicus Climate Data Store (ERA5) https://cds.climate.copernicus.eu	Historical reanalysis for training site power models.
7	ENGIE La Haute Borne Wind Farm Open Data https://opendata-renewables.engie.com	Real SCADA data from wind turbines used to prototype power curve and availability modelling.
8	Renewable Energy Planning Database https://www.gov.uk/government/publications/renewable-energy-planning-database-monthly-extract	Locations and capacities of GB renewable projects for regional context.

Deliverables (17)

- D1 Forecast accuracy and imbalance baseline.** Current forecast error and imbalance costs by site, horizon and weather regime.

- D2 Winter resilience plan.** Immediate actions to remove single points of failure before winter.
- D3 Build, buy and hybrid appraisal.** A comparison of vendor forecasting, in-house platform and hybrid options with costs.
- D4 Weather data ingestion pipeline.** Automated, monitored download and subsetting of weather model outputs.
- D5 Forecast archive.** A store of every issued weather and power forecast with issue time for training.
- D6 Site data integration.** Near-real-time actual output and availability from site systems with minimal load.
- D7 Availability data capture.** Replacement of the manual spreadsheet with a controlled availability feed.
- D8 Market data ingestion.** Integration of system prices, imbalance volumes and operator forecasts.
- D9 Scheduled model execution.** Orchestrated forecast runs with monitoring, retries and version control.
- D10 Fallback forecasting.** Automatic fallback to vendor or simple forecasts if the primary pipeline fails.
- D11 Short-term real-time correction.** Models adjusting near-term forecasts using recent actual output.
- D12 Probabilistic forecast outputs.** Forecast ranges from ensembles to support risk-based trading.
- D13 Forecast value analytics.** Attribution of imbalance costs and savings to forecast performance.
- D14 Data usage agreement with asset management.** Agreed terms for using site data, respecting manufacturer restrictions.
- D15 Monitoring and alerting.** Alerts on stale data, failed runs and large forecast deviations for traders.
- D16 Research environment.** A reproducible environment for quants to train and backtest models on archived data.
- D17 Return on investment tracker.** Ongoing measurement of accuracy improvements and financial benefits.

CASE STUDY 54 · OIL AND GAS (UPSTREAM PRODUCTION)

Offshore Well Sensor Historian Migration and Subsurface Data Platform

Industry	Oil and Gas (Upstream Production)
Organisation	Northgate Energy, a North Sea oil and gas operator with four producing fields and a late-life decommissioning programme
Your role	Lead Data Engineer, Operations Data and Digital
Core themes	Industrial historians, OT/IT convergence, regulatory data retention, late-life asset economics

The Scenario

Northgate Energy operates four oil and gas fields in the central and northern North Sea. Two are mature fields that have been producing for more than thirty years and are approaching the end of their economic lives; one is a mid-life field with a programme of infill drilling; and the newest, a subsea tieback to one of the older platforms, came on stream three years ago. Across these assets, around eighty thousand sensors measure pressures, temperatures, flow rates, valve positions, vibration and dozens of other variables on wells, pipelines, separators, compressors and power generation equipment. Their readings are stored in process historians, specialised time-series databases that have been the backbone of industrial operations for decades. You are the lead data engineer for operations data and digital, and the historians have become the centre of an argument about the company's future.

The immediate trigger is technical. Northgate runs three different historian products, inherited from different field developers and acquisitions. The oldest, installed on one of the mature platforms in the late 1990s, runs on an operating system that is no longer supported and is maintained by a single contractor who is planning to retire. Its data is stored in a proprietary compressed format, and exporting it requires the vendor's tools. The second historian, used on the mid-life field and the tieback, is a modern commercial product with a large licence cost that has just risen sharply at renewal. The third is a smaller product used at the onshore terminal. Each historian names tags differently: the same type of measurement, such as a wellhead pressure, might be called by a cryptic code on one platform and a descriptive name on another. Nobody has a single list of what every tag means.

The operations director, Callum Fraser, wants to consolidate onto one historian, preferably the modern commercial product, and to migrate the old platform's data before its contractor retires. He regards the historians as operational technology: systems that exist to keep the platforms running safely and that should be managed by the control systems engineers who understand them. He is wary of anything that would connect operational networks more closely to corporate IT, because cyber security on offshore installations is a matter of safety, and incidents elsewhere in the energy sector have shown what can happen when attackers reach control systems.

The head of digital, Dr Ingrid Solberg, sees a larger opportunity. She wants a cloud-based operations data platform that ingests historian data alongside maintenance records, production accounting, well test results, subsurface models and inspection data, so that engineers can analyse them together. Her team has built prototypes showing how data-driven models could detect early signs of compressor failure, optimise gas lift in wells and reduce flaring. She argues that the historian choice is secondary: whichever product is used should be treated as a source feeding a broader platform, not as the destination. She also wants to adopt emerging industry data standards, such as the open subsurface data platform that many operators have backed, so that Northgate's data can be exchanged more easily with partners and contractors.

The finance director, Moira Campbell, has the hardest numbers. Two of the four fields are expected to cease production within five to seven years. Every pound spent on digital infrastructure for those fields must pay back before they close. Decommissioning will then take several more years and will generate its own data needs, but the revenue will have stopped. She is sceptical of platform investments with long payback periods and wants any spending tied to specific value: reduced downtime, lower operating costs, reduced emissions or avoided licence fees.

Regulation adds obligations that outlast production. The North Sea Transition Authority requires operators to report production, flaring and venting data, and it has issued expectations on data and information management, including the preservation and publication of well and field data after cessation of production. When fields are sold or decommissioned, data must be handed over to new owners or retained for the regulator. The Health and Safety Executive expects operators to

maintain records relevant to the integrity of safety-critical equipment. And the environmental regulator requires accurate measurement and reporting of emissions, which increasingly means continuous data rather than periodic estimates. Losing decades of sensor data from the mature platforms because a historian could not be migrated would be not only a technical failure but a regulatory problem.

There is also a precedent in the industry that shapes expectations. A major operator released the complete data set from a decommissioned field, including well logs, seismic surveys, production data and real-time drilling data, for research and education. It has been widely used by universities and startups and has become a reference case for what well-managed field data looks like. Ingrid has proposed that Northgate might one day do something similar for its own late-life fields, which would add reputational value and support industry skills, but only if the data is organised well enough to be released.

The control systems engineers, led by a veteran offshore instrument engineer named Duncan Reid, are cautious about moving data off the platforms. The historians sit within carefully segmented networks, and data currently leaves them through a single, tightly controlled link to an onshore replica. Any additional data flows would need to pass through a demilitarised zone and be designed to be strictly one-directional, so that nothing on the corporate or cloud side can send commands or data back into the operational network. Duncan has seen digital projects before that promised much, generated firewall change requests for months and delivered dashboards nobody used. He also points out that offshore bandwidth is limited, especially on the older platforms, which rely on a microwave link and a backup satellite connection.

Data quality is another honest problem. Many sensors on the mature platforms have not been calibrated for years, because the instruments are not safety-critical and maintenance budgets have been cut. Some tags have been flatlined for months because the instrument failed and nobody replaced it. Others were repurposed when equipment was modified, so the tag name no longer describes what is being measured. The historians' compression settings, chosen to save storage decades ago, discard readings that change less than a threshold, which is fine for operators watching trends but can remove the fine detail that data scientists need for vibration and anomaly analysis.

You have a team of three data engineers, access to a cloud platform already used for corporate systems, and a budget that Moira has approved for the historian migration only. Ingrid wants you to use the migration as the first step towards the broader platform. Callum wants the migration done as quickly and safely as possible, with nothing else attached. The old platform's contractor has given notice that he will stop supporting it in eight months.

You believe there is a middle path: treating the migration as a data engineering project rather than purely a historian replacement, building a tag catalogue and context model that make the data understandable, extracting the full history into an open, long-term archive format as well as into the new historian, and designing a secure one-way data flow that could later feed a broader platform without committing to it now. But each of those additions costs time and money, and each could be seen by one side or the other as scope creep.

THE DECISION IN FRONT OF YOU

Do you execute a narrow historian consolidation, build the broader cloud operations data platform now, or design a migration that preserves data and context for future use without committing to the platform, and how do you satisfy cyber security, regulatory retention and late-life economics at the same time?

Data Sources

#	Source and link	How it is used in this project
1	Equinor Volve Field Data Village https://www.equinor.com/energy/volve-data-sharing	Complete open data set from a decommissioned North Sea field used as a reference for data organisation, retention and release.
2	OSDU Forum (Open Subsurface Data Universe) https://osduforum.org	Industry-backed open data platform standard for subsurface and well data.
3	North Sea Transition Authority Data and Reporting https://www.nstauthority.co.uk/data-and-insights/	Regulatory reporting requirements, production and flaring data, and data management expectations.
4	NSTA National Data Repository https://ndr.nstauthority.co.uk	Repository for well and field data submissions and retention obligations.
5	US EIA Petroleum and Natural Gas Data https://www.eia.gov/petroleum/data.php	Production and price data for benchmarking and economic context.

#	Source and link	How it is used in this project
6	ISA/IEC 62443 Industrial Cybersecurity Standards https://www.isa.org/standards-and-publications/isa-standards/isa-iec-62443-series-of-standards	Security standards for operational technology networks and one-way data flows.
7	3W Dataset (Petrobras Oil Well Events) https://github.com/petrobras/3W	Real and simulated well sensor time series with labelled undesirable events for anomaly detection prototyping.
8	Apache Parquet https://parquet.apache.org/docs/	Open columnar archive format for long-term preservation of historian data.

Deliverables (17)

- D1 Historian estate assessment.** Inventory of historians, tag counts, data volumes, retention, compression settings and support risk.
- D2 Migration options appraisal.** A comparison of narrow consolidation, broader platform and preservation-first approaches with costs.
- D3 Tag catalogue and naming standard.** A unified catalogue mapping every tag to equipment, measurement type, units and status.
- D4 Asset context model.** A hierarchy linking tags to wells, systems, equipment and facilities.
- D5 Legacy historian extraction.** Full-history export from the unsupported historian with validation of completeness.
- D6 Long-term open archive.** Preservation of all historian data in an open format with metadata for regulatory retention.
- D7 Consolidated historian migration.** Migration of tags and history into the chosen historian with reconciliation.
- D8 Secure one-way data flow design.** A data diode or equivalent architecture compliant with OT security standards.
- D9 Bandwidth-aware replication.** Data transfer design suited to offshore microwave and satellite links.
- D10 Sensor health and data quality report.** Identification of flatlined, uncalibrated and repurposed tags.
- D11 Compression and resolution policy.** Revised settings preserving high-frequency data for critical equipment.
- D12 Regulatory retention schedule.** Retention requirements for production, emissions, integrity and well data.
- D13 Value case for analytics use cases.** Quantified value of compressor, gas lift and flaring analytics against late-life timelines.
- D14 Decommissioning data handover plan.** Procedures for data transfer to regulators or buyers at cessation of production.

- D15 Data release feasibility study.** An assessment of releasing late-life field data openly, modelled on industry precedent.
- D16 Governance and ownership model.** Clear responsibilities between control systems engineering and digital teams.
- D17 Cutover and contingency plan.** A sequenced migration plan protecting operations and meeting the contractor deadline.

CASE STUDY 55 • MINING

Haul Truck Telematics and Mine Safety Analytics Platform

Industry	Mining
Organisation	Ridgeback Minerals, an operator of two open-pit copper mines with a fleet of 140 haul trucks
Your role	Senior Data Engineer, Mine Technology
Core themes	Fleet telematics integration, safety event analytics, worker monitoring ethics, remote-site infrastructure

The Scenario

Ridgeback Minerals operates two open-pit copper mines in remote locations. At each, enormous haul trucks, each carrying more than two hundred tonnes of ore or waste rock, run continuously along steep, narrow haul roads between shovels at the pit face and crushers or waste dumps. The trucks are the core of the operation's economics: fuel, tyres and maintenance for the fleet make up a large share of operating costs, and every minute a truck spends waiting at a shovel or queuing at the crusher is lost productivity. They are also the core of its safety risk. Collisions between haul trucks and light vehicles, vehicles leaving haul roads, and fatigue-related incidents are among the most serious hazards in surface mining. Last year, a haul truck at Ridgeback's northern mine ran over a pickup truck that had stopped in its blind spot. The pickup driver escaped with minor injuries by jumping clear, but the incident was classed as a potential fatality, and the chief executive, Helen Marsh, has made safety data the company's top technology priority. You are the senior data engineer in mine technology.

The data available is abundant but fragmented. Each haul truck carries several onboard systems: the manufacturer's vehicle health monitoring system, which records engine, transmission, brake and tyre data; a fleet management system that tracks position, load status, cycle times and dispatch assignments; a collision avoidance system that detects nearby vehicles and people and alerts the driver; and, on some trucks, a fatigue monitoring system that uses a camera to detect signs of drowsiness such as closed eyes and head nodding. Light vehicles carry simpler tracking devices. Shovels, drills, graders and water carts each have their own

systems. Each system was bought from a different vendor, at a different time, and each stores its data in its own database or cloud portal.

After the incident, the investigation team spent three weeks assembling data from five systems to reconstruct what had happened. The fleet management system showed the haul truck's position and speed; the collision avoidance system had logged an alert, but its clock was forty seconds out of sync with the fleet management system, so it was initially unclear whether the alert preceded or followed the collision; the pickup's tracker reported position only every thirty seconds; and the fatigue system on that truck had been disabled weeks earlier because of a camera fault that had not been reported. The investigation concluded that the pickup had entered an area it should not have been in, that the collision avoidance alert had been raised but the driver had habituated to frequent false alerts, and that no single system gave supervisors a picture of near misses leading up to the incident.

The general manager of the northern mine, Carlos Mendes, wants an integrated safety analytics platform. He wants every alert from every system, synchronised in time and located on a map, so that the safety team can identify patterns of near misses, high-risk locations on haul roads, drivers with frequent fatigue events and shifts with elevated risk. He also wants proactive monitoring: a live view that shows supervisors when a light vehicle enters a restricted zone or when a truck repeatedly triggers alerts. The collision avoidance vendor has a newer version of its system that supports interaction with the truck's controls, slowing or stopping the vehicle automatically, and regulators in several mining jurisdictions have encouraged or required such systems for large vehicles.

The unions representing the mine workers have a different view. The workforce has accepted collision avoidance and fatigue monitoring as safety measures, but there is deep suspicion that the data will be used to discipline individuals, to monitor productivity or to build cases for dismissal. The union's safety representative, Jo Nkosi, has pointed out that fatigue is often a consequence of rosters, long commutes to remote sites and camp conditions, not of individual failings, and that using fatigue data to blame drivers would discourage reporting and lead workers to disable systems, as apparently happened with the faulty camera. The union wants a formal agreement on how safety data will be used, who can see individual-level data and what protections workers have.

The operations team has its own interest in the same data. The fleet management system's cycle time data shows where trucks wait and where roads are slow, and the mine planning team wants to combine it with vehicle health data to optimise haul routes, reduce fuel consumption and plan maintenance. The maintenance manager, Pieter du Plessis, estimates that tyre failures alone cost several million dollars a year, and that tyre pressure and temperature data combined with haul road condition and payload data could extend tyre life significantly. Carlos sees the productivity and safety uses as complementary; Jo sees the productivity uses as the thin end of a surveillance wedge.

Infrastructure is a hard constraint. The mines are hours from the nearest town. Each has a private wireless network covering the pit, but coverage in deep parts of the pit and on some haul roads is patchy, and bandwidth is shared with dispatch, voice and autonomous drilling systems. Connectivity to the corporate network and the cloud is via a single fibre link at one mine and satellite at the other. Onboard systems buffer data when out of coverage and upload later, sometimes hours later. Any real-time safety monitoring must work locally at the mine, not depend on the cloud.

The regulatory environment is demanding. Mining safety regulators require reporting of serious incidents and near misses, and they expect operators to manage principal hazards through documented controls, including the monitoring of their effectiveness. The regulator's inspectors will want to see how the company detects and responds to near misses, and they may request data after incidents. Data protection law in the mines' jurisdictions also applies to workers' personal data, including biometric data from fatigue cameras, which may be classed as sensitive.

Your team is two data engineers based at head office, plus a mine technology technician at each site. Vendors have offered integrated safety platforms, but each is tied to that vendor's own hardware. Helen wants visible progress before the anniversary of the incident, seven months away, and a sustainable platform within eighteen months.

You believe that the foundational work is time synchronisation and a common event model: every alert, position report and state change from every system, on a common clock, with a common vehicle and person identity, located on a common map of the mine. Without it, neither investigation nor analytics nor live monitoring works reliably. You also believe that the governance agreement with the workforce is not a side

issue but a precondition: without worker trust, systems will be disabled and the data will be worthless.

THE DECISION IN FRONT OF YOU

Do you buy a vendor's integrated safety platform, build a vendor-neutral safety event platform with local real-time monitoring, or start with retrospective near-miss analytics while negotiating a data use agreement, and how do you separate safety uses from productivity uses so that workers trust the systems enough to keep them switched on?

Data Sources

#	Source and link	How it is used in this project
1	MSHA Mine Data Retrieval System https://www.msha.gov/data-and-reports/mine-data-retrieval-system	US mine accident, injury and inspection data used to analyse haulage incident patterns and near-miss categories.
2	MSHA Open Government Data Sets https://arlweb.msha.gov/OpenGovernmentData/OGIMSHA.asp	Downloadable accident and employment data for statistical analysis of haulage risk.
3	USGS Mineral Commodity Summaries (Copper) https://www.usgs.gov/centers/national-minerals-information-center/copper-statistics-and-information	Industry production context for copper mining operations.
4	ICMM Innovation for Cleaner, Safer Vehicles https://www.icmm.com/en-gb/our-work/innovation-for-cleaner-safer-vehicles	Industry programme on collision avoidance and vehicle interaction levels.
5	EMESRT Vehicle Interaction Guidance https://emesrt.org	Industry guidance on vehicle interaction controls and data requirements.
6	ISO 21815 Earth-moving Machinery Collision Warning and Avoidance https://www.iso.org/standard/76553.html	Standard for collision avoidance systems and interfaces between systems and machines.
7	ISO 15143 Worksite Data Exchange (AEMP 2.0) https://www.iso.org/standard/67556.html	Standard telematics data exchange for mixed fleets.
8	Vehicle Energy Dataset (VED) https://github.com/gsoh/VED	Real vehicle telemetry used to prototype ingestion, synchronisation and event extraction.

Deliverables (17)

- D1 Incident data reconstruction review.** A review of the investigation's data problems with root causes.
- D2 Source system inventory.** A catalogue of all onboard and site systems, data fields, clocks, buffering and access methods.
- D3 Time synchronisation standard.** Clock alignment across systems with drift monitoring and correction.

- D4 Common safety event model.** A unified schema for alerts, positions, zone entries and vehicle states.
- D5 Vehicle and person identity register.** Consistent identifiers across trucks, light vehicles, equipment and operators.
- D6 Mine map and zone model.** A spatial model of haul roads, restricted zones and pit areas updated as the pit changes.
- D7 Edge data platform design.** Local processing at each mine for real-time monitoring independent of cloud connectivity.
- D8 Near-miss analytics.** Detection of close-proximity events, repeated alerts and high-risk locations.
- D9 Alert quality and habituation analysis.** Measurement of false alert rates by system and location to reduce alarm fatigue.
- D10 Fatigue data governance.** Rules for fatigue data use focused on roster and system factors rather than individual blame.
- D11 Workforce data use agreement.** A negotiated agreement with unions on purposes, access, retention and protections.
- D12 Safety and productivity data separation.** Technical and organisational separation of safety data from productivity uses.
- D13 Live supervisor dashboard.** Real-time map of vehicles, alerts and zone breaches at each mine.
- D14 Tyre and haul road analytics.** Linking tyre data, payload and road conditions for maintenance optimisation.
- D15 System health monitoring.** Detection of disabled or faulty safety devices with maintenance escalation.
- D16 Regulator reporting pack.** Near-miss trends and control effectiveness evidence for safety regulators.
- D17 Vendor evaluation.** An assessment of integrated safety platforms against vendor-neutral requirements.

CASE STUDY 56 • TELECOMMUNICATIONS

Mobile Network Performance Monitoring and Customer Experience Data Platform

Industry	Telecommunications
Organisation	Vantage Mobile, a UK mobile network operator with 9 million subscribers and 18,000 cell sites
Your role	Principal Data Engineer, Network Analytics
Core themes	Network counters and probes at scale, crowdsourced measurement, coverage obligations, experience metrics

The Scenario

Vantage Mobile is the smallest of the UK's mobile network operators, with around nine million subscribers and eighteen thousand cell sites. It competes on price and customer service, but in independent network tests published by consumer organisations and measurement companies, it consistently ranks last or second to last for data speeds and reliability. The marketing director says this costs the company tens of thousands of customers a year. The network director, Dr Rajesh Patel, disputes the rankings: he believes they reflect sampling in locations where Vantage is weak and that Vantage's own data tells a more nuanced story. The chief executive, Emma Lawson, wants the argument settled with data, and she wants a platform that tells the company, continuously and credibly, what experience its customers actually receive. You are the principal data engineer in network analytics.

A modern mobile network generates data at extraordinary scale. Every cell site produces performance counters every fifteen minutes: thousands of measures of traffic, connection attempts, dropped calls, handovers, throughput, resource utilisation and interference. The core network generates records for every session and call. Probes on network interfaces capture signalling and user-plane data, which can be analysed to estimate the experience of each subscriber, such as video stalls or slow web page loads. Customers' devices can report measurements through the operator's app, with consent. Third-party measurement companies collect crowdsourced tests from millions of devices across all networks, and some publish open data sets. The regulator, Ofcom, publishes coverage and performance data for all operators.

Vantage's network data is managed in silos that reflect its organisational history. Radio counters are collected by the network management systems of the two equipment vendors whose kit Vantage uses, each with its own counter names, definitions and formats. The radio engineering team exports counters to a performance management tool that stores three months of detail. Core network data is in a separate system managed by the core network team. Probe data is analysed by a vendor tool that produces dashboards but does not easily allow raw data export. The customer experience team uses complaint data and app ratings. None of these teams share a common definition of what good performance means.

Rajesh wants a network data lake that consolidates radio, core and probe data with long retention, harmonised counter definitions and the ability to analyse performance at cell, area and subscriber level. His engineers want to use it for capacity planning, fault detection and optimisation, and they are particularly keen on anomaly detection that spots degrading cells before customers complain. The volumes are daunting: probe data alone runs to many terabytes a day, and storing it in full would be extremely expensive.

The customer experience director, Fiona Grant, argues that network metrics are the wrong starting point. Customers do not experience counters; they experience whether a video plays, whether a call connects in a basement, whether their banking app works on the train. She wants experience-centric metrics built from the customer's perspective, and she wants them linked to complaints, churn and customer value so the company can prioritise investment where it matters most to customers. She points out that the crowdsourced measurements used in public rankings measure what customers see, whatever their sampling biases.

The regulatory affairs manager, David Owusu, has another concern. Vantage's spectrum licences include coverage obligations, and the government's rural coverage programme, under which operators committed to extending 4G coverage to more of the country's landmass, has specific targets that are verified using coverage predictions and drive testing. Vantage's coverage predictions, generated by propagation models, have been challenged by rural communities who say the maps overstate coverage. David wants measured data to validate predictions, identify discrepancies and prepare for regulatory scrutiny. He also notes

that Ofcom has been considering how to present more realistic, experience-based coverage information to consumers.

Privacy is a significant issue. Subscriber-level experience data derived from probes and device measurements reveals where people are, what applications they use and when. Telecommunications data is subject to specific privacy regulation in addition to general data protection law, which restricts the use of traffic and location data without consent or anonymisation. Vantage's data protection officer, Nadia Karim, has made clear that subscriber-level analytics must be justified, minimised and, where possible, aggregated or anonymised, and that the use of location data for marketing would require explicit consent. The probe vendor's tool, she notes, currently retains subscriber identifiers for longer than she is comfortable with.

The engineering team you lead has five data engineers. The company has recently adopted a cloud data platform for IT analytics and billing. The network teams are wary of moving network data to the cloud, partly for cost reasons and partly because network data has traditionally stayed within network systems. The equipment vendors each offer analytics platforms that work well with their own kit but not with each other's.

You see several tensions. Engineering wants depth and retention; customer experience wants breadth and customer linkage; regulatory wants credible coverage validation; privacy wants minimisation. The volumes make storing everything impractical, so decisions about what to aggregate, sample or discard at ingestion will determine what questions can be answered later. And the public rankings argument will only be settled if Vantage's own measurements can be reconciled with the crowdsourced data, which requires understanding the biases in both.

The organisation has also learned some lessons the hard way. Two years ago, a software upgrade from one of the equipment vendors changed the definition of a dropped call counter without clear notice, and Vantage's dropped call rate appeared to improve by a third overnight. The improvement was reported to the board and cited in marketing before an engineer noticed that the change coincided exactly with the upgrade. The episode left the executive team suspicious of network statistics and the engineers embarrassed. It also illustrated a deeper problem: counter definitions are controlled by vendors, change with software releases and are documented in lengthy technical manuals that few people read. Any

serious data platform must track counter definitions and software versions over time, flag discontinuities and prevent definitional changes from masquerading as real performance changes. Rajesh accepts this but notes that doing it properly requires continuous engagement with both vendors, who are not always forthcoming about their changes.

There is also a question of pace. Emma wants a credible experience scorecard within three months, before the next round of public rankings is published. A full network data lake would take a year or more. A scorecard built quickly from crowdsourced data and app measurements could be ready in time but would not satisfy the engineers. A scorecard built from network counters would be more complete but slower and would reflect the network's view rather than the customer's.

THE DECISION IN FRONT OF YOU

Do you build an engineering-led network data lake, a customer experience-centric data platform, or a phased design that delivers an experience scorecard first, and how do you decide what to retain, aggregate or discard at ingestion while meeting coverage verification and telecommunications privacy obligations?

Data Sources

#	Source and link	How it is used in this project
1	Ookla Open Data (Speedtest by Ookla Global Performance Tiles) https://github.com/teamookla/ookla-open-data	Open crowdsourced fixed and mobile performance data aggregated into map tiles for benchmarking.
2	Measurement Lab (M-Lab) Data https://www.measurementlab.net/data/	Open internet performance measurement data used to understand crowdsourced measurement methods and biases.
3	OpenCelliD https://opencellid.org	Open database of cell tower locations used for site mapping and coverage context.
4	Ofcom Connected Nations and Mobile Coverage Data https://www.ofcom.org.uk/research-and-data/multi-sector-research/infrastructure-research/connected-nations	Official coverage and performance data for all UK operators.
5	Shared Rural Network Programme https://www.gov.uk/guidance/shared-rural-network	Government programme defining rural coverage obligations and verification.
6	3GPP TS 32.425 and TS 28.552 (Performance Measurements) https://www.3gpp.org/specifications-technologies	Standard definitions of radio performance counters for harmonising vendor data.

#	Source and link	How it is used in this project
7	Privacy and Electronic Communications Regulations (ICO Guidance) https://ico.org.uk/for-organisations/direct-marketing-and-privacy-and-electronic-communications/guide-to-pecr/	Rules on traffic and location data processing in telecoms.
8	Telecom Italia Big Data Challenge Dataset https://dataverse.harvard.edu/dataverse/bigdatachallenge	Real aggregated mobile network activity data used to prototype spatial and temporal analytics.

Deliverables (17)

- D1 Network data landscape assessment.** An inventory of counters, probe data, core records and external measurement sources with volumes.
- D2 Experience metric framework.** Customer-centric definitions of experience such as video, browsing, voice and reliability.
- D3 Counter harmonisation layer.** Mapping of both vendors' counters to standard definitions.
- D4 Ingestion tiering policy.** Decisions on what to retain in full, aggregate, sample or discard at ingestion.
- D5 Radio and core data pipeline.** Scalable ingestion of performance counters and core records with long retention of aggregates.
- D6 Probe data minimisation.** Removal or pseudonymisation of subscriber identifiers with retention limits.
- D7 Crowdsourced data integration.** Ingestion of open and licensed crowdsourced measurements for all operators.
- D8 Measurement bias analysis.** A comparison of crowdsourced, app and network measurements explaining ranking discrepancies.
- D9 Experience scorecard.** A three-month deliverable showing experience by area, time and service.
- D10 Coverage validation analysis.** Comparison of predicted and measured coverage with discrepancy mapping.
- D11 Degrading cell detection.** Anomaly detection on cell performance with alert routing to engineers.
- D12 Complaint and churn linkage.** Aggregated linkage of experience metrics to complaints and churn by area.
- D13 Investment prioritisation model.** Ranking of areas for investment by customer impact and value.
- D14 Privacy impact assessment.** Assessment of subscriber and location data processing under telecoms privacy rules.

- D15 Platform architecture decision.** Cloud, on-premises or hybrid design for network data with cost modelling.
- D16 Regulatory evidence pack.** Measured coverage and performance evidence for Ofcom and rural programme verification.
- D17 Cross-team data ownership model.** Agreed ownership and definitions across radio, core, customer and regulatory teams.

CASE STUDY 57 • SOFTWARE AS A SERVICE

Usage-Based Billing Data Pipeline for a B2B SaaS Platform

Industry	Software as a Service
Organisation	Stackform, a developer platform company offering hosted CI/CD and observability services to 6,500 business customers
Your role	Staff Data Engineer, Billing and Revenue Data
Core themes	Metering accuracy, idempotent event processing, revenue recognition, customer trust in invoices

The Scenario

Stackform sells tools that software teams use to build, test, deploy and monitor their applications. For its first six years, it charged customers a simple monthly fee per user seat, with tiers for different feature sets. Last year, under pressure from investors to grow revenue faster and from customers who wanted to pay for what they used, the company launched usage-based pricing for its two most popular products: build minutes for its continuous integration service, charged by the minute of compute consumed and varying by machine size, and data ingestion for its observability service, charged by the gigabyte of logs, metrics and traces ingested and retained. Usage-based revenue now makes up forty per cent of the total and is growing fast. It has also created a crisis of trust. You are the staff data engineer for billing and revenue data, and the metering pipeline is at the centre of that crisis.

The metering pipeline was built quickly by the product engineering teams responsible for each service. Build minutes are recorded by the build orchestration service, which emits an event when each build job starts and ends. Ingestion volumes are measured by the observability ingestion gateway, which counts bytes per customer per minute. Both services publish usage events to a message queue, from which a metering service aggregates them into hourly usage records per customer, per product and per pricing dimension. At the end of each billing period, the aggregated usage is sent to a third-party billing platform, which applies pricing, discounts and commitments and generates invoices.

Problems have emerged at every stage. Build end events are sometimes lost when a build machine is terminated abruptly, leaving builds that started but never finished; the metering service currently ignores them,

which under-bills customers but also means usage dashboards and invoices disagree. During a message queue incident three months ago, events were redelivered, and the metering service, which was not idempotent, counted some usage twice. Several hundred customers were overcharged, some substantially. Customer success spent weeks issuing credits, and three large customers escalated to their executive sponsors. The observability gateway's byte counts measure compressed data at one point in the pipeline, while the pricing page says customers are charged for uncompressed data, which is higher; nobody is sure which the contracts actually specify. Clock differences between services mean that usage near the end of a billing period sometimes lands in the wrong month.

The chief financial officer, Rachel Kim, has two concerns. The first is revenue recognition. Under accounting standards, Stackform must recognise revenue when it satisfies performance obligations, and for usage-based contracts with committed spend, prepaid credits and overage charges, the timing and amount of revenue depend on accurate usage data. The company is preparing for an audit ahead of a potential public listing in two years, and auditors will test the completeness and accuracy of usage data as a key input to revenue. The second is controls. There is currently no reconciliation between usage recorded by the services, usage aggregated by the metering service and usage invoiced by the billing platform. Rachel wants a system that can prove, for any customer and any period, that what was invoiced equals what was used.

The vice president of customer success, Tom Harding, has a different emphasis. Customers want real-time visibility of their usage, not a surprise at month end. Several have been shocked by large bills after a misconfigured build pipeline ran thousands of unnecessary builds, or after a logging change multiplied ingestion volumes. Tom wants usage dashboards that update within minutes, budget alerts that warn customers before they overspend and the ability to cap usage. Competitors offer all of these, and their absence is costing Stackform deals.

The head of pricing, Ana Ribeiro, wants flexibility. She is planning new pricing models, including committed-use discounts, tiered rates, free allowances that differ by plan and separate pricing for premium machine types and longer data retention. Each change currently requires engineering work in the metering service and configuration in the billing

platform, and the two sometimes disagree about how a rule should be applied. She would like pricing logic to live in one place.

The engineering leadership is divided about architecture. The platform engineering lead, Marcus Webb, wants to build a dedicated usage data platform: an append-only event log of every usage event with unique identifiers, stream processing for real-time aggregation, and batch reconciliation for billing, all owned by a new billing engineering team. The product engineering teams, which own the services that emit usage events, are reluctant to take on stricter event contracts and testing obligations. The finance systems team prefers to buy a specialised usage-based billing product that handles metering, pricing and invoicing together, arguing that billing is not Stackform's core business. Some such products ingest usage events directly and provide customer dashboards, but migrating to one would mean replacing the current billing platform mid-growth.

Your team is four engineers, recently formed. The company processes around two billion usage events a day across both products, a figure that is doubling every year.

You believe the fundamental requirement is that usage events be treated as financial records: uniquely identified, immutable, complete and reconcilable from the moment they are emitted to the moment they appear on an invoice. That requires changes in the services that emit events, which you do not own, as well as in the pipeline, which you do. It also requires clarity on contractual definitions, such as compressed versus uncompressed bytes, which is a legal and commercial question rather than an engineering one.

There is a further complication in the way enterprise contracts are structured. Stackform's largest customers negotiate bespoke terms: committed annual spend with drawdown, custom rates for particular machine types, excluded workloads such as builds triggered by security scans, and in some cases caps written into the contract. These terms are recorded in the customer relationship management system and in signed order forms, but they are configured in the billing platform by hand, by a small revenue operations team, and nobody systematically checks that the configuration matches the contract. An internal spot check of twenty large contracts found configuration errors in four, two in the customer's favour and two against. As usage-based revenue grows, the risk that contract

terms and billing logic diverge grows with it. Any solution needs to treat contract terms as structured data linked to usage and pricing, rather than as free-text notes interpreted by whoever sets up the account.

You are also conscious that the next incident could be more damaging than the last. If a large customer is overcharged again, it could become public, and trust in usage-based pricing, now a strategic pillar, could collapse. On the other hand, a long rebuild that freezes pricing changes would frustrate Ana and slow revenue growth.

THE DECISION IN FRONT OF YOU

Do you build an in-house usage data platform, adopt a specialised usage-based billing product, or harden the existing pipeline while designing the long-term solution, and how do you make usage events auditable financial records, resolve contractual metering definitions and deliver real-time customer visibility without another billing incident?

Data Sources

#	Source and link	How it is used in this project
1	Stripe Billing and Usage-Based Billing Documentation https://docs.stripe.com/billing/subscriptions/usage-based	Reference for meter events, aggregation, pricing models and invoicing APIs.
2	Stripe API Reference https://docs.stripe.com/api	API design patterns including idempotency keys for reliable financial operations.
3	GH Archive https://www.gharchive.org	Public GitHub event stream used as a realistic high-volume event source to test metering, deduplication and aggregation.
4	OpenTelemetry Specification https://opentelemetry.io/docs/specs/	Standard for traces, metrics and logs defining observability usage units.
5	OpenMeter https://openmeter.io	Open-source usage metering reference architecture for event-based billing.
6	IFRS 15 Revenue from Contracts with Customers https://www.ifrs.org/issued-standards/list-of-standards/ifrs-15-revenue-from-contracts-with-customers/	Revenue recognition requirements for usage-based and committed contracts.
7	Apache Kafka Exactly-Once Semantics https://kafka.apache.org/documentation/#semantics	Delivery guarantees relevant to deduplication and idempotent processing.
8	FinOps Foundation FOCUS Specification https://focus.finops.org	Open specification for cloud usage and cost data, useful as a model for usage records.

Deliverables (17)

- D1 Billing incident retrospective.** An analysis of the duplicate usage incident with root causes and customer impact.
- D2 Contractual metering definitions review.** Clarification of billable units such as compressed versus uncompressed bytes with legal and pricing.
- D3 Usage event contract.** A mandatory schema with unique event identifiers, timestamps, customer and product dimensions.
- D4 Emitter conformance tests.** Automated tests that producing services must pass before deploying changes.
- D5 Idempotent ingestion.** Deduplication by event identifier across redeliveries and retries.
- D6 Orphaned build handling.** Detection and billing policy for builds without end events.
- D7 Immutable usage ledger.** An append-only store of all usage events retained for audit.
- D8 Billing period cut-off rules.** Consistent assignment of usage to periods using event time and late data policies.
- D9 Real-time usage aggregation.** Stream processing producing near-real-time usage per customer.
- D10 Customer usage dashboards and alerts.** Data services powering usage views, budget alerts and caps.
- D11 Three-way reconciliation.** Reconciliation of emitted, aggregated and invoiced usage for every customer and period.
- D12 Centralised pricing rules.** A single source of pricing logic shared by metering and invoicing.
- D13 Revenue recognition data feed.** Usage and commitment data supporting revenue recognition and audit testing.
- D14 Build versus buy evaluation.** An assessment of usage-based billing products against in-house platform options.
- D15 Incident response runbook.** Procedures for detecting, halting and correcting billing errors.
- D16 Audit readiness pack.** Controls documentation and evidence for pre-listing auditors.
- D17 Scaling plan.** Capacity planning for doubling event volumes each year.

CASE STUDY 58 • UTILITIES (CUSTOMER SERVICE)

Contact Centre Interaction Analytics and Complaint Root Cause Platform

Industry	Utilities (Customer Service)
Organisation	Brightwater Energy Supply, a UK household energy supplier with 1.3 million customer accounts
Your role	Lead Data Engineer, Customer Operations Analytics
Core themes	Speech and text analytics, complaint regulation, vulnerable customer identification, agent performance ethics

The Scenario

Brightwater Energy Supply sells gas and electricity to around one point three million households. Like every energy supplier in Britain, it has lived through years of turmoil: wholesale price spikes that drove dozens of suppliers out of business, a price cap that limits what can be charged, customers in genuine financial hardship, and a regulator, Ofgem, that has intensified its scrutiny of how suppliers treat customers, particularly those in vulnerable circumstances. Brightwater's customer service operation handles around two hundred and fifty thousand contacts a month across phone, email, web chat and social media. Its complaint volumes are above the industry average, and it was recently named in a regulatory review of suppliers' handling of customers in payment difficulty. You lead data engineering for customer operations analytics, and the chief customer officer, Joanna Hale, has asked you to help her understand why customers are unhappy and what to do about it.

Brightwater's contact data sits in several systems. Phone calls are handled by a cloud contact centre platform that records every call and produces metadata such as queue, wait time, handle time and transfer path. Email and chat are handled by a separate customer service ticketing platform. Social media contacts are managed through a third tool. Complaints are logged in the customer relationship management system when agents categorise a contact as a complaint, which they do inconsistently: some agents log every expression of dissatisfaction as a complaint, as the regulations technically require, while others log only formal complaints. Billing and account data is in the billing system. Smart meter data, payment records and debt collection records are in yet other systems. The customer satisfaction survey is run by an external vendor.

The contact centre platform vendor offers speech analytics: automatic transcription of calls, sentiment detection and topic classification. Joanna is keen to use it. She believes that transcripts would reveal the real reasons customers call, which the agents' disposition codes do not capture, and would allow the company to find complaints that were never logged, identify customers in vulnerable circumstances who mentioned illness, bereavement or financial hardship but were not flagged, and monitor whether agents follow required procedures, such as offering payment plans to customers in debt.

The head of customer operations, Mike Donovan, is more cautious. He manages four hundred agents across two contact centres and a team of home workers, many of whom are on modest pay and under considerable stress. Speech analytics has been used in some companies to score every call automatically and to drive performance management, which agents experience as oppressive and which has been linked to high staff turnover. Mike is not opposed to analytics, but he wants it used to understand and fix problems in processes and systems, not to punish individual agents for the consequences of those problems. He points out that most complaints are caused by billing errors, smart meter faults and confusing letters, not by agents.

The regulatory compliance manager, Priya Sandhu, has specific concerns. Ofgem's rules require suppliers to identify and support customers in vulnerable situations, to record their needs on a priority services register where appropriate and to treat customers in payment difficulty fairly. The complaint handling rules require every expression of dissatisfaction to be recorded and handled. If speech analytics reveals that Brightwater has systematically failed to log complaints or identify vulnerable customers, the company will have to address past failures, potentially including redress. Priya supports the analytics precisely because the regulator expects suppliers to know about such failures, but she wants the implications understood before the results are presented to the board.

The data protection officer, Laura Watts, has concerns of her own. Call recordings and transcripts contain personal data, including sensitive information about health, disability and financial difficulties that customers disclose. Using that information to flag vulnerability is arguably in the customer's interest and supported by regulatory obligations, but it still requires a lawful basis for processing special category data,

transparency with customers and strict controls. Transcripts also contain payment card details when customers pay by phone, which must be excluded under card industry security standards. Retention of recordings is currently set by the contact centre platform's default, which is longer than the company's retention policy.

The technical challenges are substantial. Transcription accuracy varies with accents, background noise and call quality, and the vendor's model has not been tested on Brightwater's customer base, which includes many regional accents and customers for whom English is a second language. Topic classification requires a taxonomy relevant to energy supply, and the vendor's default taxonomy is generic. Linking a call to the customer's account, billing history, meter data and previous contacts requires reliable customer identification, which is complicated because many customers call from numbers not registered on their account. Linking contact reasons to root causes, such as a specific billing error affecting a batch of customers, requires joining contact data with operational data at scale.

Your team comprises three engineers and a data scientist. The contact centre platform's speech analytics module can be switched on for an additional licence fee. Alternatively, recordings could be exported and transcribed using other services, giving more control but requiring more engineering. The billing system is old and difficult to extract from.

Joanna wants to present insights to the board within three months. Priya wants a clear view of regulatory exposure before anything goes to the board. Mike wants agents consulted before anything is deployed. Laura wants a data protection impact assessment before any transcripts are analysed.

There is a commercial undercurrent too. Brightwater's cost to serve is higher than its competitors', largely because of repeat contacts: customers who call about a billing problem, are told it will be fixed, and call again when it is not. Internal estimates suggest that around a third of calls are repeats about an issue already raised. The finance director sees analytics primarily as a way to reduce contact volumes and therefore staff costs, and has already modelled savings that assume a significant reduction in headcount over two years. Mike is aware of this model and it colours his view of the whole programme. If agents believe that analytics exists to replace them, they are unlikely to cooperate with tagging, feedback or process redesign, and some of the most experienced agents, whose

knowledge of the billing system's quirks is invaluable, may leave. Joanna wants the programme framed around better customer outcomes, with cost savings following, but she has not yet persuaded the finance director to present it that way.

You see a way to structure the work around root causes rather than agents: using contacts, of all channels, as a signal of where the company's processes and systems fail customers, joined to the operational data that explains why. That framing might win Mike's support and give Joanna the insights she wants. But it does not remove the need to confront the vulnerability and complaint logging findings, and it does not resolve whether individual-level agent analytics should ever be allowed.

THE DECISION IN FRONT OF YOU

Do you deploy the vendor's speech analytics for rapid insight, build a controlled root cause analytics platform joining contacts to operational data, or begin with a limited regulatory review of complaints and vulnerability, and how do you set boundaries on agent monitoring and handle sensitive customer disclosures lawfully?

Data Sources

#	Source and link	How it is used in this project
1	CFPB Consumer Complaint Database https://www.consumerfinance.gov/data-research/consumer-complaints/	Large open complaint narrative dataset used to prototype complaint classification and root cause taxonomy.
2	FCA Firm Complaints Data https://www.fca.org.uk/data/complaints-data	Published complaint statistics illustrating regulatory complaint reporting structures.
3	Ofgem Consumer Vulnerability Strategy https://www.ofgem.gov.uk/energy-policy-and-regulation/policy-and-regulatory-programmes/consumer-vulnerability-strategy	Regulatory expectations for identifying and supporting vulnerable customers.
4	Ofgem Complaints Handling Standards https://www.ofgem.gov.uk/energy-regulation/domestic-and-non-domestic/energy-supplier-guidance/complaints-handling	Complaint recording and handling requirements for suppliers.
5	Energy Ombudsman Data https://www.energyombudsman.org	Complaint outcomes and trends in the energy sector.
6	Zendesk API Reference https://developer.zendesk.com/api-reference/	Ticketing platform API for extracting email and chat interactions and metadata.
7	OpenAI Whisper https://github.com/openai/whisper	Open-source speech recognition model for evaluating transcription accuracy on regional accents.

#	Source and link	How it is used in this project
8	PCI DSS Document Library https://www.pcisecuritystandards.org/document_library/	Card data security requirements affecting call recording and transcript handling.

Deliverables (17)

- D1 Contact data inventory.** Channels, systems, volumes, metadata and retention settings across the operation.
- D2 Data protection impact assessment.** A DPIA for transcription, vulnerability detection and analytics.
- D3 Payment card redaction.** Detection and removal of card data from recordings and transcripts.
- D4 Transcription accuracy evaluation.** Measured accuracy on representative calls across accents and call quality.
- D5 Energy contact reason taxonomy.** A domain-specific taxonomy of contact reasons and complaint causes.
- D6 Omnichannel contact model.** A unified model of phone, email, chat and social contacts linked to customers.
- D7 Customer identification and linkage.** Methods linking contacts to accounts where phone numbers are unregistered.
- D8 Root cause linkage.** Joining contacts to billing, metering and payment events to identify systemic causes.
- D9 Unlogged complaint detection.** Identification of expressions of dissatisfaction not recorded as complaints.
- D10 Vulnerability signal detection.** Detection of disclosed vulnerability with referral to priority services processes.
- D11 Regulatory exposure assessment.** Quantification of historic complaint and vulnerability failures with redress implications.
- D12 Agent analytics principles.** An agreed policy with agents and managers on permitted and prohibited uses of analytics.
- D13 Process improvement dashboard.** Contact drivers and root causes by volume, trend and customer impact.
- D14 Retention policy alignment.** Recording and transcript retention aligned with company policy.
- D15 Build versus vendor assessment.** A comparison of the vendor module and an exported transcription pipeline.
- D16 Board insight pack.** A summary of findings, root causes and remediation priorities.
- D17 Monitoring and continuous improvement plan.** Ongoing measurement of contact drivers and complaint reductions.

CASE STUDY 59 • COMMERCIAL REAL ESTATE

Smart Building Energy and Occupancy Data Platform for a Commercial Property Portfolio

Industry	Commercial Real Estate
Organisation	Albion Estates, owner and manager of 64 office buildings in London, Manchester and Edinburgh
Your role	Senior Data Engineer, Building Performance
Core themes	Building management system integration, semantic data models, occupancy privacy, net zero reporting

The Scenario

Albion Estates owns sixty-four office buildings, from grand Victorian blocks to glass towers completed in the last decade. It leases space to around four hundred tenants, from global banks to small creative agencies. The pandemic changed how those tenants use their offices: occupancy on a typical day is now around half of what it was before, heavily concentrated on midweek days, while buildings are still heated, cooled, lit and ventilated as if everyone were there. Meanwhile, energy prices have soared, investors demand credible net zero plans, and minimum energy efficiency standards for commercial buildings are tightening. The chief executive, Victoria Hamilton, has committed publicly to making the portfolio net zero in operation by the end of the decade, and she wants data to show where energy is wasted and how buildings are actually used. You are the senior data engineer in the building performance team.

Every modern office building has a building management system that controls heating, ventilation, air conditioning and often lighting. These systems collect data from thousands of points: temperatures, set points, valve and damper positions, fan speeds, chiller outputs and alarms. Albion's buildings use building management systems from six different manufacturers, installed at different times by different contractors. Point naming follows no standard: the supply air temperature of an air handling unit might be called one thing in one building and something entirely different in another, and sometimes the same name means different things. Many systems are old, with limited ability to export data. Energy meters record electricity, gas and heat consumption, some at half-hourly intervals via smart meters and some only monthly. Newer buildings have occupancy

sensors, access control systems and desk booking systems, all of which indicate how many people are present.

The head of sustainability, Dr Rachel Stone, needs data for reporting and for planning investments. She must report energy use and emissions for the portfolio annually, following industry reporting frameworks and investor benchmarks, and must calculate operational energy intensity per square metre. She also needs to prioritise capital investments, such as replacing gas boilers with heat pumps, improving insulation and upgrading controls, based on evidence of each building's performance. She has been working with consultants who produce spreadsheet models from monthly bills.

The head of facilities management, Gary Bennett, manages a team of building engineers and a contracted maintenance provider. He is interested in operational improvements: finding faults such as heating and cooling running simultaneously, valves stuck open, schedules not matching occupancy and sensors out of calibration. Fault detection and diagnostics software exists that can analyse building management data for such faults, and studies suggest that fixing them can reduce energy use significantly at low cost. Gary's concern is that his engineers are already overloaded and that a system generating hundreds of fault alerts would not help unless it was prioritised and linked to maintenance workflows.

The leasing director, Alexander Price, sees occupancy data as commercially valuable. Tenants want to know how their space is used to right-size their leases; Albion could offer occupancy insights as a service and use portfolio-wide data to design flexible workspace products. But occupancy data raises privacy questions. Access control systems record individuals entering the building, and desk booking and Wi-Fi data can reveal where individual employees are. Tenants are employers with their own obligations to their staff, and several have made clear that they do not want their employees' movements tracked by the landlord. The data protection officer, Sanjay Mehta, insists that Albion should process only aggregated, anonymous occupancy data unless there is a clear legal basis and agreement with tenants.

Data ownership in multi-let buildings is complicated. Albion controls the base building systems, such as central plant and common areas, but tenants often control their own floors' fit-out systems, lighting and sometimes their own metering. Energy consumed by tenants may be

recharged through service charges or billed directly. Green lease clauses, which require landlords and tenants to share energy data, exist in some leases but not most. To understand whole-building performance and report emissions, Albion needs tenant data, which it can obtain only with tenants' cooperation.

The technology landscape is crowded. Many smart building platform vendors offer integration of building management systems, analytics and dashboards, each with its own approach to normalising data. Open semantic standards for describing building systems and their relationships have emerged, allowing applications to be written once and applied to many buildings. Albion's IT team, led by Fiona Clarke, is concerned about cyber security: building management systems are often insecure, and connecting them to the internet has led to incidents elsewhere. She wants any integration designed with strict network segmentation.

Your team comprises two data engineers and an energy analyst. The budget allows either a vendor platform subscription across the portfolio or an in-house platform built incrementally with some vendor components. Victoria wants a portfolio energy dashboard within six months and a credible net zero pathway within a year.

You recognise that the hardest part of the problem is not collecting data but making it meaningful: mapping thousands of inconsistently named points to a common semantic model, understanding which meter serves which area and which system, and relating occupancy to energy use. That tagging and modelling work must be done building by building, and it requires knowledge that often lives only with the engineers on site. Whether you buy or build, someone must do it.

Past experience shapes attitudes. Five years ago, Albion commissioned an energy monitoring system for twelve of its buildings. A contractor installed sub-meters and a dashboard, and for a few months the facilities team looked at it regularly. Then the contractor's support lapsed, several meters stopped communicating, and the dashboard showed gaps that nobody investigated. Within two years, the system had been forgotten, and when Rachel's consultants asked for sub-metered data, the only reliable records were the monthly utility bills. Gary remembers the episode as proof that technology without operational ownership decays quickly. He wants any new platform to come with clear responsibilities for keeping data flowing, a process for fixing broken meters and sensors, and a way for his engineers

to see that their fixes produce measurable results. Otherwise, he predicts, the new platform will follow the old one into obscurity once the initial enthusiasm fades.

There is also a sequencing question. The largest energy savings often come from a small number of buildings with the worst performance or the most obvious faults. Starting there would produce quick wins and credibility. But the net zero pathway and investor reporting require portfolio-wide coverage, even at a coarser level, and the dashboard Victoria wants covers everything.

THE DECISION IN FRONT OF YOU

Do you adopt a vendor smart building platform across the portfolio, build an in-house platform on open semantic standards, or combine portfolio-level metering with deep analytics on priority buildings, and how do you obtain tenant energy data and use occupancy information without tracking individuals?

Data Sources

#	Source and link	How it is used in this project
1	Building Data Genome Project 2 https://github.com/buds-lab/building-data-genome-project-2	Open meter data from more than 1,600 non-residential buildings used to prototype benchmarking and anomaly detection.
2	ASHRAE Great Energy Predictor III https://www.kaggle.com/c/ashrae-energy-prediction	Building energy consumption and weather data for baseline modelling.
3	Brick Schema https://brickschema.org	Open semantic model for building systems, points and relationships.
4	Project Haystack https://project-haystack.org	Open tagging standard for building and equipment data.
5	UK Government Energy Performance of Buildings Data (Non-Domestic EPC) https://epc.opendatacommunities.org	Energy performance certificates for commercial buildings used for regulatory compliance tracking.
6	Non-Domestic Private Rented Sector Minimum Energy Efficiency Standard https://www.gov.uk/guidance/non-domestic-private-rented-property-minimum-energy-efficiency-standard-landlord-guidance	Minimum standards affecting investment priorities.
7	CRREM Decarbonisation Pathways https://www.crrem.eu	Science-based carbon pathways for real estate used in net zero planning.
8	GRESB Real Estate Assessment https://www.gresb.com/nl-en/real-estate-assessment/	Investor ESG benchmark defining reporting data requirements.

Deliverables (17)

- D1 Portfolio data inventory.** Building management systems, meters, occupancy sources and data access by building.
- D2 Metering hierarchy model.** Mapping of meters to buildings, areas, systems and tenants.
- D3 Semantic point model.** Mapping of building management points to a common semantic standard.
- D4 Tagging methodology and tooling.** Semi-automated tagging supported by on-site engineer validation.
- D5 Secure integration architecture.** Segmented, read-only data collection from building systems.
- D6 Energy data pipeline.** Ingestion of half-hourly and monthly meter data with gap handling and estimation flags.
- D7 Portfolio energy dashboard.** Energy use, intensity and emissions by building against benchmarks.
- D8 Weather-normalised baselines.** Baseline models for measuring savings and comparing buildings.
- D9 Fault detection and diagnostics.** Rules and analytics for common faults in priority buildings.
- D10 Fault prioritisation and work order integration.** Ranking of faults by energy and comfort impact linked to maintenance systems.
- D11 Aggregated occupancy model.** Anonymous occupancy counts by building, floor and time.
- D12 Occupancy privacy framework.** Rules agreed with tenants on occupancy data collection and use.
- D13 Tenant data sharing programme.** Green lease clauses and voluntary data sharing agreements with tenants.
- D14 Net zero pathway model.** Building-level pathways combining performance data, investments and carbon targets.
- D15 Investor and regulatory reporting feed.** Data for annual sustainability reports and investor benchmarks.
- D16 Vendor versus in-house evaluation.** A comparison of platform options on cost, openness and capability.
- D17 Priority building programme.** A plan for deep analytics and quick-win savings in the worst-performing buildings.

CASE STUDY 60 · AVIATION MAINTENANCE, REPAIR AND OVERHAUL

Aircraft Maintenance Records Digitisation and Reliability Analytics

Industry	Aviation Maintenance, Repair and Overhaul
Organisation	Skyward Technics, an MRO provider maintaining regional turboprop and jet fleets for six airlines
Your role	Principal Data Engineer, Engineering Data Services
Core themes	Airworthiness records integrity, unstructured maintenance text, reliability programmes, regulatory audit

The Scenario

Skyward Technics maintains aircraft. From hangars at two regional airports, its engineers carry out scheduled checks, repair defects, replace components and embody modifications on around one hundred and twenty regional turboprops and small jets operated by six airlines. Every task performed on an aircraft must be recorded: what was done, by whom, using which parts, according to which approved data, and certified by an appropriately licensed engineer. These records are not paperwork in the ordinary sense; they are the legal proof of an aircraft's airworthiness, and they must be retained, often for the life of the aircraft and beyond. You are the principal data engineer in engineering data services, and Skyward's records have become both its greatest liability and a potential source of new business.

Much of Skyward's recordkeeping is still on paper. Task cards are printed for each check, filled in by hand, stamped and signed by engineers, and then scanned into a document management system. Defects found during checks are written on non-routine cards in free text, such as "found corrosion on lower wing skin at station 425, blended out within limits per SRM 57-10-05". Parts are recorded by part and serial number, often handwritten. The airlines' own maintenance information systems hold the aircraft's configuration, maintenance programme, flight hours and cycles, and deferred defects, and Skyward receives work packages from them and returns completed records. Each airline uses a different system. Component shops that repair parts send certificates back with the parts.

Two recent events have focused attention. First, during a lease return, when an aircraft was being returned from one of Skyward's airline customers to its owning lessor, the lessor's records auditors found gaps

and inconsistencies in the back-to-birth traceability of several life-limited parts, such as landing gear components and engine discs, whose remaining life depends on complete records of every cycle since manufacture. The aircraft could not be accepted until the records were reconstructed, which took eleven weeks and cost the airline a large sum in penalties and lost lease revenue. Some of the missing records had been Skyward's responsibility. Second, the aviation regulator's audit of Skyward's approval found that the organisation's system for analysing recurring defects was weak: engineers were repeatedly fixing the same defects on the same aircraft without the root cause being identified, which is exactly what a reliability programme should prevent.

The accountable manager, Richard Holmes, who carries personal responsibility for the organisation's compliance, wants both problems fixed. The commercial director, Sophie Ward, sees opportunity as well as risk. Lessors and airlines increasingly want digital records: complete, searchable, verified records that make aircraft transitions faster and cheaper. An MRO that could offer digital records and reliability insights as part of its service could win contracts. Several technology companies offer digital records platforms for aviation, some using optical character recognition and language models to extract structured data from scanned records.

The quality manager, Dr Kavita Rao, is cautious. Airworthiness records are legally significant. If a digital record is created by extracting data from a scan, which record is authoritative, the scan or the extracted data? If extraction makes an error, such as misreading a serial number, and that error propagates into a lease return or a life-limited part calculation, the consequences could be serious. She insists that any digitisation must be verified, that the original signed records remain the authoritative source unless and until the regulator accepts electronic records, and that electronic signatures, if introduced, must meet the regulator's requirements for authentication and integrity.

The chief engineer, Tom Fletcher, is most interested in reliability. He wants to analyse defect text across all aircraft to find recurring problems, correlate them with components, environmental conditions and maintenance actions, and feed findings into the airlines' reliability programmes and the manufacturers' service data. The free text of defect reports, written by hundreds of engineers in abbreviated technical

language, is rich but difficult to analyse. Standard chapter coding of defects exists across the industry, but engineers apply it inconsistently.

The airlines themselves are a complication. They own their aircraft's records, and Skyward holds records on their behalf. Some airlines are enthusiastic about digital records and analytics; others are protective of their data and suspicious of an MRO building analytics across multiple airlines' fleets. Contracts are silent on whether Skyward may use aggregated defect data across customers for reliability analysis. One airline's technical director has said informally that he would welcome fleet comparisons as long as his airline could not be identified; another has said that any use of its data beyond performing the contracted work would be a breach of confidence. Manufacturers, for their part, would value aggregated in-service defect data for type certificate holders' continuing airworthiness responsibilities, and have offered data sharing arrangements that could benefit Skyward commercially.

Your team is three engineers and a technical records specialist, Brian Doyle, who has worked with aircraft records for thirty years and can read a handwritten task card that would baffle anyone else. The document management system holds around four million scanned pages. The hangars have tablets for some tasks but most work is still on paper.

The workforce dimension deserves attention. Skyward's licensed engineers are proud professionals whose signatures carry legal weight, and many have worked with paper records for decades. Some are enthusiastic about tablets; others see electronic task cards as an imposition that will slow them down, especially when working in cramped spaces inside wings or engine nacelles, wearing gloves, in hangars where wireless coverage is unreliable. Previous attempts to introduce tablets for some tasks produced mixed results: engineers used them for reading manuals but continued to record work on paper because the software required too many screens and did not work offline. Brian points out that records quality depends on the people writing them at the end of a long night shift, and that any digital system must make correct recording easier than incorrect recording, not merely possible. He also notes that apprentices joining the company expect digital tools and find the paper system baffling, which is becoming a recruitment issue.

You face two linked but distinct challenges. The backward-looking one is making existing records usable: extracting and verifying key data from

millions of scanned pages, prioritising those that matter most, such as life-limited parts and airworthiness directive compliance. The forward-looking one is changing how records are created so that they are digital, structured and verified at the source, which requires electronic task cards, signatures and integration with airline systems. Doing the first without the second means digitising paper forever. Doing the second without the first leaves the historical liability unaddressed.

THE DECISION IN FRONT OF YOU

Do you prioritise extracting and verifying historical records for high-risk items, introducing digital task cards and electronic signatures for new work, or building defect reliability analytics, and how do you ensure digitised records remain legally authoritative and use multi-airline data only with appropriate permission?

Data Sources

#	Source and link	How it is used in this project
1	FAA Service Difficulty Reporting System https://sdrs.faa.gov	Real defect reports with free-text descriptions and ATA chapter codes used to prototype defect text classification.
2	NASA Aviation Safety Reporting System https://asrs.arc.nasa.gov	Voluntary safety reports used to study free-text analysis of maintenance-related events.
3	NASA Open Data Portal (Turbofan Degradation Simulation) https://data.nasa.gov	Engine degradation data used for reliability and remaining life analytics prototyping.
4	FAA Airworthiness Directives https://drs.faa.gov	Airworthiness directive data for compliance tracking.
5	EASA Part-145 and Part-CAMO Regulations https://www.easa.europa.eu/en/regulations	Requirements for maintenance organisations, records and reliability programmes.
6	UK CAA Continuing Airworthiness Guidance https://www.caa.co.uk/commercial-industry/aircraft/airworthiness/	UK regulator guidance including electronic records and signatures.
7	ATA Spec 2000 https://publications.airlines.org/collections/spec-2000	Industry standard for aviation data exchange including records and reliability data.
8	OCRmyPDF https://ocrmypdf.readthedocs.io	Open-source OCR for scanned maintenance records.

Deliverables (17)

- D1 Records risk assessment.** Identification of record types and aircraft with the highest airworthiness and commercial risk.

- D2 Lease return lessons learned.** An analysis of record gaps in the recent transition and their causes.
- D3 Life-limited part record extraction.** Extraction and verification of part histories from scanned records.
- D4 Airworthiness directive compliance dataset.** Structured compliance records linked to aircraft and components.
- D5 Verification workflow.** Human verification of extracted data with dual checks for critical fields.
- D6 Authoritative record policy.** Rules defining the status of scanned, extracted and electronic records.
- D7 Electronic task card pilot.** A pilot of digital task cards in one hangar with usability evaluation.
- D8 Electronic signature design.** Authentication and integrity controls meeting regulatory requirements.
- D9 Airline system integration.** Structured exchange of work packages and completed records with airline systems.
- D10 Defect text normalisation.** Expansion of abbreviations and standardisation of technical language.
- D11 Defect classification model.** Assignment of ATA chapters and defect types to free-text reports.
- D12 Recurring defect detection.** Identification of repeat defects by aircraft, component and location.
- D13 Reliability programme reporting.** Reports supporting airline reliability programmes and regulator expectations.
- D14 Customer data use agreements.** Contract terms for aggregated multi-airline reliability analysis.
- D15 Digital records service proposition.** A commercial offer for lessors and airlines based on verified digital records.
- D16 Records retention and preservation.** Long-term storage, integrity checking and retrieval of records.
- D17 Regulator engagement plan.** A plan for approval of electronic records and responses to audit findings.

CASE STUDY 61 · RAIL TRANSPORT

Railway Punctuality and Delay Attribution Data Platform

Industry	Rail Transport
Organisation	Pennine Rail, a train operating company running 1,100 daily services across northern England
Your role	Lead Data Engineer, Performance and Planning
Core themes	Real-time train movement feeds, delay attribution disputes, passenger-centric metrics, industry data sharing

The Scenario

Pennine Rail runs around eleven hundred passenger trains a day across northern England, on routes ranging from busy commuter lines into Leeds and Manchester to long rural lines through the hills. Its punctuality is poor, even by recent industry standards, and passengers let it know: complaints, social media anger and pointed questions from local politicians follow every bad week. The company operates under a national contract with the government that includes performance incentives and penalties, and it shares the railway with other operators and with the infrastructure manager, which owns and maintains the track, signals and stations. You are the lead data engineer in the performance and planning team, and your job is to help the company understand why its trains are late and what it can do about it.

Train movement data in Britain is remarkably rich. The infrastructure manager's systems record when each train passes signalling points, stations and junctions, producing a stream of movement events that is available in near real time through industry data feeds. Timetable data defines where each train should be and when. The national passenger information system combines movements and timetables to produce live departure information. Delay attribution, the industry's formal process for assigning responsibility for each minute of delay, is recorded in a separate system, where delays above a threshold are investigated and attributed to causes, such as a signal failure, a train fault, late crew, a trespass incident or congestion caused by another operator's train. Attribution matters financially: under the performance regime, parties responsible for delay pay compensation to those affected.

Pennine's internal data is fragmented. Train crew rosters and actual crew assignments are in the crew management system. Fleet availability, faults and maintenance are in the fleet engineering system. Passenger loadings are estimated from ticket data and automatic passenger counting equipment on some trains. The control room, which manages disruption in real time, keeps logs in a separate incident system. The performance team uses a combination of industry reporting tools and spreadsheets, and analysts spend much of their time producing weekly performance reports rather than analysing causes.

The performance director, Andrew Kelly, has a specific frustration. He believes Pennine is being attributed delay that is not its fault. Attribution is negotiated between operators and the infrastructure manager, often under time pressure, and disputes can take weeks. Pennine's attribution team lacks the data to challenge attributions effectively: they need to reconstruct what happened, when, and in what sequence, combining movement data, crew and fleet records and control logs. He wants a platform that makes this reconstruction fast and evidence-based. He estimates that better attribution could be worth significant sums each year.

The head of operations planning, Michelle Adams, has a different priority. She wants to understand structural causes of delay so the timetable can be made more robust. Some delays are caused by timetables that are simply not achievable: insufficient time for passengers to board at busy stations, tight turnarounds at terminals, conflicts at junctions. She wants to analyse historic running times at every point on the network to find where the timetable is unrealistic, and to test proposed timetable changes against historic data. That requires detailed movement data at signalling-point level over several years.

The customer experience director, Jennifer Walsh, points out that the industry's standard punctuality measures do not reflect what passengers experience. A train that runs a few minutes late at every station but arrives on time at its destination may count as punctual. A cancelled train counts in some measures but not in the way passengers feel it. Crowding matters as much as lateness. The industry has moved towards measuring punctuality at every station stop, but Jennifer wants to go further: measuring passenger-weighted journey delay, combining lateness with

how many passengers were affected, and linking to compensation claims under the delay repay scheme, which are processed by a separate system.

There are industry constraints. Some data feeds are open, published under licences that allow reuse, while others are available only to industry parties under data sharing agreements. The industry has been working towards a more open and integrated data landscape, and a new industry data marketplace has been launched to make feeds easier to access. The infrastructure manager is being restructured into a new public body that will bring together track and train, and its data systems and governance may change. Any platform Pennine builds must be adaptable to changes in the industry's data architecture over the next few years.

Your team comprises four engineers and three analysts. The company has a small cloud data platform used for commercial and revenue data. Real-time movement feeds are received by a message subscription that was set up years ago by a contractor and forwards data to a database that fills up and crashes every few months. Each crash loses several hours of movement data, and nobody has ever gone back to fill the gaps, so the historical record the planners want to analyse is riddled with holes.

The human side of performance management matters too. Pennine's train crew and control staff are acutely conscious of how performance data is used. Drivers' unions have resisted the use of on-train data recorders for anything other than safety investigations, and there is suspicion that detailed movement analysis could be used to blame individual drivers for slow running or late departures. Station dispatch staff worry about being blamed for delays caused by passengers boarding late or by crowding that the timetable does not allow for. Andrew wants the platform to support improvement, not blame, and has promised the unions that it will focus on systems and processes, but he also knows that some delays are caused by avoidable individual actions, such as crew arriving late for duty. Designing analytics that respect these sensitivities while remaining honest about causes will require agreement with staff representatives on how data is aggregated and reported.

You see a need to consolidate: one reliable ingestion of movement, timetable and incident feeds, joined with internal crew, fleet, passenger and control data, with long-term retention at detailed level. On that foundation, attribution reconstruction, timetable analysis and passenger-

centric metrics are all possible. But each stakeholder wants their use case first, and the attribution opportunity has the clearest financial case.

You are also aware that attribution analysis, done aggressively, could damage relationships with the infrastructure manager and other operators, with whom Pennine must cooperate daily in running a shared railway. The industry's culture is shifting from blame towards joint performance improvement. A platform that makes Pennine better at winning disputes might win money but lose goodwill, whereas one that helps all parties understand delays might improve performance for everyone.

THE DECISION IN FRONT OF YOU

Do you prioritise delay attribution evidence, timetable robustness analysis or passenger-centric performance measurement, and how do you build a movement data platform that is reliable, retains detailed history and adapts to industry restructuring while supporting collaborative rather than adversarial performance management?

Data Sources

#	Source and link	How it is used in this project
1	Network Rail Open Data Feeds https://publicdatafeeds.networkrail.co.uk	Real-time train movements, TD berth steps, schedules and VSTP data used for movement ingestion.
2	National Rail Darwin Data Feeds https://www.nationalrail.co.uk/developers/	Live running information and forecasts for passenger-facing performance measures.
3	Rail Data Marketplace https://raildata.org.uk	Industry marketplace for rail data feeds including performance and timetable data.
4	ORR Data Portal https://dataportal.orr.gov.uk	Official statistics on punctuality, cancellations, delay causes and passenger journeys.
5	Delay Attribution Principles and Rules https://www.delayattributionboard.co.uk	Industry rules governing attribution of delay minutes.
6	Great British Railways Transition https://www.gov.uk/government/collections/great-british-railways	Industry restructuring affecting future data architecture and governance.
7	Open Rail Data Wiki https://wiki.openraildata.com	Community documentation for rail data feeds, codes and formats.
8	UK Rail Timetable (CIF) Documentation https://wiki.openraildata.com/index.php/CIF_File_Format	Format specification for timetable data.

Deliverables (17)

- D1 Use case prioritisation.** A scored comparison of attribution, timetable and passenger metric use cases.
- D2 Feed ingestion rebuild.** Resilient subscription and ingestion of movement, TD and schedule feeds with monitoring.
- D3 Movement history store.** Long-term detailed retention of train movements at signalling-point level.
- D4 Timetable model.** Structured schedules including planned timings at every point.
- D5 Train service reconstruction.** Joined timelines of each service combining movements, timetable and incidents.
- D6 Internal data integration.** Crew, fleet, passenger count and control log data linked to services.
- D7 Attribution evidence workbench.** Tools reconstructing incident sequences to support attribution decisions.
- D8 Attribution accuracy analysis.** Analysis of historic attributions and disputes to identify misattribution patterns.
- D9 Running time analysis.** Historic sectional running and dwell times against the timetable.
- D10 Timetable robustness testing.** Simulation of proposed timetable changes against historic performance.
- D11 Passenger-weighted delay metric.** A measure combining lateness, cancellations and passenger numbers.
- D12 Delay repay linkage.** Linking compensation claims to service performance.
- D13 Weekly performance automation.** Automated weekly reports replacing manual analyst work.
- D14 Joint performance data sharing.** A proposal for shared analysis with the infrastructure manager and other operators.
- D15 Industry change adaptability plan.** An architecture that accommodates industry restructuring and new data sources.
- D16 Data licence compliance.** Documentation of licence terms for each feed and permitted uses.
- D17 Performance insight dashboard.** Views of delay causes, hotspots and trends for managers.

CASE STUDY 62 • BANKING (CORPORATE AND COMMERCIAL)

Corporate Customer KYC and Beneficial Ownership Data Platform

Industry	Banking (Corporate and Commercial)
Organisation	Harcourt Commercial Bank, serving 48,000 business customers from sole traders to mid-sized corporate groups
Your role	Lead Data Engineer, Financial Crime Data
Core themes	Entity resolution against public registers, ownership graph construction, periodic review automation, sanctions ownership rules

The Scenario

Harcourt Commercial Bank provides accounts, lending, payments and trade finance to around forty-eight thousand businesses. Before it can open an account for a business, and periodically thereafter, the bank must understand who it is dealing with: the legal entity, its directors, its ultimate beneficial owners, the nature of its business and the source of its funds. This process, known as know your customer, is a core requirement of anti-money laundering regulation. It is also the bane of the bank's business customers, who complain that onboarding takes weeks and that periodic reviews bombard them with requests for documents the bank already has. You lead data engineering for financial crime data, and the bank has just received a letter from the Financial Conduct Authority that has transformed a nagging problem into an urgent one.

The letter followed a supervisory visit. The regulator found that a significant proportion of the bank's corporate customer files were incomplete or out of date: beneficial ownership information had not been verified, periodic reviews were overdue for around eleven thousand customers, and in several cases, customers' ownership structures had changed since onboarding without the bank noticing. In one case, a customer's controlling shareholder had become the subject of sanctions after the bank last reviewed the file, and the bank had continued to process payments for the business for four months. The bank reported the breach, froze the account and is now under investigation. The regulator has required the bank to remediate its backlog of reviews within eighteen months and to demonstrate a sustainable approach to keeping customer information current.

The bank's know-your-customer data sits in a case management system where analysts record information collected during onboarding and reviews, with documents attached as scanned files. Ownership structures are drawn as diagrams in documents rather than recorded as structured data. Customer entities are recorded with names and registration numbers, but registration numbers are missing for older customers, and overseas entities are identified inconsistently. The core banking system holds account data with its own customer records, which do not always match the case management records. There is no systematic monitoring of changes in customers' ownership or directors between reviews.

The head of financial crime, Dr Miriam Shaw, has a clear vision: perpetual know your customer. Instead of reviewing every customer on a fixed cycle, the bank would continuously monitor public registers and other sources for changes to customers' ownership, directors, addresses, filings and adverse media, and trigger reviews only when something material changes or when risk indicators suggest attention is needed. Public company registers in the UK and many other jurisdictions now publish structured, machine-readable data on companies, officers and persons with significant control. Global legal entity identifiers provide a standard way to identify entities and their parent relationships. Open data projects consolidate sanctions lists and ownership information from many countries. Miriam believes that with this data, much of the review backlog could be cleared automatically for low-risk customers whose information can be verified against authoritative sources.

The chief operating officer, Robert Chen, is under pressure from the business to reduce onboarding times and customer friction. He supports automation but is concerned about cost: the remediation programme alone will require dozens of additional analysts if done manually. He wants technology to reduce the manual effort, but he is also wary of a large platform build that might not be ready in time to meet the regulator's deadline.

The compliance risk director, Alan Pierce, sounds a note of caution. Public registers are not always accurate. UK company register data is largely self-reported and, until recently, was not verified; reforms are introducing identity verification for directors and persons with significant control, but historic data contains errors, false entries and missing information. Overseas registers vary widely in quality and accessibility, and some

jurisdictions do not publish beneficial ownership at all. Automating verification against public data could create false assurance. Alan also points out that sanctions regimes apply not only to named persons but to entities they own or control, directly or indirectly, above certain thresholds, which requires calculating aggregated ownership through complex, multi-layered structures, a task that public data alone may not support reliably.

The data engineering challenges are formidable. Matching the bank's forty-eight thousand customers to entities in public registers requires entity resolution across inconsistent names, addresses and identifiers, particularly for older and overseas customers. Building ownership graphs requires linking entities to their shareholders and controlling persons, who may themselves be entities in other jurisdictions, recursively. Monitoring requires regular ingestion of register updates and change detection. Sanctions ownership calculations require traversing graphs and aggregating ownership percentages, with careful handling of incomplete data. And every automated decision, such as clearing a review, must be explainable and auditable.

Your team has five engineers and two analysts. The bank uses a commercial screening vendor for sanctions and adverse media, and several vendors offer know-your-customer data platforms that consolidate registry data and ownership structures. The remediation programme has budget for analysts and technology but not unlimited funds.

You see three paths. You could support the remediation programme with targeted tooling, such as automated pre-population of review cases with register data, while building the perpetual monitoring platform in parallel. You could buy a vendor platform to deliver both quickly. Or you could build an in-house entity and ownership graph that becomes the bank's authoritative view of corporate customers, feeding both remediation and ongoing monitoring. Each has different implications for the regulator's deadline, the bank's long-term capability and the risk of automation errors.

The sanctions breach casts a long shadow. The regulator will want to see not only that the backlog is cleared but that the bank can detect, within days, when a customer's owner becomes sanctioned. That capability depends on knowing customers' ownership structures accurately and on monitoring both sanctions designations and ownership changes

continuously. It is the single most important outcome, and it depends on the data foundations more than on any other part of the programme.

There is also the matter of customers themselves. The bank's business customers are tired of repeated document requests, and relationship managers report that some have moved their banking elsewhere after particularly burdensome reviews. A system that uses public data to pre-fill and verify information would reduce that burden, but only if the bank is confident enough in the data to stop asking customers for things it can find itself. If automation is too cautious, customers still receive requests; if it is too trusting, the bank repeats its compliance failures. Getting the calibration right, and being able to explain it to the regulator, is at the heart of the problem.

THE DECISION IN FRONT OF YOU

Do you build targeted remediation tooling, buy a vendor KYC data platform or build an in-house entity and ownership graph for perpetual monitoring, and how do you calibrate automated verification against imperfect public registers while guaranteeing timely detection of sanctioned ownership?

Data Sources

#	Source and link	How it is used in this project
1	Companies House Public Data API https://developer.company-information.service.gov.uk	UK company profiles, officers, filings and persons with significant control for matching and monitoring.
2	Companies House Bulk Data Products https://download.companieshouse.gov.uk/en_output.html	Bulk company and PSC snapshots for initial matching and graph construction.
3	GLEIF LEI Data https://www.gleif.org/en/lei-data	Global legal entity identifiers with level 2 parent relationship data.
4	OpenSanctions https://www.opensanctions.org	Consolidated sanctions, politically exposed persons and related entity data.
5	Open Ownership Register and Beneficial Ownership Data Standard https://www.openownership.org	Data standard and resources for representing beneficial ownership structures.
6	OFSI Consolidated List of Financial Sanctions Targets https://www.gov.uk/government/publications/financial-sanctions-consolidated-list-of-targets	UK sanctions list and ownership and control guidance.

#	Source and link	How it is used in this project
7	FATF Guidance on Beneficial Ownership of Legal Persons https://www.fatf-gafi.org/en/publications/Fatfrecommendations/Guidance-Beneficial-Ownership-Legal-Persons.html	International standards on beneficial ownership verification.
8	Splink Record Linkage https://moj-analytical-services.github.io/splink/	Probabilistic linkage for matching customers to register entities.

Deliverables (17)

- D1 Regulatory remediation data plan.** A plan aligning data work with the regulator's eighteen-month deadline.
- D2 Customer data quality baseline.** Completeness and accuracy of entity, ownership and review data across customers.
- D3 Register entity matching.** Probabilistic matching of customers to company registers and LEIs with confidence scores.
- D4 Structured ownership model.** Conversion of ownership diagrams into structured, versioned ownership data.
- D5 Ownership graph.** A multi-jurisdiction graph of entities, persons and control relationships.
- D6 Aggregated ownership calculation.** Traversal and aggregation of direct and indirect ownership percentages.
- D7 Sanctions ownership screening.** Detection of customers owned or controlled by sanctioned persons above thresholds.
- D8 Register change monitoring.** Daily ingestion of register updates with detection of material changes.
- D9 Event-driven review triggers.** Rules triggering reviews on ownership, officer, filing and sanctions events.
- D10 Review pre-population.** Automatic pre-filling of review cases with verified register data.
- D11 Automated clearance rules.** Criteria for clearing low-risk reviews with audit trails and sampling checks.
- D12 Register data reliability assessment.** Evaluation of register accuracy by jurisdiction and field.
- D13 Vendor platform evaluation.** An assessment of KYC data vendors against in-house requirements.
- D14 Explainability and audit trail.** Recording of data sources, rules and decisions for every review outcome.
- D15 Customer request reduction metrics.** Measurement of document requests avoided through verified data.

- D16 Regulator progress reporting.** Regular evidence of backlog reduction and monitoring capability.
- D17 Operating model for perpetual KYC.** Roles, processes and quality assurance for continuous monitoring.

CASE STUDY 63 • WEALTH MANAGEMENT

Wealth Management Portfolio Consolidation and Instrument Reference Data

Industry	Wealth Management
Organisation	Kingsley Rowe Wealth, a wealth manager overseeing £9.4 billion for 12,000 private clients
Your role	Senior Data Engineer, Investment Data
Core themes	Security master data, multi-custodian reconciliation, performance calculation, consumer duty reporting

The Scenario

Kingsley Rowe Wealth manages money for about twelve thousand private clients: entrepreneurs, professionals, retirees and families with substantial assets. Its investment managers build portfolios of shares, bonds, funds and alternative investments, and its financial planners advise clients on pensions, tax and estate planning. Client assets are held with three custodians, the institutions that hold securities in safekeeping and settle trades, because the firm has grown by acquiring smaller wealth managers, each of which came with its own custodian arrangements. Clients receive quarterly reports showing holdings, transactions, performance and fees. They also increasingly expect a digital portal where they can see their wealth consolidated in one place, including assets held elsewhere. You are the senior data engineer in investment data, and the quarterly reports are, in the words of the chief investment officer, "an embarrassment waiting to happen".

The core problem is reference data. Every security held in a client portfolio must be identified and described: its name, type, currency, issuer, country, sector, asset class, identifiers, pricing source and, for funds, the underlying holdings and ongoing charges. Each custodian identifies securities with its own internal codes, alongside international identifiers that are sometimes missing or wrong. The firm's portfolio management system has its own security master, maintained by a small team who create new securities manually when a custodian reports a holding the system does not recognise. There are duplicate records for the same security, securities misclassified by asset class, and funds whose share classes have been confused. Last quarter, a client with a large holding in a bond fund received a report showing it as an equity fund, because two share classes

with similar names had been merged into one record, and the portfolio's asset allocation was misreported. The client, a retired actuary, noticed and complained.

Positions and transactions arrive daily from each custodian in different file formats. The operations team reconciles them against the portfolio management system, investigating breaks where quantities or cash balances differ. Corporate actions, such as dividends, stock splits, mergers and rights issues, are processed by each custodian slightly differently and on slightly different dates, causing breaks that take days to resolve. Prices come from a market data vendor, but some assets, such as unlisted funds and structured products, are priced manually from statements. Performance is calculated by the portfolio management system, but for accounts that moved between custodians during an acquisition, historical performance was not migrated correctly, and some clients' reports show performance only since the migration.

The chief investment officer, Dr Anthony Blake, wants a clean, authoritative security master and consistent, correct performance for every client. The head of client services, Charlotte Hughes, wants the client portal to show consolidated wealth, including assets held outside the firm, such as pensions with other providers and property, entered by clients or their planners. The chief operating officer, Gavin Mercer, wants to reduce operational risk and cost, and is considering consolidating onto one custodian, which would simplify data flows but would require migrating thousands of accounts, with all the data risks that entails.

Regulatory pressure is increasing. The Financial Conduct Authority's consumer duty requires firms to demonstrate that their products and services deliver good outcomes for customers and represent fair value, and that communications are clear. Clients must receive accurate information about costs and charges, including the ongoing charges of funds they hold, aggregated with the firm's own fees. The firm's annual costs and charges disclosures have been calculated in spreadsheets, and a recent internal audit found errors in the fund charges data used. The compliance director, Lucy Barnes, wants the charges data sourced reliably and the calculation automated and auditable.

Data sources are a mixture of commercial and public. The firm licenses reference and pricing data from a market data vendor at considerable cost; the licence restricts redistribution and counts users. Open identifiers and

public data sources exist for some reference data: an open mapping service links many identifiers to a common instrument identifier; regulatory filings databases provide fund and company information; central banks and statistics agencies publish interest rates and exchange rates; and fund managers publish charges and holdings in standard templates. Using open data where possible could reduce vendor costs, but coverage is incomplete and quality varies.

Your team is three engineers and a reference data analyst. The portfolio management system is a commercial product with a database the vendor discourages customers from modifying directly. The firm has a data warehouse used for management information but not for client reporting.

You see a data mastering problem at the centre: an authoritative instrument master that reconciles vendor, custodian and open data sources, with clear rules for which source wins for each attribute, full history of changes, and stewardship processes for exceptions. On top of that, positions and transactions from all custodians can be normalised, reconciled and used for consistent performance and charges calculations. The custodian consolidation question matters but is separate: whatever the firm decides, it needs clean reference data to migrate successfully.

The history of the acquisitions shapes everything. Each acquired firm had its own conventions: one classified absolute return funds as alternatives, another as bonds; one recorded fees as a deduction from the portfolio, another as a separate charge; one calculated performance net of fees, another gross. When accounts were migrated onto the group's portfolio management system, these conventions were mapped as well as the migration team could manage under time pressure, but clients' historical reports were produced under the old conventions and their new reports under the new ones. Several long-standing clients have noticed that their asset allocation appears to have changed although nothing was bought or sold, and their investment managers have had to explain that only the classification changed. The investment committee has never formally agreed a single classification scheme, and different committee members hold strong views about how certain assets should be categorised, particularly multi-asset funds and structured products.

The client portal adds a different dimension. Showing externally held assets means accepting client-entered or planner-entered data of variable quality, possibly obtained through open finance connections to other

providers, and presenting it alongside the firm's own authoritative data without implying the firm has verified it. Charlotte wants the portal launched within nine months to match competitors; Anthony wants nothing shown to clients until the core data is right.

The team also knows that the firm's reputation rests on trust. Wealthy clients are often financially sophisticated and quick to spot errors; a single misreported allocation or performance figure can prompt a client to question everything else the firm tells them. Getting the foundations right is not merely an operational concern but a matter of client retention.

THE DECISION IN FRONT OF YOU

Do you prioritise an authoritative instrument master and multi-custodian normalisation, support custodian consolidation first, or launch the client portal with consolidated wealth views, and how do you balance commercial and open reference data sources while meeting consumer duty obligations on costs and charges?

Data Sources

#	Source and link	How it is used in this project
1	OpenFIGI API https://www.openfigi.com/api	Open mapping of third-party identifiers to FIGIs for instrument identification and deduplication.
2	SEC EDGAR Application Programming Interfaces https://www.sec.gov/search-filings/edgar-application-programming-interfaces	Company and fund filings data for issuer and fund reference information.
3	ECB Data Portal https://data.ecb.europa.eu	Reference exchange rates and interest rates for valuation and performance.
4	FRED Economic Data (St. Louis Fed) https://fred.stlouisfed.org	Benchmark rates and economic series for performance comparison.
5	GLEIF LEI Data https://www.gleif.org/en/lei-data	Issuer identification and corporate hierarchies.
6	European MiFID Template (EMT) https://findatex.eu	Standard template for fund costs and charges data from manufacturers.
7	FCA Consumer Duty https://www.fca.org.uk/firms/consumer-duty	Requirements on fair value, communications and outcome monitoring.
8	GIPS Standards (CFA Institute) https://www.gipsstandards.org	Performance calculation principles for consistent reporting.

Deliverables (17)

- D1** **Reference data quality assessment.** Duplicates, misclassifications and missing identifiers across the security master.
- D2** **Instrument master design.** An authoritative model with source precedence rules by attribute.
- D3** **Identifier mapping service.** Mapping of custodian, vendor and international identifiers to a common key.
- D4** **Share class disambiguation.** Rules and checks preventing confusion between fund share classes.
- D5** **Data stewardship workflow.** Exception handling and approval for new and changed instruments.
- D6** **Custodian feed normalisation.** Standardised positions, transactions and cash from all custodians.
- D7** **Automated reconciliation.** Daily reconciliation of custodian and portfolio records with break categorisation.
- D8** **Corporate actions harmonisation.** Consistent processing of corporate actions across custodians.
- D9** **Manual pricing controls.** Governance and evidence for manually priced assets.
- D10** **Performance history repair.** Restoration of pre-migration performance histories for affected accounts.
- D11** **Consistent performance calculation.** Standardised calculation methodology across all client accounts.
- D12** **Costs and charges data pipeline.** Sourcing of fund charges from standard templates with automated aggregation.
- D13** **Consumer duty reporting.** Fair value and outcome monitoring data for compliance.
- D14** **Open versus licensed data review.** Evaluation of open sources to reduce vendor dependency with coverage gaps.
- D15** **Custodian consolidation data readiness.** Data requirements and risks for any custodian migration.
- D16** **Client portal data services.** Consolidated wealth views with clear labelling of verified and client-entered data.
- D17** **Client report quality controls.** Pre-release validation checks on quarterly reports.

CASE STUDY 64 · RETAIL (DEPARTMENT STORES AND ONLINE)

Retail Loyalty Customer 360 and Identity Resolution Platform

Industry	Retail (Department Stores and Online)
Organisation	Hartfield & Lowe, a department store group with 42 stores, an online shop and a loyalty scheme with 5.5 million members
Your role	Principal Data Engineer, Customer Data
Core themes	Identity resolution, consent-aware personalisation, retail media data, measurement integrity

The Scenario

Hartfield & Lowe has been selling clothing, beauty, homeware and gifts from its department stores for more than a century. Its forty-two stores still generate most of its revenue, but online sales now account for nearly a third, and the company's future depends on combining the two into a seamless experience. Its loyalty scheme, relaunched three years ago with a mobile app, has five and a half million members, and the marketing director, Olivia Carter, believes it is the company's most valuable asset. She wants to know each customer: what they buy in store and online, what they browse, how they respond to offers, and what they are likely to want next. You are the principal data engineer for customer data, and the reality is that the company does not know who most of its customers are.

The identity problem is severe. In store, a transaction is linked to a customer only if they scan their loyalty card or app, which happens for around forty per cent of sales. Card payments carry a tokenised card reference that can link transactions from the same card, but not to a person. Online, customers may shop as guests, as registered users, or as loyalty members, sometimes with different email addresses. The beauty counters in stores, operated partly by concession brands, collect their own customer details. The wedding gift list service has its own customer database. The customer service centre has records of contacts. Marketing emails are sent from a platform with its own subscriber list, which includes people who signed up years ago and have not shopped since. The company estimates that it holds around eleven million customer records across systems, representing perhaps six million distinct people, but nobody knows the true number.

Olivia wants a customer data platform that unifies these records into a single profile per person, with a complete history of purchases, browsing, communications and preferences, and that feeds personalised marketing across email, app, website and in-store clienteling tools used by sales associates. Several vendors of customer data platforms have pitched products that promise identity resolution out of the box. The commercial director, Daniel Frost, has a further idea: retail media. Many retailers now sell advertising to their suppliers, using customer data to target shoppers with brand promotions on their websites, apps and even external channels, and to measure whether the advertising drove sales. Daniel believes Hartfield & Lowe's affluent customer base would be attractive to beauty and fashion brands, and that retail media could become a significant profit stream.

The data protection officer, Emma Richardson, has concerns about both. Unifying records from different sources requires a lawful basis and transparency: customers who gave their email to a beauty concession may not expect it to be linked to their loyalty profile. Consent for marketing was collected differently across channels and over time, and much of it may not meet current standards. Tokenised card references, used to link unidentified in-store transactions to known customers, are personal data, and using them to build profiles of people who have not joined the loyalty scheme requires careful justification. Retail media raises further questions: sharing customer data with brands, even in aggregated or pseudonymised form through clean room technologies, must be explained to customers and must not amount to selling their data without their knowledge.

The head of analytics, Dr Sanjay Gupta, raises the question of measurement. Much of the business case for personalisation and retail media depends on proving that marketing activity causes additional sales, rather than simply reaching customers who would have bought anyway. The company's current measurement relies on attributing sales to the last campaign a customer received, which overstates impact. He wants the platform to support controlled experiments with holdout groups, so that incremental impact can be measured properly. Without that, retail media will be sold to brands on inflated claims, which will backfire when brands measure results themselves.

The store operations director, Rebecca Stone, cares about the in-store experience. Sales associates in beauty, fashion and personal shopping use a clienteling app to recognise loyal customers and recommend products. They want to see customer purchase histories and preferences, but some customers find it intrusive when associates know too much, and associates themselves have mixed views about recording customer information.

Identity resolution itself is technically challenging. Deterministic matching on email addresses and loyalty numbers is reliable but covers only part of the population. Probabilistic matching on names, addresses and phone numbers can link more records but introduces errors: two members of the same household might be merged, or one person's records might remain split. Errors matter differently for different uses: a false merge in marketing might send a man's name on a mailing to his wife, which is embarrassing; in a subject access request, it could disclose one person's data to another, which is a breach.

Your team has five engineers. The company's data warehouse holds transactions and loyalty data. The email platform, e-commerce platform and clienteling app each hold their own customer data. Olivia wants a unified customer view within nine months to support the Christmas season.

You recognise that the choice of customer data platform is less important than decisions about identity rules, consent, data use boundaries and measurement, which must be made regardless of technology. A vendor platform that resolves identities aggressively could create privacy and accuracy problems; one that resolves conservatively might disappoint marketing. Retail media multiplies the stakes, because data leaves the company's direct control.

The company's own organisation reflects the fragmentation of the data. Store operations, e-commerce, marketing, the beauty concessions team, customer service and the gift list service each report to different directors and each regards its customer relationships as its own. The e-commerce team has already bought a personalisation engine that recommends products on the website using browsing data, and it has its own identity logic based on cookies and logins. The beauty team has a separate arrangement with several concession brands to share customer contact details for brand marketing, which was agreed years ago and has never been reviewed by the data protection team. Emma discovered it only

recently and is concerned that customers were not properly informed. Any customer data platform will have to bring these teams together and, in some cases, unwind arrangements that individual teams regard as successful and valuable.

There is also a question of what customers want. A recent survey of loyalty members found that most welcomed relevant offers but disliked feeling watched, and that trust in the brand was one of the main reasons they shopped at Hartfield & Lowe rather than cheaper competitors. A heavy-handed approach to personalisation could erode the very trust that makes the brand valuable.

THE DECISION IN FRONT OF YOU

Do you buy a customer data platform with built-in identity resolution, build an in-house identity graph with explicit rules, or start with a consent and identity foundation before personalisation and retail media, and how do you design identity matching, consent and measurement so that personalisation and retail media are accurate, lawful and trusted?

Data Sources

#	Source and link	How it is used in this project
1	dunnhumby The Complete Journey https://www.dunnhumby.com/source-files/	Household-level retail transaction and campaign data used to prototype customer 360 and campaign measurement.
2	H&M Personalized Fashion Recommendations https://www.kaggle.com/competitions/h-and-m-personalized-fashion-recommendations	Real fashion retail transactions, customers and articles for recommendation and profile modelling.
3	Instacart Market Basket Analysis https://www.kaggle.com/c/instacart-market-basket-analysis	Order history data for repeat purchase and propensity modelling.
4	Splink Record Linkage https://moj-analytical-services.github.io/splink/	Open-source probabilistic identity resolution for building the identity graph.
5	ICO Direct Marketing Code of Practice https://ico.org.uk/for-organisations/direct-marketing-and-privacy-and-electronic-communications/direct-marketing-guidance/	Guidance on consent, profiling and marketing communications.
6	IAB Tech Lab Data Clean Rooms Guidance https://iabtechlab.com/standards/data-clean-rooms/	Standards for privacy-preserving data collaboration in retail media.
7	Royal Mail Postcode Address File (PAF) https://www.poweredbypaf.com	Address standardisation for matching.

#	Source and link	How it is used in this project
8	Online Retail II Dataset (UCI) https://archive.ics.uci.edu/dataset/502/online+retail+ii	Real online retail transactions for testing customer-level aggregation and deduplication.

Deliverables (17)

- D1 Customer record inventory.** Sources, volumes, identifiers and consent status of all customer records.
- D2 Identity resolution rules.** Deterministic and probabilistic rules with thresholds by use case.
- D3 Identity graph.** A graph of identifiers, records and resolved persons with confidence and history.
- D4 Household modelling.** Separate person and household entities to avoid inappropriate merges.
- D5 Match quality evaluation.** Measured precision and recall of identity resolution against labelled samples.
- D6 Consent and preference model.** Consolidated consent by channel and purpose with history and source.
- D7 Consent remediation plan.** Actions for records lacking valid marketing consent.
- D8 Transparency and privacy notice updates.** Customer-facing explanations of profiling, linking and retail media.
- D9 Unified customer profile.** Purchases, interactions and preferences across channels per resolved customer.
- D10 Clienteling data service.** Appropriate profile information for store associates with customer controls.
- D11 Experimentation and holdout framework.** Controlled measurement of campaign and personalisation impact.
- D12 Retail media data governance.** Rules for brand collaboration, aggregation thresholds and clean room use.
- D13 Clean room implementation design.** A privacy-preserving environment for brand measurement.
- D14 Subject access safeguards.** Controls preventing disclosure of wrongly merged records.
- D15 Platform build versus buy evaluation.** An assessment of CDP vendors against identity and consent requirements.
- D16 Christmas readiness plan.** Scoped deliverables for the peak season with fallback options.
- D17 Customer trust metrics.** Monitoring of opt-outs, complaints and survey feedback on personalisation.

CASE STUDY 65 · RETAIL (GROCERY AND GENERAL MERCHANDISE)

Competitive Price Intelligence Data Pipeline for a Grocery and General Merchandise Retailer

Industry	Retail (Grocery and General Merchandise)
Organisation	Fairway Stores, a UK discount grocery and general merchandise chain with 310 stores and online delivery
Your role	Lead Data Engineer, Pricing Analytics
Core themes	Web data collection ethics and legality, product matching, pricing decision governance, competition law

The Scenario

Fairway Stores built its business on a simple promise: lower prices than the big supermarkets on the products people buy every week. It sells groceries, household essentials and a rotating range of general merchandise from three hundred and ten stores and a growing online delivery service. The promise has become harder to keep. Food inflation drove prices up sharply, competitors launched loyalty-card pricing that gives members lower prices, and discounters from continental Europe have expanded aggressively. The pricing team, led by pricing director Martin Hughes, sets prices for around twelve thousand products and needs to know, reliably and quickly, how Fairway's prices compare with competitors'. You lead data engineering in pricing analytics, and the price intelligence that the team relies on has serious problems.

Fairway currently obtains competitor prices from two sources. A market research firm sends store auditors to competitor stores weekly to record prices for a basket of about two thousand products, and delivers the data in spreadsheets several days later. A price comparison data vendor supplies online prices scraped from competitors' websites daily for around eight thousand products. The two sources often disagree. Online prices may differ from in-store prices; loyalty prices are shown online only to logged-in customers; promotions such as multibuy are recorded inconsistently; and product matches are sometimes wrong, comparing Fairway's own-label product with a competitor's premium brand, or a four-pack with a six-pack. The pricing team spends much of each week cleaning the data and arguing about which figure to believe.

The commercial director, Laura Pearson, wants a comprehensive price intelligence platform with daily coverage of all comparable products

across all major competitors, in store and online, including loyalty and promotional prices, and automated recommendations for price changes. She has seen presentations from dynamic pricing vendors that claim to optimise prices using competitor data and demand models. The chief executive, David Morgan, wants to be able to say publicly, with evidence, that Fairway is cheaper than its rivals, because price perception drives footfall.

The legal director, Susan Clarke, has raised several issues. Scraping competitors' websites may breach their terms of use, and while the legal position on scraping publicly available prices is generally considered defensible, the methods used matter: circumventing technical measures, logging in with accounts to see loyalty prices, or placing heavy loads on competitors' sites could create legal risk. The vendor's methods are not fully transparent. Competition law is another concern. Using competitors' published prices to set your own is normal and lawful. But sharing pricing intentions with competitors, or using algorithms that respond to each other in ways that could facilitate tacit collusion, is an area of increasing regulatory attention. The competition authority has also been examining grocery pricing, unit pricing and loyalty pricing practices, and any public claim about being cheapest must be substantiated under advertising rules, which require fair comparisons of like-for-like products.

The head of own label, Priya Desai, points out that product matching is the core difficulty. Branded products can be matched across retailers using barcodes, although the same product may have different barcodes for different pack sizes or promotional packs. Own-label products, which make up more than half of Fairway's sales, have no direct equivalent: Fairway's own-label baked beans must be compared with each competitor's own-label beans, which may differ in size, quality tier and recipe. Matching requires judgement about equivalence, and different judgements can change the outcome of a price comparison dramatically. She wants matching rules agreed and documented, not left to the vendor.

The data science team, led by Dr James Obi, has been building demand models to understand how price changes affect sales. They need historical competitor prices aligned with Fairway's own prices and sales at a daily, product and region level. The current data, with its gaps, inconsistencies and changing product matches, makes modelling unreliable. They also want to understand the effect of competitor promotions on Fairway's sales.

Public data sources offer some help. The official statistics agency publishes consumer price indices and, increasingly, experimental statistics based on scanner and web-scraped data, with published methods for product classification and matching. Open product databases provide barcodes, product attributes and nutritional information for many grocery products, and some community projects collect prices from receipts and user submissions. These cannot replace commercial price intelligence but can support product matching and classification and provide an independent sense check.

Your team is three engineers and an analyst. The pricing team is eight people, most of them experienced in retail but with limited data skills, and they currently spend nearly half their time reconciling data rather than analysing it or making decisions. Two members of the team have been recording competitor promotions manually from store visits and leaflets, because neither source captures them reliably. Laura wants the new platform in place before the next major pricing review in five months. The budget allows either an expanded vendor contract or an in-house collection capability with some vendor data, but not both at full scale.

You see that the heart of the problem is not collection but matching and governance: an agreed, documented, auditable set of product equivalences, including judgements about own-label tiers and pack sizes, maintained by the business with data support. Without that, more data simply produces more disagreements. You also see that automated price recommendations raise governance questions: who approves a price change, how the algorithm's logic is documented, and how the company would demonstrate to a regulator that its pricing is independent.

Internally, pricing decisions carry a history of friction. Category managers, who buy products from suppliers and are measured on margin, often resist price cuts that the pricing team recommends to maintain competitiveness. Regional managers argue that competitor prices vary by region and that national price comparisons miss local competition, particularly where a discounter has opened nearby. The online team sets delivery charges and minimum order values separately, which affect the real cost of a basket for online customers. Martin has asked for a single, trusted view of competitive position that all these parties accept, so that arguments focus on decisions rather than on whose numbers are right. He acknowledges that the pricing team itself has sometimes chosen whichever data source

supported the conclusion it preferred, and he wants a platform that removes that temptation by making data quality and match confidence visible alongside every comparison.

There is also a tension between internal and external uses. Price intelligence used internally to set prices can tolerate some imprecision. Price comparisons used publicly in advertising must meet a much higher evidential standard, with like-for-like products, verified prices and documented methods. A single platform must serve both, with clear distinctions.

THE DECISION IN FRONT OF YOU

Do you expand the vendor's scraped data and adopt dynamic pricing recommendations, build an in-house collection and matching capability, or focus first on product matching governance and data quality, and how do you manage the legal risks of collection, the competition law implications of algorithmic pricing and the evidential standard for public price claims?

Data Sources

#	Source and link	How it is used in this project
1	ONS Consumer Price Inflation and Methodology https://www.ons.gov.uk/economy/inflationandpriceindices	Official price indices and published methods for scanner and web-scraped price data.
2	Open Prices (Open Food Facts) https://prices.openfoodfacts.org	Crowdsourced open product price data for independent sense checks.
3	Open Food Facts https://world.openfoodfacts.org/data	Open product database with barcodes, attributes and categories for product matching.
4	UCI Online Retail II https://archive.ics.uci.edu/dataset/502/online+retail+ii	Real transactional sales data used to prototype price-sales alignment and elasticity pipelines.
5	CMA Grocery Unit Pricing and Loyalty Pricing Work https://www.gov.uk/cma-cases/loyalty-pricing-in-the-groceries-sector	Competition authority work on grocery pricing practices.
6	CMA Algorithms and Competition Guidance https://www.gov.uk/government/publications/algorithms-how-they-can-reduce-competition-and-harm-consumers	Regulatory analysis of algorithmic pricing risks.
7	ASA Guidance on Price Comparisons https://www.asa.org.uk/advice-online/prices-comparisons-with-competitors.html	Advertising rules for substantiating competitive price claims.

#	Source and link	How it is used in this project
8	Price Marking Order Guidance https://www.gov.uk/guidance/pricing-practices-guide	Rules on unit pricing and price display relevant to comparison methodology.

Deliverables (17)

- D1 Current data quality audit.** Gaps, disagreements and match errors between auditor and vendor data.
- D2 Product matching framework.** Documented rules for branded, own-label, pack size and tier equivalence.
- D3 Match governance workflow.** Business ownership, review and approval of product matches.
- D4 Product attribute enrichment.** Barcode, size, category and attribute data supporting matching.
- D5 Unit price normalisation.** Comparable per-unit prices across pack sizes and multibuy.
- D6 Promotion and loyalty price model.** Consistent representation of promotions, multibuy and member prices.
- D7 Collection legal review.** An assessment of collection methods, terms of use and technical measures.
- D8 Collection strategy decision.** A choice between expanded vendor, in-house collection or hybrid.
- D9 Price intelligence pipeline.** Daily ingestion, validation and history of competitor prices.
- D10 Historical price-sales dataset.** Aligned competitor prices, Fairway prices and sales for modelling.
- D11 Price index and position metrics.** Basket-level price position by competitor, category and region.
- D12 Recommendation governance.** Approval, documentation and override rules for any automated price suggestions.
- D13 Competition law safeguards.** Controls ensuring independent price-setting and documented algorithm logic.
- D14 Public claim substantiation process.** Evidence standards and audit trails for advertised price comparisons.
- D15 Independent benchmark checks.** Comparison of results with official and open price data.
- D16 Pricing team dashboard.** Competitive position views with data quality and match confidence indicators.
- D17 Pricing review readiness plan.** Deliverables and scope for the five-month deadline.

CASE STUDY 66 • FASHION E-COMMERCE

Fashion E-Commerce Returns and Reverse Logistics Data Platform

Industry	Fashion E-Commerce
Organisation	Loom & Thread, an online fashion retailer selling 900 brands to customers in 14 European countries
Your role	Senior Data Engineer, Operations Data
Core themes	Returns data quality, reverse logistics tracking, customer behaviour analytics, sustainability reporting

The Scenario

Loom & Thread sells clothing, shoes and accessories online to customers across Europe. It carries its own labels and around nine hundred third-party brands, and its customers expect free delivery, free returns and generous return windows. Those expectations have made returns the company's largest operational problem. Around thirty-eight per cent of items sold are returned, rising above fifty per cent for some categories such as dresses and shoes. Each return costs money: the outbound shipping that was wasted, the inbound shipping, the handling at the returns centre, the inspection, the repackaging or refurbishment, and the markdown or write-off of items that cannot be resold at full price. The chief financial officer, Karin Johansson, estimates that returns consume more than the company's entire operating profit. You are a senior data engineer in the operations data team, and the chief operating officer, Marco Bellini, has asked you to help the company understand and reduce returns.

The data about returns is surprisingly poor. When customers initiate a return online, they choose a reason from a list: too small, too large, not as described, poor quality, changed my mind, arrived damaged and so on. Many choose the first option or the vaguest one, and some choose reasons strategically, such as claiming damage to avoid return fees in markets where the company has introduced charges for certain reasons. Returns arrive at two returns centres, in Germany and Poland, where operatives scan items, inspect them and grade them as resellable, needing refurbishment, sellable only at discount, or unsellable. Grading decisions are recorded in the warehouse management system, but the grading criteria differ between the two centres, and a third-party logistics provider that handles returns from Scandinavian countries uses its own grading

codes. Items returned in bundles often cannot be matched to specific orders if the customer uses the wrong returns label. Refunds are processed by the order management system when items are scanned, but refunds for lost or disputed returns are handled manually by customer service.

The head of customer insight, Dr Aisha Bello, believes that returns are primarily a product information problem. Sizing is inconsistent across brands; product photographs and descriptions sometimes misrepresent fit, colour and fabric; and customers buy multiple sizes deliberately because they are uncertain. She wants to link returns data with product attributes, size guides, customer reviews and customer body measurements from the fit recommendation tool, to identify products and brands with high return rates for fixable reasons and to improve size recommendations. The brand partnerships team would share findings with brands, some of which are eager for such data.

The head of logistics, Tomasz Nowak, sees returns as a processing efficiency problem. Items spend an average of eleven days between the customer sending them and being available for resale, during which time their value falls, particularly for seasonal fashion. He wants to track every return from the moment the customer initiates it, through carrier handovers, arrival, inspection and restocking, to identify bottlenecks and predict incoming volumes so he can plan staffing. Carriers provide tracking events, but in different formats and with different levels of detail across fourteen countries.

The fraud and risk team has noticed patterns of abuse. Some customers wear items and return them, a practice known as wardrobing; some return different or damaged items; some claim non-arrival of returns that were never sent. A small number of customers account for a disproportionate share of returns and losses. The team wants a customer-level returns risk model that could restrict free returns for abusive customers. The legal and customer experience teams are cautious: consumer protection law in the European Union gives customers a statutory right to withdraw from distance purchases within fourteen days, and restricting returns must be handled carefully, fairly and transparently. Profiling customers and treating them differently based on automated decisions also raises data protection issues.

The sustainability director, Freya Lindqvist, points out that returns have a significant environmental footprint: transport emissions, packaging waste

and, for items that cannot be resold, disposal. New European regulations on sustainable textiles, including restrictions on destroying unsold goods and extended producer responsibility for textiles, will require the company to report on what happens to returned and unsold items. She wants a complete record of each item's fate, from resale to donation, recycling or destruction.

The data landscape includes the order management system, the warehouse management systems at both centres, the third-party provider's system, carrier tracking feeds, the product information management system, customer reviews, the fit recommendation tool and the customer relationship management system. Each uses its own identifiers for orders, items and products, and the link between a returned physical item and the original order line is often broken.

Your team comprises three engineers and a data analyst. Marco wants a returns data platform that serves all these stakeholders and a measurable reduction in return rates within a year.

You recognise that the foundation is a reliable, item-level returns event model: every returned item linked to its original order line, product, customer and brand, with a full history of events from initiation to final disposition, captured consistently across centres and countries. With it, product insight, logistics efficiency, abuse detection and sustainability reporting all become possible. Without it, each team will continue to build partial analyses on inconsistent data.

But each stakeholder wants their use case prioritised, and they carry different risks. Product and fit insights are commercially valuable and relatively low risk. Logistics tracking is operationally valuable. Abuse detection could save money but touches consumer rights and profiling. Sustainability reporting is increasingly mandatory. You must decide the sequence, and you must decide how far to go with customer-level risk models before legal and ethical questions are resolved.

Brands are also changing the terms of engagement. Several of the larger third-party brands now insist on receiving detailed returns data, including reasons and grades, as a condition of supply, and some want the right to recover returned items themselves rather than have them resold at discount on Loom & Thread's outlet channel. Others have begun to question the company's grading, claiming that items marked as damaged and charged back to them were in fact resellable. These disputes are

resolved by email and spreadsheets, and the brand partnerships team has no reliable evidence to support its position. A consistent, photograph-backed grading record would help, but it would also expose inconsistencies in current grading practice that the returns centres would prefer not to reveal.

There is a strategic question too. Some competitors have started charging for returns, which has reduced return rates but also reduced sales, as customers buy fewer items speculatively. Loom & Thread's leadership is debating whether to follow. Any such decision should be informed by data on the true cost of returns by category, customer segment and country, and on how customers respond to changes in returns policy, which the company has never measured rigorously. The chief executive has asked for that analysis within three months, before the next strategy review.

THE DECISION IN FRONT OF YOU

Do you prioritise product and fit insights, reverse logistics tracking, returns abuse detection or sustainability disposition reporting, and how do you build a reliable item-level returns data model while handling customer profiling and statutory return rights fairly?

Data Sources

#	Source and link	How it is used in this project
1	Olist Brazilian E-Commerce Public Dataset https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce	Real e-commerce orders, deliveries, reviews and sellers used to prototype order-to-return linkage and delivery event modelling.
2	H&M Personalized Fashion Recommendations https://www.kaggle.com/competitions/h-and-m-personalized-fashion-recommendations	Fashion transactions, articles and customers for product attribute and purchase behaviour analysis.
3	Rent the Runway Clothing Fit Dataset https://www.kaggle.com/datasets/rmisra/clothing-fit-dataset-for-size-recommendation	Fit feedback and body measurements for size recommendation prototyping.
4	EU Consumer Rights Directive https://commission.europa.eu/law/law-topic/consumer-protection-law/consumer-contract-law/consumer-rights-directive_en	Statutory withdrawal rights governing returns policy and restrictions.
5	EU Strategy for Sustainable and Circular Textiles https://environment.ec.europa.eu/strategy/textiles-strategy_en	Policy framework including destruction restrictions and producer responsibility.

#	Source and link	How it is used in this project
6	Ecodesign for Sustainable Products Regulation https://commission.europa.eu/energy-climate-change-environment/standards-tools-and-labels/products-labelling-rules-and-requirements/ecodesign-sustainable-products-regulation_en	Requirements on disclosure and destruction of unsold consumer products.
7	GS1 EPCIS Standard https://www.gs1.org/standards/epcis	Event data standard for tracking items through reverse logistics.
8	EDPB Guidelines on Automated Decision-Making and Profiling https://www.edpb.europa.eu/our-work-tools/our-documents/guidelines_en	Guidance relevant to customer returns risk scoring.

Deliverables (17)

- D1 Returns cost model.** The true cost of returns by category, country, reason and customer segment.
- D2 Item-level returns event model.** A model linking returned items to order lines and tracking all events to disposition.
- D3 Order-to-return linkage.** Matching logic for returns with wrong labels, bundles and partial returns.
- D4 Grading harmonisation.** Common grading criteria and code mapping across centres and the third-party provider.
- D5 Carrier tracking integration.** Normalised carrier events across fourteen countries.
- D6 Returns cycle time analytics.** Measurement of time from initiation to resale availability with bottleneck identification.
- D7 Inbound volume forecasting.** Predicted returns arrivals by centre and day for staffing.
- D8 Return reason quality analysis.** Assessment of reason reliability and redesign of reason capture.
- D9 Product and brand return insights.** Return rates by product, size and fit issue for merchandising and brands.
- D10 Size recommendation data feed.** Returns and fit data feeding the recommendation tool.
- D11 Brand data sharing framework.** Rules and formats for sharing returns insights with brands.
- D12 Returns abuse analytics.** Aggregate abuse patterns with a legally reviewed approach to any customer-level action.
- D13 Profiling legal assessment.** A review of automated returns restrictions under consumer and data protection law.
- D14 Disposition and sustainability reporting.** A record of every item's final fate for regulatory and sustainability reporting.

- D15 Returns policy experiment design.** A controlled test of policy changes with measurement of sales and returns.
- D16 Returns dashboard.** Operational and commercial views of returns volumes, costs and causes.
- D17 Delivery roadmap.** A sequenced plan across stakeholders with a one-year reduction target.

CASE STUDY 67 · AUTOMOTIVE RETAIL (ONLINE MARKETPLACE)

Used Vehicle Listings Aggregation and Valuation Data Platform

Industry	Automotive Retail (Online Marketplace)
Organisation	MotorMarket, a UK online marketplace listing 420,000 used vehicles from dealers and private sellers
Your role	Principal Data Engineer, Marketplace Data
Core themes	Vehicle data enrichment from government APIs, listing quality and fraud, valuation model data, dealer data rights

The Scenario

MotorMarket is one of the UK's larger online marketplaces for used cars and vans. Around four hundred and twenty thousand vehicles are listed at any time, most by franchised and independent dealers who pay subscription fees, and some by private sellers. Buyers search by make, model, price, mileage and location, and increasingly rely on the site's price indicator, which labels each listing as a great, good, fair or high price based on a valuation model. The indicator is popular with buyers and resented by some dealers, who say it misjudges their vehicles. You are the principal data engineer for marketplace data, and a recent complaint from a large dealer group has put the valuation data under scrutiny.

The dealer group, which lists several thousand vehicles, complained that MotorMarket's price indicator systematically labelled its vehicles as high priced, reducing enquiries. When your team investigated, it found several issues. The valuation model relied on the vehicle specification recorded in the listing, but dealers' stock management systems often send incomplete or generic specifications: a high-specification trim level might be listed as the base model, making it appear overpriced. Some listings had mileage recorded in the wrong units or with typographical errors. Vehicles that had been previously written off, which should be valued lower, were not always identified. And the model's training data included listings that never sold, whose asking prices do not reflect market values.

Vehicle data in Britain is unusually accessible. Government services allow lookup of a vehicle's registration details, including make, colour, fuel type, engine size, date of first registration, tax status and emissions, through an official enquiry service. The MOT history service provides records of every annual roadworthiness test, including the date, the recorded mileage at

each test and any failures and advisory notes. Official statistics on vehicle licensing provide national counts by make, model and age. Commercial data providers offer vehicle history checks covering finance outstanding, insurance write-off records, theft markers and detailed specification data decoded from vehicle identification numbers, at a cost per lookup.

The head of product, Nathan Price, wants to enrich every listing automatically from these sources, filling in missing specification, verifying mileage against MOT history and flagging vehicles with history issues. He believes this would improve search, trust and valuations at once, and could be marketed as a verified listing feature. The finance director, Samantha Reid, notes that commercial history checks cost money per lookup, and running them on every listing, with listings refreshed daily, would be expensive. She wants to know which enrichments deliver the most value.

The trust and safety manager, Kieran Walsh, is concerned about fraud. Fraudsters post fake listings, often cloning genuine vehicles' details and photographs, offering attractive prices and asking for deposits before disappearing. Others clock mileage, rolling back odometers, which MOT history can help detect. Some dealers misrepresent vehicles, such as failing to disclose write-off status. Kieran wants enrichment data used to detect suspicious listings before they go live. But he also warns that fraud detection must be careful: legitimate reasons exist for anomalies, such as replacement instrument clusters, and wrongly blocking a genuine dealer's listing damages the relationship.

The dealer relations director, Hannah Fielding, has a different perspective. Dealers are MotorMarket's paying customers, and they are sensitive about how their data is used. Their stock feeds contain not just listings but pricing strategies, stock ages and days-to-sell, which competing dealers would love to know. MotorMarket already sells market insight reports to dealers and manufacturers using aggregated listing data, and some dealers have questioned whether their data is being used to help their competitors. Hannah wants clear principles on dealer data use and wants dealers to be able to see and correct the enrichment applied to their listings, including specification and valuation.

The data science team, led by Dr Chen Wei, wants to rebuild the valuation model. The current model uses asking prices of all listings, which is flawed. A better model would use evidence of transaction prices, but MotorMarket does not know the prices at which vehicles actually sell. Proxies include the

final asking price before a listing is removed, time on market and dealer-reported sales, where available. Chen also wants richer features: specification, condition signals from MOT advisories, history markers and regional demand.

The regulatory context matters. Consumer protection law requires that information presented to buyers is not misleading. If MotorMarket labels a vehicle as a great price or verified, and the label proves wrong, the platform could face complaints or regulatory interest. The advertising regulator has also examined price indicators and verification claims on marketplaces. The data protection position is nuanced: vehicle registration data and MOT histories are not in themselves personal data, but combined with private seller listings and locations, they can relate to identifiable individuals.

Your team has six engineers. Listings arrive through dealer feeds from dozens of stock management systems, in several formats, updated hourly or daily, and through the private seller listing form. The government services have rate limits and terms of use that restrict certain uses of their data.

You see the foundation as a vehicle entity, separate from the listing: a canonical record for each vehicle, keyed by registration and vehicle identification number, enriched from official and commercial sources with provenance, and linked to its listings over time, including relistings and changes of seller. That would support verification, fraud detection and valuation. But it raises questions about how long MotorMarket retains vehicle histories after listings end, how dealer data is separated from the platform's own enrichment, and what buyers and dealers are told.

The marketplace's own history complicates matters. MotorMarket grew by acquiring two smaller listing sites, each with its own vehicle taxonomy for makes, models and derivatives. The three taxonomies were merged hastily, and some model variants now appear under several names, which fragments search results and weakens the valuation model, because similar vehicles are treated as different ones. Private seller listings are entered through a form that relies on sellers choosing from dropdown lists, and many choose the nearest match rather than the correct derivative. Dealers' stock management systems map their own codes to MotorMarket's taxonomy through a translation table maintained by a single analyst. Any enrichment strategy must also address the taxonomy,

since enrichment from official sources returns data in different vocabularies again, and reconciling them is essential for search, valuation and insight products alike.

There is also the question of sequencing and cost. Enrichment from free official sources can be done for every listing quickly. Commercial history checks might be applied selectively, such as for higher-value vehicles, vehicles flagged by fraud rules, or as a paid add-on for dealers. The valuation model rebuild depends on better specification data, which depends on enrichment, so the order of work matters. Meanwhile, the dealer group is threatening to withdraw its listings unless the price indicator is fixed quickly.

THE DECISION IN FRONT OF YOU

Do you prioritise listing enrichment and verification, fraud detection or the valuation model rebuild, and how do you use official and commercial vehicle data within their terms and cost constraints while respecting dealers' data rights and avoiding misleading buyers?

Data Sources

#	Source and link	How it is used in this project
1	DVLA Vehicle Enquiry Service API https://developer-portal.driver-vehicle-licensing.api.gov.uk	Official vehicle registration details used to enrich and verify listing specifications.
2	MOT History API https://documentation.history.mot.api.gov.uk	Test history with mileage and advisories used for mileage verification and condition signals.
3	DfT Vehicle Licensing Statistics https://www.gov.uk/government/collections/vehicles-statistics	National vehicle stock data by make, model and age for market context.
4	Used Cars Dataset (Craigslist, Kaggle) https://www.kaggle.com/datasets/austinreese/craigslist-carstrucks-data	Large real used vehicle listing dataset used to prototype valuation features and listing quality checks.
5	NHTSA vPIC VIN Decoder API https://vpic.nhtsa.dot.gov/api/	VIN decoding for specification enrichment and validation methods.
6	ASA Guidance on Price Claims and Marketplaces https://www.asa.org.uk/advice-online/pricing-general.html	Advertising standards relevant to price indicators and verification labels.
7	CMA Consumer Protection Guidance for Online Platforms https://www.gov.uk/government/collections/consumer-protection-law-for-online-businesses	Rules on misleading information presented to consumers.

#	Source and link	How it is used in this project
8	Action Fraud Vehicle Fraud Guidance https://www.actionfraud.police.uk/a-z-of-fraud/vehicle-fraud	Typologies of vehicle-related fraud informing detection rules.

Deliverables (17)

- D1 Price indicator complaint investigation.** A root cause analysis of valuation errors for the complaining dealer group.
- D2 Canonical vehicle entity model.** A vehicle record keyed by registration and VIN linked to listings over time.
- D3 Dealer feed normalisation.** Standardised ingestion of stock feeds from many systems with validation.
- D4 Official data enrichment.** Automatic enrichment of listings from registration and MOT sources within rate limits.
- D5 Specification decoding.** Improved trim and specification identification using VIN and registration data.
- D6 Mileage verification.** Comparison of listed mileage with MOT history and anomaly flags.
- D7 Commercial history check strategy.** A cost-benefit model for selective use of paid history checks.
- D8 Enrichment provenance.** Recording of the source and date of every enriched attribute.
- D9 Dealer correction workflow.** A process for dealers to review and correct enrichment and specification.
- D10 Fraud detection rules.** Detection of cloned, duplicate and suspicious listings before publication.
- D11 Fraud review process.** Human review and appeal to avoid wrongly blocking genuine sellers.
- D12 Valuation training data redesign.** Proxies for transaction prices using listing outcomes and dealer data.
- D13 Valuation model rebuild.** A model using verified specification, condition and history features.
- D14 Price indicator transparency.** Buyer-facing explanations and confidence levels for price labels.
- D15 Dealer data use principles.** Rules on aggregated insight products and protection of dealer-specific data.
- D16 Terms of use compliance review.** Assessment of official data use against service terms.

- D17 Marketplace quality dashboard.** Listing completeness, verification rates and fraud metrics.

CASE STUDY 68 • TOURISM AND DESTINATION MANAGEMENT

Regional Visitor Economy Data Observatory for a Tourism Board

Industry	Tourism and Destination Management
Organisation	Visit Lakes & Dales, the destination management organisation for a national park region welcoming 19 million visitors a year
Your role	Data Engineering Lead, Visitor Insight
Core themes	Combining official statistics with alternative data, small-business data sharing, overtourism, open data publication

The Scenario

Visit Lakes & Dales promotes and manages tourism in a region of mountains, lakes, market towns and national park landscapes that attracts around nineteen million visitors a year. Tourism is the backbone of the local economy, supporting thousands of small businesses: hotels, guest houses, holiday cottages, campsites, pubs, cafés, outdoor activity providers and attractions. It also strains the region. On summer weekends, narrow roads jam, car parks overflow, verges are damaged by illegal parking, popular paths erode and villages are overwhelmed. Meanwhile, the growth of short-term holiday lets has hollowed out some communities, with local families priced out of housing. You lead data engineering in the visitor insight team, a small unit funded jointly by local authorities, the national park authority and business membership fees.

The organisation's current evidence base is thin and slow. The main source of visitor numbers and spending is an economic impact model run annually by consultants, which combines national survey data with local accommodation occupancy surveys and attraction visitor counts. Results arrive a year after the period they describe and cannot say much about where visitors go, when or why. Accommodation occupancy is measured by a monthly survey of volunteer businesses, whose response rate has fallen below thirty per cent. Attractions report visitor numbers voluntarily. Car park occupancy is measured by the councils' parking systems, but only in pay-and-display car parks. Pedestrian and cycle counters on some popular paths record footfall. The national park authority's rangers record incidents such as wild camping, fires and litter.

The chief executive, Dr Karen Whitfield, wants a visitor economy data observatory: timely, granular information on visitor volumes, flows,

spending and impacts, available to the organisation, its partners and its members. She has several purposes in mind: making the case to government for investment and for powers such as a visitor levy, which some regions have introduced; helping businesses plan; managing visitor pressure by encouraging visits to quieter places and times; and informing the housing debate with evidence on short-term lets.

Alternative data sources promise much. Mobile network operators sell anonymised, aggregated data on the movements of devices, which can estimate visitor numbers, origins and dwell times at fine spatial and temporal resolution. Card payment data from banks and payment processors, aggregated by merchant category and area, can estimate visitor spending and its origins. Online accommodation platforms publish listings and availability, and independent projects scrape and publish short-term let data for research. Search and booking trends from travel platforms indicate demand. Each source has limitations: mobile data depends on the operator's market share and methodology, card data covers only card spending, and scraped listings data raises legal and ethical questions.

The finance director, Gareth Lloyd, is wary of cost. Mobile and card data products are expensive, typically sold as annual licences with restrictions on sharing. The organisation's budget is modest and depends on partners who each want different things. He wants a clear justification before committing to any commercial data licence.

The head of business engagement, Sally Roberts, is concerned about members. Small tourism businesses are asked for data constantly by surveys and platforms and see little in return. If the observatory is to include business-level data, such as occupancy and bookings, businesses need confidence that their individual data will not be shared with competitors and that they will get something useful back, such as benchmarks comparing their performance with similar businesses nearby. Some property management systems used by holiday cottage agencies and hotels can share data automatically, which would reduce the burden, but only with owners' consent.

The national park authority's planning officer, Dr Ian Harper, has a policy interest. Planning decisions on new tourism development, car parks and visitor facilities require evidence of visitor numbers and pressures. The authority is also under pressure from residents to control short-term lets,

and the government has introduced a registration scheme and planning changes for holiday lets in England. Ian wants reliable counts of short-term lets by settlement over time, and he is aware that scraped data may be challenged if used in formal planning decisions.

Privacy and ethics feature in several ways. Aggregated mobile data is designed to prevent identification of individuals, but at very fine resolution in sparsely populated areas, small counts could reveal patterns about residents. Short-term let data scraped from platforms includes host names and property locations, which relate to individuals. Publishing detailed visitor pressure information could be misused, for example by directing crowds to fragile sites that are currently quiet. And the organisation's own public messaging about avoiding busy places must be informed by data that is accurate enough not to mislead.

Your team is two engineers and an analyst, with support from a university research partnership that has offered to help with methods. The organisation has no data platform beyond spreadsheets and a statistics dashboard hosted by the consultants.

You see an opportunity to build something genuinely useful by combining a modest set of reliable sources rather than buying everything. Official statistics provide benchmarks and national context. Counters, car parks and business data provide ground truth at specific places. One commercial source, chosen carefully, could provide the spatial and temporal granularity that nothing else offers. Open data on short-term lets could inform the housing debate if handled responsibly. But combining sources with different definitions, coverage and biases into a coherent picture is methodologically hard, and the observatory's credibility depends on getting the methods right and being transparent about uncertainty.

Seasonality and weather make everything harder. The region's visitor numbers swing dramatically between a quiet winter midweek and a sunny bank holiday weekend, and the busiest days can bring several times the average number of visitors to a handful of honeypot locations. Decisions about managing pressure, such as deploying rangers, opening overflow car parks, running park-and-ride buses or issuing public messages, are made days or even hours in advance. Annual statistics are useless for such decisions; what the operational teams need is a short-term forecast of pressure at key sites, driven by weather forecasts, school holidays and events, and validated against counters and car parks. That operational use

case sits alongside the strategic ones and has quite different requirements for timeliness and accuracy, yet it would draw on many of the same sources.

There is also the visitor levy debate, which is politically charged. Some businesses fear a levy would deter visitors; others see it as funding for infrastructure. Data from the observatory will inevitably be used by both sides. Karen wants the observatory to be seen as independent and trusted, not as an advocate for a particular position. That requires published methods, clear caveats and governance that includes partners with different views.

THE DECISION IN FRONT OF YOU

Do you license commercial mobile or card data for granular visitor insight, build the observatory primarily from official statistics, counters and member data, or combine one carefully chosen commercial source with open and local data, and how do you protect business and resident privacy and maintain independence in contested debates over levies and short-term lets?

Data Sources

#	Source and link	How it is used in this project
1	ONS Travel Trends and Leisure and Tourism Statistics https://www.ons.gov.uk/peoplepopulationandcommunity/leisureandtourism	Official visitor and spending statistics for national benchmarks.
2	VisitBritain/VisitEngland Research and Insights https://www.visitbritain.org/research-insights	Domestic and inbound tourism surveys and accommodation occupancy data.
3	Eurostat Tourism Statistics https://ec.europa.eu/eurostat/web/tourism/database	European tourism data for comparison with other destinations.
4	Inside Airbnb https://insideairbnb.com/get-the-data/	Open short-term let listing data used for research on holiday let concentration.
5	Short-Term Lets Registration Scheme (England) https://www.gov.uk/government/consultations/introduction-of-a-use-class-for-short-term-lets-and-associated-permitted-development-rights	Policy context for short-term let regulation and future official data.
6	DfT Road Traffic Statistics https://roadtraffic.dft.gov.uk	Traffic counts on routes into the region for visitor flow estimation.
7	Ordnance Survey Open Data https://osdatahub.os.uk/downloads/open	Open mapping of paths, roads and places for spatial analysis.

#	Source and link	How it is used in this project
8	Scottish Visitor Levy Act 2024 https://www.legislation.gov.scot/asp/2024/1/contents	Legislative precedent for visitor levies informing the policy debate.

Deliverables (17)

- D1 Stakeholder needs assessment.** Purposes, questions and data needs of each partner and member group.
- D2 Source evaluation matrix.** Coverage, granularity, cost, licence terms and bias for each candidate source.
- D3 Commercial data selection.** A justified choice of at most one commercial source with licence terms negotiated.
- D4 Official statistics integration.** Ingestion of national and regional tourism statistics as benchmarks.
- D5 Ground truth data pipeline.** Integration of counters, car parks and attraction data with quality checks.
- D6 Member business data sharing scheme.** Consent-based collection of occupancy data with benchmarking returned to members.
- D7 Business data confidentiality rules.** Aggregation thresholds preventing identification of individual businesses.
- D8 Visitor volume estimation method.** A documented method combining sources with uncertainty ranges.
- D9 Visitor pressure indicators.** Timely indicators of congestion at key sites for management and messaging.
- D10 Short-term let monitoring.** Counts of short-term lets by settlement over time with ethical handling of host data.
- D11 Resident privacy safeguards.** Minimum counts and spatial aggregation rules for sparse areas.
- D12 Visitor spending estimates.** Estimates of spending by area and visitor origin with limitations stated.
- D13 Observatory methods publication.** Public documentation of sources, methods and caveats.
- D14 Independent governance group.** A cross-partner group overseeing methods and outputs.
- D15 Public dashboard.** Accessible, caveated indicators for partners and the public.
- D16 Levy evidence briefing.** Neutral evidence on visitor economy scale and costs to inform the levy debate.
- D17 Sustainability and funding plan.** A funding model for maintaining the observatory beyond initial grants.

CASE STUDY 69 • SOCIAL HOUSING

Social Housing Repairs, Damp and Mould Risk Data Platform

Industry	Social Housing
Organisation	Riverside Homes Group, a housing association managing 34,000 homes across the North West of England
Your role	Lead Data Engineer, Asset and Tenant Data
Core themes	Stock condition data, repairs data quality, predictive risk and tenant trust, regulatory consumer standards

The Scenario

Riverside Homes Group provides affordable rented homes to around seventy thousand people in thirty-four thousand properties, from post-war terraces and 1960s tower blocks to new-build flats. Like every social landlord in England, it is operating under intensified scrutiny after the death of a two-year-old boy in 2020 from a respiratory condition caused by prolonged exposure to mould in his family's housing association flat. The case led to new legislation requiring social landlords to investigate and fix hazards such as damp and mould within fixed timescales, new consumer standards enforced by the Regulator of Social Housing through inspections, and a tougher approach from the Housing Ombudsman. Riverside's own record is mixed: its tenant satisfaction measures are below the sector median for repairs, and the Ombudsman recently made a severe maladministration finding against it in a damp and mould case where a family's repeated complaints were closed without effective action. You lead data engineering for asset and tenant data, and the chief executive, Sarah Okafor, has made damp and mould the organisation's top priority.

Riverside's data about its homes is patchy. A stock condition survey programme aims to inspect every home every five years, recording the condition of components such as roofs, windows, heating systems and kitchens, but the programme fell behind during the pandemic, and around a third of homes have not been surveyed for more than five years. Survey data is held in an asset management system. Energy performance certificates exist for most homes but many are old and were based on assumptions rather than inspection. Repairs are reported by tenants through a contact centre, online portal and app, recorded in a repairs system, and completed by an in-house workforce and several contractors,

each of which records completion in its own system and sends updates back, not always accurately. Damp and mould reports are recorded as repairs, but categorisation is inconsistent: a report of condensation on windows might be logged as a window repair, a plastering job or a general inspection.

Tenant data is in a housing management system: tenancy details, household composition, rent accounts, vulnerabilities and contact history. Vulnerability information, such as a household member with a respiratory condition, a disability or a young child, is recorded inconsistently, often in free-text notes rather than structured fields.

The director of property, Mark Jennings, wants a damp and mould risk model that predicts which homes are most likely to have problems, so that surveyors can inspect them proactively rather than waiting for complaints. Other landlords have built such models using property characteristics, repair history, energy efficiency, occupancy and even data from environmental sensors installed in homes. Mark has seen a vendor's model that claims to identify high-risk homes with impressive accuracy. He also wants sensors in a sample of homes to monitor humidity and temperature.

The director of tenant services, Joy Mensah, supports proactive action but is concerned about how it will be experienced by tenants. Many tenants feel that when they report damp and mould, they are blamed for causing it through their lifestyle, such as drying clothes indoors or not heating their homes, which may reflect the cost of energy rather than choice. A model that uses household composition and behaviour data to predict risk could reinforce that perception. Sensors in homes raise privacy concerns: humidity and temperature data can reveal when people are home, when they shower and cook, and how many people live there. Joy wants tenants involved in designing any such programme and wants the focus kept on the building, not the people.

The compliance and assurance director, Anthony Blake, has a regulatory focus. The new legal requirements set deadlines for investigating reported hazards and completing repairs. The regulator's inspections will test whether the landlord knows the condition of its homes, whether it has accurate data about tenants' needs and whether it responds effectively to complaints. The tenant satisfaction measures, which every landlord must publish, include repairs satisfaction, and some of them draw on management information that must be accurate. Anthony wants reliable

reporting on every damp and mould case: when it was reported, when it was inspected, what was found, what was done and whether it was resolved, with evidence that deadlines were met.

The repairs data is the most immediate problem. When your team analysed a sample of damp and mould cases, it found that many were closed after a first visit in which an operative wiped mould and applied treatment, without investigating or addressing the cause, such as inadequate ventilation, cold bridging or leaks. Cases were reopened when the mould returned, but the repairs system treated each as a new job, so repeat cases were invisible in reporting. Contractors' completion notes were free text, often brief. Photographs were stored on operatives' devices and sometimes not uploaded.

Your team has four engineers and two analysts. The housing management and repairs systems are commercial products with limited integration capabilities. The organisation has recently created a data warehouse but it holds only financial and rent data so far.

You see that the risk model Mark wants, the tenant-centred approach Joy wants and the regulatory assurance Anthony wants all depend on the same foundations: a reliable record of each home's condition and characteristics, a case-based view of damp and mould that links reports, inspections, works and outcomes over time, and accurate, structured information about tenants' vulnerabilities. Without those, a risk model would be trained on inconsistent labels and would miss homes that have never been surveyed, and regulatory reporting would continue to hide repeat cases.

The workforce has its own perspective. Riverside's in-house repairs operatives and surveyors are under pressure: job volumes have risen sharply since the new rules, and they are measured on jobs completed per day. Several operatives told your analysts that they know a mould wash will not fix the underlying problem, but they are not trained or authorised to diagnose ventilation or structural issues, and raising a follow-on job requires a supervisor's approval that often takes weeks. Surveyors say they are sent to inspections with no information about previous visits, so they cannot see that a home has had the same problem five times. The data problems, in other words, are not only in the systems but in the work processes and incentives that generate the data. Fixing the data will

require changes to how jobs are raised, closed and followed up, which the repairs managers must own.

There is urgency in both directions. Tenants are living in homes with damp and mould now, and some are vulnerable. Waiting for perfect data before acting would be indefensible. But acting on a flawed model could send surveyors to the wrong homes while missing the most serious cases, and could damage trust if tenants feel profiled. A pragmatic approach might combine an immediate programme of inspections based on simple, transparent criteria, such as homes with any damp report in the past two years, homes with vulnerable occupants and homes of construction types known to be at risk, with a longer-term investment in data foundations and modelling.

THE DECISION IN FRONT OF YOU

Do you deploy a predictive damp and mould risk model and sensor programme now, build the case, condition and vulnerability data foundations first, or run a transparent rules-based proactive inspection programme alongside foundation work, and how do you keep the focus on buildings rather than blaming tenants while meeting new legal deadlines and regulatory scrutiny?

Data Sources

#	Source and link	How it is used in this project
1	English Housing Survey https://www.gov.uk/government/collections/english-housing-survey	National data on housing condition including damp prevalence by tenure and property type.
2	Energy Performance of Buildings Data (Domestic EPC) https://epc.opendatacommunities.org	EPC records with property characteristics used as risk features and for stock data validation.
3	Housing Ombudsman Spotlight Report on Damp and Mould https://www.housing-ombudsman.org.uk/reports/spotlight-reports/	Findings and recommendations on landlord handling of damp and mould.
4	Regulator of Social Housing Consumer Standards and Tenant Satisfaction Measures https://www.gov.uk/government/organisations/regulator-of-social-housing	Regulatory standards and required measures for landlords.
5	Awaab's Law Guidance https://www.gov.uk/government/consultations/awaabs-law-consultation-on-timescales-for-repairs-in-the-social-rented-sector	Legal requirements and timescales for addressing hazards.

#	Source and link	How it is used in this project
6	Housing Health and Safety Rating System Guidance https://www.gov.uk/government/publications/housing-health-and-safety-rating-system-guidance-for-landlords-and-property-related-professionals	Framework for assessing damp and mould hazards.
7	Met Office HadUK-Grid Climate Data https://www.metoffice.gov.uk/research/climate/maps-and-data/data/haduk-grid/haduk-grid	Regional temperature and rainfall data used in risk analysis.
8	Housing Associations' Charitable Trust (HACT) UK Housing Data Standards https://hact.org.uk/tools-and-services/uk-housing-data-standards/	Sector data standards for housing, assets and repairs.

Deliverables (17)

- D1 Damp and mould case audit.** An analysis of historic cases revealing repeat visits, closures without cause resolution and deadline performance.
- D2 Case-based data model.** A model linking reports, inspections, works, photographs and outcomes into cases.
- D3 Repeat case detection.** Identification of recurring problems at the same property across separate jobs.
- D4 Repairs categorisation standard.** Consistent classification of damp, mould and related repairs across channels and contractors.
- D5 Contractor data requirements.** Mandatory structured completion data and photograph uploads from contractors.
- D6 Stock condition gap analysis.** Homes without recent surveys prioritised for inspection.
- D7 Property attribute enrichment.** Integration of EPC, construction type and component data.
- D8 Vulnerability data remediation.** Structured, consented recording of household vulnerabilities from free-text notes.
- D9 Rules-based proactive inspection list.** Transparent criteria identifying homes for immediate inspection.
- D10 Risk model development plan.** A staged plan for predictive modelling focused on building characteristics.
- D11 Tenant co-design panel.** Tenant involvement in risk modelling and sensor programme design.
- D12 Sensor pilot privacy framework.** Consent, data minimisation and access rules for environmental sensors.
- D13 Statutory deadline tracking.** Monitoring of investigation and repair timescales for every hazard case.

- D14 Tenant satisfaction measures data assurance.** Validation of data used in published measures.
- D15 Regulatory assurance reporting.** Board and regulator reporting on damp and mould performance.
- D16 Ombudsman learning integration.** Tracking of Ombudsman findings and resulting process changes.
- D17 Implementation roadmap.** A plan balancing immediate action with foundational data work.

CASE STUDY 70 · EMERGENCY MEDICAL SERVICES

Emergency Call and Ambulance Dispatch Performance Data Platform

Industry	Emergency Medical Services
Organisation	The Central Counties Ambulance Service, an NHS ambulance trust serving 4.8 million people
Your role	Principal Data Engineer, Performance and Clinical Intelligence
Core themes	Real-time operational data, response time standards, hospital handover delays, clinical outcome linkage

The Scenario

The Central Counties Ambulance Service answers around four thousand emergency calls a day. Each call is triaged by call handlers using a structured clinical assessment system that assigns a category, from category one for immediately life-threatening conditions such as cardiac arrest, to category four for less urgent cases that may be managed by telephone advice or referred elsewhere. National standards set target response times for each category. For several years, the service has missed most of them, sometimes by a wide margin. The main reason is not a shortage of ambulances on paper but the time ambulances spend queuing outside hospital emergency departments, unable to hand over their patients, sometimes for many hours. You are the principal data engineer in performance and clinical intelligence, and the service's data has become central to a fierce argument between the ambulance service, local hospitals and the national bodies that oversee them.

The operational data is generated by the computer-aided dispatch system, which records every call, its triage category and every timestamp in the life of an incident: call connect, category assignment, resource allocation, mobilisation, arrival on scene, departure from scene, arrival at hospital, patient handover and resource clear. The electronic patient care record, completed by crews on tablets, records clinical observations, treatments and outcomes on scene. Hospitals record handover times in their own systems, and there is a regional handover tool where crews and hospital staff are meant to record handover jointly, but its use is inconsistent. Vehicle telematics record location, speed and status. Staff rosters and sickness absence are recorded in a workforce system.

The chief executive, Dr Andrew Moss, uses data every day in difficult conversations. He wants to show hospital chief executives, commissioners and national leaders exactly how many hours his crews lose outside each hospital, what that means for response times and, crucially, what it means for patients waiting at home for ambulances. There have been cases of patients who died while waiting for an ambulance that was stuck outside a hospital, and coroners have issued reports calling for action. Andrew wants a platform that shows, in near real time, where every ambulance is, what it is doing and how long it has been doing it, with historical analysis that attributes lost hours to specific causes.

The hospitals dispute the ambulance service's figures. Handover times recorded by the ambulance service, based on crews pressing a button on their terminals, often differ from those recorded by hospitals, which measure from when the patient was formally accepted. Hospitals argue that crews sometimes delay pressing the clear button after handover, inflating the apparent delay, and that the ambulance service's data does not reflect the pressure hospitals themselves are under because they cannot discharge patients into social care. The regional director of operations, Claire Bennett, wants a single, agreed dataset for handover times, jointly governed by the ambulance service and the hospitals, so that arguments focus on solutions rather than numbers.

The medical director, Dr Farah Ahmed, has a different priority. Response times are a proxy for outcomes, not outcomes themselves. What matters is whether patients survive and recover. She wants to link ambulance data with hospital outcome data, such as survival to discharge after cardiac arrest, stroke treatment times and mortality after major trauma, to understand how delays affect patients and whether clinical care on scene is effective. National clinical audits and registries collect some of this data, but linkage requires matching patients across organisations using NHS numbers, which crews do not always record, and it requires information governance approvals.

The head of emergency operations centres, Paul Kirby, is focused on the call handling side. Call answer times have deteriorated, and a significant proportion of calls are from patients calling back because their ambulance has not arrived, which ties up call handlers and creates duplicate incidents. He wants analysis of call volumes, call backs, triage outcomes and the effectiveness of clinical validation, in which clinicians in the

control room review lower-category calls and may arrange alternative care. He believes that better targeting of clinical validation could safely divert many patients away from ambulances and emergency departments.

The data engineering challenges are considerable. The dispatch system was designed for operations, not analytics, and its reporting database is refreshed nightly, so near-real-time analysis requires a different approach. Timestamps can be missing or out of sequence when crews press buttons in the wrong order or forget. Incidents may have several resources assigned, cancelled and reassigned, and national rules define precisely how response times should be calculated from these events, rules that have been misapplied in the past. The patient care record system is separate and uses its own identifiers. Hospital data comes from several trusts with different systems. And the platform must be highly reliable, because operational managers depend on it during periods of extreme pressure.

Information governance is significant. Dispatch and patient care records contain confidential patient information, including sensitive clinical details. Linking with hospital data requires a legal basis, typically under provisions for direct care or with approval for purposes beyond direct care, and a data protection impact assessment. Sharing handover data with hospitals is less sensitive if it is aggregated, but hospitals want patient-level detail to investigate specific delays.

Your team has five engineers and four analysts. The trust's data warehouse is on premises and was built fifteen years ago. The trust has access to a national NHS cloud environment and to a regional shared care record, but using them requires agreement with other organisations.

Winter is five months away, and every winter brings the worst handover delays. Andrew wants the real-time view in place before winter. Claire wants the jointly governed handover dataset agreed before winter so that the arguments do not repeat. Farah wants outcome linkage, which will take longer because of approvals. Paul wants call handling analytics.

You see that the most contested data, handover times, is also the most consequential, and that agreeing definitions and a single source with the hospitals would reduce conflict more than any technology. But agreement depends on trust between organisations that are currently blaming each other in public. You also see that a real-time operational view is achievable

technically but must be built on correct event logic, or it will produce misleading numbers at the worst possible time.

You are also conscious of the human stakes. Crews waiting outside hospitals for hours are exhausted and demoralised, and they know that patients in the community are waiting for them. Call handlers take calls from distressed relatives asking where the ambulance is. Data that reveals the scale of the problem accurately can help drive change; data that is disputed or wrong can make things worse by distracting leaders into arguments about numbers.

THE DECISION IN FRONT OF YOU

Do you prioritise a real-time operational view, a jointly governed handover dataset with hospitals, outcome linkage or call handling analytics before winter, and how do you ensure national response time rules are applied correctly and that disputed data becomes a shared basis for action rather than blame?

Data Sources

#	Source and link	How it is used in this project
1	NHS England Ambulance Quality Indicators https://www.england.nhs.uk/statistics/statistical-work-areas/ambulance-quality-indicators/	National response time, handover and outcome indicators with published methodology.
2	San Francisco Fire Department Calls for Service https://data.sfgov.org/Public-Safety/Fire-Department-and-Emergency-Medical-Services-Dispatched-Calls-for-Service/nuek-vuh3	Real dispatch event timestamps used to prototype incident timeline reconstruction and response time calculation.
3	NEMSIS National EMS Data https://nemsis.org	National EMS data standard and research dataset for patient care record structure and analytics.
4	NHS Data Dictionary https://www.datadictionary.nhs.uk	Standard definitions for health data items used in linkage.
5	Out of Hospital Cardiac Arrest Outcomes Registry (Warwick) https://warwick.ac.uk/fac/sci/med/research/ctu/trials/ohcao/	Cardiac arrest outcome registry illustrating outcome linkage.
6	NHS Emergency Care Data Set https://digital.nhs.uk/data-and-information/data-collections-and-data-sets/data-sets/emergency-care-data-set-ecds	Hospital emergency department data used for handover and outcome linkage.
7	Sentinel Stroke National Audit Programme https://www.strokeaudit.org	Stroke pathway timing and outcome data for linkage with ambulance response.

#	Source and link	How it is used in this project
8	NHS Data Security and Protection Toolkit https://www.dsptoolkit.nhs.uk	Information governance standards for patient data processing and sharing.

Deliverables (17)

- D1 Incident event model.** A canonical model of calls, incidents, resources and timestamped events.
- D2 Response time calculation engine.** Implementation of national rules with test cases and validation against published figures.
- D3 Timestamp quality checks.** Detection of missing, out-of-sequence and implausible event times.
- D4 Near-real-time data feed.** Streaming or frequent extraction from the dispatch system for operational views.
- D5 Operational status dashboard.** Live view of resources, incidents, waits and hospital queues.
- D6 Lost hours attribution.** Quantification of hours lost to handover delays by hospital, day and time.
- D7 Joint handover data definitions.** Agreed definitions and a single source for handover times with hospitals.
- D8 Handover data governance group.** A joint group overseeing handover data and resolving disputes.
- D9 Patients waiting analysis.** Measurement of patients waiting in the community and associated risk.
- D10 Call handling analytics.** Analysis of call answer times, call backs and duplicate incidents.
- D11 Clinical validation effectiveness.** Outcomes of validated calls and opportunities for safe diversion.
- D12 Patient care record linkage.** Linking dispatch and on-scene clinical records.
- D13 Hospital outcome linkage design.** Legal basis, approvals and matching method for outcome linkage.
- D14 Information governance documentation.** DPIAs and data sharing agreements for all flows.
- D15 Winter readiness plan.** Deliverables prioritised for the coming winter.
- D16 Platform modernisation plan.** A route from the legacy warehouse to a modern platform.
- D17 Board and system reporting pack.** Consistent reporting to the board, commissioners and national bodies.

CASE STUDY 71 · TRANSPORT PLANNING (LOCAL GOVERNMENT)

Regional Traffic Monitoring and Congestion Data Platform for Transport Planning

Industry	Transport Planning (Local Government)
Organisation	The Westmoor Transport Partnership, the transport authority for a city region of 2.3 million people
Your role	Senior Data Engineer, Transport Analytics
Core themes	Sensor and probe data fusion, counter data quality, scheme evaluation, commercial data licensing

The Scenario

The Westmoor Transport Partnership plans and funds transport across a city region of two point three million people. It manages traffic signals, bus priority, cycle infrastructure and major road schemes, and it bids for national funding against other regions. Every scheme it proposes must be justified by evidence of current traffic conditions and forecasts of future benefits, and every scheme it delivers must be evaluated to show whether it worked. Its current flagship programme involves bus lanes, segregated cycle routes and low-traffic neighbourhoods across the city's inner suburbs, which has provoked intense public debate. You are a senior data engineer in the transport analytics team, and the data on which the debate rests is thinner and more contested than anyone would like.

Traffic data comes from many sources. Automatic traffic counters, loops buried in the road, record vehicle counts, speeds and sometimes vehicle classes at around four hundred permanent sites, but around a quarter are faulty at any time, and repairs take months. Temporary counters, pneumatic tubes and cameras, are deployed for scheme studies, typically for two weeks at a time. Traffic signal systems record detector data at junctions, intended for signal control rather than analysis. Automatic number plate recognition cameras on key corridors measure journey times between points. Bus operators' vehicle location data shows bus journey times. Cycle counters on major routes record cycle flows. National statistics on road traffic provide annual estimates by road link. Commercial probe data from navigation and connected vehicle providers offers speeds and journey times across the whole network, at a price.

The director of transport strategy, Dr Helen Morris, needs evidence for a major funding bid due in six months, which requires a transport model

calibrated against observed traffic data and a clear baseline of conditions. The scheme delivery director, Richard Evans, needs evaluation of the low-traffic neighbourhoods, which have been in place for a year: did traffic reduce inside them, did it displace onto boundary roads, and did bus journey times and cycling levels change? Local councillors, residents' groups and campaign groups on both sides are all asking for data, and some have commissioned their own counts, producing conflicting numbers that are shared widely on social media.

The evaluation problem is methodologically difficult. Traffic varies with season, weather, school holidays, roadworks and long-term trends, including changes in working patterns since the pandemic. Isolating the effect of a scheme requires a before-and-after comparison with control areas and careful adjustment. The partnership's counts inside the neighbourhoods were collected with temporary counters for two weeks before and two weeks after implementation, at different times of year. Boundary road counts are from permanent counters, several of which were faulty during the evaluation period. Journey times on boundary roads come from number plate cameras on some corridors and bus data on others. The data does not fit together neatly.

The head of network management, Gary Holt, runs the traffic control centre and cares about real-time operations. He wants a platform that shows current network conditions, detects incidents quickly and informs signal timing changes. He points out that detector data from signals is the richest source of continuous traffic information the partnership owns, but it is currently discarded after a few days because nobody has built a way to store and clean it.

The procurement of commercial probe data has divided the team. It would provide network-wide speeds and journey times, filling gaps that counters cannot, and several neighbouring authorities have licensed it. But it measures speeds, not volumes; its coverage of minor roads is thin; its methodology is proprietary; and the licence is expensive and restricts publication of detailed data. The partnership's commitment to open data, strongly supported by its political leadership, sits awkwardly with a licence that prevents sharing the evidence behind decisions.

Public trust is a major issue. The low-traffic neighbourhood debate has been polarised, and each side accuses the other of cherry-picking data. The partnership's credibility depends on its evidence being seen as rigorous

and neutral. The head of communications, Amira Patel, wants all data used in evaluation to be published with methods, so that anyone can check it, and wants the partnership to be transparent about uncertainty. Some colleagues worry that publishing messy data with caveats will simply give ammunition to critics.

Your team is three engineers and two analysts. Counter data is held in a vendor's system with a web interface and limited export capabilities. Detector data is in the signal control system. Camera data is in another vendor's system. There is no central data platform, and analysts currently assemble evaluation datasets by hand, exporting files from each system and joining them in spreadsheets, a process that is slow and difficult to reproduce.

You see a need for a traffic data platform that consolidates all sources, cleans and validates counter data systematically, imputes missing data with documented methods, retains detector data long term, and supports both operational monitoring and evaluation. But you also see that the funding bid and the evaluation have fixed deadlines and cannot wait for a platform. Some of the data needed for the evaluation cannot be recovered: counts that were never collected cannot be created retrospectively, and faulty counters cannot be retrospectively repaired. The evaluation will have to be honest about its limitations.

The politics of the evaluation are pressing. The partnership's board includes the leaders of five councils, two of whom have publicly supported the low-traffic neighbourhoods and one of whom has promised to remove them if re-elected. A local newspaper has published a story based on a residents' group's own counts, claiming that traffic on a boundary road doubled after the scheme was introduced; the partnership's permanent counter on that road was faulty for most of the period in question, so it cannot directly refute or confirm the claim. National government has also issued guidance expressing scepticism about low-traffic neighbourhoods and emphasising the need for local support and evidence, and future funding may depend on how evaluations are conducted. Richard has been asked to present the evaluation to the board in four months, and he is under pressure from different directions to reach different conclusions. He wants the analysis to be unimpeachable, which means clear methods set out before the results are known, and independent review.

There is also the question of what the partnership should measure in future. Traffic counts measure vehicles, not people. The partnership's policy goals are about moving people sustainably, reducing emissions and improving safety and health. Measuring those outcomes requires counting pedestrians, cyclists and bus passengers as well as cars, and linking to air quality, road safety and health data. A platform designed around vehicle counts would perpetuate the bias the partnership is trying to move away from.

THE DECISION IN FRONT OF YOU

Do you license commercial probe data, build a consolidated traffic data platform on the partnership's own sensors, or focus on delivering a defensible evaluation and funding bid with existing data while designing the platform, and how do you maintain public trust and open data commitments in a polarised debate?

Data Sources

#	Source and link	How it is used in this project
1	DfT Road Traffic Statistics https://roadtraffic.dft.gov.uk	National traffic count points and annual average daily flows used for model calibration and control comparisons.
2	Caltrans Performance Measurement System (PeMS) https://pems.dot.ca.gov	Large loop detector dataset with documented imputation methods used to prototype counter cleaning.
3	DfT Transport Analysis Guidance (TAG) https://www.gov.uk/guidance/transport-analysis-guidance-tag	Official guidance for appraisal and evaluation of transport schemes.
4	Bus Open Data Service https://data.bus-data.dft.gov.uk	Bus timetable and vehicle location data for bus journey time analysis.
5	DfT Road Safety Data (STATS19) https://www.gov.uk/government/statistical-data-sets/road-safety-open-data	Collision data for safety outcome evaluation.
6	DEFRA UK-AIR https://uk-air.defra.gov.uk	Air quality monitoring data for environmental outcomes of schemes.
7	National Travel Survey https://www.gov.uk/government/collections/national-travel-survey-statistics	National travel behaviour data for context and mode share.
8	Active Travel England Guidance https://www.gov.uk/government/organisations/active-travel-england	Guidance on active travel schemes and monitoring expectations.

Deliverables (17)

- D1 Traffic data source inventory.** Sources, coverage, quality, retention and licensing for all traffic data.
- D2 Counter health monitoring.** Automated detection of faulty counters with maintenance escalation.
- D3 Counter data cleaning and imputation.** Documented methods for validating and filling gaps in counts.
- D4 Signal detector data retention.** Long-term storage and cleaning of junction detector data.
- D5 Journey time integration.** Combination of camera, bus and probe journey times with consistent segments.
- D6 Low-traffic neighbourhood evaluation dataset.** A consolidated before-and-after dataset with control areas.
- D7 Evaluation methodology.** A method adjusting for seasonality, trends and confounders with stated uncertainty.
- D8 Evaluation limitations statement.** A transparent account of data gaps and their implications.
- D9 Funding bid baseline.** Observed traffic conditions and model calibration data for the bid.
- D10 Probe data evaluation.** An assessment of commercial probe data coverage, accuracy and licence terms.
- D11 Open data publication.** Publication of counts, methods and evaluation data with licences.
- D12 Real-time network monitoring.** Current conditions and incident detection for the control centre.
- D13 People-movement metrics.** Measures of walking, cycling and bus use alongside vehicle counts.
- D14 Outcome linkage.** Integration of safety and air quality data into scheme evaluation.
- D15 Public data engagement plan.** Guidance and forums for sharing data with residents and campaign groups.
- D16 Platform architecture.** A design for a consolidated traffic data platform.
- D17 Monitoring framework for future schemes.** Standard data collection requirements before and after every scheme.

CASE STUDY 72 · WASTE MANAGEMENT (LOCAL AUTHORITY SERVICES)

Smart Waste Collection and Recycling Performance Data Platform

Industry	Waste Management (Local Authority Services)
Organisation	Greystone Borough Council, a unitary authority collecting waste from 140,000 households and 2,000 public litter bins
Your role	Data Engineering Lead, Environmental Services
Core themes	IoT fill-level sensors, route optimisation, statutory waste reporting, national policy change

The Scenario

Greystone Borough Council collects household waste and recycling from around one hundred and forty thousand homes, empties two thousand litter bins in parks and town centres, and runs two household waste recycling centres. Waste services are among the most visible things a council does: when collections are missed or bins overflow, residents notice immediately and complain loudly. They are also among the most expensive, and Greystone, like most councils, faces severe budget pressure. At the same time, national policy is transforming waste collection in England. New requirements for consistent collections oblige councils to collect a standard set of recyclable materials, including weekly food waste collections, from every household. Extended producer responsibility for packaging will make packaging producers pay for the costs of collecting and managing packaging waste, with payments to councils based on their efficiency and effectiveness. A deposit return scheme for drinks containers will divert some materials from council collections. You lead data engineering for environmental services, and the director, Simon Adeyemi, has asked you to prepare the service's data for this new world.

The service's data is operational rather than analytical. Collection rounds are planned in a routing system and carried out by crews using in-cab devices that record completed collections, missed bins and reasons, such as contamination or bins not presented. Weighbridge data records the weight of each vehicle's load at the transfer station and treatment facilities. The council reports tonnages quarterly to the national waste data system, which calculates recycling rates. Resident complaints and missed collection reports come through the contact centre and the council's

website. A pilot scheme installed fill-level sensors in two hundred litter bins and fifty communal bins at flats, reporting how full they are several times a day.

The sensor pilot has produced mixed results. Its vendor promised that emptying bins only when full would reduce collections by a third. In practice, the sensors reported fill levels reliably in some bins but gave erratic readings in others, affected by the shape of waste, moisture and temperature. Crews continued to empty bins on fixed schedules because the routing system could not easily incorporate sensor data into daily routes, and the vendor's dashboard sat unused. The vendor now wants to expand the pilot to all litter bins, at a substantial cost, and has offered route optimisation software as part of the deal.

The operations manager, Karen Blackwood, is sceptical. She has run the service for fifteen years and knows its rounds intimately. She believes the bigger efficiency gains lie in redesigning household rounds to accommodate the new food waste collections, which will require new vehicles, crews and routes, and in reducing missed collections, which generate return trips and complaints. She wants accurate data on round performance, missed collections and their causes, not more sensors.

The finance manager, Tom Riley, is focused on extended producer responsibility. Payments to councils will depend on data: the tonnages of packaging collected, the composition of waste, and assessments of efficiency and effectiveness against benchmarks. Councils will need to provide evidence of their costs and performance, and the scheme administrator will use data from the national waste data system and compositional analyses. Tom wants the council's data to be accurate and complete enough to maximise legitimate payments, which could be substantial. He has heard that councils with poor data may receive lower payments.

The waste strategy officer, Dr Lucy Chen, wants to understand recycling behaviour. Greystone's recycling rate is below the national average, and contamination of recycling bins, where residents put the wrong materials in, causes loads to be rejected at the recycling facility. She wants to identify areas with high contamination or low participation so that communication campaigns can be targeted. The crews' in-cab devices record contamination at individual properties, which could support targeted feedback, but using property-level data to identify and contact

residents raises questions about fairness and privacy, and some councillors worry about being seen to police residents.

The data quality problems are real. In-cab device data depends on crews recording events correctly, which varies. Weighbridge data records vehicle loads, which may combine several rounds or materials. Tonnages reported to the national system are sometimes estimated rather than measured, particularly for materials handled by third-party contractors. Round definitions change over time, so historical comparisons are difficult. Property data in the routing system does not always match the council's address gazetteer.

Your team is two engineers and an analyst, shared with other environmental services. The council's corporate data platform is a small cloud data warehouse used for finance and social care reporting. The waste service's systems are commercial products with limited integration.

The national policy timetable is fixed. Food waste collections must start within eighteen months. Extended producer responsibility payments have started, with data requirements evolving. The deposit return scheme will change material flows. Each of these changes affects the data the service needs, and building a platform around current arrangements risks it becoming obsolete.

You see several competing priorities: expanding sensors, supporting round redesign for food waste, strengthening tonnage and cost data for producer responsibility payments, and analysing recycling behaviour. You also see a common foundation: a reliable model of properties, rounds, collections, vehicles, tonnages and materials, linked to the address gazetteer and the national reporting system, which would support all of them.

The workforce angle matters here too. Collection crews work physically demanding jobs in all weathers, often starting before dawn, and many have done the same rounds for years. They have been paid on a task-and-finish basis, which rewards completing rounds quickly and has led to practices, such as skipping hard-to-reach bins and recording them as not presented, that affect data quality as well as service quality. Trade union representatives have agreed to discuss new round designs for food waste, but they are wary of any technology that monitors crews' speed or location more closely, and they point out that the in-cab devices were introduced with an assurance that they would not be used for disciplinary purposes. Any use of collection data to evaluate crews, rather than rounds, would

breach that assurance and could lead to industrial action at a time when the service can least afford disruption.

There is also the question of what data residents should see. Several councils publish live maps of collection progress so that residents know whether their bin has been emptied or whether the crew is running late, which reduces calls to the contact centre. Others publish recycling performance by ward. These services are popular, but they require accurate, timely data, and publishing inaccurate data would generate more complaints, not fewer. Karen worries that live tracking would expose crews to criticism for delays that are beyond their control, such as blocked roads or vehicle breakdowns.

THE DECISION IN FRONT OF YOU

Do you expand the fill-level sensor programme with the vendor's route optimisation, prioritise round redesign and missed collection data for the food waste rollout, or focus on tonnage and cost data for producer responsibility payments, and how do you use property-level recycling data fairly and build a platform that survives national policy changes?

Data Sources

#	Source and link	How it is used in this project
1	WasteDataFlow https://www.wastedataflow.org	National local authority waste reporting system defining tonnage and material data requirements.
2	DEFRA Local Authority Collected Waste Statistics https://www.gov.uk/government/statistical-data-sets/env18-local-authority-collected-waste-annual-results-tables	Published recycling rates and tonnages by authority for benchmarking.
3	Packaging Extended Producer Responsibility Guidance https://www.gov.uk/guidance/extended-producer-responsibility-for-packaging-what-local-authorities-need-to-know	Requirements and payment mechanisms affecting council data.
4	Simpler Recycling Guidance (England) https://www.gov.uk/government/publications/simpler-recycling-in-england-policy-update	Consistent collection requirements including food waste.
5	WRAP Local Authority Resources https://wrap.org.uk/resources	Guidance on collection systems, contamination and compositional analysis.
6	OS AddressBase and GeoPlace https://www.geoplace.co.uk/addresses-streets	Authoritative address gazetteer for linking properties to rounds.
7	Google OR-Tools Vehicle Routing https://developers.google.com/optimization/routing	Open-source routing optimisation for round redesign scenarios.

#	Source and link	How it is used in this project
8	Smart Bin Fill-Level Sensor Dataset (Kaggle) https://www.kaggle.com/datasets/tolgakurtulus/smart-waste-bin-dataset	Sensor readings used to prototype fill-level ingestion and validation.

Deliverables (17)

- D1** **Sensor pilot evaluation.** An assessment of sensor accuracy, reliability, operational use and cost-benefit.
- D2** **Waste service data model.** A model of properties, rounds, collections, vehicles, tonnages and materials.
- D3** **Address gazetteer linkage.** Matching of routing system properties to the authoritative gazetteer.
- D4** **In-cab data quality improvement.** Standards and feedback to crews on event recording.
- D5** **Missed collection analytics.** Analysis of missed collections by cause, round and crew conditions.
- D6** **Round performance dashboard.** Completion, timing and exception views for operations managers.
- D7** **Food waste round redesign scenarios.** Modelling of new rounds, vehicles and crews for food waste collections.
- D8** **Tonnage reconciliation.** Reconciliation of weighbridge, contractor and national reporting tonnages.
- D9** **Producer responsibility data pack.** Cost, tonnage and performance evidence for EPR payments.
- D10** **Material flow tracking.** Tracking of materials from collection to treatment and end destination.
- D11** **Contamination and participation analysis.** Area-level recycling behaviour analysis with fair use principles.
- D12** **Resident feedback policy.** Rules for using property-level contamination data for engagement.
- D13** **Sensor data integration design.** Integration of reliable sensor data into routing where it adds value.
- D14** **Live collection status service.** A resident-facing collection progress feature with data quality safeguards.
- D15** **Policy change impact assessment.** Expected effects of food waste, EPR and deposit return on data and operations.
- D16** **Platform roadmap.** A flexible design accommodating national policy changes.
- D17** **Vendor contract recommendation.** A decision on the sensor and routing vendor's expansion offer.

CASE STUDY 73 · BIOTECHNOLOGY (DRUG DISCOVERY)

Laboratory Information and Instrument Data Integration for Early Drug Discovery

Industry	Biotechnology (Drug Discovery)
Organisation	Catalyst Biosciences, a drug discovery company with 180 scientists working on small molecule oncology programmes
Your role	Lead Data Engineer, Research Informatics
Core themes	Instrument data capture, FAIR data principles, assay data standardisation, machine learning readiness

The Scenario

Catalyst Biosciences discovers small molecule drugs for cancer. Its chemists design and synthesise new compounds; its biologists test them in assays measuring how strongly they bind to protein targets, how they affect cancer cells and how they behave in models of absorption, metabolism and toxicity. Each design, make, test and analyse cycle generates data that informs the next round of design. The company's investors, and increasingly its pharmaceutical partners, believe that machine learning could accelerate this cycle dramatically, by predicting which compounds are worth making before they are synthesised. The company has hired a small machine learning team and announced an AI-driven discovery strategy. You lead data engineering in research informatics, and you have discovered that the company's experimental data is not in a state that machine learning can use.

The laboratory runs dozens of instruments: plate readers, liquid handlers, high-content imaging systems, mass spectrometers, nuclear magnetic resonance spectrometers and automated synthesis platforms. Each instrument produces data in its own format, often proprietary, saved to a local computer attached to the instrument. Scientists copy files to shared drives, process them in spreadsheets or vendor analysis software, and record summary results, such as an inhibitory concentration for each compound, in an electronic laboratory notebook. A laboratory information management system tracks samples and compound registration. A compound registration system holds chemical structures. There is no consistent link between raw instrument files, processed results, the compounds tested and the experimental conditions. Several instrument

computers still run outdated operating systems because the vendor software will not run on newer ones.

The head of the machine learning team, Dr Priya Raman, has been trying to build predictive models for the company's main programme. She found that compound potency values for the same compound in the same assay varied widely between experiments, sometimes by more than an order of magnitude, and that the reasons were impossible to reconstruct: different assay protocols, reagent batches, cell passage numbers, plate layouts and curve-fitting settings had been used, but none of these were recorded consistently. Some results had been excluded from the notebook by scientists who judged them unreliable, without recording why. Negative results, compounds that showed no activity, were often not recorded at all, which biases any model trained on the data. Priya estimates that more than half of the historical data is unusable for modelling without extensive manual curation.

The chief scientific officer, Dr Martin Hayes, is frustrated. He wants the machine learning strategy to succeed, and he knows that investors are watching. But he also knows that his scientists are overloaded and resistant to changes that slow them down. Many regard data recording as administrative overhead. The head of biology, Dr Sarah Lin, supports better data but warns that imposing rigid data entry requirements on bench scientists will fail unless the systems make their work easier, for example by automating data capture from instruments and curve fitting.

The head of chemistry, Dr James Okoro, has a related concern. Chemical structures in the registration system are mostly reliable, but salt forms, stereochemistry and batch information are sometimes recorded inconsistently, and some compounds bought from suppliers were registered with structures that later proved incorrect. Structure-activity models depend on accurate structures.

A pharmaceutical partner, which funds one of Catalyst's programmes, has contractual rights to receive data from that programme, and has asked for it in a standardised, well-annotated form that its own data scientists can use. The partnership agreement also restricts use of that programme's data for other purposes, including training models for Catalyst's other programmes, which complicates Priya's plans to build models across all the company's data.

Public data offers a partial remedy. Large public databases of bioactivity data, curated from scientific literature and patents, contain millions of measurements across thousands of targets. They could supplement Catalyst's own data for pre-training models, but they suffer from similar inconsistency problems at larger scale, and their relevance to Catalyst's specific targets and chemical series varies. Community principles for making research data findable, accessible, interoperable and reusable have been widely adopted in academia and are increasingly expected by partners and funders.

Your team is two engineers and a data scientist. The company's IT is modest, with most systems hosted in the cloud but no research data platform. Vendors offer scientific data management platforms that promise to capture instrument data automatically, standardise it and link it to experiments, but they are expensive, require significant configuration and have mixed reputations. Two peer companies that bought such platforms told you privately that implementation took twice as long as planned and that adoption by bench scientists remained patchy a year later.

You see a need to change how data is captured at the point of generation, so that future data is complete, consistent and linked, and a separate need to rescue historical data that is valuable but messy. The first requires working with scientists to automate capture and standardise protocols without slowing them down. The second requires curation, which consumes expert time. Both compete with the machine learning team's desire for data now.

There is also the question of scope. The laboratory runs many assay types; standardising all of them at once would be impossible. Focusing on the few assays that matter most for the lead programme, such as the primary biochemical potency assay and key cellular assays, would produce usable data quickly but leave other areas untouched. The partner programme has its own data requirements that might take precedence for contractual reasons.

There is also a cultural divide between the machine learning team and the laboratory scientists. The machine learning team joined recently from technology companies and academic computational groups; they speak a different language, work to different rhythms and have occasionally frustrated scientists by asking for data in formats that make no sense at the bench. At a recent programme meeting, a model prediction that a set of

compounds would be highly potent was presented with confidence, and the compounds turned out to be inactive when synthesised; the chemists have not forgotten. Some senior scientists now regard the AI strategy as investor theatre. Priya is acutely aware of this and wants the data programme to demonstrate value to scientists directly, for example by giving them faster, more reliable access to their own results, rather than being seen as a service to the machine learning team alone.

The company's cash runway adds pressure. Catalyst has about two years of funding, and its next financing round will depend on demonstrating progress in its lead programme and credibility in its AI strategy. A data programme that delivers value only after eighteen months would be of limited use to that round.

THE DECISION IN FRONT OF YOU

Do you buy a scientific data management platform, build automated instrument capture and assay data standards in-house for priority assays, or focus on curating historical data for immediate machine learning use, and how do you respect partner data restrictions while building models across the company's data?

Data Sources

#	Source and link	How it is used in this project
1	ChEMBL https://www.ebi.ac.uk/chembl/	Curated bioactivity database used to prototype assay data models and pre-train activity prediction models.
2	PubChem BioAssay https://pubchem.ncbi.nlm.nih.gov	Large public repository of screening results including inactive compounds.
3	Allotrope Foundation Data Standards https://www.allotrope.org	Standards for instrument data formats and semantic annotation in laboratories.
4	FAIR Guiding Principles (GO FAIR) https://www.go-fair.org/fair-principles/	Principles for findable, accessible, interoperable and reusable research data.
5	BioAssay Ontology https://bioportal.bioontology.org/ontologies/BAO	Ontology for describing assay protocols and conditions consistently.
6	RDKit https://www.rdkit.org	Open-source cheminformatics toolkit for structure standardisation and validation.
7	SiLA 2 Standard https://sila-standard.com	Laboratory automation interoperability standard for instrument integration.
8	Therapeutics Data Commons https://tdcommons.ai	Curated machine learning datasets for drug discovery benchmarking.

Deliverables (17)

- D1 Data landscape and readiness assessment.** Instruments, systems, formats and data quality across research data.
- D2 Priority assay selection.** A selection of assays for first standardisation based on programme value.
- D3 Assay metadata standard.** Required conditions, protocols, reagents and plate layouts recorded per experiment.
- D4 Automated instrument capture.** Collection of raw data from priority instruments directly into a managed store.
- D5 Curve-fitting standardisation.** Consistent, versioned analysis methods for potency calculations.
- D6 Experiment-compound-result linkage.** Linking raw files, processed results, compounds and conditions.
- D7 Negative and excluded result capture.** Recording of inactive compounds and exclusion reasons.
- D8 Compound structure standardisation.** Validation of structures, salts and stereochemistry in registration.
- D9 Historical data curation programme.** Prioritised curation of historical results with quality flags.
- D10 Machine learning ready datasets.** Curated, versioned training datasets with documented provenance.
- D11 Public data integration.** Selected public bioactivity data aligned to internal standards.
- D12 Partner data segregation.** Controls enforcing partnership data use restrictions.
- D13 Partner data delivery.** Standardised, annotated data packages meeting contractual requirements.
- D14 Scientist workflow design.** Capture processes co-designed with scientists to reduce effort.
- D15 Vendor platform evaluation.** An assessment of scientific data management platforms.
- D16 FAIR compliance roadmap.** A plan for aligning research data with FAIR principles.
- D17 Financing milestone plan.** Deliverables aligned with the next financing round.

CASE STUDY 74 · ANIMAL HEALTH (GOVERNMENT AGENCY)

Livestock Disease Surveillance and Animal Movement Data Platform

Industry	Animal Health (Government Agency)
Organisation	The National Animal Health Agency, responsible for surveillance and control of notifiable livestock diseases
Your role	Principal Data Engineer, Epidemiology and Surveillance
Core themes	Animal movement tracing, outbreak preparedness, laboratory and field data integration, farm data sensitivity

The Scenario

The National Animal Health Agency exists to protect the country's livestock from serious diseases and, through them, to protect public health, trade and the rural economy. It investigates suspected cases of notifiable diseases such as foot-and-mouth disease, avian influenza, African swine fever, bluetongue and bovine tuberculosis; runs surveillance programmes; and leads the response when outbreaks occur. The country's worst outbreak of foot-and-mouth disease, a generation ago, led to the slaughter of millions of animals, cost billions and devastated rural communities. Its lessons shaped today's contingency plans, including the critical importance of tracing animal movements quickly. You are the principal data engineer for epidemiology and surveillance, and a recent national exercise simulating a foot-and-mouth outbreak exposed serious weaknesses in the agency's data.

During the exercise, epidemiologists needed to identify all farms that had sent or received animals from the first infected premises in the previous three weeks, all farms within control zones around it, all vehicles, markets and abattoirs connected to it, and the contacts of those farms in turn. The data existed in principle. Cattle movements are recorded in a national cattle tracing system in which every animal has a passport and every movement is reported. Sheep, goats and pigs are recorded in separate movement reporting systems that record batch movements rather than individual animals. Farm locations and holdings are recorded in a register of agricultural holdings, but many holdings consist of multiple parcels of land, sometimes miles apart, and the registered address may be the farmhouse rather than where animals are kept. Markets and abattoirs report movements, but with delays. Laboratory results from the agency's

own laboratories and from private veterinary laboratories are held in a laboratory information system. Field investigation records are completed by veterinary officers on paper forms and later typed into a case management system.

In the exercise, assembling a complete picture of the first infected premises' connections took almost two days, because each system had to be queried separately, holdings were identified differently across systems, sheep batch movements could not be traced to individual flocks reliably, and land parcels used for grazing were not linked to holdings. In a real outbreak of a disease that spreads as quickly as foot-and-mouth, two days could allow the disease to reach dozens more farms. The exercise report recommended an integrated tracing capability that could produce a complete contact picture within hours.

The chief veterinary officer, Professor Helen Carr, has made this a priority. She also wants the agency's routine surveillance improved, particularly for avian influenza, which has caused unprecedented outbreaks in poultry and wild birds in recent years, and for new threats such as bluetongue, spread by midges and linked to climate change. Surveillance requires combining laboratory results, farm data, movement data, wild bird findings, environmental data and international disease reports.

The head of epidemiology, Dr Rajesh Kumar, wants a platform that supports disease spread modelling. During outbreaks, models are used to predict how the disease might spread under different control strategies, informing decisions such as the size of control zones and whether to vaccinate. Models need accurate data on farm locations, animal populations, contact networks and movements. The agency's modellers currently spend days preparing input data from extracts before they can run models.

The IT director, Mark Stevens, is concerned about the age and fragility of the systems. Several are more than fifteen years old and are maintained by contractors under contracts that will expire soon. A replacement programme for the livestock information systems has been under way for years, led by another government body, and will eventually replace the movement reporting systems with a single new service. Mark does not want the agency to build a tracing platform on systems that are about to be replaced, but the replacement timetable has slipped repeatedly and nobody can say when it will be complete.

The data protection officer, Janet Morris, reminds colleagues that farm data includes personal data about farmers, who are often sole traders or family businesses, and that some data, such as disease status, is commercially sensitive: a farm under restrictions cannot sell animals, and publication of its identity could harm its business and the farmer's reputation. During past outbreaks, farmers' details were leaked, and some were harassed. Sharing data with other bodies, such as local authorities, police and the armed forces, which help during major outbreaks, must be planned in advance with clear agreements.

The farming industry has a stake too. Farmers' organisations support effective disease control but are wary of government data gathering. They have raised concerns about how data collected for disease control might be used for other purposes, such as enforcement of environmental rules or subsidy compliance. Trust matters because disease control depends on farmers reporting suspicion of disease promptly, which they may be reluctant to do if they fear the consequences.

Your team is four engineers and two analysts. The agency has a modern data platform used for some reporting, but the movement and holdings data is accessed through nightly extracts from the source systems. Helen wants a tracing capability demonstrably faster than the exercise within six months, before the next national exercise.

You see several design choices. Building an integrated tracing data model that links holdings, land parcels, animals, batches, movements, markets, vehicles and laboratory results, with daily or more frequent updates from each source, would address the exercise findings directly. Doing so on the current systems risks rework when they are replaced, but waiting for the replacement risks being unprepared for a real outbreak, which could happen at any time. A tracing model built around stable concepts, with adapters for each source system that can be replaced later, might reduce the risk of rework.

There is an operational reality on the ground that data systems often ignore. During an outbreak, veterinary officers work in protective clothing on infected premises, with limited connectivity and under great stress. The forms they complete, and the information they record about animals, clinical signs, estimated ages of lesions and contacts reported by farmers, form the evidence base for epidemiological judgements about when infection entered a premises and where it may have spread. In the

exercise, some of this information was relayed by telephone to the incident centre and typed into spreadsheets, where it was hard to link to the movement data. Field staff asked for simple mobile tools that work offline and synchronise when a connection is available, and they asked not to be sent long forms that take time away from the animals and the farmers, who are often in shock at the prospect of losing their herds.

There is also a tension between routine and emergency use. In routine times, a platform might be used for surveillance analysis by a small team. In an emergency, it must support hundreds of users, including field staff and partner organisations, under intense pressure, around the clock, and must continue to work if parts of the infrastructure fail. Designing for emergency scale and resilience is expensive and may seem excessive in quiet periods, but under-designing could fail when it matters most.

THE DECISION IN FRONT OF YOU

Do you build an integrated tracing and surveillance data platform on current source systems, wait for the national livestock system replacement, or build a source-agnostic tracing model with replaceable adapters, and how do you design for emergency scale, protect farmers' data and maintain the trust that disease reporting depends on?

Data Sources

#	Source and link	How it is used in this project
1	FAO EMPRES Global Animal Disease Information System (EMPRES-i) https://empres-i.apps.fao.org	International animal disease outbreak data used to prototype event ingestion and international surveillance.
2	WOAH World Animal Health Information System (WAHIS) https://wahis.woah.org	Official international notifiable disease reports.
3	APHA Animal Disease Surveillance Reports https://www.gov.uk/government/collections/animal-disease-surveillance-reports	UK surveillance outputs and disease situation data.
4	Avian Influenza Findings in Wild Birds (GOV.UK) https://www.gov.uk/government/publications/avian-influenza-in-wild-birds	Wild bird surveillance data for avian influenza risk assessment.
5	Livestock Information Service https://www.livestockinformation.org.uk	National livestock movement reporting service replacing legacy systems.
6	Cattle Tracing System Guidance https://www.gov.uk/guidance/cattle-tracing-system-cts	Cattle movement reporting requirements and data structures.

#	Source and link	How it is used in this project
7	Rural Payments Agency Land Parcel Data https://environment.data.gov.uk/dataset/7a9fd4b4-4f4f-4dde-bcf2-8e0fb3e33b88	Agricultural land parcel boundaries used to link holdings to land.
8	Foot and Mouth Disease 2001 Lessons Learned Inquiry https://www.gov.uk/government/publications/foot-and-mouth-disease-2001-lessons-to-be-learned-inquiry-report	Historical lessons on tracing and data in outbreak response.

Deliverables (17)

- D1 Exercise findings data analysis.** A detailed analysis of data delays and gaps in the national exercise.
- D2 Tracing data model.** A source-agnostic model of holdings, land, animals, batches, movements and premises.
- D3 Holding and land parcel linkage.** Linking holdings to all land parcels and locations where animals are kept.
- D4 Movement data integration.** Frequent ingestion of cattle, sheep, goat and pig movements through replaceable adapters.
- D5 Batch movement tracing logic.** Methods for tracing batch movements to origin and destination flocks and herds.
- D6 Market and abattoir integration.** Timely ingestion of market and slaughterhouse movement records.
- D7 Laboratory result integration.** Linkage of laboratory submissions and results to holdings and cases.
- D8 Field investigation digitisation.** Digital capture of field investigation forms.
- D9 Rapid contact tracing service.** Queries producing complete contact networks within hours.
- D10 Control zone analysis.** Identification of premises and animals within control zones.
- D11 Modelling input datasets.** Ready-to-use population, location and contact data for epidemiological models.
- D12 Surveillance analytics.** Integrated surveillance views for avian influenza, bluetongue and other priorities.
- D13 Emergency scale and resilience design.** Capacity and failover for outbreak conditions.
- D14 Partner data sharing agreements.** Pre-agreed data sharing with local authorities, police and other responders.
- D15 Farm data protection controls.** Access restrictions and disclosure rules for disease status and farmer data.
- D16 Farmer engagement statement.** A public statement on data use limited to disease control purposes.

D17 **Exercise readiness plan.** Deliverables and testing for the next national exercise.

CASE STUDY 75 • COMMODITY TRADING AND FOOD MANUFACTURING

Deforestation-Free Supply Chain Monitoring with Earth Observation Data

Industry	Commodity Trading and Food Manufacturing
Organisation	Ashby Commodities, a trader and processor of cocoa, palm oil, soy and coffee supplying European food manufacturers
Your role	Lead Data Engineer, Sustainability and Traceability
Core themes	Satellite imagery at scale, geolocation data from smallholders, regulatory due diligence, false positive management

The Scenario

Ashby Commodities buys cocoa from West Africa, palm oil from Southeast Asia, soy from South America and coffee from East Africa and Latin America, processes some of it and sells it to food manufacturers across Europe. It works with tens of thousands of farms, most of them smallholdings of a few hectares, through networks of cooperatives, local traders and processing mills. Its business has been transformed by the European Union's regulation on deforestation-free products, which requires companies placing certain commodities on the European market to demonstrate, through due diligence, that the products were not produced on land deforested after the end of 2020 and that they comply with the laws of the country of production. Crucially, the regulation requires geolocation of every plot of land where the commodities were produced. You lead data engineering for sustainability and traceability, and the company's ability to keep trading in Europe depends on what you build.

The regulation's requirements are demanding. For each shipment placed on the market, Ashby must submit a due diligence statement to an information system run by the European Commission, including geolocation coordinates or polygons for all plots of land that contributed to the shipment. It must assess the risk that any plot was deforested after the cut-off date, using satellite imagery and other information, and it must mitigate any risk found, which may mean excluding the plot's production. Competent authorities in member states will check a proportion of statements, and penalties for non-compliance include fines and confiscation of goods. Customers, the food manufacturers, want assurance

that Ashby's supply is compliant, because they are also subject to the regulation.

Ashby's traceability data is incomplete. For palm oil, it knows the mills from which it buys and has partial data on the plantations and smallholders that supply those mills. For cocoa, it has farm lists from its direct sourcing programmes with cooperatives, covering about sixty per cent of its volume, with GPS points for most farms and polygons for some; the rest comes through local traders with little farm-level information. For soy, it buys from large exporters that provide aggregated origin data. For coffee, traceability is mixed. Where farm data exists, quality varies: GPS points are sometimes recorded at the farmer's house rather than the farm, polygons were drawn by field staff walking boundaries with handheld devices and are sometimes self-intersecting or implausibly large, and the same farm appears in several cooperatives' lists.

Earth observation data is the main tool for assessing deforestation. Global forest monitoring platforms publish annual maps of tree cover loss derived from satellite imagery, near-real-time deforestation alerts and maps of primary forest. The European Commission has published a global map of forest cover at the end of 2020 as a reference. Satellite programmes provide freely available imagery at resolutions sufficient to see changes at the scale of small farms. Commercial providers offer higher-resolution imagery and analysis services. But satellite-based assessment has well-known limitations: distinguishing forest from tree crops such as cocoa agroforestry and rubber is difficult; cloud cover obscures tropical regions for much of the year; and alerts produce false positives, such as seasonal changes or harvesting of plantation trees, as well as false negatives, such as degradation under canopy.

The head of sustainability, Dr Grace Mensah, has worked on deforestation issues for years and understands the complexity. She is concerned that a blunt approach will exclude smallholders who cannot provide accurate geolocation or whose farms are flagged by imperfect satellite analysis, pushing them out of supply chains that serve Europe and into less scrupulous markets. Smallholders are among the poorest people in the supply chain, and excluding them could worsen poverty without protecting forests. She wants Ashby to invest in helping cooperatives map farms accurately and to investigate flagged plots on the ground rather than excluding them automatically.

The head of trading, Philippe Durand, is focused on commercial continuity. Shipments must move. If due diligence statements cannot be produced on time, goods will be held at ports. He wants a system that produces compliant statements quickly and reliably, and he is willing to shift sourcing towards suppliers with better data if necessary, even if that means dropping some smallholder programmes. Several competitors have already announced that they will source only from fully mapped supply chains.

The legal counsel, Anna Fischer, warns that the regulation also requires compliance with local laws, including land rights, labour rights and environmental laws, which satellite data cannot assess. She also notes that geolocation data about individual farmers is personal data, and that sharing it with customers, authorities and the European information system requires a legal basis and appropriate protections, particularly for smallholders who may not understand how their data will be used. Some producing countries have their own data sovereignty rules and national traceability systems.

The data engineering challenges are substantial. Farm geolocation data must be collected from many sources, standardised, validated and deduplicated. Plot polygons must be checked for geometric validity and plausibility. Supply chain links must be established from plots to cooperatives, mills, warehouses, shipments and customer contracts, with mass balance or segregation rules reflecting how commodities are mixed during processing. Satellite analysis must be run on every plot, repeatedly, with results stored and reviewed. Due diligence statements must be generated and submitted through the European system's interface, with references returned and passed to customers.

Your team is four engineers and two geospatial analysts. The company's existing systems include a trading platform, a sustainability data collection tool used by field staff and spreadsheets. Several technology vendors offer compliance platforms that combine traceability, satellite monitoring and statement submission. The regulation's application date is approaching, and Ashby must be ready.

You see that the core challenge is data quality at the plot level and linkage through the supply chain, not satellite analysis, which is increasingly commoditised. You also see an ethical choice embedded in the technical

design: how the system treats uncertain or flagged plots determines whether smallholders are supported or excluded.

The competitive dynamics add pressure. Some of Ashby's customers have begun asking for plot-level data themselves, so that they can run their own satellite checks, and at least one large manufacturer has said it will reduce purchases from suppliers that cannot provide full traceability. Other customers are less demanding but want contractual warranties that shipments are compliant, shifting the legal risk onto Ashby. Meanwhile, some producing-country governments and cooperatives have expressed frustration that European buyers demand farm-level data without paying for the cost of collecting it or sharing the value created. The regulation has also been subject to political debate within the European Union, with changes to implementation timelines and simplifications for low-risk countries, which makes planning harder: the system must be ready for the strictest interpretation while remaining adaptable to changes in the rules and in country risk classifications.

There is also a practical question of volume and repeated analysis. Tens of thousands of plots must be assessed against multiple data layers, reassessed as new imagery and alerts arrive, and the results linked to specific shipments. The satellite data volumes are large, but open platforms and cloud-based analysis tools have made processing feasible for a team of Ashby's size. The harder part is the workflow for reviewing flagged plots: who investigates, with what evidence, and how decisions are recorded for auditors.

THE DECISION IN FRONT OF YOU

Do you buy an integrated compliance platform, build in-house plot data management and satellite risk assessment, or combine a vendor for monitoring with in-house supply chain linkage, and how do you design the treatment of uncertain and flagged plots so that compliance does not come at the cost of excluding smallholders unfairly?

Data Sources

#	Source and link	How it is used in this project
1	Global Forest Watch Open Data Portal https://data.globalforestwatch.org	Tree cover loss, primary forest and deforestation alert datasets used for plot-level risk assessment.

#	Source and link	How it is used in this project
2	Copernicus Data Space Ecosystem (Sentinel-2) https://dataspace.copernicus.eu	Free satellite imagery for visual verification and change detection.
3	EU Regulation on Deforestation-free Products https://environment.ec.europa.eu/topics/forests/deforestation/regulation-deforestation-free-products_en	Regulatory requirements for due diligence, geolocation and statements.
4	JRC Global Forest Cover Map 2020 https://forest-observatory.ec.europa.eu	The EU reference map of forest cover at the regulatory cut-off date.
5	Trase Supply Chain Data https://trase.earth	Commodity supply chain and deforestation exposure data by origin.
6	Accountability Framework Initiative https://accountability-framework.org	Guidance on deforestation-free and conversion-free supply chain commitments.
7	Google Earth Engine https://earthengine.google.com	Cloud geospatial analysis platform for large-scale satellite processing.
8	Open Geospatial Consortium GeoJSON and Simple Features https://www.ogc.org/standards/	Standards for geolocation polygon formats and validation.

Deliverables (17)

- D1 Traceability gap assessment.** Plot-level data coverage and quality by commodity, origin and supplier.
- D2 Plot data model.** A model of plots, producers, geolocation, ownership and supplier links with provenance.
- D3 Geolocation validation.** Geometric validity, plausibility and duplicate checks on points and polygons.
- D4 Supplier data collection standard.** Requirements and tools for cooperatives and traders to provide plot data.
- D5 Farm mapping support programme.** Field mapping assistance for smallholder cooperatives.
- D6 Supply chain linkage.** Linking plots to cooperatives, mills, warehouses, shipments and contracts.
- D7 Mass balance and segregation logic.** Rules reflecting how commodities are mixed in processing.
- D8 Satellite risk assessment pipeline.** Automated assessment of each plot against reference and loss datasets.
- D9 Alert monitoring.** Ongoing ingestion of deforestation alerts with plot matching.
- D10 Flagged plot review workflow.** Investigation, evidence recording and decision process for flagged plots.

- D11 Smallholder treatment policy.** Principles for handling uncertain and flagged plots fairly.
- D12 Legality risk assessment.** Assessment of land rights and legal compliance beyond satellite data.
- D13 Due diligence statement generation.** Automated statement preparation and submission to the EU system.
- D14 Customer assurance data feed.** Statement references and compliance data for food manufacturers.
- D15 Producer data protection controls.** Legal basis, consent and protection for farmer geolocation data.
- D16 Vendor platform evaluation.** An assessment of compliance platforms against requirements.
- D17 Audit readiness pack.** Evidence and documentation for competent authority checks.

CASE STUDY 76 • REINSURANCE

Catastrophe Exposure Data Pipeline for a Specialist Reinsurer

Industry	Reinsurance
Organisation	Meridian Re, a Bermuda- and London-based property catastrophe reinsurer with \$4.2 billion of annual premium
Your role	Head of Exposure Data Engineering
Core themes	Cedant exposure data quality, catastrophe model inputs, post-event loss estimation, open hazard data

The Scenario

Meridian Re sells reinsurance to insurance companies around the world, protecting them against catastrophic losses from events such as hurricanes, earthquakes, floods and wildfires. When a hurricane strikes Florida or an earthquake strikes Japan, insurers that have written many policies in the affected area face huge claims, and reinsurers like Meridian pay a share of them. Pricing this protection, and managing the reinsurer's own accumulated exposure to catastrophes, depends on catastrophe models: complex simulations that estimate the probability and severity of losses from thousands of possible events. Catastrophe models are only as good as the exposure data fed into them, the details of the properties insured: their locations, values, construction, occupancy, age and policy terms. You are the head of exposure data engineering, and the quality of that data has become a board-level concern.

Every year, around the reinsurance renewal dates, Meridian receives exposure data from its clients, the ceding insurers, known as cedants. Cedants submit their portfolios in spreadsheets or in formats specific to the major commercial catastrophe modelling platforms. Some submissions are detailed, with each property's precise coordinates, construction class, number of storeys, year built, roof type and insured values for buildings, contents and business interruption. Others are aggregated to postcode or county level, with limited information on construction. Some are missing values or contain obvious errors: properties geocoded to the middle of lakes, buildings with a thousand storeys, values in the wrong currency, or entire portfolios duplicated because a cedant sent the same file twice under different names. Meridian's exposure analysts clean and enhance the data, run it through catastrophe models and pass results to

underwriters, all within the intense time pressure of renewal season, when hundreds of submissions arrive within weeks.

Last year's hurricane season exposed the consequences. A major hurricane caused losses to several of Meridian's cedants far above what the catastrophe models had predicted. When the exposure team investigated, they found that some cedants' exposure data had understated property values significantly, because values had not been updated for construction cost inflation; some had misclassified construction types; and some had excluded exposures written through managing agents. The models had also underestimated losses from storm surge and from post-event demand surge, when repair costs rise because of the scale of damage. Meridian's loss from the hurricane was well above its own estimates, and the board wants to know why and what will change.

The chief underwriting officer, Charles Whitmore, wants better exposure data quality assessment at the point of submission, so that underwriters can price for data uncertainty and push back on cedants with poor data. He wants each submission scored for quality and completeness, with automatic flags for anomalies, and he wants historical comparisons that show how a cedant's portfolio has changed from year to year.

The head of catastrophe research, Dr Mei Tanaka, wants Meridian to develop its own view of risk rather than relying entirely on commercial catastrophe models. She wants to use open hazard data, such as historical hurricane tracks, earthquake catalogues, flood maps and wildfire perimeters, along with claims data from past events, to validate and adjust the commercial models. Industry initiatives have produced open catastrophe modelling frameworks and open exposure data standards, which could reduce dependence on a small number of commercial model vendors. She also wants the exposure data to include new attributes that affect vulnerability, such as roof age and mitigation features.

The chief risk officer, David Okafor, is focused on accumulation management. Meridian's aggregate exposure to particular regions and perils must stay within risk appetite limits set by the board and must satisfy regulators and rating agencies. After an event, David needs to estimate Meridian's likely loss quickly, within days, to inform the board, investors and regulators. Last year, post-event loss estimation took two weeks because exposure data from hundreds of contracts had to be gathered and rerun through models.

The operations director, Sophie Grant, points out that the exposure team is small and exhausted. During renewal season, analysts work very long hours cleaning data by hand. Much of the cleaning is repetitive: standardising formats, geocoding addresses, mapping construction codes between modelling platforms, checking values. She wants automation to reduce the burden, not new tasks.

The data landscape involves several commercial catastrophe modelling platforms, each with its own exposure schema; an open exposure data standard supported by an industry body; Meridian's underwriting system, which records contracts and terms; and a data warehouse holding historical submissions inconsistently. Geocoding services, property data providers and hazard data sources, both open and commercial, could enhance submissions.

Your team is four engineers and works alongside eight exposure analysts. Renewal season begins in six months. In the last renewal, the team processed more than four hundred submissions in seven weeks, and two experienced analysts resigned shortly afterwards, citing burnout; their replacements are still learning the idiosyncrasies of each cedant's data. Charles wants quality scoring by then. Mei wants a research platform. David wants faster post-event estimation. Sophie wants the analysts' workload reduced.

You see a common foundation: a canonical exposure data model, aligned with the open standard, into which every submission is converted, validated, enhanced and stored with full history. On that foundation, quality scoring, year-on-year comparison, accumulation management, post-event loss estimation and research all become feasible. But converting hundreds of cedant formats automatically is hard, and the renewal deadline leaves little room for a large build.

The commercial platforms add their own friction. Meridian licenses two catastrophe modelling platforms, and each uses its own exposure schema, construction and occupancy codes, and geocoding engine. The same property can be modelled differently depending on which platform's codes it is mapped to, and analysts spend considerable time translating between them. Model version changes, which vendors release periodically, can shift loss estimates materially for reasons unrelated to exposure, and when underwriters see a cedant's modelled loss change from one year to the next, they often cannot tell whether the change reflects the portfolio, the

data quality or the model. Separating these effects requires keeping exposure data, model versions and settings under version control, so that any result can be reproduced and any change decomposed into its causes. That discipline does not exist today; results are stored as spreadsheets attached to underwriting files.

There is also a question of how far to go in challenging cedants. Poor data quality affects pricing, but cedants are clients, and in a competitive market, demanding better data may push them to reinsurers that ask fewer questions. On the other hand, the market as a whole has been moving towards higher data standards after recent losses, and some cedants would welcome clear feedback that helps them improve.

THE DECISION IN FRONT OF YOU

Do you prioritise submission quality scoring for the renewal season, faster post-event loss estimation, or a research platform for an independent view of risk, and how do you build a canonical exposure model that reduces analyst workload while handling hundreds of inconsistent cedant formats?

Data Sources

#	Source and link	How it is used in this project
1	Oasis Loss Modelling Framework and Open Exposure Data (OED) https://oasislmf.org	Open catastrophe modelling framework and open exposure data standard used as the canonical model.
2	FEMA OpenFEMA Data Sets https://www.fema.gov/about/openfema/data-sets	Disaster declarations and National Flood Insurance Program claims for event validation.
3	NOAA Storm Events Database https://www.ncdc.noaa.gov/stormevents/	Historical storm event records for hazard and loss validation.
4	NOAA IBTrACS Hurricane Tracks https://www.ncei.noaa.gov/products/international-best-track-archive	Global tropical cyclone track data for hazard analysis.
5	USGS Earthquake Catalog https://earthquake.usgs.gov/earthquakes/search/	Earthquake event data for hazard validation and post-event footprints.
6	Global Earthquake Model (GEM) https://www.globalquakemodel.org	Open earthquake hazard and risk models and exposure data.
7	Lloyd's Realistic Disaster Scenarios https://www.lloyds.com/conducting-business/market-oversight/realistic-disaster-scenarios	Market stress scenarios for accumulation testing.
8	Microsoft Global Building Footprints https://github.com/microsoft/GlobalMLBuildingFootprints	Open building footprints for geocoding validation and exposure enhancement.

Deliverables (17)

- D1 Hurricane loss post-mortem.** An analysis of exposure data and model factors behind the loss.
- D2 Canonical exposure model.** A model aligned with the open exposure standard with Meridian extensions.
- D3 Submission format converters.** Automated conversion of common cedant and platform formats.
- D4 Validation rules library.** Automated checks for location, value, construction and policy anomalies.
- D5 Geocoding and location enhancement.** Improved geocoding with building footprint validation.
- D6 Value adequacy checks.** Comparison of insured values with construction cost benchmarks and inflation.
- D7 Submission quality scoring.** A standard score for completeness and plausibility per submission.
- D8 Year-on-year portfolio comparison.** Detection of portfolio changes and anomalies versus previous submissions.
- D9 Cedant feedback reports.** Structured feedback to cedants on data quality.
- D10 Exposure history store.** Versioned storage of every submission and enhancement.
- D11 Accumulation management views.** Aggregated exposure by region, peril and risk appetite limit.
- D12 Post-event rapid loss estimation.** A workflow producing loss estimates within days of an event.
- D13 Hazard data integration.** Open hazard footprints and event data for validation and research.
- D14 Model validation framework.** Comparison of model outputs with historical event losses.
- D15 Analyst workflow automation.** Measured reduction in manual cleaning effort.
- D16 Renewal readiness plan.** Priorities and scope for the coming renewal season.
- D17 Research platform roadmap.** A plan for developing an independent view of risk.

CASE STUDY 77 • WORKFORCE DEVELOPMENT AND RECRUITMENT

Labour Market Intelligence from Job Postings and Skills Taxonomies

Industry	Workforce Development and Recruitment
Organisation	SkillsBridge, a national skills body advising government, employers and education providers on labour market needs
Your role	Lead Data Engineer, Labour Market Intelligence
Core themes	Online job posting data, skills extraction and taxonomy mapping, representativeness and bias, public data products

The Scenario

SkillsBridge is a public body that advises government, employers, colleges and universities on the skills the economy needs. Its reports shape funding decisions for further education courses, apprenticeship standards and regional skills plans. For years, its analysis relied on official labour market statistics, employer surveys and occupational projections, all of which are authoritative but slow and coarse: they describe broad occupations over years, not the specific skills employers are asking for now. Ministers have asked SkillsBridge to provide more timely, granular labour market intelligence, particularly on emerging skills such as artificial intelligence, green technologies and digital skills, and on regional variation. You lead data engineering in the labour market intelligence team, and the obvious source of such intelligence is online job postings.

Millions of job advertisements are posted online every year on job boards, company career sites and recruitment agency websites. Each describes a role, often including job title, employer, location, salary, required qualifications, experience and skills. Aggregated and analysed, postings can reveal which skills are in demand, where, and how demand is changing, almost in real time. Several commercial providers collect, deduplicate and enrich job postings at scale and sell the data, and some job boards offer public interfaces with limited data. Official statistics agencies have experimented with online job advert data as an indicator of labour demand, publishing methods and caveats. Standard taxonomies of occupations and skills, such as the European multilingual classification of skills and occupations and the American occupational information database, allow postings to be classified consistently.

The chief executive, Dr Laura Bennett, wants a labour market intelligence platform that ingests job posting data continuously, extracts and classifies skills, maps them to taxonomies aligned with the national occupational classification and with qualifications, and publishes regular analyses and an interactive tool for colleges and employers. She has a meeting with ministers in four months and wants to show early results.

The head of research, Professor Alan Hughes, is an economist who worries about misleading conclusions. Job postings are not the same as jobs. Some roles are advertised many times on multiple sites; some are never advertised online, especially in sectors such as construction, care and hospitality, where recruitment often happens through word of mouth or agencies; some postings are for roles that are never filled; and recruitment agencies post speculative advertisements to attract candidates. Postings over-represent professional, digital and urban jobs. Skills mentioned in postings reflect employers' language and aspirations, which may differ from the skills actually used. Alan insists that any outputs must be benchmarked against official statistics, weighted or caveated for representativeness, and never presented as counts of jobs.

The head of partnerships, Sophie Chen, is enthusiastic about the potential for colleges and training providers, who struggle to know which courses to offer. But she notes that the platform's outputs will influence funding decisions affecting real people: if the data overstates demand for some skills and understates others, colleges may shift courses in the wrong direction. She wants the tool to be used alongside local employer engagement, not instead of it.

The data governance lead, David Oyelaran, has raised legal and ethical questions. Commercial job posting data comes with licences that may restrict publication of detailed data. Scraping job boards directly raises terms-of-use issues. Job postings sometimes contain personal data, such as recruiter names and contact details. More subtly, analysis of salary data by region or occupation could be misinterpreted, and skills classification models may encode biases, for example, associating certain skills with certain genders or treating skills expressed in different ways by different sectors inconsistently.

The skills extraction challenge is technically hard. Job descriptions are free text, written in varied styles. Skills may be named explicitly, such as "Python" or "forklift licence", or described implicitly, such as "able to

manage a small team". The same skill may be expressed in many ways, and similar phrases may mean different things in different contexts. Mapping extracted skills to a taxonomy requires handling synonyms, ambiguity and emerging skills that the taxonomy does not yet include. Language models have made extraction more feasible, but they introduce their own inconsistencies and must be evaluated rigorously. Official skills taxonomies are large, hierarchical and updated periodically; aligning them with the national occupational classification and with qualification frameworks requires mappings that are only partly available.

Your team is three engineers and two data scientists, working with Alan's economists. The body has a modest budget: enough for one commercial data licence or for building and running collection from public sources, not both at scale. It has a cloud data platform used for other statistical work, and a small team of statisticians who produce official-standard outputs and are used to rigorous quality assurance processes that can slow publication but protect credibility. They have made clear that they will not put their names to outputs whose methods they cannot scrutinise.

You see a tension between speed and rigour. A commercial data licence would provide deduplicated, enriched postings immediately, allowing early results for the ministerial meeting, but its methods are proprietary and its publication restrictions may conflict with SkillsBridge's role as a public body publishing open evidence. Building collection in-house would give control and transparency but would take longer and face legal and technical obstacles. Either way, skills extraction, taxonomy mapping and representativeness adjustment must be done carefully, and they take time.

The policy context raises the stakes further. The government has introduced a new skills funding regime that gives regions more say over adult education budgets and requires local skills improvement plans to be informed by employer demand. Colleges have told SkillsBridge that they are overwhelmed by competing sources of labour market information, from consultancies, local enterprise bodies and commercial dashboards, which often disagree. One college principal described receiving three reports in a month showing three different lists of the most in-demand skills in the same region. SkillsBridge's authority as a public body gives it an opportunity to provide a trusted reference point, but only if its methods are demonstrably sound and consistent. Laura is aware that a headline-

grabbing but flawed analysis in front of ministers could undermine that authority for years.

There is also a question of what to publish and how. Ministers want headline findings, such as the fastest-growing skills. Colleges want local detail. Researchers want methods and data. Each audience needs different outputs with different caveats. Publishing without adequate caveats could mislead; publishing with extensive caveats might be ignored. And once SkillsBridge publishes a measure, it will be used in funding and policy debates, so errors will be costly and consistency over time will matter.

THE DECISION IN FRONT OF YOU

Do you license commercial job posting data for rapid results, build in-house collection from public sources and interfaces, or combine a limited licence with open methods and benchmarking, and how do you ensure skills intelligence is representative, transparent and fair enough to guide funding decisions that affect learners and workers?

Data Sources

#	Source and link	How it is used in this project
1	Adzuna API https://developer.adzuna.com	Job advertisement search and statistics API used to prototype posting ingestion, deduplication and salary analysis.
2	ESCO European Skills, Competences, Qualifications and Occupations https://esco.ec.europa.eu	Multilingual skills and occupations taxonomy with downloadable classification and API.
3	O*NET Database https://www.onetcenter.org/database.html	Detailed occupational information including skills, knowledge and technologies for taxonomy mapping.
4	ONS Online Job Advert Estimates https://www.ons.gov.uk/economy/economicoutputandproductivity/output/datasets/onlinejobadvertestimates	Official experimental statistics on online job adverts with methods and caveats.
5	ONS Standard Occupational Classification (SOC 2020) https://www.ons.gov.uk/methodology/classificationsandstandards/standardoccupationalclassification/soc/soc2020	National occupational classification for aligning postings with official statistics.
6	ONS Labour Force Survey and Vacancy Statistics https://www.ons.gov.uk/employmentandlabourmarket	Official labour market statistics for benchmarking representativeness.
7	Nesta UK Skills Taxonomy and Skills Extractor https://github.com/nestauk/ojd_daps_skills	Open-source skills extraction and taxonomy work on UK job adverts.

#	Source and link	How it is used in this project
8	Lightcast Open Skills Taxonomy https://lightcast.io/open-skills	Open skills library used for comparison with ESCO mappings.

Deliverables (17)

- D1 Data source options appraisal.** A comparison of commercial licences, public interfaces and in-house collection.
- D2 Licence and legal review.** An assessment of terms of use, publication rights and personal data in postings.
- D3 Posting ingestion pipeline.** Ingestion with deduplication across boards and repostings.
- D4 Posting quality filters.** Detection of speculative, duplicate and agency adverts.
- D5 Occupation classification.** Assignment of postings to national occupational codes.
- D6 Skills extraction pipeline.** Extraction of explicit and implicit skills from job descriptions.
- D7 Taxonomy mapping.** Mapping of extracted skills to ESCO and national frameworks with confidence.
- D8 Emerging skills detection.** Identification of skills not yet in official taxonomies.
- D9 Extraction evaluation framework.** Human-labelled benchmarks measuring extraction and mapping accuracy.
- D10 Representativeness benchmarking.** Comparison of posting volumes with official vacancies and employment by sector and region.
- D11 Weighting and caveat methodology.** Adjustments and published caveats for coverage bias.
- D12 Bias and fairness review.** Assessment of classification bias by sector, region and demographic language.
- D13 Regional skills demand indicators.** Caveated indicators for local skills planning.
- D14 Ministerial briefing pack.** Early, carefully framed findings for the ministerial meeting.
- D15 Interactive tool for colleges.** A data product for providers with guidance on appropriate use.
- D16 Methods publication.** Public documentation of sources, methods and limitations.
- D17 Time series consistency plan.** Versioning and revision policies to keep measures comparable.

CASE STUDY 78 • HIGHER EDUCATION

University Admissions Funnel and Widening Participation Analytics Platform

Industry	Higher Education
Organisation	The University of Northfield, a research-intensive university with 32,000 students and 58,000 undergraduate applications a year
Your role	Senior Data Engineer, Planning and Student Insight
Core themes	Admissions data integration, contextual admissions fairness, regulatory access targets, predictive model ethics

The Scenario

The University of Northfield receives around fifty-eight thousand undergraduate applications a year for about seven thousand places. Admissions is a high-stakes operation: the university must fill its places with students who will succeed, meet its financial targets, which depend heavily on the mix of home and international students, and meet commitments to widen participation among students from under-represented backgrounds. Those commitments are set out in an access and participation plan agreed with the higher education regulator, the Office for Students, which monitors progress against targets and can impose conditions on universities that fall short. You are a senior data engineer in the planning and student insight team, and admissions data has become the focus of an uncomfortable debate.

Applications arrive through the national admissions service for most undergraduate courses, which provides data on applicants, their qualifications, predicted grades, personal statements, references and contextual information. International applicants may apply directly or through agents. The university's student record system holds applications and offers once they are imported, and admissions tutors make decisions in an admissions module of that system, recording offers, conditions and reasons. Contextual data, indicators of disadvantage such as living in an area with low participation in higher education, attending a school with low attainment, having been in care or receiving free school meals, comes partly from the national admissions service and partly from external datasets that the university matches to applicants' postcodes and schools. Confirmation, when results arrive in August and offers are confirmed or declined, is handled under intense time pressure over a few days.

The university operates a contextual admissions policy: applicants with certain markers of disadvantage receive offers with lower grade requirements. The policy was introduced to widen participation and is supported by research showing that contextual offer holders perform well once admitted. But its implementation is inconsistent. Different departments apply it differently; some admissions tutors apply additional judgement; and the data flags used to identify eligible applicants have known flaws. Area-based measures, which classify applicants by the neighbourhood they live in, include some advantaged applicants living in disadvantaged areas and miss disadvantaged applicants living in advantaged areas. School-level measures depend on accurate school identification, which is missing for some applicants. Individual measures such as free school meal eligibility are more accurate but are not available for all applicants.

The pro-vice-chancellor for education, Professor Amanda Clarke, has asked for a review. The university's latest monitoring report to the regulator showed that it was falling behind its targets for students from the most under-represented areas, and the regulator has asked for an explanation. At the same time, a newspaper has run a story suggesting that the university's contextual offers are being used to fill places with applicants from wealthy families who happen to live in areas classified as disadvantaged. Amanda wants to know what the data actually shows.

The director of admissions, Dr Richard Holt, wants a predictive model to manage the admissions funnel. Every year, the university must decide how many offers to make for each course, anticipating how many applicants will accept and how many will meet their conditions. Over-recruitment strains teaching capacity and accommodation; under-recruitment loses income. The current approach relies on historical conversion rates and tutors' judgement. Richard wants a model that predicts, for each applicant, the probability of accepting an offer and meeting conditions, allowing the university to calibrate offer-making more precisely.

The head of widening participation, Dr Nadia Osei, is alarmed by the idea. She fears that a model predicting conversion would learn that applicants from disadvantaged backgrounds are less likely to accept offers or meet conditions, and that using it would lead to fewer offers for exactly the applicants the university is committed to supporting. She points out that lower conversion among disadvantaged applicants is partly a result of the

university's own practices, such as the timing of scholarship announcements and accommodation guarantees, and that the right response is to change those practices, not to predict around them.

The data protection officer, Karen Walsh, has additional concerns. Admissions data includes special category data, such as ethnicity and disability, collected for monitoring purposes. Using it in models or decisions would require clear justification. Applicants must be told how their data is used, and automated decision-making that significantly affects them is restricted. The university's privacy notice for applicants has not been updated in years.

The data landscape is fragmented. Admissions data is in the student record system, with daily imports from the national admissions service. Contextual datasets are held in spreadsheets maintained by the widening participation team. Outreach programme data, which records applicants who took part in summer schools and other access activities, is in a separate customer relationship management system. Continuation and attainment data, showing how admitted students progress, is in the student record system but is rarely linked back to admissions characteristics. Regulatory returns are prepared by a statutory returns team using its own extracts.

Your team is three engineers and two analysts. Amanda wants an evidence review within three months to inform the response to the regulator. Richard wants a funnel model before the next admissions cycle, which starts in six months. Nadia wants the university to understand whether its contextual policy works and for whom.

You see several needs converging: a reliable, integrated view of applicants from application through offer, acceptance, confirmation, enrolment and outcomes; accurate, consistently applied contextual flags; linkage with outreach participation; and analyses of how each stage of the funnel affects different groups. A predictive model might be useful for capacity planning at aggregate level without being used for individual decisions, but the boundaries would need to be clear and enforced.

Academic departments add their own complexity. Admissions decisions for many courses are devolved to academic selectors, who guard their autonomy and have different views about contextual offers. The medical school runs a separate, highly competitive process with interviews and aptitude tests, and has its own widening participation schemes. Some

departments with fewer applicants than places make offers to almost everyone and see contextual data as irrelevant; others, heavily oversubscribed, rely on it to differentiate between applicants with similar predicted grades. Any central analysis that implies departments are applying policy inconsistently will be read as criticism, and academics have long memories. Amanda wants the review to be framed as institutional learning rather than department league tables, but the regulator will want to see where performance varies and why, and the data will inevitably show that some departments contribute far more to the gap against targets than others.

There is also a question of transparency. The university's contextual admissions policy is public, but how it is applied and what effect it has are not. Publishing clear data on contextual offers and outcomes would support accountability and could rebut misleading press coverage, but it could also expose inconsistencies that the university would rather fix quietly.

THE DECISION IN FRONT OF YOU

Do you prioritise the evidence review of contextual admissions, the predictive funnel model, or an integrated admissions-to-outcomes data foundation, and how do you ensure that any predictive modelling supports rather than undermines widening participation commitments?

Data Sources

#	Source and link	How it is used in this project
1	UCAS Data and Analysis https://www.ucas.com/data-and-analysis	National undergraduate application, offer and acceptance statistics by applicant characteristics.
2	HESA Data and Analysis https://www.hesa.ac.uk/data-and-analysis	Official higher education student and outcomes data for benchmarking.
3	OfS Access and Participation Data Dashboard https://www.officeforstudents.org.uk/data-and-analysis/access-and-participation-data-dashboard/	Regulatory data on access, continuation and attainment by student group.
4	OfS POLAR and TUNDRA Area Classifications https://www.officeforstudents.org.uk/data-and-analysis/young-participation-by-area/	Area-based participation measures used as contextual flags.
5	English Indices of Deprivation https://www.gov.uk/government/collections/english-indices-of-deprivation	Area deprivation measures used in contextual data.

#	Source and link	How it is used in this project
6	DfE School Performance Data https://www.compare-school-performance.service.gov.uk/download-data	School-level attainment data for school-based contextual indicators.
7	ICO Guidance on Automated Decision-Making https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/individual-rights/automated-decision-making-and-profiling/	Rules restricting automated decisions affecting individuals.
8	Higher Education Access Tracker (HEAT) https://heat.ac.uk	Outreach tracking service linking access activity participants to outcomes.

Deliverables (17)

- D1** **Contextual admissions evidence review.** Analysis of how contextual offers are applied, to whom and with what outcomes.
- D2** **Contextual flag quality assessment.** Accuracy and coverage of area, school and individual markers.
- D3** **Contextual flag standardisation.** Consistent derivation and application of flags across departments.
- D4** **Integrated admissions funnel model.** A data model from application to enrolment and outcomes.
- D5** **Outreach participation linkage.** Linking outreach programme participants to applications and outcomes.
- D6** **Funnel stage analysis by group.** Conversion rates at each stage by characteristic and course.
- D7** **Outcome linkage.** Continuation and attainment of admitted students by admissions route.
- D8** **Regulator response evidence pack.** Data supporting the explanation of progress against access targets.
- D9** **Aggregate capacity forecasting.** Course-level recruitment forecasts without individual scoring.
- D10** **Predictive modelling boundaries policy.** Rules on permitted uses of predictive models in admissions.
- D11** **Fairness impact assessment.** Analysis of how any model would affect under-represented groups.
- D12** **Special category data controls.** Restrictions on use of ethnicity and disability data to monitoring purposes.
- D13** **Applicant privacy notice update.** Updated transparency information for applicants.
- D14** **Practice change recommendations.** Evidence on scholarships, accommodation and communications affecting conversion.

- D15** **Contextual admissions transparency report.** Public reporting on contextual offers and outcomes.
- D16** **Confirmation period data readiness.** Reliable data flows for August confirmation decisions.
- D17** **Admissions analytics dashboard.** Funnel views for admissions and widening participation teams.

CASE STUDY 79 · ARTIFICIAL INTELLIGENCE (MODEL DEVELOPMENT)

Large Language Model Training Data Governance and Deduplication Pipeline

Industry	Artificial Intelligence (Model Development)
Organisation	Lumen AI, a European startup training open-weight language models for European languages
Your role	Staff Data Engineer, Pretraining Data
Core themes	Web-scale text processing, copyright opt-outs, personal data in training corpora, data documentation and provenance

The Scenario

Lumen AI trains large language models specialised for European languages, released with open weights so that businesses, public bodies and researchers can run and adapt them. The company raised a substantial funding round on the promise of models that perform well in languages that larger models handle poorly, such as Czech, Hungarian, Greek and the Baltic languages, and that respect European values on privacy and copyright. Its next model, significantly larger than its predecessor, is scheduled to begin pretraining in four months. Pretraining requires trillions of tokens of text. You are the staff data engineer responsible for the pretraining data pipeline, and the pipeline has become the company's most scrutinised asset.

The core of the training data is web text. Public web crawl archives provide petabytes of pages collected over more than a decade, and several open projects have processed them into cleaned, deduplicated, filtered datasets for language model training, publishing their methods. Lumen's previous model used a combination of these datasets, supplemented by its own crawl of European websites, public domain books, Wikipedia and other encyclopaedic sources, parliamentary proceedings, open-licensed code and some licensed content from publishers. For the new model, the team wants much more text in lower-resource languages, which are poorly represented in existing datasets.

The processing pipeline involves several stages. Raw pages are extracted from archive files and converted from HTML to text. Language identification assigns each document a language. Quality filtering removes spam, boilerplate, machine-generated text and low-quality pages, using heuristics and classifiers. Deduplication removes exact and near-duplicate

documents, which are extremely common on the web and cause models to memorise repeated content. Toxic and harmful content is filtered. Personal data, such as email addresses, phone numbers and identification numbers, is detected and removed or masked. The resulting corpus is tokenised and mixed according to target proportions for each language and source.

Regulation has transformed the context. The European Union's artificial intelligence act imposes obligations on providers of general-purpose AI models, including preparing and publishing a sufficiently detailed summary of the content used for training according to a template provided by the AI Office, and putting in place a policy to comply with European copyright law, including respecting rights holders' reservations of rights to text and data mining. A code of practice for general-purpose AI providers elaborates these obligations. Rights holders can reserve their rights through machine-readable means, such as robots exclusion protocols and emerging opt-out standards, and the extent to which crawlers must honour them, including retrospectively for archived pages, is a live legal question. Data protection authorities have issued opinions on training models with personal data scraped from the web, requiring a lawful basis, measures to minimise personal data and respect for data subjects' rights.

The chief executive, Dr Elena Novak, wants the new model to be both best-in-class for European languages and exemplary on compliance, because the company's brand depends on it. The head of research, Dr Tomas Keller, is focused on model quality. His experiments show that aggressive filtering improves quality up to a point but that removing too much data, particularly in low-resource languages, harms performance. He is worried that compliance measures, such as removing all content from domains that have opted out of AI training and aggressive personal data removal, will disproportionately reduce data in small languages, where the pool of text is already limited.

The general counsel, Sofia Marchetti, wants provenance for every document in the training corpus: where it came from, when it was collected, under what terms, whether the source had reserved its rights and what processing was applied. She wants the ability to remove content from future training runs if a rights holder objects, and the ability to answer regulators' questions about what the model was trained on. She is also concerned about licensed publisher content, whose contracts restrict use to specific model versions and require deletion after the licence ends.

The trust and safety lead, Amira Haddad, is concerned about harmful content and about personal data. Web text includes personal information about private individuals, such as names linked to addresses, health information in forum posts and content about children. Filtering such content at scale is imperfect. She wants measurement of how much personal data remains after filtering and testing of whether the model can be induced to reproduce it.

The engineering challenges are formidable. Processing petabytes of web archives requires distributed computing at large scale, and the company's compute budget is shared with model training. Near-duplicate detection across billions of documents requires efficient algorithms such as MinHash and locality-sensitive hashing, and choices about thresholds affect both quality and quantity. Recording provenance for billions of documents requires careful design so that the metadata does not become unmanageable. Checking opt-outs requires collecting robots files and other signals for millions of domains, at the time of crawling and, arguably, at the time of training. Personal data detection in dozens of languages requires multilingual models and rules, which are less accurate for lower-resource languages.

Your team is five engineers. The pipeline used for the previous model was built quickly by researchers and is not reproducible: nobody can regenerate the exact corpus used. Tomas wants the new corpus ready in four months so pretraining can start on schedule, since compute has been reserved and delays are expensive. Sofia wants the corpus to meet a documented compliance standard before training begins.

You see a need to rebuild the pipeline as a reproducible, versioned system with provenance at the document level, opt-out checking, personal data filtering and documentation generation built in. That is a substantial undertaking. You also see a need to make explicit trade-offs between quality, quantity, language coverage and compliance, with evidence, rather than leaving them implicit in code written by researchers under deadline pressure.

Evaluation of data decisions is itself a challenge. The effect of a filtering choice on model quality can only be measured reliably by training models, and training even small experimental models on candidate corpora consumes compute that is scarce and expensive. Tomas's team has developed a practice of training small proxy models on samples to compare

data recipes, but results at small scale do not always hold at large scale, and evaluation benchmarks for lower-resource European languages are limited, sometimes consisting of machine-translated English benchmarks that do not reflect how native speakers use the language. The research team is building new evaluation sets with native speakers, but they will not be complete before pretraining starts. Decisions about the corpus will therefore have to be made with incomplete evidence, and the pipeline must make it possible to revisit them later by rebuilding the corpus with different settings.

There is also a question of openness. Lumen's brand is open models, and the research community values open data. Publishing the training corpus, or at least detailed documentation and the processing code, would support transparency and reproducibility, and some competitors have done so. But publishing the corpus would redistribute content whose rights status is uncertain, and publishing too much detail could expose the company to legal claims. The summary required by regulation must be sufficiently detailed without revealing trade secrets.

THE DECISION IN FRONT OF YOU

Do you prioritise corpus scale and language coverage to meet the training schedule, a reproducible pipeline with document-level provenance and compliance controls, or a phased approach, and how do you make explicit, evidence-based trade-offs between model quality, copyright opt-outs, personal data minimisation and coverage of low-resource languages?

Data Sources

#	Source and link	How it is used in this project
1	FineWeb and FineWeb-2 Datasets (Hugging Face) https://huggingface.co/datasets/HuggingFaceFW/fineweb	Openly documented web-scale training corpora with published filtering and deduplication methods.
2	Common Crawl https://commoncrawl.org	Petabyte-scale public web archive used as the raw source for web text.
3	Wikimedia Dumps https://dumps.wikimedia.org	Encyclopaedic text in many European languages under open licences.
4	EU AI Act and General-Purpose AI Code of Practice https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai	Obligations for GPAI providers on training data summaries and copyright policy.

#	Source and link	How it is used in this project
5	EU Directive on Copyright in the Digital Single Market (Article 4) https://eur-lex.europa.eu/eli/dir/2019/790/oj	Text and data mining exception and rights reservation mechanism.
6	EDPB Opinion on AI Models and Personal Data https://www.edpb.europa.eu/our-work-tools/our-documents/opinion-board-art-64/opinion-282024-certain-data-protection-aspects_en	Data protection guidance on training models with personal data.
7	datatrove Processing Library https://github.com/huggingface/datatrove	Open-source large-scale text processing and deduplication library.
8	Datasheets for Datasets https://arxiv.org/abs/1803.09010	Documentation framework for recording dataset motivation, composition and processing.

Deliverables (17)

- D1 Previous corpus audit.** A reconstruction and assessment of the prior model's training data.
- D2 Pipeline architecture.** A reproducible, versioned, distributed processing design.
- D3 Document-level provenance model.** Metadata recording source, collection time, terms and processing for every document.
- D4 Opt-out signal collection.** Collection and versioning of robots and rights reservation signals by domain.
- D5 Opt-out enforcement policy.** Documented rules on honouring reservations at collection and training time.
- D6 Language identification evaluation.** Accuracy measurement for target languages with improvements for low-resource ones.
- D7 Quality filtering framework.** Configurable, evaluated filters with ablation results per language.
- D8 Near-duplicate detection.** Scalable MinHash-based deduplication with threshold analysis.
- D9 Personal data filtering.** Multilingual detection and removal with measured residual rates.
- D10 Harmful content filtering.** Detection of toxic and harmful content with category-level measurement.
- D11 Licensed content management.** Tracking and removal of licensed content according to contract terms.
- D12 Removal request process.** A mechanism to exclude content from future training runs on request.
- D13 Trade-off analysis.** Evidence on quality, quantity and language coverage effects of compliance measures.

- D14 Data mixture specification.** Target proportions by language and source with rationale.
- D15 Training data summary.** The regulatory summary of training content following the official template.
- D16 Dataset documentation.** A datasheet describing the corpus for internal and external use.
- D17 Schedule and compute plan.** A plan reconciling pipeline work with the reserved training window.

CASE STUDY 80 • ENGINEERING CONSULTANCY

Enterprise Knowledge Ingestion for Retrieval-Augmented Generation

Industry	Engineering Consultancy
Organisation	Halden Partners, a global engineering and infrastructure consultancy with 14,000 staff in 30 countries
Your role	Lead Data Engineer, Knowledge Platforms
Core themes	Document ingestion at scale, access control propagation, content freshness, evaluation of generative answers

The Scenario

Halden Partners designs bridges, tunnels, water treatment plants, hospitals, airports and power infrastructure. Its fourteen thousand engineers, planners, environmental scientists and project managers work on thousands of projects in thirty countries. The firm's value lies in its accumulated expertise: design standards, technical guidance, lessons learned, past proposals, project reports, calculations and specifications. That expertise is scattered across a document management system holding tens of millions of files, a collaboration platform with thousands of project sites, an intranet of technical guidance, a bid library, email archives and the heads of senior engineers approaching retirement. The chief executive, Sir Robert Grant, has sponsored an initiative to build an internal AI assistant that allows staff to ask questions and receive answers grounded in the firm's knowledge. You lead data engineering for knowledge platforms, and the ingestion and retrieval pipeline is yours.

A pilot built by the innovation team six months ago showed promise and exposed problems. It indexed the technical guidance intranet and a sample of project reports, used a language model to answer questions, and was trialled by two hundred engineers. Users liked it when it worked, but they reported that answers sometimes cited superseded standards, mixed guidance from different countries with different codes, and occasionally quoted from project documents that the user should not have been able to see. In one case, an engineer on a competitive bid received an answer drawing on a confidential report from a previous project for a client that was now a competitor of the bid's client. The pilot was paused. Investigation showed that the pilot's index had been built using a service account with broad read access, and that permission filtering relied on a

nightly synchronisation that had failed silently for three weeks. Nobody had been monitoring it, because the pilot was treated as an experiment rather than a production system. The episode damaged confidence among the partners who lead client relationships, several of whom now insist that their projects be excluded from any future version of the tool.

The chief technology officer, Dr Aisha Rahman, wants the assistant rebuilt properly and rolled out across the firm within a year. She is under pressure from the board, which has seen competitors announce similar tools, and from staff who want it back. The general counsel, Martin Blake, is focused on confidentiality. Client contracts contain confidentiality clauses, some prohibiting use of project information for any other purpose, and some clients, particularly in defence and critical infrastructure, have security requirements that forbid their data being processed by external services. Ethical walls exist between teams working for competing clients. Martin wants assurance that the assistant cannot reveal information that a user is not permitted to see and that restricted projects are excluded entirely.

The chief engineer, Professor Diane Moore, is focused on technical accuracy. Engineering design must follow the codes and standards applicable in each jurisdiction and at the time of design. An assistant that confidently cites the wrong version of a standard, or applies a British code to a project in Australia, could contribute to design errors with safety consequences. She wants every answer to show sources with versions and jurisdictions, and she wants the assistant to decline when it is uncertain. She also wants technical guidance owners to be able to mark content as authoritative, superseded or draft.

The head of knowledge management, Sarah Lin, points out that much of the firm's content is duplicated, outdated or of poor quality. The document management system contains multiple versions of the same document, drafts that were never finalised and files with meaningless names. Metadata is sparse. The technical guidance intranet has owners for each page, but many owners have left the firm. Sarah wants the ingestion effort to be an opportunity to improve content governance, not simply to index everything.

The IT security manager, Chris Evans, raises infrastructure concerns. The document management system's permissions are complex, with inheritance, groups, exceptions and project-level restrictions. The collaboration platform uses a different permission model. Replicating

permissions accurately into a retrieval index, and keeping them synchronised as people join and leave projects, is technically challenging. He favours querying source systems' permissions at retrieval time, which is more reliable but slower.

The data engineering challenges are extensive. Documents come in many formats: word processing files, spreadsheets, presentations, PDFs including scanned drawings, CAD files and email. Extracting text, tables and figures reliably is difficult, particularly for engineering documents with complex layouts. Chunking documents for retrieval must respect their structure, so that a clause of a standard is retrieved with its context. Embeddings must be generated for tens of millions of chunks and updated as documents change. Metadata, such as project, client, country, discipline, document type, version and status, must be extracted or inferred to support filtering. Evaluation requires a test set of questions with expected answers and sources, developed with engineers, to measure retrieval and answer quality.

Your team is five engineers. The firm has a cloud platform with access to enterprise-hosted language models with contractual guarantees on data use. Budget is available but the board expects results within a year.

You see a need to start from governed, high-value content and expand deliberately, rather than indexing everything and hoping. Technical guidance and standards, with clear ownership and versioning, are the foundation. Project documents are valuable but carry confidentiality risk and quality variation; they might be included selectively, with explicit client permissions and restrictions enforced. You also see that permissions, versioning and jurisdiction metadata are as important as retrieval quality.

The global footprint complicates the architecture. Halden operates in countries with data residency requirements, including some where client and government contracts require that project data remain within national borders. The firm's document management system is deployed regionally, with separate instances for the Middle East, Asia-Pacific, Europe and the Americas, each with its own permission structure and administrators. A single global index would be simplest to build but might breach residency obligations; regional indexes would respect them but fragment the firm's knowledge, which is precisely what the initiative was meant to overcome. Technical guidance, most of which is not client-confidential, could be

shared globally, while project content might be indexed regionally. Staff also work in many languages, and much of the older project documentation in some regions is not in English, which affects extraction, retrieval and the language models' ability to answer accurately.

There is also a question of what the assistant should do. A general question-answering tool is appealing but hard to evaluate. Focused use cases, such as finding relevant past projects for a bid, answering questions about design standards or summarising lessons learned for a project type, might be more measurable and safer to deliver first.

THE DECISION IN FRONT OF YOU

Do you rebuild the assistant with broad indexing of all repositories, start with governed technical guidance and standards and expand selectively, or focus on specific use cases such as bid support, and how do you enforce confidentiality, ethical walls and version correctness while keeping the index fresh and answers accurate?

Data Sources

#	Source and link	How it is used in this project
1	arXiv Bulk Data Access https://info.arxiv.org/help/bulk_data.html	Large corpus of technical documents in PDF and source form used to test extraction, chunking and retrieval at scale.
2	Wikipedia Dumps https://dumps.wikimedia.org	Encyclopaedic text for retrieval pipeline development and benchmarking.
3	SQuAD 2.0 Dataset https://rajpurkar.github.io/SQuAD-explorer/	Question answering benchmark including unanswerable questions for evaluating abstention.
4	BEIR Benchmark https://github.com/beir-cellar/beir	Heterogeneous retrieval benchmark for evaluating embedding and retrieval approaches.
5	Unstructured Open Source Library https://github.com/Unstructured-IO/unstructured	Document partitioning and extraction for many file formats.
6	Microsoft Graph API (SharePoint and OneDrive Permissions) https://learn.microsoft.com/en-us/graph/api/resources/permission	API for reading document permissions to enforce access control in retrieval.
7	RAGAS Evaluation Framework https://docs.ragas.io	Metrics for evaluating retrieval-augmented generation faithfulness and relevance.
8	NIST AI Risk Management Framework https://www.nist.gov/itl/ai-risk-management-framework	Framework for managing risks of AI systems in organisations.

Deliverables (17)

- D1 Pilot incident review.** An analysis of the confidentiality incident and accuracy problems.
- D2 Content inventory and quality assessment.** Repositories, volumes, duplication, ownership and quality.
- D3 Content scope and phasing plan.** Phased inclusion of content types based on value and risk.
- D4 Client restriction register.** Structured records of confidentiality and processing restrictions by client and project.
- D5 Restricted project exclusion.** Mechanisms excluding restricted projects from ingestion entirely.
- D6 Permission enforcement design.** Retrieval-time permission checks with synchronisation strategy.
- D7 Ethical wall enforcement.** Controls preventing retrieval across walled teams.
- D8 Document extraction pipeline.** Robust extraction of text, tables and figures from engineering formats.
- D9 Structure-aware chunking.** Chunking that preserves clauses, sections and context.
- D10 Metadata enrichment.** Extraction of project, client, jurisdiction, discipline, version and status.
- D11 Version and supersession handling.** Prioritisation of current, authoritative content and flagging of superseded versions.
- D12 Incremental indexing.** Timely updates as documents are added, changed or deleted.
- D13 Evaluation test set.** Engineer-authored questions, answers and sources for measurement.
- D14 Answer provenance and abstention.** Source citations with versions and declining when uncertain.
- D15 Use case prioritisation.** Focused initial use cases with success metrics.
- D16 Content governance improvements.** Ownership assignment and status marking for technical guidance.
- D17 Rollout and monitoring plan.** Staged rollout with usage, quality and incident monitoring.

CASE STUDY 81 · DIVERSIFIED INDUSTRIAL AND CONSUMER CONGLOMERATE

Data Mesh Migration for a Diversified Conglomerate

Industry	Diversified Industrial and Consumer Conglomerate
Organisation	Castleford Group, a conglomerate with seven operating companies in building materials, logistics, consumer products and financial services
Your role	Head of Data Platform Engineering
Core themes	Centralised versus federated data architecture, domain ownership, platform self-service, federated governance

The Scenario

Castleford Group owns seven operating companies: a building materials manufacturer, a road haulage and warehousing business, a consumer household products company, a garden centre chain, a specialist insurer, a consumer finance lender and a small software business. Each company has its own management, systems and culture. The group centre provides finance, legal, treasury, some shared services and, for the past six years, a central data platform and team. You are the head of data platform engineering, and the central model is under attack from several directions at once.

The central data platform was built to serve group reporting and to give the operating companies analytics capabilities they could not build alone. A central team of forty data engineers and analysts ingests data from the companies' systems into a cloud data warehouse, models it and produces dashboards and reports. For group finance, it works reasonably well: consolidated financial reporting is faster and more reliable than before. For the operating companies, it has become a bottleneck. Requests for new data sources, models and reports queue for months. The central team does not understand each company's business deeply, so its models often misrepresent operational realities. When a source system changes, pipelines break, and the central team learns about the change only when a report fails. Some companies have given up and built their own data capabilities outside the platform, creating shadow data teams, duplicated costs and inconsistent numbers.

The group's new chief data officer, Dr Rebecca Hughes, joined from a large technology company and has proposed a data mesh: a model in which each operating company, or each domain within it, owns its data as products,

publishing well-documented, reliable datasets for others to use, supported by a self-service data platform provided centrally, and governed by federated standards agreed across the group. She argues that the companies understand their data best and should be accountable for it, and that the central team should become a platform team enabling them rather than a bottleneck doing everything.

The group chief financial officer, Martin Clarke, is sceptical. Group financial reporting, regulatory reporting for the insurer and lender, and group risk management all depend on consistent, reliable data. He fears that devolving ownership to operating companies, several of which have limited data maturity, will fragment definitions and quality. He also points out that the financial services companies have regulatory obligations, including risk data aggregation principles, that require strong central control.

The operating company leaders have mixed views. The chief executive of the logistics business, Paul Evans, is enthusiastic: his company has a strong technology team and wants control over its data. The garden centre chain's managing director, Susan Price, is alarmed: her business has no data engineers and relies entirely on the central team. The consumer finance lender's chief risk officer, Dr James Okafor, wants autonomy for credit modelling but insists that regulatory data remain tightly controlled. The software business already operates independently and sees little value in either model.

The central team itself is anxious. Engineers fear that a mesh would disperse them into operating companies, where they would be isolated and have fewer career opportunities, or make them redundant. Several of the best engineers have been approached by recruiters. Morale is fragile.

The technical dimensions are substantial. A self-service platform requires infrastructure provisioning, data pipeline tooling, data product templates, a catalogue where products are discoverable, lineage tracking across products, access control, observability and cost allocation. Federated governance requires agreed standards for metadata, identifiers, quality, security and interoperability, with automated enforcement where possible. Cross-domain analytics, such as understanding customers who buy from both the garden centres and the consumer products business, requires shared identifiers and semantics. Migrating hundreds of existing pipelines and reports from the central warehouse into domain-owned products

would be a multi-year effort, during which both models would run in parallel.

Rebecca has proposed starting with two pilot domains, logistics and consumer products, and building the self-service platform alongside them. Martin wants any change to leave group finance and regulatory reporting untouched until the model is proven. The board has asked for a recommendation within three months and wants to see measurable improvement in time to insight for operating companies within a year.

You see merit in the diagnosis but also in the concerns. Data mesh principles address real problems of scale and ownership, but they assume a level of data maturity and engineering capacity in domains that some of Castleford's companies do not have. A uniform model across such different businesses may be wrong; a hybrid in which mature domains own their products while the central team continues to serve less mature ones, and critical cross-group data remains centrally governed, might be more realistic. But hybrids can combine the worst of both models if responsibilities are unclear.

History matters as well. Castleford's central data team was formed six years ago by consolidating the separate analytics teams of the operating companies, a decision that was controversial at the time. Some operating company leaders felt that their best people had been taken from them and that the promised benefits never materialised for their businesses. The leaders who lost their teams are now among the most vocal supporters of a mesh, partly because it would return control to them. The central team's leadership, for its part, points to genuine achievements, such as a consolidated view of group cash, a single customer credit exposure across the two financial services businesses and a common supplier view that saved millions in procurement. Both narratives are partly true, and the recommendation will need to acknowledge what the central model achieved as well as where it has failed, if it is to carry people on both sides.

The operating companies also differ in their technology estates. The logistics business runs modern cloud systems with event streams and well-documented interfaces; the building materials manufacturer runs an old enterprise resource planning system on premises with batch extracts; the garden centre chain uses a packaged retail system managed by a vendor; and the financial services companies run regulated core systems with strict change controls. A self-service platform that works elegantly for

logistics may be unusable for the garden centres without substantial support, and any standards for data products will have to accommodate very different starting points.

There is also a cost question. Running a self-service platform with multiple domain teams may cost more in total than a central team, even if it delivers more value. The board expects efficiency, and duplicated effort across domains could undermine the case.

THE DECISION IN FRONT OF YOU

Do you implement a full data mesh with domain ownership across all operating companies, retain and reform the central platform, or adopt a hybrid with domain ownership for mature companies and central stewardship for others and for regulated data, and how do you design the self-service platform, federated governance and people transition so the change improves outcomes rather than fragmenting them?

Data Sources

#	Source and link	How it is used in this project
1	TPC-DS Benchmark https://www.tpc.org/tpcds/	Standard retail decision-support schema and data generator used to prototype domain data products and cross-domain queries.
2	dbt Documentation https://docs.getdbt.com	Transformation framework supporting modular, tested data products with documentation.
3	OpenLineage https://openlineage.io	Open standard for lineage across pipelines and domains.
4	Open Data Contract Standard (Bitol) https://bitol-io.github.io/open-data-contract-standard/	Specification for data contracts between producers and consumers.
5	DataHub Open-Source Metadata Platform https://datahubproject.io	Catalogue and metadata platform for data product discovery and governance.
6	Data Mesh Principles (Zhamak Dehghani) https://martinfowler.com/articles/data-mesh-principles.html	Foundational description of data mesh principles and architecture.
7	BCBS 239 Principles https://www.bis.org/publ/bcbs239.htm	Risk data aggregation principles constraining regulated financial data.
8	DAMA-DMBOK Data Management Body of Knowledge https://www.dama.org/cpages/body-of-knowledge	Framework for data governance roles and practices.

Deliverables (17)

- D1** **Current state assessment.** Central platform usage, backlog, shadow data teams, costs and satisfaction by company.
- D2** **Domain maturity assessment.** Data capability and readiness of each operating company.
- D3** **Operating model options paper.** Full mesh, reformed central and hybrid models with trade-offs.
- D4** **Target operating model.** Roles, responsibilities and decision rights across central and domain teams.
- D5** **Data product standard.** Required documentation, contracts, quality checks and service levels for products.
- D6** **Data contract implementation.** Contracts between producers and consumers with automated enforcement.
- D7** **Self-service platform blueprint.** Provisioning, pipeline templates, catalogue, lineage, access and observability.
- D8** **Federated governance council.** A cross-company body agreeing standards and resolving disputes.
- D9** **Group shared identifiers and semantics.** Common customer, product, location and organisation keys for cross-domain analytics.
- D10** **Regulated data carve-out.** Central stewardship arrangements for regulatory and financial reporting data.
- D11** **Pilot domain plan.** Delivery plan and success measures for two pilot domains.
- D12** **Pipeline migration strategy.** Prioritised migration of central pipelines to domain products.
- D13** **Cost allocation model.** Transparent platform and domain cost attribution.
- D14** **People transition plan.** Career paths, secondments and communication for central team members.
- D15** **Shadow data consolidation.** Bringing shadow data activities into the governed model.
- D16** **Time to insight metrics.** Measurement of delivery speed and quality before and after change.
- D17** **Board recommendation.** A decision paper with a phased roadmap and risks.

CASE STUDY 82 • BANKING (REGULATORY REPORTING)

Regulatory Reporting Data Quality Incident and Risk Data Aggregation Remediation

Industry	Banking (Regulatory Reporting)
Organisation	Sterling Northern Bank, a UK-headquartered bank with £85 billion in assets and operations in the US and Europe
Your role	Principal Data Engineer, Regulatory Reporting Data
Core themes	Regulatory misreporting, end-to-end reconciliation, BCBS 239 compliance, remediation under supervisory scrutiny

The Scenario

Sterling Northern Bank submits hundreds of regulatory reports every quarter to supervisors in the UK, the United States and the European Union. They cover capital adequacy, liquidity, large exposures, leverage, credit risk, market risk and much more. Each report is built from data flowing through dozens of systems, from trading and lending platforms through risk engines and finance ledgers to the reporting tools that produce the final templates. Two months ago, the bank discovered that its reported liquidity coverage ratio, a key measure of its ability to withstand a short-term funding stress, had been overstated for five consecutive quarters. The error was not catastrophic, the bank remained above its minimum requirement, but the overstatement was material, and it had to be reported to the Prudential Regulation Authority. The regulator's response was swift: it has required an independent review under its powers to commission a skilled person's report, and it has made clear that it will look at the bank's risk data aggregation and reporting capabilities as a whole. You are the principal data engineer for regulatory reporting data, and you are at the centre of the remediation.

The root cause of the liquidity error was, on the surface, simple. A product classification in the treasury system had been changed when the bank launched a new type of corporate deposit. The reporting logic, implemented in a transformation layer that mapped treasury products to liquidity categories, had not been updated, so the new deposits were treated as more stable funding than they were, reducing the stressed outflows in the calculation. The change had passed through the treasury system's change control, but nobody had assessed its impact on regulatory reporting. The reporting team's reconciliation checks compared total

deposits between the treasury system and the general ledger, which matched, but did not check the classification of deposits into liquidity categories. For five quarters, nobody noticed.

The deeper issue, as the chief risk officer, Dr Catherine Moore, has acknowledged to the board, is that the bank's regulatory reporting data architecture is fragile. Data flows through many systems with transformations applied at each stage, often in spreadsheets or undocumented code. Lineage is documented at a high level for some reports and not at all for others. Data ownership is unclear: the treasury team owns the treasury system, but nobody owns the mapping of treasury products to regulatory categories. Controls focus on totals rather than on the classifications that drive regulatory calculations. And the bank's self-assessment against the Basel Committee's principles for effective risk data aggregation and risk reporting, completed two years ago, rated the bank as largely compliant, an assessment that now looks optimistic.

The chief financial officer, Andrew Stone, owns regulatory reporting and is under the most pressure. The regulator's letter asked for a remediation plan within six weeks, and the skilled person's review will examine the plan's credibility. Andrew wants quick, visible fixes: correcting the liquidity mapping, resubmitting affected reports, adding reconciliation checks on classifications, and establishing change impact assessment for systems feeding regulatory reports. He also knows that the regulator will not be satisfied with tactical fixes alone.

The chief operating officer, Laura Kent, sponsors a multi-year data transformation programme that is migrating the bank's data infrastructure to a modern cloud platform. She sees the remediation as an opportunity to accelerate the programme and to build regulatory reporting on a new foundation with automated lineage, data contracts and quality monitoring. Andrew fears that tying remediation to a large, risky transformation programme will delay fixes and expose the bank to further regulatory criticism if the programme slips.

The head of regulatory reporting, Nicola Evans, leads a team of sixty people who produce the reports. Her team is exhausted: the incident has required restatement of five quarters of liquidity reports, extensive investigation and a flood of questions from the regulator, all while continuing to produce routine reports. She points out that much of her team's time is spent on manual adjustments and reconciliations that

compensate for poor upstream data, and that any remediation that adds controls without fixing upstream data will increase the burden.

The head of internal audit, Richard Hale, has been asked by the audit committee why audit did not detect the problem. Audit's reviews of regulatory reporting had focused on the reporting process and controls within the reporting team, not on upstream data flows and product classifications. Richard wants an end-to-end view of critical data flows for every material report, so that audit can test them.

Your team has eight data engineers. Regulatory reporting data is currently assembled in a data warehouse fed by nightly extracts from source systems, with transformations in SQL and a commercial regulatory reporting tool that applies templates and validation rules. There are more than two hundred distinct source feeds.

You see several streams of work that must proceed simultaneously. Immediate remediation of the liquidity mapping and resubmission of affected reports. A rapid assessment of whether similar classification risks exist in other reports, such as capital and large exposures, which would be even more serious if found. Strengthening controls: classification-level reconciliations, change impact assessment for upstream systems, and ownership of mappings. And a longer-term programme to achieve genuine compliance with risk data aggregation principles, including lineage, data quality measurement and governance, which may or may not be aligned with the cloud transformation.

The prioritisation question is critical. The regulator's attention is now on the whole reporting estate. If a second error is found in another report during the skilled person's review, the consequences could include a formal enforcement investigation, capital add-ons and personal accountability for senior managers under the senior managers regime. Finding such errors first, and disclosing them, is far better than having them found. But searching for errors across hundreds of reports with an exhausted team, while fixing known issues and designing long-term remediation, requires careful allocation of scarce people.

The organisational dynamics are delicate. Under the senior managers regime, specific executives hold prescribed responsibilities for regulatory reporting and risk data, and each is personally accountable. The incident has made them cautious, and some decisions that would once have been taken quickly now go through several layers of review. The treasury team,

whose product change triggered the error, feels unfairly blamed, since it followed its own change process correctly; the gap lay between teams. Several source system owners have begun to resist any new obligation to assess regulatory impact, arguing that they lack the knowledge to do so. Meanwhile, the regulatory reporting team has lost two experienced managers since the incident. Any remediation plan must address not only systems and controls but accountability across organisational boundaries, which requires executive sponsorship that goes beyond the data function.

There is also the question of honesty in the self-assessment. The previous assessment rated the bank as largely compliant with risk data aggregation principles. A candid reassessment will almost certainly downgrade that rating. Some executives worry that admitting widespread weaknesses will invite harsher regulatory treatment. You believe that credibility with the regulator depends on candour, but the decision about what to disclose is not yours alone.

THE DECISION IN FRONT OF YOU

How do you sequence immediate remediation, a proactive search for similar errors across other reports, control strengthening and long-term risk data aggregation compliance, and do you build the remediation on the existing warehouse or on the cloud transformation programme, given the regulator's scrutiny and an exhausted reporting team?

Data Sources

#	Source and link	How it is used in this project
1	BCBS 239 Principles for Effective Risk Data Aggregation and Risk Reporting https://www.bis.org/publ/bcbs239.htm	The principles against which the bank's capabilities will be assessed.
2	BCBS Progress Reports on BCBS 239 Adoption https://www.bis.org/bcbs/publ/d559.htm	Supervisory findings on common weaknesses in banks' implementation.
3	PRA Skilled Persons Reviews https://www.bankofengland.co.uk/prudential-regulation/supervision/skilled-persons-reviews	Regulatory framework for independent reviews commissioned by the PRA.
4	Bank of England Liquidity Coverage Ratio Reporting https://www.bankofengland.co.uk/prudential-regulation/regulatory-reporting/regulatory-reporting-banking-sector	Reporting requirements and templates for UK banks.
5	FFIEC Central Data Repository Public Data https://cdr.ffiec.gov/public/	Public US bank call report data used to prototype reporting data models and cross-period consistency checks.

#	Source and link	How it is used in this project
6	EBA Reporting Framework and Data Point Model https://www.eba.europa.eu/risk-and-data-analysis/reporting-frameworks	European regulatory reporting taxonomy and validation rules.
7	Great Expectations Documentation https://docs.greatexpectations.io	Data quality testing framework for classification-level reconciliation checks.
8	OpenLineage https://openlineage.io	Open standard for capturing lineage across pipelines to support end-to-end traceability.

Deliverables (17)

- D1 Liquidity error root cause report.** A full analysis of the mapping failure, control gaps and impact.
- D2 Corrected liquidity mapping and resubmissions.** Fixed product-to-category mapping and restated reports.
- D3 Similar-risk sweep.** A prioritised review of classification mappings across capital, large exposures and other reports.
- D4 Material report inventory.** A catalogue of reports ranked by materiality and data risk.
- D5 Critical data element register.** Key data elements for material reports with owners and quality thresholds.
- D6 End-to-end lineage for material reports.** Lineage from source systems to report cells for priority reports.
- D7 Classification reconciliation controls.** Automated checks on product classifications and category totals.
- D8 Upstream change impact assessment.** A process requiring regulatory impact review for changes to feeding systems.
- D9 Mapping ownership model.** Accountable owners for product-to-regulatory mappings.
- D10 Manual adjustment inventory.** Documentation and reduction plan for manual adjustments in reporting.
- D11 Data quality measurement.** Metrics on accuracy, completeness and timeliness for critical data.
- D12 Candid BCBS 239 reassessment.** An honest self-assessment against each principle with evidence.
- D13 Remediation plan for the regulator.** A credible, time-bound plan with milestones and owners.
- D14 Platform alignment decision.** A decision on building remediation on the existing warehouse or the cloud programme.
- D15 Team capacity plan.** Resourcing and workload relief for the reporting team.
- D16 Audit testing support.** Materials enabling internal audit to test end-to-end controls.

- D17 Skilled person review support.** Evidence packs and coordination for the independent review.

CASE STUDY 83 · ONLINE MARKETPLACE AND CONSUMER SERVICES

Right to Erasure and Data Subject Rights Across a Data Lake

Industry	Online Marketplace and Consumer Services
Organisation	Pantry, a UK online grocery and meal kit service with 2.4 million customers
Your role	Staff Data Engineer, Data Platform
Core themes	Deletion in immutable storage, personal data discovery, backups and machine learning models, regulatory complaint response

The Scenario

Pantry delivers groceries and meal kits to around two point four million customers across the UK. Its data platform, built over eight years, is a lakehouse in cloud object storage: raw data from the website, apps, order management, delivery, customer service and marketing systems lands in a raw zone, is cleaned and modelled into curated tables, and feeds analytics, reporting, recommendation models, demand forecasting and marketing personalisation. The platform holds several petabytes of data. For most of its life, the platform team focused on getting data in; nobody thought carefully about getting it out. You are the staff data engineer on the data platform, and a complaint to the data protection regulator has changed that.

The complaint came from a former customer who had asked Pantry to delete all her personal data under her right to erasure. Pantry's customer service team processed the request through the customer relationship management system, which deleted her account, and confirmed to her that her data had been erased. Months later, she received a marketing email from Pantry. It turned out that her email address had persisted in a marketing audience table in the lakehouse, which had been exported to the email platform when a new campaign was built. She complained to the Information Commissioner's Office, which wrote to Pantry asking how it handles erasure requests across its systems, how many requests it receives and whether it can demonstrate compliance.

When the data protection officer, Hannah Price, asked the platform team to explain, the answers were uncomfortable. Personal data exists in hundreds of tables across raw, curated and analytics zones. The raw zone holds immutable copies of every event and record ever received, stored as files partitioned by date, which are not designed to be modified. Curated

tables are rebuilt from raw data periodically, so deleting data from a curated table without deleting it from raw would see it reappear on the next rebuild. Analysts have created thousands of derived tables and extracts in personal workspaces. Customer identifiers are propagated into event logs, clickstream data, delivery routes, customer service transcripts and model training datasets. Backups and snapshots retain data for months. Machine learning models were trained on data including the complainant's orders. And data has been shared with third parties, including the email platform, an advertising partner and a logistics analytics provider.

The chief technology officer, Olivia Harper, wants the problem fixed properly. She has asked for a plan that ensures erasure requests are honoured across the entire platform, demonstrably and at reasonable cost. The platform processes around four hundred erasure requests a month, plus thousands of subject access requests, where customers ask for a copy of their data, which raise the related problem of finding all of a customer's data.

The head of analytics, Tom Bradley, is worried about the effect on analysis. Deleting customers' data from historical tables would change historical metrics, such as customer counts, order volumes and cohort retention. Forecasting models depend on long, consistent histories. He proposes anonymisation rather than deletion: replacing identifiers with irreversible tokens so that the records remain for aggregate analysis but can no longer be linked to the individual. Hannah is sympathetic but cautious. Anonymisation must be genuine: if the remaining records, such as a delivery address, order history and timing, could still identify the person, they are not anonymous. The regulator's guidance on anonymisation sets a high bar.

The head of data science, Dr Anika Shah, raises the question of models. If a customer's data was used to train a recommendation model, does erasure require retraining the model without her data? Legal opinion is unsettled, but the regulator has indicated that models can contain personal data in some circumstances. Retraining every model for every erasure request would be impractical, but retraining periodically on data that excludes erased customers might be reasonable.

The engineering options are complex. Modern table formats for lakehouses support row-level deletes and updates on object storage, with features for

time travel and snapshot expiry that allow deleted data to be physically removed after a retention period. Migrating hundreds of tables from plain file partitions to such formats is a substantial undertaking. An alternative approach, sometimes called crypto-shredding, encrypts each customer's personal data with a customer-specific key and deletes the key upon erasure, rendering the data unreadable wherever it resides, including in backups. This requires encrypting data at ingestion, managing millions of keys and changing how analytics access personal fields. A third approach separates personal data into a small number of tightly controlled tables, with the rest of the platform using pseudonymous identifiers, so that erasure requires deletion only in those tables plus breaking the link.

Each approach requires knowing where personal data is. The platform has a data catalogue, but it is incomplete and personal data fields are not consistently tagged. Automated scanning tools can detect personal data patterns in tables but produce false positives and miss indirect identifiers.

Your team is six engineers. The regulator expects a response within a month describing Pantry's approach and a remediation plan. Hannah wants the response to be honest and credible. Olivia wants a sustainable solution, not a one-off clean-up. Tom wants analytics protected. Anika wants clarity on models.

You see that the immediate response must include tracing and erasing the complainant's data everywhere it persists, including third parties, and a candid description of the platform's gaps. The long-term solution requires architectural change: minimising where personal data lives, pseudonymising by default, adopting table formats that support deletion, managing retention and backups, and automating erasure across the platform and to third parties with verification. That is a programme of a year or more.

Commercial pressures push in the opposite direction. The marketing team has just launched a personalisation programme that relies on long purchase histories, and the commercial director has argued that deleting or anonymising historical data will weaken recommendations and reduce revenue. The data science team is building a new demand forecasting model that uses customer-level order histories going back five years. The finance team uses customer-level data to reconcile refunds and chargebacks for up to six years, to satisfy tax and accounting requirements, which constitutes a legitimate reason to retain some

personal data even after an erasure request. Different teams therefore hold different, sometimes legitimate, claims on the same data, and the erasure design must distinguish between data that must be deleted, data that may be retained for a specific legal purpose with restricted access, and data that can be anonymised and kept for aggregate analysis.

You also recognise a cultural change is required. The platform grew under a philosophy of keeping everything forever because storage was cheap and data might be useful. Data protection law's principles of data minimisation and storage limitation run against that philosophy, and applying them will mean deleting data that some teams value.

THE DECISION IN FRONT OF YOU

Do you pursue row-level deletion with a modern table format, crypto-shredding with per-customer keys, or architectural separation of personal data with pseudonymisation, and how do you balance erasure obligations with analytics continuity, model retraining and a candid response to the regulator?

Data Sources

#	Source and link	How it is used in this project
1	ICO Right to Erasure Guidance https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/individual-rights/individual-rights/right-to-erasure/	Regulatory requirements for erasure requests, exemptions and third-party notification.
2	ICO Anonymisation, Pseudonymisation and Privacy Enhancing Technologies Guidance https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/data-sharing/anonymisation/	Standards for effective anonymisation versus pseudonymisation.
3	Delta Lake Documentation https://docs.delta.io	Table format supporting deletes, updates, time travel and vacuuming of removed files.
4	Apache Iceberg Documentation https://iceberg.apache.org/docs/latest/	Table format with row-level deletes and snapshot expiration for physical removal.
5	Microsoft Presidio https://microsoft.github.io/presidio/	Open-source personal data detection and anonymisation for scanning tables and text.
6	Instacart Market Basket Analysis https://www.kaggle.com/c/instacart-market-basket-analysis	Grocery order data used to prototype pseudonymisation and aggregate-preserving deletion.
7	NIST Privacy Framework https://www.nist.gov/privacy-framework	Framework for managing privacy risk across data processing.

#	Source and link	How it is used in this project
8	ICO Guidance on AI and Data Protection https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/artificial-intelligence/guidance-on-ai-and-data-protection/	Regulatory position on personal data in AI models and individual rights.

Deliverables (17)

- D1 Complainant data trace and erasure.** Identification and removal of the complainant's data from all locations including third parties.
- D2 Regulator response.** A candid description of gaps, immediate actions and a remediation plan.
- D3 Personal data discovery.** Scanning and cataloguing of personal data fields across all zones.
- D4 Personal data inventory and tagging.** A complete catalogue with field-level tags and owners.
- D5 Erasure architecture options appraisal.** A comparison of row-level deletion, crypto-shredding and separation approaches.
- D6 Pseudonymisation by default.** Replacement of direct identifiers with pseudonymous keys outside controlled tables.
- D7 Personal data vault.** A small set of controlled tables holding directly identifying data.
- D8 Table format migration.** Migration of priority tables to a format supporting row-level deletes.
- D9 Raw zone retention redesign.** Retention limits and deletion capability for immutable raw data.
- D10 Backup and snapshot policy.** Retention periods and erasure handling for backups and snapshots.
- D11 Analyst workspace governance.** Controls and clean-up for personal data in personal workspaces and extracts.
- D12 Third-party erasure propagation.** Automated notification and confirmation of erasure with data recipients.
- D13 Model retraining policy.** Periodic retraining excluding erased customers with documented rationale.
- D14 Anonymisation assessment.** Testing of whether retained records are genuinely anonymous.
- D15 Automated erasure workflow.** End-to-end orchestration of erasure requests with verification and audit logs.
- D16 Subject access request automation.** Automated discovery and compilation of a customer's data.

- D17 Data minimisation programme.** Reduction of unnecessary personal data collection and retention.

CASE STUDY 84 · BROADCASTING AND NEWS MEDIA

Election Night Live Results Data Service

Industry	Broadcasting and News Media
Organisation	The National Broadcasting Corporation's News Division, a public service broadcaster reaching 30 million people on election night
Your role	Lead Data Engineer, Elections Data
Core themes	Extreme reliability under peak load, data verification, projection models, public trust and neutrality

The Scenario

On the night of a general election, tens of millions of people turn to the National Broadcasting Corporation to find out who will govern the country. Its television coverage, radio programmes, website and app present results as they are declared in each of the six hundred and fifty constituencies, along with analysis, projections of the final outcome and maps that update through the night. The data behind all of this must be fast, accurate and resilient. A wrong result shown on screen, even briefly, would be a national embarrassment and would damage the broadcaster's reputation for impartiality and accuracy. A system failure at the peak of the night would be worse. You are the lead data engineer for elections data, and a general election is expected within the next nine months, though the exact date is not yet known.

The results data service has several components. Before election night, the team assembles reference data: constituencies and their boundaries, candidates and parties, previous results and notional results for constituencies whose boundaries have changed. On the night, results are collected from counting centres, where the broadcaster's reporters and a network of stringers relay declared figures by phone and through a mobile app. Results are also published by returning officers and, later, compiled by official bodies. A verification desk checks each result before it is released to output. The verified results feed a central database, from which graphics systems, the website, the app, radio and partner organisations draw data. A team of statisticians and political scientists runs a projection model that forecasts the final seat totals as results come in, and on some elections, the broadcaster jointly commissions an exit poll with other broadcasters, published when polls close.

The current system was built a decade ago and has been patched for each election since. It worked at the last general election, but not without incident. The collection app crashed for twenty minutes at the busiest point of the night, when results were being declared at a rate of more than one a minute, and the verification desk fell behind, leading to a backlog of unreleased results that competitors displayed first. On one occasion, a result was released with the vote shares of two candidates transposed, and it appeared on screen for ninety seconds before being corrected. The website slowed under load. The post-election review recommended significant investment.

The head of elections, Deborah Hart, a veteran journalist who has run election coverage for twenty years, has clear priorities: accuracy first, then speed, then everything else. She would rather be second with a correct result than first with a wrong one. She wants a verification process that catches every error, a system that never goes down and graphics that never show wrong data. She is sceptical of new technology introduced close to an election and wants any new system tested exhaustively, including in full-scale rehearsals.

The head of digital news, Marcus Reid, wants more. Audiences increasingly follow election night on phones, and they expect personalised, interactive experiences: results for their own constituency, live maps, swing calculators and alerts. Digital competitors, including social media platforms and newer news organisations, have raised expectations. Marcus wants the results data to be available through a modern, scalable interface that can serve millions of simultaneous users and support new features, and he wants the development team to be able to build and test them against realistic data well in advance.

The head of political research, Dr Samuel Okoye, runs the projection model. He needs clean, structured results data in real time, along with reference data on notional results, demographics and past voting patterns. His model has been developed over many elections, but it runs on a single analyst's laptop, and the data it consumes is pasted in from the results system. He wants it integrated properly, with automated data feeds, logging and the ability to run on robust infrastructure. He is cautious about changing the model itself close to an election.

The editorial policy adviser, Jennifer Walsh, reminds everyone of the broadcaster's obligations. Election coverage is subject to strict rules on

impartiality. There are rules about what may be published while polls are open, including a prohibition on publishing exit poll results before polls close. Projections must be presented with appropriate caveats and must not be framed in ways that could be perceived as favouring any party. Errors must be corrected transparently. The broadcaster also provides results data to other organisations under agreements, and any failures affect them too.

The security team, led by Ian Clarke, warns that election systems are attractive targets for disruption and manipulation, including by hostile state actors. A cyber attack that took down the results service or, worse, injected false results would be a major incident with political consequences. The collection app, used by hundreds of stringers on personal devices, is a potential attack surface. Phone-based reporting could be subject to impersonation. The team wants strong authentication, monitoring for anomalies and the ability to operate in a degraded mode if systems are compromised.

Your team is six engineers, supported by developers in the digital and graphics teams. You have nine months at most, possibly less if the election is called early. The budget allows a significant rebuild but not unlimited development.

You see a fundamental choice. You could rebuild the results data service as a modern, cloud-based, event-driven system with an interface for all consumers, which would meet Marcus's ambitions and address scalability, but which would be new and untested, a risk Deborah is loath to accept. Or you could harden the existing system, fixing the known failures, adding capacity and improving verification, while building a modern interface layer on top for digital consumers. Or you could run both in parallel with the old system as a fallback, which doubles operational complexity on the night.

There is a history of hard lessons. At one past election, a constituency result was declared, reported and displayed, and then declared again after a recount with a different winner; the system had no clean way to supersede a verified result, and the correction took twenty minutes to reach every output. At another, an early projection was widely criticised for overstating one party's likely majority, and the broadcaster faced complaints that it had shaped the national conversation before enough results were in. Deborah wants the new system to treat recounts,

corrections and projection uncertainty as first-class features rather than afterthoughts.

Whatever the architecture, the verification process is the heart of accuracy. Results arrive through multiple channels, sometimes with conflicts: a reporter's phoned result may differ from the returning officer's published figure because of a mishearing or a recount. The system must detect inconsistencies, such as vote totals that do not match turnout, shares that differ implausibly from expectations or results for the wrong constituency, and route them for human review without creating a bottleneck. It must also support corrections cleanly, propagating them to every output.

THE DECISION IN FRONT OF YOU

Do you rebuild the results service on a modern event-driven platform, harden the existing system with a new digital interface layer, or run both in parallel, and how do you guarantee verification accuracy, resilience under peak load and security against manipulation within an uncertain election timetable?

Data Sources

#	Source and link	How it is used in this project
1	Electoral Commission Election Results Data https://www.electoralcommission.org.uk/research-reports-and-data/our-reports-and-data-past-elections-and-referendums	Official UK election results and turnout data for reference data and validation rules.
2	Democracy Club Candidates and Elections Data https://democracyclub.org.uk	Open data on candidates, elections and polling stations for reference data and API design.
3	House of Commons Library Election Results https://commonslibrary.parliament.uk/research-briefings/cbp-10009/	Detailed historical results and notional results after boundary changes.
4	MIT Election Data and Science Lab https://electionlab.mit.edu/data	Large historical US election results datasets for load testing and validation logic development.
5	ONS Westminster Parliamentary Constituency Boundaries https://geoportal.statistics.gov.uk	Constituency boundaries for maps and geographic lookups.
6	Ofcom Broadcasting Code (Section Six: Elections) https://www.ofcom.org.uk/tv-radio-and-on-demand/broadcast-codes/broadcast-code	Rules on election coverage including exit poll publication.

#	Source and link	How it is used in this project
7	NCSC Guidance on Election Security https://www.ncsc.gov.uk/collection/election-guidance	Security guidance for organisations involved in elections.
8	Representation of the People Act 1983 https://www.legislation.gov.uk/ukpga/1983/2/contents	Legal framework including restrictions on publishing exit polls before polls close.

Deliverables (17)

- D1 Last election incident review.** An analysis of the app crash, verification backlog and transposition error.
- D2 Architecture options appraisal.** A comparison of rebuild, harden-and-extend and parallel approaches with risks.
- D3 Reference data pipeline.** Constituencies, candidates, parties, boundaries and notional results with validation.
- D4 Results event model.** A canonical model of results, sources, versions, verification status and corrections.
- D5 Multi-channel collection design.** Resilient collection from reporters, stringers and official sources with authentication.
- D6 Verification rules engine.** Automated consistency checks routing anomalies to human reviewers.
- D7 Verification desk workflow.** A review process and interface designed for peak throughput.
- D8 Correction propagation.** Clean handling of corrections across every output with audit trail.
- D9 Results distribution interface.** A scalable interface for graphics, digital, radio and partners.
- D10 Peak load engineering.** Capacity for declaration rates and audience traffic with load tests.
- D11 Degraded mode operations.** Fallback procedures if systems or channels fail.
- D12 Security controls.** Authentication, anomaly detection and threat monitoring for manipulation attempts.
- D13 Projection model integration.** Automated data feeds and robust infrastructure for the projection model.
- D14 Embargo and exit poll controls.** Technical enforcement of publication restrictions.
- D15 Rehearsal programme.** Full-scale rehearsals simulating election night with injected faults.
- D16 Early election contingency plan.** Minimum viable readiness if the election is called early.

- D17** **Post-election evaluation framework.** Metrics and review process for accuracy, speed and resilience.

CASE STUDY 85 • SCIENTIFIC RESEARCH (ASTRONOMY)

Astronomical Survey Data Pipeline and Archive Modernisation

Industry	Scientific Research (Astronomy)
Organisation	The Northern Sky Observatory Consortium, a university-led partnership operating a wide-field survey telescope
Your role	Lead Data Engineer, Survey Data Management
Core themes	Scientific data pipelines at scale, real-time transient alerts, archive interoperability, reproducibility and open science

The Scenario

The Northern Sky Observatory Consortium, a partnership of seven universities and a national research institute, operates a wide-field survey telescope on a mountain site overseas. Every clear night, the telescope's large camera images a swathe of the sky, returning to each field every few nights, building up a map of hundreds of millions of stars and galaxies and detecting anything that changes: supernovae, variable stars, asteroids, flaring black holes and the rare, fleeting counterparts of gravitational wave events. Its data supports hundreds of researchers and has been cited in thousands of papers. The consortium has just secured funding for a major camera upgrade that will increase the data rate fivefold, and the funding agency has attached conditions: the consortium must modernise its data management, deliver transient alerts to the global community within minutes, and make its archive openly accessible with standard interfaces within a defined period. You are the lead data engineer for survey data management.

The current pipeline was written over fifteen years by astronomers and research software engineers, largely in scripts and legacy code maintained by a handful of people. Each night, raw images are transferred from the telescope site over a limited network link to a data centre at one of the universities. Processing includes calibration, removal of instrument signatures, astrometric and photometric calibration against reference catalogues, source detection and measurement, and difference imaging, which subtracts a reference image from each new image to reveal changes. Detected changes are classified, and interesting candidates are posted to a web page that astronomers check each morning. The processed

catalogues and images are loaded into an archive database that researchers query through a custom web interface.

The system is creaking. Processing a night's data currently takes most of the following day, meaning that transient candidates are reported twelve to eighteen hours after observation, far too slow for many science cases, such as catching supernovae in their first hours or following up gravitational wave events. The archive database is struggling with its current size, and queries that span the whole survey take hours or time out. Data releases, when processed catalogues are frozen and published for the community, have been delayed repeatedly because reprocessing the full survey with improved calibrations takes months on the available computing. Provenance is incomplete: it is sometimes impossible to determine which version of the pipeline software and calibration files produced a given catalogue entry.

The principal investigator, Professor Eleanor Marsh, is responsible to the funding agency and the consortium's board. She wants the upgrade to succeed scientifically and to meet the agency's conditions. She knows that the agency is watching other surveys that have invested heavily in modern data systems and that funding renewal will depend on the consortium's data being usable and used.

The survey scientist, Dr Rahul Mehta, leads the transient science programme. He wants alerts delivered within minutes of observation, in the standard formats and through the community alert brokers that other surveys use, so that astronomers worldwide can filter them and trigger follow-up observations. That requires near-real-time processing at the telescope site or in the cloud, reliable data transfer and a streaming alert infrastructure.

The archive scientist, Dr Claire Donovan, is responsible for data releases and the archive. She wants the archive to comply with international virtual observatory standards, so that researchers can query it with standard tools alongside other archives, cross-matching the consortium's catalogues with those from space missions and other surveys. She also wants a sustainable approach to reprocessing data releases, which requires scalable computing and well-managed calibration and software versioning.

The research software engineering lead, Dr Tom Hughes, maintains the current pipeline and is concerned about continuity. The pipeline contains decades of accumulated astronomical expertise encoded in its algorithms,

and rewriting it risks introducing subtle errors in calibration that would take years to discover. Some consortium astronomers have published results that depend on the pipeline's specific behaviour, and changes could make new results inconsistent with old ones. Tom favours incremental modernisation: containerising existing code, improving orchestration and provenance, and replacing components one at a time with thorough validation.

Computing is a constraint and an opportunity. The university data centre has limited capacity and is shared with other research groups. National research computing facilities offer significant capacity but through allocations awarded competitively. Commercial cloud offers elastic capacity, and several cloud providers have research credit programmes, but long-term storage of petabytes of data and egress to researchers could be expensive, and the agency's funding rules restrict ongoing commercial commitments. The telescope site's network link will be upgraded with the camera but will still limit how much raw data can be transferred in real time, which may require processing at the site.

The consortium's governance adds complexity. Each university contributes funding and has data rights: consortium members have proprietary access to data for a period before public release. The open access condition requires the consortium to shorten or remove proprietary periods, which some members resist because their researchers invested years in the survey and want first opportunity to publish. Alerts, by community convention, are public immediately.

Your team is four data engineers, working with Tom's three research software engineers. The upgraded camera will be installed in eighteen months. Alerts must be flowing within six months of first light with the new camera, and the first open data release under the new conditions is due within two years.

The people who make the survey work are stretched thin. Much of the pipeline's operation depends on two research software engineers who are employed on short-term grant contracts, and one of them has accepted a permanent post in industry starting in six months. Observing staff at the telescope site handle nightly operations, but data problems are escalated by email to whoever happens to be awake in Europe. Early-career astronomers, who do much of the day-to-day checking of transient candidates, are rewarded for publications rather than for maintaining

infrastructure. Eleanor is acutely aware that the funding agency's conditions imply a level of operational maturity, including monitoring, on-call support and service levels for alerts, that the consortium's academic structure has never been designed to provide. Building the systems is only half the challenge; sustaining them requires a different kind of organisation.

You see the work in three strands: a real-time alert pipeline, which is new and time-critical; a scalable batch processing and reprocessing system with full provenance, which modernises the core pipeline; and an interoperable, scalable archive with open access. Each depends on shared foundations such as data models, calibration management, software versioning and computing infrastructure. You also see a tension between building new systems, which is exciting and attracts funding, and preserving the scientific integrity of existing algorithms, which is essential.

THE DECISION IN FRONT OF YOU

Do you rewrite the processing pipeline on modern infrastructure, modernise it incrementally while building a new alert system alongside, or adopt community software from other surveys, and how do you choose between site, national facility and commercial cloud computing and resolve consortium data rights with the funder's open access conditions?

Data Sources

#	Source and link	How it is used in this project
1	MAST Mikulski Archive for Space Telescopes https://archive.stsci.edu	Major astronomical archive providing interoperable access to mission data for cross-matching and archive design reference.
2	Gaia Archive (ESA) https://gea.esac.esa.int/archive/	Astrometric and photometric reference catalogue used for calibration and cross-matching.
3	IVOA Standards (Virtual Observatory) https://www.ivoa.net/documents/	Standards such as TAP, ADQL, SIA and VOEvent for interoperable archives and alerts.
4	Rubin Observatory LSST Science Pipelines https://pipelines.lsst.io	Community-developed survey processing software considered as a potential basis for modernisation.
5	ZTF Public Data and Alert Stream https://www.ztf.caltech.edu/ztf-public-releases.html	Real survey alert stream and data releases for alert format and broker integration prototyping.
6	Apache Avro Specification https://avro.apache.org/docs/	Serialisation format used by community alert streams.

#	Source and link	How it is used in this project
7	PLAsTiCC Astronomical Classification Challenge https://www.kaggle.com/c/PLAsTiCC-2018	Simulated transient light curves for developing and testing classification of alerts.
8	Astropy Project https://www.astropy.org	Core Python astronomy library for data handling, coordinates and standards support.

Deliverables (17)

- D1** **Current pipeline assessment.** Code, performance, provenance gaps and key-person dependencies.
- D2** **Modernisation strategy paper.** A comparison of rewrite, incremental modernisation and community software adoption.
- D3** **Data model and provenance standard.** Consistent data products with pipeline version, calibration and processing history.
- D4** **Containerised pipeline components.** Packaging of existing algorithms for reproducible execution.
- D5** **Workflow orchestration.** Scalable scheduling of nightly processing and reprocessing campaigns.
- D6** **Calibration management.** Versioned calibration files and reference catalogues with traceability.
- D7** **Validation framework.** Regression tests comparing new and existing pipeline outputs on reference fields.
- D8** **Real-time alert pipeline.** Processing producing alerts within minutes of observation.
- D9** **Alert distribution.** Standard-format alert streams integrated with community brokers.
- D10** **Site processing design.** On-site computing for real-time processing given network limits.
- D11** **Computing strategy.** Allocation across site, university, national facilities and commercial cloud with costs.
- D12** **Scalable archive.** A redesigned archive supporting whole-survey queries and growth.
- D13** **Virtual observatory interfaces.** Standard query and image access interfaces for interoperability.
- D14** **Data release process.** A repeatable reprocessing and release workflow with documentation.
- D15** **Data rights and access policy.** A consortium agreement reconciling proprietary periods with open access conditions.
- D16** **Funding agency compliance plan.** Milestones demonstrating alert, archive and open access commitments.

- D17 Knowledge transfer plan.** Documentation and training to reduce dependence on key individuals.